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Comparative Neurobiology in Chiroptera

Macromorphology, Brain Structures, Tables and Atlases

Volume 1

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Table 1. List of abbreviations used for brain structures and key to tables containing data on these structures.
Size is given in mm^3 if not indicated otherwise.
N = number of species.

			Table numbers			
		N	Size	Index	Aver. Index	%
LIN	linear brain measures (mm)	303	2	3	4	
BrW	brain weight (mg)	337	5,6	6	7	6,7
NET	net brain volume	272	8	6	7	8,9
VENT	ventricles	272	8	-	-	8,9
REST	rests	272	8	-	-	8,9
OBL	medulla oblongata	272	10	11	12	
OBL	in species with OBL subdivision	150	16	17	18	
TR	complexus sensorius trigeminalis	150	13	14	15	
TSO	nucleus tractus solitarii	150	13	14	15	
FUN	funicular nuclei	150	13	14	15	
FGR	nucleus fascicularis gracilis	150	13	14	15	
FCM	nucleus fascicularis cuneatus medialis	150	13	14	15	
FCE	nucleus fascicularis cuneatus externus	150	13	14	15	
REL	nucleus reticularis lateralis	150	16	17	18	
INO	complexus olivaris inferior	150	16	17	18	
PRP	nucleus prepositus hypoglossi	150	16	17	18	
MOT	motor nuclei	150	16	17	18	
X	nucleus dorsalis motorius nervi vagi	150	16	17	18	
XII	nucleus nervi hypoglossi	150	16	17	18	
VC	vestibular complex	150	19	20	21	
VM	nucleus vestibularis medialis	150	19	20	21	
VI	nucleus vestibularis inferior	150	19	20	21	
VL	nucleus vestibularis lateralis	150	19	20	21	
VS	nucleus vestibularis superior	150	19	20	21	
AUD	auditory (acoustic) nuclei	150	22	23	24	
CON	cochlear nuclei	150	22	23	24	
DCO	nucleus cochlearis dorsalis	150	22	23	24	
VCO	nucleus cochlearis ventralis	150	22	23	24	
OLS	complexus olivaris superior	150	22	23	24	

			Table numbers		
		N	Size	Index	Aver. Index
MES		272	10	11	12
MES	in species with MES subdivision	150	25	26	27
MES	in species with NLL measurements	19	30	31	31
MTG	tegmentum mesencephali	150	25	26	27
NLL	nuclei lemnisci lateralis	19	30	31	31
MTC	tectum mesencephali	150	25	26	27
INC	colliculi inferiores	150	25	26	27
SUC	colliculi superiores	150	25	26	27
CER	cerebellum	272	10	11	12
CER	in species with CER nuclei measurements	18	28	29	29
TCN	total cerebellar nuclei	18	28	29	29
MCN	nucleus medialis cerebelli	18	28	29	29
ICN	nucleus interpositus cerebelli	18	28	29	29
LCN	nucleus lateralis cerebelli	18	28	29	29
DIE	diencephalon	272	10	11	12
DIE	in species with DIE subdivision	19	30	31	31
CGM	corpus geniculatum mediale	19	30	31	31
CGL	corpus geniculatum laterale	19	30	31	31
GLD	dorsal part of CGL	19	30	31	31
GLV	ventral part of CGL	19	30	31	31
TEL	telencephalon	272	10	11	12
BOL	olfactory bulbs	272	32	33	34
PAL	paleocortex	272	32	33	34
STR	striatum	272	32	33	34
SEP	septum	272	32	33	34
AMY	amygdala	272	32	33	34
HIP	hippocampus	272	32	33	34
SCH	schizocortex	272	32	33	34
NEO	neocortex	272	32	33	34
AOB	accessory olfactory bulb	63	35	35	35
Surface area measurements		N	mm^2	%	Index
Schneider (1957b)		14	36	37	38
Lütgemeier (1962)		6	39	40	41
" "	Semicortex (SEM)	6	42	43	44
" "	"Bicortex" (BIC)	6	45	46	47
Brodmann (1913) (+ 14 non-Chiroptera species)		1	48	49	50
			mm^3		
Pirlot's **volume** data and resulting indices		6	51		52
Summarized size characteristics of all structures					53

Table 2. Linear measures of brains in mm

For proportion indices in species see Table 3
For average proportion indices in supraspecific taxonomic units see Table 4

	n	BL	HL	HW	HH	n	CCL
PTEROPODIDAE							
PTEROPODINAE							
Eidolon helvum	13	27.5	20.6	20.3	15.0	3	5.43
Rousettus aegyptiacus	17	23.3	15.8	16.0	12.0	2	4.00
R. amplexicaud. brachyotis §)	6	21.0	13.2	13.4	10.5	3	3.52
R. a. stresem. + R.leschen. §)	23	23.0	14.4	14.5	11.2	3	3.88
Lissonycteris angolensis	8	21.1	14.7	14.9	11.2	-	-.-
Myonycteris torquata	4	20.2	12.9	12.9	9.9	3	2.80
Pteropus alecto	4	38.8	26.2	22.8	18.0	2	8.88
Pteropus conspicillatus	1	41.3	28.2	24.1	19.4	1	8.88
Pteropus hypomelanus	1	35.5	23.3	20.7	16.3	1	7.98
Pteropus lylei	7	36.9	25.8	21.7	17.7	4	7.58
Pteropus macrotis	1	36.2	25.7	22.6	16.7	-	-.--
Pteropus mahaganus	2	32.8	23.6	21.0	16.8	2	7.17
Pteropus neohibernicus	2	46.3	31.8	24.5	19.3	2	10.63
Pteropus poliocephalus	2	39.8	25.7	23.9	18.0	2	8.21
Pteropus samoensis	1	35.8	25.6	20.4	17.3	1	7.08
Pteropus scapulatus	3	33.4	23.3	22.2	16.2	2	6.46
Pteropus temmincki	2	33.6	23.1	20.3	16.3	2	5.62
Pteropus tonganus	9	36.7	25.2	21.2	17.3	3	8.58
Pteropus vampyrus	5	42.8	28.7	24.8	19.0	2	9.95
Dobsonia inermis	7	27.0	17.1	16.2	13.9	3	5.09
D. moluccensis magna §)	4	35.2	22.7	20.5	16.8	2	7.02
D. moluccensis anderseni §)	7	30.1	19.4	18.5	15.5	3	5.56
Dobsonia praedatrix	7	28.7	18.2	16.8	13.8	3	5.40
Hypsignathus monstrosus	1	30.2	18.3	19.9	15.7	1	4.12
Epomops franqueti	10	25.9	15.6	16.7	11.6	3	3.93
Epomophorus labiatus	21	21.4	14.1	14.4	10.4	3	3.21
Epomophorus wahlbergi	10	20.7	14.4	15.7	10.7	-	-.--
Micropteropus pusillus	8	15.2	10.9	11.7	8.9	3	2.87
Scotonycteris zenkeri	3	16.1	10.1	11.2	8.7	2	2.84
Casinycteris argynnis	12	16.0	10.8	12.1	9.5	4	2.60
Cynopterus brachyotis	21	18.0	11.6	12.1	9.8	3	3.08
Cynopterus horsfieldi	14	21.0	13.3	13.2	10.7	3	3.14
Cynopterus sphinx sphinx	16	19.5	12.2	13.1	10.1	3	3.12
Cynopterus sph. angulatus	2	17.5	11.6	12.2	9.3	-	-.--
Megaerops ecaudatus	10	16.5	10.3	11.3	9.1	2	1.97
Megaerops niphanae	3	18.1	11.8	11.6	9.7	-	-.--
Chironax melanocephalus	10	15.0	9.4	10.2	8.3	3	2.26

§) See addendum at end of Voume. 3, pp. 1592-1595

	n	BL	HL	HW	HH	n	CCL
PTEROPODIDAE (cont.)							
PTEROPODINAE (cont.)							
Sphaerias blanfordi	11	17.0	10.8	11.8	9.0	4	2.34
Balionycteris maculata	3	13.7	8.8	9.7	8.0	2	2.16
NYCTIMENINAE							
Nyctimene albiventer	2	17.2	10.9	11.5	9.2	2	2.59
Nyctimene robinsoni	8	20.4	12.8	12.6	10.5	2	3.20
Nyctimene vizcaccia	6	18.7	11.7	11.8	9.9	3	3.12
Paranyctimene raptor	7	16.5	10.4	10.9	8.9	3	2.47
MACROGLOSSINAE							
Eonycteris spelaea	11	20.2	12.8	14.0	10.2	3	3.57
Megaloglossus woermanni	10	16.6	10.4	11.4	8.0	3	3.01
Macroglossus minimus	17	15.0	9.7	10.5	7.7	4	2.77
Macroglossus sobrinus	21	15.8	10.2	11.3	8.1	3	2.82
Syconycteris a. australis §)	1	15.2	9.7	10.6	8.0	1	2.64
S. a. papuana + finschi §)	16	15.4	10.2	10.7	8.0	3	2.82
Melonycteris melanops	9	20.4	14.3	13.2	9.9	3	4.73
Nesonycteris woodfordi	7	18.8	13.0	12.5	9.3	3	3.71
Notopteris macdonaldi	10	21.3	14.2	13.2	10.6	3	4.82
RHINOPOMATIDAE							
Rhinopoma hardwickei	11	12.1	6.8	8.0	5.3	3	0.78
Rhinopoma microphyllum	10	13.9	7.6	8.9	5.8	3	0.72
EMBALLONURIDAE							
EMBALLONURINAE							
Emballonura monticola	15	9.5	5.5	6.6	5.0	3	0.76
Emballonura raffrayana	6	10.0	5.9	6.8	5.1	3	0.91
Emballonura semicaudata	14	9.8	5.7	6.9	5.2	3	0.82
Coleura afra	9	10.1	6.4	7.7	5.8	3	1.13
Rhynchonycteris naso	11	8.3	4.6	5.8	4.9	3	0.65
Saccopteryx bilineata	15	10.9	6.3	7.6	5.8	4	1.05
Saccopteryx canescens	1	8.8	4.9	6.2	-.-	1	0.58
Saccopteryx leptura	7	9.6	5.4	6.8	5.0	3	0.68
Cormura brevirostris	3	10.6	6.1	7.6	5.5	3	0.77
Peropteryx macrotis	17	9.5	5.4	6.6	5.0	3	0.60
Peropteryx trinitatis	6	8.7	5.3	6.3	4.8	3	0.75
Taphozous australis	3	14.5	7.3	9.9	6.9	2	1.00
Taphozous hildegardeae	7	12.8	7.6	9.7	7.4	3	0.83
Taphozous longimanus	2	13.6	7.3	9.1	6.8	-	-.--
Taphozous mauritianus	2	13.1	7.5	9.9	7.4	2	0.85
Taphozous melanopogon	10	14.4	7.7	9.7	7.2	3	1.09
Taphozous theobaldi	8	16.1	8.1	10.6	7.4	3	1.29
Taphozous saccolaimus	13	16.0	8.5	10.8	7.6	3	0.92
Taphozous saccolaimus ssp.	10	16.6	8.9	11.4	7.2	-	-.--
DICLIDURINAE							
Cyttarops alecto	1	9.2	5.5	6.2	5.3	1	0.72

> Table 2 <

	n	BL	HL	HW	HH	n	CCL
CRASEONYCTERIDAE							
Craseonycteris thonglongyai	5	7.2	4.4	5.5	3.9	3	0.67
NYCTERIDAE							
Nycteris arge	2	12.0	7.6	8.3	6.7	2	2.20
Nycteris grandis	10	15.7	9.7	10.5	8.1	4	2.67
Nycteris hispida	1	10.3	6.9	7.8	6.2	1	1.95
Nycteris javanica	1	13.7	8.4	9.0	-.-	1	2.16
Nycteris macrotis	3	11.7	8.0	8.7	6.8	3	2.48
Nycteris nana	4	9.7	6.4	6.9	5.7	3	1.35
Nycteris thebaica	3	11.0	7.5	8.1	6.4	3	2.15
Nycteris tragata	2	13.3	8.3	8.8	7.0	2	2.10
MEGADERMATIDAE							
Megaderma lyra	16	17.9	9.8	12.1	9.3	4	2.43
Megaderma spasma	15	16.2	9.0	10.3	8.1	3	2.40
Cardioderma cor	11	14.3	9.4	10.6	8.1	3	1.71
Lavia frons	3	14.4	8.7	10.3	8.0	2	1.43
RHINOLOPHIDAE							
Rhinolophus acuminatus	10	12.5	7.5	8.4	5.7	-	-.--
Rhinolophus affinis	18	13.1	7.4	8.7	6.2	-	-.--
Rhinolophus coelophyllus	8	11.7	6.7	7.9	5.3	-	-.--
Rhinolophus cornutus	4	10.9	6.2	6.9	5.0	2	1.07
Rhinolophus eloquens	2	12.2	7.8	9.0	6.2	2	1.58
Rhinolophus hildebrandti	1	14.9	10.3	10.2	7.4	-	-.--
Rhinolophus landeri	12	10.7	6.8	8.0	5.7	3	1.30
Rhinolophus lepidus	6	10.3	6.2	7.0	4.9	3	1.24
Rhinolophus luctus	3	17.0	9.6	10.6	6.9	3	1.67
Rhinolophus macrotis	7	10.8	6.4	7.0	5.1	2	1.13
Rhinolophus malayanus	8	10.8	6.2	7.1	4.8	2	1.20
Rhinolophus megaphyllus	7	11.8	6.4	7.9	5.6	2	1.03
Rhinolophus megaphyllus ssp.	3	11.1	6.6	7.6	5.3	3	1.24
Rhinolophus paradoxolophus	1	12.4	7.5	8.1	5.1	1	1.26
Rhinolophus pearsoni	7	13.1	7.9	8.8	5.7	1	1.21
Rhinolophus pusillus	8	9.7	5.7	6.6	4.8	3	1.28
Rhinolophus refulgens	1	10.0	6.0	7.1	4.9	-	-.--
Rhinolophus robinsoni	2	12.0	7.0	8.1	5.4	-	-.--
Rhinolophus stheno ssp.	7	11.2	6.8	7.6	5.4	-	-.--
Rhinolophus stheno stheno	6	11.7	7.3	8.2	5.6	-	-.--
Rhinolophus thomasi	8	11.3	6.6	7.6	5.2	-	-.--
Rhinolophus trifoliatus	9	13.4	7.1	8.9	5.9	3	1.11
Rhinolophus yunanensis	2	14.4	8.8	9.8	6.4	2	1.51
HIPPOSIDERIDAE							
Hipposideros armiger	10	18.5	10.2	11.0	8.4	3	2.24
Hipposideros ater	5	10.5	5.6	7.1	5.2	2	1.43
Hipposideros a. aruensis	2	10.4	5.9	6.8	5.1	2	1.33
Hipposideros ater saevus	4	10.4	5.7	6.6	5.1	3	1.61
Hipposideros ater ssp.	1	11.2	6.4	7.5	5.6	-	-.--

	n	BL	HL	HW	HH	n	CCL
HIPPOSIDERIDAE (cont.)							
Hipposideros bicolor gentilis	6	11.3	6.4	7.4	5.5	-	-.--
Hipposideros bicolor atrox	3	11.5	6.6	7.6	5.7	-	-.--
Hipposideros caffer	14	11.0	6.8	8.0	5.9	3	1.52
Hipposideros caffer ssp.	5	12.3	7.0	8.4	5.9	-	-.--
Hipposideros c. calcaratus §)	7	14.2	7.7	8.6	6.5	3	2.49
Hipposideros c. cupidus §)	6	12.8	7.3	8.0	6.2	3	2.28
Hipposideros cervinus	17	11.1	6.2	7.1	5.3	3	1.76
Hipposideros commersoni	10	15.4	9.9	11.1	7.9	3	2.17
Hipposideros diadema	21	17.7	9.5	10.9	7.6	4	1.96
Hipposideros fulvus	6	11.3	6.5	7.5	5.5	2	1.50
Hipposideros galeritus	3	11.9	6.8	7.9	5.8	-	-.--
Hipposideros halophyllus	2	9.8	5.6	6.1	4.5	2	1.46
Hipposideros lankadiva	9	18.2	9.8	11.0	7.5	3	2.37
Hipposideros larvatus	20	14.3	7.9	9.0	6.5	3	1.72
Hipposideros lekaguli	10	16.1	8.6	10.1	7.5	2	1.85
Hipp. m. maggietaylorae §)	1	16.5	9.3	10.0	7.6	1	2.92
Hipp. m. erroris §)	4	14.4	7.8	8.7	7.1	3	2.48
Hipposideros ridleyi	10	11.6	5.9	7.8	5.8	3	1.33
Hipposideros ruber	1	10.0	6.9	7.9	5.6	-	-.--
Hipposideros speoris	6	12.5	7.1	8.1	5.8	2	1.57
Hipposideros turpis	4	16.3	9.0	9.9	7.6	3	1.93
Aselliscus stoliczkanus	3	9.1	5.2	5.9	4.7	3	1.11
Aselliscus tricuspidatus	10	9.5	5.5	5.9	4.7	3	1.38
Triaenops persicus	10	10.9	6.2	7.0	5.9	3	1.89
NOCTILIONIDAE							
Noctilio albiventris	10	12.7	8.6	11.1	8.4	3	2.67
Noctilio leporinus	9	15.9	10.8	13.6	9.9	3	4.28
MORMOOPIDAE							
Pteronotus gymnonotus	6	11.2	6.1	8.4	7.1	3	2.03
Pteronotus personatus	17	9.6	5.5	7.3	6.2	3	1.79
Pteronotus parnelli	15	13.3	7.6	10.4	8.0	3	2.11
Mormoops megalophylla	9	9.8	5.2	8.6	7.7	3	1.34
PHYLLOSTOMIDAE							
PHYLLOSTOMINAE							
Micronycteris megalotis	9	11.1	6.8	7.2	6.7	3	2.03
Micronycteris minuta	1	11.3	6.7	7.2	6.5	1	2.00
Micronycteris schmidtorum	10	11.5	6.9	7.6	7.0	3	2.34
Micronycteris brachyotis	1	12.9	7.2	8.3	7.2	1	2.68
Macrophyllum macrophyllum	10	11.2	7.1	7.8	6.7	3	2.54
Tonatia bidens	10	16.6	10.1	10.2	9.4	3	3.70
Tonatia schulzi	1	13.6	8.2	9.2	8.3	1	3.27
Tonatia sylvicola	1	15.0	10.2	10.4	9.4	1	3.52
Mimon crenulatum	2	12.5	7.4	8.1	6.7	2	2.23

	n	BL	HL	HW	HH	n	CCL
PHYLLOSTOMIDAE (cont.)							
PHYLLOSTOMINAE (cont.)							
Phyllostomus discolor	15	18.2	11.8	12.0	10.0	3	4.26
Phyllostomus elongatus	9	17.2	10.6	11.0	9.3	3	3.94
Phyllostomus hastatus	16	21.0	12.9	13.5	11.1	3	4.69
Phylloderma stenops	2	19.9	12.1	12.5	11.6	2	4.71
Trachops cirrhosus	2	17.8	11.2	11.0	10.4	2	3.84
Vampyrum spectrum	2	25.9	15.6	15.1	13.2	2	5.31
GLOSSOPHAGINAE							
Glossophaga longirostris	2	13.5	8.2	8.6	7.2	1	2.41
Glossophaga soricina	19	12.8	7.8	8.4	7.0	3	2.34
Monophyllus plethodon	2	14.4	8.6	9.0	6.6	2	3.40
Leptonycteris curasoae	1	16.0	9.3	9.5	7.4	1	3.56
Leptonycteris nivalis	4	15.6	9.6	9.6	7.6	3	3.91
Lonchophylla mordax	1	14.3	8.4	8.7	6.7	1	2.79
Lonchophylla thomasi	1	11.9	7.4	8.2	6.7	1	2.43
Lionycteris spurrelli	4	12.3	7.5	8.0	6.8	3	2.02
Anoura caudifer	8	13.5	8.2	8.5	6.7	3	2.87
Anoura geoffroyi	12	15.0	9.5	9.7	7.7	3	3.15
Choeroniscus minor	1	12.8	7.3	8.4	6.9	1	2.58
CAROLLIINAE							
Carollia castanea	9	13.2	8.2	8.9	7.7	5	2.31
Carollia perspicillata	25	14.4	8.8	9.5	8.1	3	2.74
Rhinophylla pumilio	10	12.2	7.6	8.0	7.2	3	2.57
STURNIRINAE							
Sturnira lilium	29	14.6	9.3	10.0	8.5	3	3.00
Sturnira ludovici	4	15.0	9.5	10.4	8.6	2	3.11
Sturnira tildae	7	15.3	9.8	10.6	8.7	3	3.40
STENODERMINAE							
Uroderma bilobatum	22	15.3	9.4	9.5	8.3	3	2.70
Vampyrops brachycephalus	4	14.3	9.6	9.4	8.6	2	2.85
Vampyrops brachyceph. ssp.	2	14.8	9.9	9.8	8.5	-	-.--
Vampyrops helleri	12	14.3	9.2	9.0	8.0	3	2.97
Vampyrops infuscus	1	19.0	11.8	11.9	10.0	1	4.03
Vampyrops lineatus	5	16.1	10.5	10.3	8.8	3	3.39
Vampyrops vittatus	2	18.7	12.3	12.2	10.3	2	3.81
Vampyrodes caraccioloi	1	17.3	11.0	11.1	9.7	1	3.51
Vampyressa pusilla	9	12.7	7.8	8.1	7.3	2	2.16
Chiroderma salvini	4	16.7	10.3	10.9	9.1	4	3.19
Chiroderma trinitatum	2	14.2	8.8	9.3	8.1	2	2.72
Chiroderma villosum	3	16.1	9.8	10.1	8.7	3	3.16
Chiroderma villosum ssp.	1	17.0	10.0	10.8	9.2	1	2.89
Ectophylla macconnelli	5	11.9	7.7	7.8	6.9	2	2.19
Artibeus cinereus	18	13.6	8.7	8.8	7.6	3	2.40
Artibeus cinereus toltecus	3	13.9	9.1	9.1	8.1	-	-.--
Artibeus concolor	2	15.4	9.3	9.9	8.3	2	2.76

	n	BL	HL	HW	HH	n	CCL
PHYLLOSTOMIDAE (cont.)							
STENODERMINAE (cont.)							
Artibeus glaucus	1	13.7	8.1	8.8	7.8	-	-.--
Artibeus jamaicensis	16	17.7	11.6	11.7	9.8	3	3.58
Artibeus lituratus	23	19.0	12.4	12.6	10.4	4	3.94
Artibeus phaeotis	8	13.5	8.4	8.6	7.5	-	-.--
Enchisthenes harti	8	13.8	8.7	9.2	8.0	4	2.58
Ardops nichollsi	3	14.5	8.8	9.4	8.7	3	2.90
Sphaeronycteris toxophyllum	1	12.9	9.3	8.8	8.7	1	1.94
BRACHYPHYLLINAE							
Brachyphylla cavernarum	6	19.1	12.5	12.6	10.1	5	4.63
DESMODONTIDAE							
Desmodus rotundus	25	16.8	11.9	11.7	10.4	3	3.99
Diphylla ecaudata	13	15.7	10.1	11.1	9.2	4	2.99
NATALIDAE							
Natalus tumidirostris	10	10.0	6.6	8.0	6.0	3	2.23
FURIPTERIDAE							
Furipterus horrens	6	7.7	5.0	5.7	5.1	5	0.97
VESPERTILIONIDAE							
VESPERTILIONINAE							
Myotis adversus	5	10.1	5.5	8.0	5.7	2	0.69
Myotis albescens	3	8.3	4.7	6.1	4.3	3	0.51
Myotis altarium	4	9.8	5.7	7.8	5.6	3	0.55
Myotis annectans	2	10.3	6.0	7.8	5.1	2	0.69
Myotis bocagei	5	9.2	5.8	7.1	5.2	3	0.63
Myotis fortidens	7	8.1	4.8	6.5	4.7	-	-.--
Myotis hasselti	1	9.6	5.9	7.2	5.0	-	-.--
Myotis horsfieldi	5	9.6	5.8	7.1	5.2	-	-.--
Myotis lucifugus	7	9.0	5.0	7.1	4.7	-	-.--
Myotis montivagus	5	9.1	5.4	6.9	4.8	3	0.60
Myotis muricola	18	8.3	4.7	6.3	4.3	-	-.--
Myotis nigricans	12	8.4	5.0	6.3	4.4	3	0.52
Myotis nigricans ssp.	7	8.7	5.0	6.7	4.7	3	0.62
Myotis riparius	4	8.8	5.4	6.8	5.0	-	-.--
Myotis siligorensis	7	7.6	4.4	5.8	4.2	3	0.53
Myotis sp.?	1	8.3	5.4	6.6	5.0	-	-.--
Pipistrellus babu	4	7.8	4.1	5.9	3.9	3	0.44
Pipistrellus ceylonicus	8	9.6	5.1	7.1	4.7	3	0.43
Pipistrellus circumdatus	10	10.0	5.8	7.8	5.4	3	0.54
Pipistrellus coromandra	7	7.8	4.5	5.9	3.9	-	-.--
Pipistrellus crassulus	4	8.6	4.7	6.9	4.1	3	0.63
Pipistrellus imbricatus	15	8.5	4.8	6.4	4.5	2	0.57
Pipistrellus javanicus	3	7.9	4.7	6.2	4.3	3	0.37
Pipistrellus kuhli	1	8.0	5.1	6.9	4.5	-	-.--
Pipistrellus mimus	10	7.6	4.3	5.7	3.7	3	0.35

	n	BL	HL	HW	HH	n	CCL
VESPERTILIONIDAE (cont.)							
VESPERTILIONINAE (cont.)							
Pipistrellus nanus	17	7.3	4.1	5.8	4.0	3	0.46
P. papuanus + angulatus §)	12	8.0	4.4	6.3	4.2	3	0.40
Pipistrellus pulveratus	6	8.8	5.0	6.6	4.4	3	0.52
Pipistrellus subflavus	2	8.3	4.6	6.5	4.4	2	0.35
Scotozous dormeri	10	9.1	4.9	6.8	4.2	3	0.47
Nyctalus noctula	7	11.5	5.9	9.3	6.0	3	0.55
Nyctalus stenopterus	6	10.5	5.9	8.1	5.0	2	0.68
Glischropus tylopus	4	7.9	3.9	6.2	3.7	3	0.49
Eptesicus brasiliensis	12	10.0	5.5	7.3	4.9	3	0.52
Eptesicus br. melanopterus	4	9.9	5.4	7.0	4.7	3	0.53
Eptesicus demissus	1	10.4	5.9	8.2	5.2	-	-.--
Eptesicus flavescens	2	8.4	4.9	7.1	5.0	2	0.48
Eptesicus fuscus	1	11.0	6.7	8.1	4.8	1	0.83
Eptesicus gaumeri	4	9.9	5.3	7.2	4.8	-	-.--
Eptesicus pachyotis	3	9.3	5.0	6.7	4.4	-	-.--
Eptesicus pumilus	4	8.3	4.5	6.2	4.6	2	0.51
Eptesicus serotinus andersoni	5	12.5	7.1	9.2	5.6	3	0.77
Ia io	1	16.2	9.9	11.8	7.6	1	0.76
Tylonycteris pachypus	28	7.7	4.3	6.4	2.8	4	0.59
Tylonycteris robustula	25	8.5	4.7	7.2	3.3	3	0.43
Hesperoptenus blanfordi	10	8.8	4.8	7.1	4.3	3	0.33
Glauconycteris humeralis	3	7.9	4.2	6.8	4.8	3	0.35
Glauconycteris poensis	1	9.0	4.8	7.7	5.6	1	0.40
Chalinolobus gouldi	1	9.8	4.5	7.7	5.7	1	0.42
Chalinolobus morio	5	8.3	4.3	7.2	5.3	2	0.37
Scotorepens sanborni	1	9.4	4.8	6.9	4.4	1	0.63
Rhogeessa parvula	8	8.3	4.5	5.7	4.0	4	0.59
Rhogeessa tumida	2	8.1	4.0	5.8	4.0	-	-.--
Scotomanes ornatus	2	13.4	7.8	9.6	5.9	2	0.70
Scotomanes sp.?	3	11.5	6.8	9.0	6.1	3	0.66
Scotophilus dinganii	13	11.2	7.1	9.3	6.6	3	0.67
Scotophilus heathi	10	12.9	7.1	9.5	6.9	3	0.58
Scotophilus kuhli	22	11.8	6.7	8.7	6.1	3	0.61
Lasiurus borealis	1	8.7	4.0	6.7	5.1	1	0.23
MINIOPTERINAE							
Miniopterus australis	6	9.3	4.8	7.2	5.9	2	1.13
Miniopterus haradai	8	9.4	5.4	7.3	5.9	3	1.14
Miniopterus inflatus	8	9.9	6.1	8.2	6.6	3	1.10
Miniopterus magnater	14	10.8	6.2	8.2	6.4	3	1.20
Miniopterus medius.	10	9.9	5.7	7.6	6.2	3	0.99
Miniopterus pusillus	10	9.5	5.4	7.2	5.7	3	1.35
Miniopterus s. arenarius	12	9.0	5.6	7.5	6.0	3	1.19
Miniopterus s. blepotis	7	10.6	5.6	8.1	6.7	2	0.91
Miniopterus tristis	8	11.2	6.6	8.3	6.6	3	1.31

	n	BL	HL	HW	HH	n	CCL
VESPERTILIONIDAE (cont.)							
MURININAE							
Murina cyclotis cyclotis	2	10.5	6.3	7.4	6.0	2	1.13
Murina c. peninsularis	2	11.2	6.3	7.6	6.3	2	1.19
Murina huttoni rubella	4	11.0	6.3	7.6	5.9	3	1.35
Murina tubinaris	2	9.8	5.7	7.4	6.0	2	1.11
Harpiocephalus harpia	1	13.6	7.7	9.1	7.4	1	1.37
KERIVOULINAE							
Kerivoula papillosa	1	10.8	6.9	8.4	6.5	1	2.16
Kerivoula pellucida	1	9.3	6.5	6.6	6.0	1	2.40
Kerivoula phalaena	1	7.9	4.7	5.6	4.9	1	1.18
Phoniscus atrox	1	8.9	6.9	7.1	5.7	1	2.18
NYCTOPHILINAE							
Nyctophilus timoriensis	2	11.0	6.1	7.9	5.7	2	0.90
MOLOSSIDAE							
Tadarida aegyptiaca	6	12.9	7.0	9.6	6.2	3	1.01
Tadarida condylura	9	10.9	7.3	10.5	6.6	3	0.89
Tadarida mops	10	13.5	7.1	10.4	6.7	3	0.98
Tadarida niveiventer	8	11.1	7.1	10.0	6.2	3	0.70
Tadarida beccarii	6	11.1	5.7	7.9	4.7	2	0.77
Tadarida jobensis	1	12.6	7.0	9.1	6.1	1	0.95
Tadarida plicata plicata	3	12.1	6.8	8.9	5.9	3	0.85
Tadarida pumila	18	9.2	5.8	8.2	5.2	3	0.56
Tadarida pumila ssp.	4	10.9	5.9	8.3	5.3	-	-.--
Otomops martiensseni	14	15.5	8.7	11.5	7.8	3	1.08
Molossops abrasus	4	13.0	7.0	9.9	6.1	3	0.74
Molossops greenhalli	1	11.9	6.2	7.9	5.5	1	0.70
Molossops planirostris	2	9.5	4.8	7.2	4.8	2	0.52
Eumops auripendulus	6	15.3	7.7	10.6	7.1	3	0.78
Eumops glaucinus	3	15.3	8.0	10.9	7.1	2	1.01
Molossus ater	18	14.0	7.8	10.4	6.5	3	0.89
Molossus molossus	24	11.4	6.4	8.8	5.6	3	0.61
Molossus trinitatis	2	13.7	7.3	9.3	6.3	2	0.80
Cheiromeles torquatus	8	18.9	11.1	14.9	8.6	3	1.63

Abbreviations:

BL	brain length	HL	hemisphere length
CCL	corpus callosum length	HW	hemisphere width
HH	hemisphere height		

§) see addendum at end of Volume 3, pp. 1592-1595

n = number of individuals

Table 3. Proportion indices (= linear measures of brains and corpus callosum given in Table 2 related to the cube roots of the standard brain volumes) (ratios comprising CCL are given in %, i.e. x 100) (individual number as in Table 2)

For linear measures in species see Table 2
For average proportion indices in supraspecific taxonomic units see Table 4

	BrV	CR	BL CR	HL CR	HW CR	HH CR	CCL CR	CCL HemL
PTEROPODIDAE								
PTEROPODINAE								
Eidolon helvum	4141	16.05	1.72	1.28	1.26	0.94	33.8	26.4
Rousettus aegyptiacus	2090	12.78	1.82	1.24	1.25	0.94	31.3	25.3
Rousettus aegyptiacus **)	----	--.--	1.98	1.20	1.24	0.90	--.-	--.-
R. amplexicaud. brachyot. §)	1305	10.92	1.92	1.21	1.23	0.96	32.3	26.8
R. a. stresem. + leschen. §)	1645	11.80	1.95	1.22	1.23	0.95	32.9	27.0
Lissonycteris angolensis	1712	11.96	1.77	1.23	1.25	0.93	--.-	--.-
Myonycteris torquata	1117	10.37	1.95	1.24	1.24	0.96	27.0	21.8
Pteropus alecto	6794	18.93	2.05	1.38	1.20	0.95	46.9	34.0
Pteropus conspicillatus	8060	20.05	2.06	1.41	1.20	0.97	44.3	31.5
Pteropus hypomelanus	5118	17.23	2.06	1.35	1.20	0.95	46.3	34.2
Pteropus lylei	5916	18.08	2.04	1.43	1.20	0.98	41.9	29.4
Pteropus macrotis	5914	18.08	2.00	1.42	1.25	0.92	--.-	--.-
Pteropus mahaganus	5097	17.21	1.91	1.37	1.22	0.97	41.6	30.4
Pteropus neohibernicus	8797	20.61	2.24	1.54	1.19	0.94	51.6	33.4
Pteropus poliocephalus	6978	19.11	2.08	1.35	1.25	0.94	42.9	31.9
Pteropus samoensis	5591	17.75	2.02	1.44	1.15	0.97	39.9	27.7
Pteropus scapulatus	5174	17.29	1.93	1.35	1.29	0.94	37.4	27.7
Pteropus temmincki	4720	16.77	2.00	1.37	1.21	0.97	33.5	24.4
Pteropus tonganus	5727	17.88	2.05	1.41	1.19	0.97	48.0	34.1
Pteropus vampyrus	8890	20.69	2.07	1.39	1.20	0.92	48.1	34.6
Dobsonia inermis	2607	13.76	1.97	1.24	1.18	1.01	37.0	29.8
D. moluccensis magna §)	5156	17.27	2.04	1.31	1.19	0.97	40.6	30.9
D. moluccensis anderseni §)	3717	15.48	1.94	1.26	1.19	1.00	35.9	28.6
Dobsonia praedatrix	2889	14.24	2.01	1.27	1.18	0.97	37.9	29.8
Hypsignathus monstrosus	3636	15.38	1.96	1.19	1.29	1.02	26.8	22.5
Epomops franqueti	2125	12.85	2.02	1.21	1.30	0.90	30.6	25.3
Epomophorus labiatus	1512	11.47	1.86	1.23	1.25	0.91	28.0	22.7
Epomophorus wahlbergi	1751	12.04	1.72	1.20	1.30	0.89	--.-	--.-
Epomophorus wahlbergi **)	----	--.--	1.91	1.37	1.38	0.97	--.-	--.-
Micropteropus pusillus	798	9.27	1.64	1.17	1.26	0.96	31.0	26.5
Scotonycteris zenkeri	686	8.82	1.83	1.15	1.27	0.98	32.2	28.0
Casinycteris argynnis	796	9.26	1.73	1.17	1.30	1.02	28.1	24.0
Cynopterus brachyotis	952	9.83	1.83	1.18	1.23	1.00	31.3	26.5
Cynopterus horsfieldi	1325	10.98	1.91	1.21	1.21	0.97	28.6	23.5
Cynopterus sphinx sphinx	1158	10.50	1.86	1.17	1.25	0.96	29.8	25.5
Cynopterus sph. angulatus	891	9.62	1.81	1.20	1.26	0.97	--.-	--.-
Megaerops ecaudatus	770	9.16	1.80	1.12	1.23	0.99	21.5	19.1
Megaerops niphanae	925	9.73	1.86	1.22	1.20	0.99	--.-	--.-
Chironax melanocephalus	590	8.39	1.79	1.12	1.21	0.99	27.0	24.0

	BrV	CR	BL CR	HL CR	HW CR	HH CR	CCL CR	CCL HemL
PTEROPODIDAE (cont.)								
PTEROPODINAE (cont.)								
Sphaerias blanfordi	803	9.29	1.83	1.16	1.27	0.96	25.2	21.8
Balionycteris maculata	494	7.90	1.73	1.12	1.23	1.01	27.3	24.5
NYCTIMENINAE								
Nyctimene albiventer	796	9.26	1.85	1.17	1.24	0.99	27.9	23.8
Nyctimene robinsoni	1189	10.59	1.92	1.21	1.19	0.99	30.2	24.9
Nyctimene vizcaccia	953	9.84	1.90	1.19	1.20	1.00	31.7	26.7
Paranyctimene raptor	706	8.90	1.86	1.17	1.22	1.00	27.7	23.8
MACROGLOSSINAE								
Eonycteris spelaea	1264	10.80	1.87	1.19	1.30	0.94	33.0	27.8
Megaloglossus woermanni	657	8.69	1.91	1.19	1.31	0.92	34.7	29.1
Macroglossus minimus	542	8.15	1.84	1.19	1.29	0.94	34.0	28.5
Macroglossus sobrinus	670	8.74	1.80	1.17	1.29	0.92	32.3	27.6
Syconycteris a. australis §)	550	8.19	1.85	1.18	1.29	0.98	32.2	27.2
S. a. papuana + finschi §)	605	8.45	1.82	1.20	1.26	0.95	33.4	27.7
Melonycteris melanops	1247	10.76	1.90	1.33	1.23	0.92	43.9	33.0
Nesonycteris woodfordi	982	9.94	1.89	1.31	1.25	0.93	37.3	28.5
Notopteris macdonaldi	1394	11.17	1.91	1.27	1.18	0.94	43.2	34.0
RHINOPOMATIDAE								
Rhinopoma hardwickei	266	6.43	1.88	1.06	1.24	0.82	12.1	11.4
Rhinopoma microphyllum	370	7.17	1.93	1.05	1.24	0.81	10.0	9.5
EMBALLONURIDAE								
EMBALLONURINAE								
Emballonura monticola	160	5.43	1.75	1.02	1.22	0.93	13.9	13.7
Emballonura raffrayana	168	5.51	1.81	1.07	1.23	0.92	16.5	15.4
Emballonura semicaudata	177	5.61	1.75	1.01	1.23	0.93	14.7	14.6
Coleura afra	249	6.29	1.61	1.01	1.23	0.92	18.0	17.8
Rhynchonycteris naso	114	4.84	1.70	0.94	1.19	1.00	13.4	14.2
Saccopteryx bilineata	224	6.07	1.80	1.04	1.25	0.95	17.2	16.5
Saccopteryx canescens	126	5.01	1.76	0.98	1.24	-.-	11.6	11.8
Saccopteryx leptura	156	5.38	1.79	1.00	1.26	0.93	12.6	12.6
Cormura brevirostris	212	5.96	1.77	1.02	1.28	0.92	12.9	12.7
Peropteryx macrotis	155	5.37	1.76	1.01	1.23	0.94	11.2	11.1
Peropteryx trinitatis	132	5.09	1.71	1.04	1.23	0.94	14.7	14.1
Taphozous australis	499	7.93	1.83	0.92	1.24	0.87	12.5	13.6
Taphozous hildegardeae	527	8.07	1.59	0.94	1.21	0.91	10.3	11.0
Taphozous longimanus	434	7.57	1.80	0.96	1.20	0.89	--.-	--.-
Taphozous mauritianus	531	8.09	1.61	0.92	1.22	0.91	10.4	11.3
Taphozous melanopogon	517	8.02	1.80	0.95	1.21	0.89	13.5	14.2
Taphozous theobaldi	657	8.69	1.86	0.93	1.22	0.85	14.9	16.0
Taphozous saccolaimus	648	8.65	1.85	0.98	1.25	0.88	10.7	10.9
Taphozous saccolaimus ssp.	649	8.66	1.92	1.02	1.32	0.83	--.-	--.-
DICLIDURINAE								
Cyttarops alecto	169	5.53	1.66	0.99	1.12	0.96	13.0	13.1

> Table 3 <

	BrV	CR	BL / CR	HL / CR	HW / CR	HH / CR	CCL / CR	CCL / HemL
CRASEONYCTERIDAE								
Craseonycteris thonglongyai	84	4.37	1.66	1.01	1.25	0.90	15.4	15.2
NYCTERIDAE								
Nycteris arge	329	6.90	1.73	1.10	1.19	0.96	31.8	28.9
Nycteris grandis	685	8.81	1.79	1.10	1.19	0.92	30.3	27.6
Nycteris hispida	274	6.50	1.59	1.06	1.20	0.95	30.0	28.3
Nycteris javanica	454	7.68	1.78	1.09	1.17	-.--	28.1	25.7
Nycteris macrotis	375	7.21	1.62	1.11	1.20	0.94	34.5	31.0
Nycteris nana	229	6.12	1.58	1.04	1.13	0.93	22.1	21.2
Nycteris thebaica	312	6.78	1.62	1.11	1.19	0.94	31.7	28.6
Nycteris tragata	444	7.62	1.74	1.08	1.15	0.91	27.5	25.5
MEGADERMATIDAE								
Megaderma lyra	881	9.58	1.87	1.02	1.26	0.97	25.3	24.7
Megaderma spasma	622	8.53	1.89	1.06	1.21	0.95	28.1	26.6
Cardioderma cor	647	8.65	1.65	1.09	1.23	0.93	19.8	18.2
Lavia frons	621	8.53	1.69	1.02	1.20	0.93	16.8	16.4
RHINOLOPHIDAE								
Rhinolophus acuminatus	292	6.63	1.89	1.13	1.26	0.86	--.-	--.-
Rhinolophus affinis	344	7.00	1.87	1.06	1.24	0.88	--.-	--.-
Rhinolophus capensis **)	---	-.--	1.85	0.95	1.21	0.85	--.-	--.-
Rhinolophus clivosus **)	---	-.--	1.77	0.88	1.14	0.82	--.-	--.-
Rhinolophus coelophyllus	250	6.30	1.86	1.06	1.25	0.84	--.-	--.-
Rhinolophus cornutus	188	5.72	1.90	1.09	1.21	0.87	18.7	17.2
Rhinolophus eloquens	390	7.30	1.66	1.06	1.23	0.84	21.6	20.3
Rhinolophus hildebrandti	608	8.47	1.76	1.22	1.20	0.87	--.-	--.-
Rhinolophus landeri	267	6.44	1.67	1.06	1.25	0.89	20.1	19.0
Rhinolophus lepidus	179	5.63	1.83	1.10	1.24	0.87	22.0	20.1
Rhinolophus luctus	599	8.41	2.02	1.14	1.26	0.82	19.9	17.5
Rhinolophus macrotis	201	5.86	1.84	1.10	1.20	0.88	19.3	17.6
Rhinolophus malayanus	195	5.80	1.87	1.07	1.23	0.83	20.6	19.4
Rhinolophus megaphyllus	252	6.32	1.87	1.01	1.24	0.88	16.3	16.2
Rhinolophus megaphyllus ssp.	222	6.06	1.83	1.10	1.26	0.87	20.5	18.7
Rhinolophus paradoxolophus	258	6.36	1.95	1.18	1.27	0.80	19.8	16.8
Rhinolophus pearsoni	320	6.84	1.92	1.15	1.28	0.83	17.7	15.4
Rhinolophus pusillus	159	5.41	1.80	1.06	1.22	0.88	23.7	22.3
Rhinolophus refulgens	183	5.68	1.76	1.06	1.25	0.86	--.-	--.-
Rhinolophus robinsoni	267	6.44	1.86	1.09	1.26	0.84	--.-	--.-
Rhinolophus stheno ssp.	235	6.17	1.82	1.10	1.23	0.87	--.-	--.-
Rhinolophus stheno stheno	275	6.50	1.80	1.12	1.26	0.87	--.-	--.-
Rhinolophus thomasi	225	6.08	1.85	1.08	1.25	0.85	--.-	--.-
Rhinolophus trifoliatus	339	6.97	1.92	1.02	1.28	0.85	15.9	15.6
Rhinolophus yunanensis	423	7.51	1.92	1.17	1.31	0.85	20.1	17.3
HIPPOSIDERIDAE								
Hipposideros armiger	772	9.17	2.01	1.11	1.20	0.91	24.4	21.9
Hipposideros ater	206	5.90	1.78	0.96	1.20	0.88	24.1	25.3
Hipposideros ater aruensis	183	5.68	1.82	1.03	1.20	0.90	23.3	22.6

	BrV	CR	BL CR	HL CR	HW CR	HH CR	CCL CR	CCL HemL
HIPPOSIDERIDAE (cont.)								
Hipposideros ater saevus	173	5.57	1.87	1.01	1.18	0.92	28.8	28.4
Hipposideros ater ssp.	237	6.18	1.81	1.03	1.21	0.91	--.-	--.-
Hipposideros bicolor gentilis	225	6.08	1.86	1.06	1.21	0.90	--.-	--.-
Hipposideros bicolor atrox	237	6.19	1.86	1.07	1.23	0.92	--.-	--.-
Hipposideros caffer	295	6.66	1.65	1.01	1.20	0.89	22.9	22.6
Hipposideros caffer ssp.	289	6.61	1.87	1.05	1.27	0.89	--.-	--.-
Hipposideros c. calcaratus §)	388	7.29	1.95	1.06	1.18	0.90	34.2	32.4
Hipposideros calc. cupidus §)	324	6.87	1.86	1.07	1.16	0.90	33.2	31.2
Hipposideros cervinus.	207	5.91	1.87	1.04	1.19	0.90	29.8	28.5
Hipposideros commersoni	720	8.95	1.72	1.10	1.23	0.88	24.2	22.0
Hipposideros diadema	681	8.79	2.01	1.08	1.23	0.87	22.3	20.6
Hipposideros fulvus	228	6.11	1.84	1.06	1.23	0.90	24.6	23.1
Hipposideros galeritus	269	6.45	1.85	1.05	1.23	0.89	--.-	--.-
Hipposideros halophyllus	146	5.27	1.86	1.06	1.15	0.84	27.6	26.0
Hipposideros lankadiva	707	8.90	2.04	1.10	1.24	0.85	26.7	24.2
Hipposideros larvatus	399	7.36	1.94	1.07	1.22	0.88	23.3	21.9
Hipposideros lekaguli	564	8.26	1.94	1.04	1.22	0.90	22.3	21.5
H. magg. maggietaylorae §)	612	8.49	1.94	1.10	1.18	0.90	34.4	31.4
H. maggietaylorae erroris §)	435	7.58	1.90	1.03	1.15	0.93	32.8	31.8
Hipposideros ridleyi	258	6.36	1.82	0.93	1.22	0.91	20.8	22.4
Hipposideros ruber	256	6.35	1.58	1.09	1.24	0.88	--.-	--.-
Hipposideros speoris	274	6.49	1.92	1.10	1.24	0.89	24.1	22.0
Hipposideros turpis	578	8.33	1.95	1.08	1.19	0.91	23.1	21.4
Aselliscus stoliczkanus	145	5.25	1.73	0.99	1.12	0.89	21.1	21.3
Aselliscus tricuspidatus	129	5.06	1.88	1.08	1.16	0.92	27.3	25.2
Triaenops persicus	261	6.39	1.71	0.96	1.09	0.92	29.6	30.8
NOCTILIONIDAE								
Noctilio albiventris	597	8.42	1.51	1.02	1.31	1.00	31.7	31.0
Noctilio leporinus	1076	10.24	1.56	1.05	1.33	0.97	41.7	39.8
MORMOOPIDAE								
Pteronotus gymnonotus	331	6.92	1.62	0.88	1.22	1.03	29.4	33.3
Pteronotus personatus	220	6.04	1.59	0.91	1.21	1.02	29.6	32.5
Pteronotus parnelli	524	8.06	1.65	0.95	1.29	0.99	26.1	27.6
Mormoops megalophylla	373	7.19	1.36	0.72	1.20	1.07	18.6	25.8
PHYLLOSTOMIDAE								
PHYLLOSTOMINAE								
Micronycteris megalotis	264	6.41	1.73	1.06	1.12	1.04	31.6	29.9
Micronycteris minuta	276	6.51	1.74	1.03	1.11	1.00	30.7	29.9
Micronycteris schmidtorum	298	6.68	1.72	1.03	1.14	1.05	35.0	34.1
Micronycteris brachyotis	394	7.33	1.76	0.98	1.13	0.98	36.6	37.2
Macrophyllum macrophyllum	309	6.76	1.66	1.05	1.16	0.99	37.6	35.6
Tonatia bidens	761	9.13	1.81	1.11	1.12	1.03	40.5	36.7
Tonatia schulzi	492	7.90	1.72	1.04	1.17	1.05	41.4	39.9
Tonatia sylvicola	730	9.00	1.67	1.13	1.16	1.04	39.1	34.5
Mimon crenulatum	314	6.80	1.84	1.08	1.19	0.98	32.7	30.3

	BrV	CR	BL CR	HL CR	HW CR	HH CR	CCL CR	CCL HemL
PHYLLOSTOMIDAE (cont.)								
PHYLLOSTOMINAE (cont.)								
Phyllostomus discolor	1052	10.17	1.79	1.16	1.18	0.98	41.9	36.2
Phyllostomus elongatus	855	9.49	1.82	1.11	1.16	0.98	41.6	37.3
Phyllostomus hastatus	1481	11.39	1.84	1.13	1.19	0.97	41.1	36.3
Phylloderma stenops	1292	10.89	1.83	1.11	1.14	1.06	43.3	38.9
Trachops cirrhosus	968	9.89	1.80	1.13	1.11	1.05	38.8	34.4
Vampyrum spectrum	2497	13.57	1.91	1.15	1.11	0.97	39.1	34.1
GLOSSOPHAGINAE								
Glossophaga longirostris	420	7.48	1.80	1.09	1.15	0.96	32.2	29.6
Glossophaga soricina	379	7.24	1.76	1.07	1.16	0.97	32.3	30.2
Monophyllus plethodon	430	7.55	1.90	1.13	1.19	0.87	45.1	39.8
Leptonycteris curasoae	589	8.38	1.91	1.11	1.13	0.88	42.5	38.3
Leptonycteris nivalis	566	8.27	1.88	1.15	1.16	0.91	47.2	40.9
Lonchophylla mordax	417	7.47	1.91	1.12	1.16	0.90	37.3	33.2
Lonchophylla thomasi	325	6.88	1.73	1.08	1.19	0.97	35.3	32.8
Lionycteris spurrelli	340	6.98	1.76	1.07	1.15	0.97	28.9	27.0
Anoura caudifer	391	7.31	1.85	1.13	1.16	0.92	39.2	34.9
Anoura geoffroyi	566	8.27	1.82	1.15	1.18	0.93	38.1	33.2
Choeroniscus minor	376	7.21	1.77	1.01	1.16	0.96	35.8	35.3
CAROLLIINAE								
Carollia castanea	436	7.58	1.74	1.09	1.17	1.02	30.4	28.0
Carollia perspicillata	527	8.07	1.78	1.08	1.17	1.00	33.9	31.3
Rhinophylla pumilio	344	7.00	1.74	1.09	1.15	1.03	36.8	33.7
STURNIRINAE								
Sturnira lilium	597	8.41	1.74	1.11	1.19	1.01	35.7	32.2
Sturnira ludovici	652	8.67	1.73	1.10	1.20	0.99	35.9	32.7
Sturnira tildae	677	8.78	1.74	1.11	1.21	0.99	38.7	34.8
STENODERMINAE								
Uroderma bilobatum	591	8.39	1.82	1.12	1.14	0.99	32.2	28.6
Vampyrops brachycephalus	562	8.25	1.73	1.16	1.14	1.04	34.5	29.6
Vampyrops brachyceph. ssp.	611	8.48	1.74	1.17	1.16	1.00	--.-	--.-
Vampyrops helleri	501	7.94	1.80	1.15	1.14	1.01	37.4	32.4
Vampyrops infuscus	1071	10.23	1.86	1.15	1.16	0.98	39.4	34.2
Vampyrops lineatus	715	8.94	1.80	1.17	1.15	0.98	37.9	32.3
Vampyrops vittatus	1134	10.43	1.79	1.17	1.17	0.98	36.5	31.1
Vampyrodes caraccioloi	914	9.71	1.78	1.13	1.14	1.00	36.2	31.9
Vampyressa pusilla	365	7.14	1.78	1.09	1.14	1.02	30.2	27.6
Chiroderma salvini	786	9.22	1.81	1.11	1.18	0.99	34.6	31.1
Chiroderma trinitatum	518	8.03	1.77	1.09	1.16	1.00	33.9	31.1
Chiroderma villosum	681	8.80	1.83	1.11	1.14	0.99	35.9	32.2
Chiroderma villosum ssp.	806	9.31	1.83	1.07	1.16	0.99	31.1	28.9
Ectophylla macconnelli	332	6.92	1.72	1.12	1.13	1.00	31.6	28.3
Artibeus cinereus	451	7.67	1.78	1.13	1.14	1.00	31.4	27.7
Artibeus cinereus toltecus	525	8.06	1.73	1.13	1.13	1.00	--.-	--.-
Artibeus concolor	611	8.49	1.81	1.10	1.17	0.97	32.5	29.7

	BrV	CR	BL CR	HL CR	HW CR	HH CR	CCL CR	CCL HemL
PHYLLOSTOMIDAE (cont.)								
STENODERMINAE (cont.)								
Artibeus glaucus	460	7.72	1.78	1.05	1.14	1.01	--.-	--.-
Artibeus jamaicensis	981	9.93	1.78	1.17	1.18	0.99	36.0	30.7
Artibeus lituratus	1192	10.60	1.79	1.17	1.19	0.98	37.2	31.7
Artibeus phaeotis	433	7.56	1.78	1.12	1.14	0.99	--.-	--.-
Enchisthenes harti	494	7.90	1.74	1.10	1.16	1.01	32.6	29.7
Ardops nichollsi	531	8.09	1.79	1.09	1.16	1.07	35.9	33.0
Sphaeronycteris toxophyllum	531	8.10	1.59	1.15	1.09	1.07	24.0	20.9
BRACHYPHYLLINAE								
Brachyphylla cavernarum	1155	10.49	1.82	1.19	1.20	0.97	44.1	37.1
DESMODONTIDAE								
Desmodus rotundus	964	9.88	1.70	1.21	1.19	1.05	40.4	33.4
Diphylla ecaudata	750	9.08	1.73	1.11	1.22	1.02	32.9	29.5
NATALIDAE								
Natalus tumidirostris	237	6.18	1.61	1.07	1.29	0.96	36.1	33.8
FURIPTERIDAE								
Furipterus horrens	122	4.97	1.54	1.00	1.15	1.03	19.5	19.4
VESPERTILIONIDAE								
VESPERTILIONINAE								
Myotis adversus	245	6.26	1.62	0.88	1.27	0.90	10.9	12.4
Myotis albescens	121	4.95	1.68	0.95	1.24	0.87	10.2	10.8
Myotis altarium	231	6.14	1.59	0.92	1.27	0.91	9.0	9.7
Myotis annectans	213	5.97	1.72	1.00	1.31	0.85	11.6	11.6
Myotis bocagei	182	5.67	1.63	1.02	1.26	0.92	11.1	10.8
Myotis fortidens	132	5.09	1.58	0.94	1.28	0.91	--.-	--.-
Myotis hasselti	182	5.66	1.70	1.04	1.27	0.88	--.-	--.-
Myotis horsfieldi	190	5.75	1.67	1.02	1.24	0.91	--.-	--.-
Myotis lucifugus	163	5.46	1.66	0.92	1.30	0.86	--.-	--.-
Myotis montivagus	164	5.48	1.66	0.99	1.26	0.88	10.9	11.0
Myotis muricola	120	4.93	1.69	0.96	1.28	0.88	--.-	--.-
Myotis nigrigans	131	5.07	1.66	0.97	1.24	0.86	10.2	10.5
Myotis nigricans ssp.	151	5.33	1.63	0.94	1.26	0.88	11.7	12.5
Myotis riparius	167	5.50	1.60	0.98	1.24	0.91	--.-	--.-
Myotis siligorensis	99	4.63	1.64	0.96	1.25	0.90	11.5	12.0
Myotis tricolor **)	---	-.--	1.53	0.86	1.21	0.87	--.-	--.-
Myotis sp.?	143	5.23	1.59	1.03	1.26	0.96	--.-	--.-
Pipistrellus babu	94	4.54	1.72	0.91	1.29	0.85	9.8	10.7
Pipistrellus ceylonicus	176	5.60	1.71	0.91	1.27	0.85	7.7	8.5
Pipistrellus circumdatus	222	6.05	1.65	0.96	1.29	0.90	9.0	9.3
Pipistrellus coromandra	101	4.66	1.68	0.97	1.28	0.84	--.-	--.-
Pipistrellus crassulus	127	5.03	1.71	0.94	1.38	0.81	12.5	13.3
Pipistrellus imbricatus	132	5.09	1.67	0.94	1.26	0.88	11.1	11.8
Pipistrellus javanicus	115	4.86	1.63	0.96	1.28	0.88	7.5	7.9
Pipistrellus kuhli	160	5.43	1.47	0.94	1.27	0.83	--.-	--.-
Pipistrellus mimus	90	4.48	1.71	0.95	1.28	0.83	7.7	8.2

	BrV	CR	BL CR	HL CR	HW CR	HH CR	CCL CR	CCL HemL
VESPERTILIONIDAE (cont.)								
VESPERTILIONINAE (cont.)								
Pipistrellus nanus	95	4.57	1.61	0.90	1.28	0.87	10.1	11.2
P. papuanus + P.angulatus §)	109	4.77	1.68	0.92	1.33	0.89	8.3	9.1
Pipistrellus pulveratus	138	5.16	1.70	0.97	1.27	0.86	10.1	10.5
Pipistrellus subflavus	120	4.94	1.67	0.92	1.31	0.88	7.0	7.6
Scotozous dormeri	138	5.16	1.77	0.94	1.31	0.82	9.1	9.7
Nyctalus noctula	354	7.07	1.63	0.83	1.31	0.85	7.7	9.3
Nyctalus stenopterus	241	6.22	1.68	0.94	1.31	0.80	10.9	11.5
Glischropus tylopus	97	4.58	1.73	0.85	1.36	0.81	10.7	12.6
Eptesicus brasiliensis	193	5.78	1.74	0.95	1.26	0.85	8.9	9.4
Eptesicus br. melanopterus	189	5.73	1.73	0.95	1.22	0.81	9.3	9.8
Eptesicus demissus	223	6.06	1.72	0.97	1.35	0.86	--.-	--.-
Eptesicus flavescens	183	5.68	1.48	0.86	1.25	0.87	8.4	9.7
Eptesicus fuscus	230	6.13	1.80	1.09	1.32	0.78	13.6	12.4
Eptesicus gaumeri	182	5.67	1.74	0.93	1.27	0.85	--.-	--.-
Eptesicus pachyotis	153	5.35	1.73	0.93	1.26	0.82	--.-	--.-
Eptesicus pumilus	118	4.90	1.69	0.92	1.26	0.94	10.4	11.3
Eptesicus serotinus andersoni	332	6.92	1.80	1.03	1.33	0.80	11.1	10.8
Ia io	735	9.02	1.80	1.10	1.31	0.84	8.4	7.7
Tylonycteris pachypus	78	4.26	1.81	1.01	1.51	0.66	13.7	13.6
Tylonycteris robustula	112	4.81	1.77	0.97	1.50	0.69	8.9	9.2
Hesperoptenus blanfordi	152	5.34	1.65	0.89	1.32	0.81	6.2	6.9
Glauconycteris humeralis	139	5.18	1.52	0.81	1.32	0.93	6.8	8.5
Glauconycteris poensis	198	5.83	1.54	0.82	1.32	0.96	6.9	8.3
Chalinolobus gouldi	231	6.13	1.60	0.73	1.26	0.93	6.8	9.3
Chalinolobus morio	178	5.62	1.48	0.77	1.29	0.94	6.5	8.5
Scotorepens sanborni	153	5.34	1.76	0.90	1.29	0.82	11.8	13.1
Rhogeessa parvula	108	4.75	1.75	0.94	1.20	0.84	12.5	13.3
Rhogeessa tumida	101	4.66	1.73	0.86	1.23	0.86	--.-	--.-
Scotomanes ornatus	371	7.18	1.86	1.09	1.33	0.81	9.7	8.9
Scotomanes sp.?	326	6.88	1.68	0.99	1.31	0.89	9.6	9.8
Scotophilus dinganii	422	7.50	1.50	0.95	1.24	0.88	8.9	9.3
Scotophilus heathi	460	7.72	1.67	0.93	1.23	0.89	7.6	8.2
Scotophilus kuhli	330	6.90	1.71	0.97	1.26	0.88	8.8	9.1
Lasiurus borealis	164	5.48	1.59	0.73	1.22	0.93	4.2	5.8
MINIOPTERINAE								
Miniopterus australis	197	5.82	1.59	0.83	1.23	1.01	19.3	23.3
Miniopterus haradai	215	5.99	1.58	0.90	1.23	0.98	19.1	21.3
Miniopterus inflatus	314	6.79	1.46	0.89	1.20	0.96	16.1	18.1
Miniopterus magnater	312	6.78	1.60	0.92	1.20	0.95	17.6	19.3
Miniopterus medius.	256	6.35	1.57	0.89	1.20	0.97	15.7	17.5
Miniopterus pusillus	208	5.93	1.60	0.91	1.21	0.97	22.8	25.0
Miniopterus schreibersi **)	---	-.--	1.58	0.80	1.20	0.95	--.-	--.-
Miniopterus s. arenarius	233	6.15	1.46	0.91	1.22	0.98	19.4	21.4
Miniopterus s. blepotis	298	6.68	1.59	0.84	1.21	1.00	13.6	16.3
Miniopterus tristis	322	6.85	1.63	0.96	1.22	0.97	19.1	19.8

	BrV	CR	BL CR	HL CR	HW CR	HH CR	CCL CR	CCL HemL
VESPERTILIONIDAE (cont.)								
MURININAE								
Murina cyclotis cyclotis	240	6.21	1.69	1.01	1.19	0.96	18.1	17.9
Murina c. peninsularis	272	6.48	1.73	0.97	1.17	0.97	18.4	19.0
Murina huttoni rubella	249	6.29	1.75	1.01	1.20	0.93	21.4	21.3
Murina tubinaris	210	5.94	1.64	0.96	1.24	1.01	18.7	19.5
Harpiocephalus harpia	427	7.53	1.81	1.02	1.21	0.98	18.2	17.8
KERIVOULINAE								
Kerivoula papillosa	299	6.69	1.61	1.03	1.26	0.97	32.3	31.3
Kerivoula pellucida	196	5.81	1.60	1.12	1.14	1.03	41.3	36.9
Kerivoula phalaena	112	4.82	1.64	0.98	1.16	1.02	24.5	25.1
Phoniscus atrox	182	5.66	1.57	1.22	1.25	1.01	38.5	31.6
NYCTOPHILINAE								
Nyctophilus timoriensis	256	6.35	1.73	0.96	1.24	0.89	14.1	14.7
MOLOSSIDAE								
Tadarida aegyptiaca	379	7.24	1.79	0.96	1.32	0.85	14.0	14.6
Tadarida condylura	441	7.61	1.43	0.96	1.38	0.87	11.7	12.1
Tadarida mops	458	7.71	1.75	0.92	1.35	0.86	12.7	13.9
Tadarida niveiventer	417	7.47	1.49	0.95	1.34	0.83	9.3	9.8
Tadarida beccarii	225	6.08	1.82	0.94	1.30	0.77	12.7	13.5
Tadarida jobensis	348	7.03	1.79	1.00	1.29	0.87	13.5	13.6
Tadarida plicata plicata	314	6.79	1.78	1.01	1.31	0.87	12.5	12.4
Tadarida pumila	234	6.16	1.49	0.94	1.33	0.85	9.1	9.7
Tadarida pumila ssp.	242	6.23	1.75	0.95	1.33	0.85	--.-	--.-
Otomops martiensseni	730	9.00	1.73	0.97	1.28	0.87	12.0	12.3
Molossops abrasus	404	7.39	1.76	0.94	1.34	0.82	10.1	10.7
Molossops greenhalli	290	6.62	1.80	0.94	1.19	0.83	10.6	11.3
Molossops planirostris	179	5.63	1.69	0.84	1.27	0.84	9.2	10.9
Eumops auripendulus	591	8.39	1.82	0.92	1.27	0.84	9.3	10.1
Eumops glaucinus	631	8.58	1.79	0.93	1.27	0.83	11.7	12.6
Molossus ater	510	7.98	1.75	0.97	1.31	0.81	11.1	11.5
Molossus molossus	306	6.74	1.69	0.95	1.30	0.83	9.1	9.5
Molossus trinitatis	419	7.49	1.82	0.98	1.24	0.84	10.7	11.0
Cheiromeles torquatus	1315	10.95	1.73	1.01	1.36	0.78	14.8	14.6

Abbreviations:

BL	brain length		
BrV	standard brain volume = BrW/1.036		
CCL	corpus callosum length		
CR	cube root of standard BrV		
HH	hemisphere height		
HL	hemisphere length	HW	hemisphere width

§) See addendum at end of Volume 3, pp. 1592-1595

**) Data taken from RTF Bernard et al. (1988)

Number of individuals as in Table 2

Table 4. Average proportion indices (= linear measures of brains and corpus callosum related to the cube root of the standard brain volumes)

For linear measures in species see Table 2
For proportion indices in species see Table 3

		BL	HL	HW	HH		CCL	CCL
	N	CR	CR	CR	CR	N	CR	HL
PTEROPODIDAE	52	1.91	1.26	1.24	0.96	47	34.9	27.6
PTEROPODINAE *	39	1.92	1.27	1.23	0.96	34	35.2	27.5
Rousettus	3	1.90	1.22	1.24	0.95	3	32.1	26.4
Pteropus	13	2.04	1.40	1.21	0.95	12	43.5	31.1
Dobsonia	4	1.99	1.27	1.18	0.99	4	37.9	29.8
Cynopterus	4	1.85	1.19	1.24	0.97	3	29.9	25.2
NYCTIMENINAE *	4	1.88	1.18	1.21	0.99	4	29.4	24.8
MACROGLOSSINAE *	9	1.87	1.23	1.27	0.94	9	36.0	29.3
RHINOPOMATIDAE *	2	1.91	1.06	1.24	0.82	2	11.1	10.5
EMBALLONURIDAE	20	1.76	0.99	1.23	0.91§	18	13.4	13.6
EMBALLONURINAE *	19	1.76	0.99	1.23	0.91§	17	13.5	13.6
NON-*TAPHOZOUS* EMB.	11	1.75	1.01	1.23	0.94§	11	14.2	14.0
Emballonura	3	1.77	1.03	1.23	0.92	3	15.0	14.5
Saccopteryx	3	1.78	1.01	1.25	0.94	3	13.8	13.7
TAPHOZOUS	8	1.78	0.95	1.23	0.88	6	12.1	12.9
DICLIDURINAE *	1	1.66	0.99	1.12	0.96	1	13.0	13.1
CRASEONYCTERIDAE *	1	1.66	1.01	1.25	0.90	1	15.4	15.2
NYCTERIDAE *	8	1.68	1.09	1.18	0.94	8	29.5	27.1
MEGADERMATIDAE *	4	1.77	1.05	1.22	0.95	4	22.5	21.5
RHINOLOPHIDAE *	25	1.85	1.09	1.24	0.86	14	19.7	18.1
HIPPOSIDERIDAE *	29	1.86	1.05	1.20	0.90	23	26.3	25.2
Hipposideros	26	1.87	1.05	1.21	0.89	20	26.4	25.1
NOCTILIONIDAE *	2	1.53	1.04	1.32	0.98	2	36.7	35.4
MORMOOPIDAE *	4	1.56	0.87	1.23	1.03	4	25.9	29.8
Pteronotus	3	1.62	0.91	1.24	1.01	3	28.4	31.1
PHYLLOSTOMIDAE	58	1.78	1.11	1.16	0.99	53	36.3	32.7
PHYLLOSTOMINAE *	15	1.78	1.09	1.14	1.01	15	38.1	35.0
Micronycteris	4	1.74	1.02	1.12	1.02	4	33.5	32.8
Tonatia	3	1.73	1.09	1.15	1.04	3	40.4	37.0
Phyllostomus	3	1.81	1.14	1.17	0.98	3	41.5	36.6
GLOSSOPHAGINAE *	12	1.82	1.10	1.16	0.94	11	37.6	34.1
CAROLLIINAE *	3	1.76	1.09	1.16	1.02	3	33.7	31.0
STURNIRINAE *	3	1.74	1.10	1.20	1.00	3	36.8	33.3
STENODERMINAE *	24	1.78	1.13	1.15	1.00	20	34.0	30.1
Vampyrops	6	1.79	1.16	1.15	1.00	5	37.1	31.9
Chiroderma	4	1.81	1.10	1.16	0.99	4	33.9	30.8
Artibeus	7	1.78	1.12	1.16	0.99	4	34.3	30.0
BRACHYPHYLLINAE *	1	1.82	1.19	1.20	0.97	1	44.1	37.1
DESMODONTIDAE *	2	1.71	1.16	1.20	1.04	2	36.6	31.4

	N	BL CR	HL CR	HW CR	HH CR	N	CCL CR	CCL HL
NATALIDAE *	1	1.61	1.07	1.29	0.96	1	36.1	33.8
FURIPTERIDAE *	1	1.54	1.00	1.15	1.03	1	19.5	19.4
VESPERTILIONIDAE	78	1.66	0.95	1.27	0.89	65	13.0	13.6
VESPERTILIONINAE *	59	1.67	0.94	1.29	0.86	46	9.5	10.1
Myotis	16	1.64	0.97	1.26	0.89	9	10.8	11.3
Pipistrellus	13	1.66	0.94	1.29	0.86	11	9.2	9.8
Eptesicus	9	1.71	0.96	1.28	0.84	6	10.3	10.6
Tylonycteris	2	1.79	0.99	1.50	0.67	2	11.3	11.4
Scotophilus	3	1.63	0.95	1.24	0.88	3	8.4	8.9
MINIOPTERINAE *	9	1.56	0.89	1.21	0.98	9	18.1	20.2
MURININAE *	5	1.72	0.99	1.20	0.97	5	19.0	19.1
KERIVOULINAE *	4	1.61	1.09	1.20	1.01	4	34.2	31.2
NYCTOPHILINAE *	1	1.73	0.96	1.24	0.89	1	14.1	14.7
MOLOSSIDAE *	19	1.72	0.95	1.30	0.84	18	11.3	11.9
Tadarida	9	1.68	0.96	1.33	0.85	8	11.9	12.4
Molossops	3	1.75	0.91	1.27	0.83	3	10.0	11.0
Molossus	3	1.75	0.97	1.28	0.83	3	10.3	10.7
CHIROPTERA (AvCS)	303	1.77	1.06	1.23	0.92§§	263	24.3	22.2
CHIROPTERA (AvCF)	NN=28	1.73	1.06	1.22	0.95		26.3	24.4
MEGACHIR. (AvMegS)	52	1.91	1.26	1.24	0.96	47	34.9	27.6
MEGACHIR. (AvMegF)	NN= 3	1.89	1.23	1.24	0.96		33.5	27.2
MICROCHIR. (AvMicS)	251	1.74	1.02	1.23	0.91§§	216	22.0	21.1
MICROCHIR. (AvMicF)	NN=25	1.71	1.04	1.21	0.95		25.5	24.1

Number of families and/or subfamilies marked by * = NN
Number of species = N
§ = N - 1; §§ = N - 2
Abbreviations:

AvCF	average of Chiroptera families and/or subfamilies
AvCS	average of Chiroptera species
AvMegF	average of Megachiroptera families and/or subfamilies
AvMegS	average of Megachiroptera species
AvMicF	average of Microchiroptera families and/or subfamilies
AvMicS	average of Microchiroptera species
BL	brain length
CCL	corpus callosum length
CR	cube root of standard BrV
HH	hemisphere height
HL	hemisphere length
HW	hemisphere width

Table 5. Average body weights (BoW) and brain weights (BrW) of 342 species and/or subspecies of bats.

	BoW in g	CV in %	BrW in mg	CV in %	n BoW	n BrW	Source of data
PTEROPODIDAE							
PTEROPODINAE							
Eidolon helvum	**262**	9.5	**4290**	7.9	13	13	S
Rousettus aegyptiacus *)	129	19.0	2274	8.4	20	20	S
males	152	7.7	2280	11.7	6	6	
females	119	18.6	2272	7.1	14	14	
median inter sexes	**136**		**2276**				
	134.3	10.1	2070	9.7	6	6	13
R. amplexicaud. brach. §)	**55.4**	14.0	**1352**	5.0	6	6	S
R. a. stresem. + leschen.) §)	88.1	13.1	1715	8.0	20	20	S
males	98.3	6.6	1780	8.6	9	9	
females	79.7	8.5	1662	6.1	11	11	
median inter sexes	**89.0**		**1721**				
Lissonycteris angolensis	**84.3**	17.5	**1774**	7.5	8	8	S
Myonycteris torquata	**36.4**	12.0	**1157**	7.4	4	4	S
Pteropus alecto	**595**	8.2	**7039**	6.4	4	4	S
Pteropus conspicillatus	**750**		**8350**		1	1	S
Pteropus hypomelanus	**390**		**5302**		1	1	S

Notes to Table 5: If there is more than one line for a species, the first line includes the pairs of body and brain weights used to establish the standards. The other lines include either single values (either body weights or brain weights), or separate data on males, females and median inter sexes in cases of sex differences, or the combined data from the preceding lines used to establish the standards, and/or data from the literature which more strongly deviate and were therefore not included in the standards.

*) Median values between males and females are given in the case of statistically significant sex differences (99% for BoW, $P < .01$; 95% for BrW, $P < .05$)

§) See addendum at end of Volume 3, pp. 1592-1595

The standards are bold-faced
CV = coefficient of variation
n = number of specimens

Key to the source of body and brain weight data:
1: M Weber (1896); **2**: Dubois (1897); **3**: Ziehen (1899); **4:** Dräseke (1903d); **5**: Welcker und Brandt (1903); **6:** Warncke (1908); **7**: Kolmer (1910); **8**: Waterlot (1912); **9:** Mangold-Wirz (1966); **10:** collection Paul Pirlot in Pirlot and Stephan (1970); **11:** collection P. Pirlot in Stephan and Pirlot (1970a, b); **12:** collection P. Pirlot in Pirlot and Pottier (1977); **13:** RTF Bernard et al. (1988); **S:** collection Heinz Stephan in cooperation with sons, many friends and colleagues (see end of this table).

	BoW in g	CV in %	BrW in mg	CV in %	n BoW	n BrW	Source of data
PTEROPODIDAE (cont.)							
PTEROPODINAE (cont.)							
Pteropus lylei	**314**	11.0	**6129**	4.3	7	7	S
Pteropus edwardsi	**361**	13.6	**6900**	7.2	7	7	1,6
Pteropus macrotis	**366**		**6127**		1	1	S
Pteropus mahaganus	**465**	1.5	**5280**	0.6	2	2	S
Pteropus neohibernicus	**840**	5.1	**9114**	18.3	2	2	S
Pteropus poliocephalus *)	**695**	27.5	**7230**	5.8	2	2	S
male	830		7527		1	1	
female	560		6932		1	1	
Pteropus samoensis	**310**		**5792**		1	1	S
Pteropus scapulatus	**375**	9.3	**5360**	7.8	3	3	S
Pteropus temmincki	**250**	14.5	**4890**	0.4	2	2	S
Pteropus tonganus *)	449	17.8	6112	5.9	8	8	S
males	488	13.1	6202	4.7	5	5	
females	383	15.9	5961	8.0	3	3	
median inter sexes	**436**		**6082**				
Pteropus vampyrus *)	1065	13.7	9531	11.1	8	8	1,S
males	1010	5.3	9817	8.3	3	3	S
presumed males	1225		10067		3	3	1
females	909	1.7	8300	9.4	2	2	S
median inter sexes	**1014**		**9121**				
Dobsonia inermis	**149**	14.1	**2700**	6.2	7	7	S
D. molucc. magna §)	**465**	5.3	**5342**	4.9	4	4	S
D. molucc. anderseni *) §)	234	14.5	3850	8.1	7	7	S
males	255	11.5	3978	8.4	4	4	
females	206	5.6	3680	5.8	3	3	
median inter sexes	**230**		**3829**				
Dobsonia praedatrix *)	178	16.0	2993	6.2	7	7	S
males	198	27.9	3109	10.5	2	2	
females	170	7.6	2947	4.3	5	5	
median inter sexes	**184**		**3028**				
Hypsignathus monstrosus	**310**	16.2	**3479**	7.3	3	3	S
Epomops franqueti *)	118	15.9	2217	4.7	11	11	S
males	128	15.5	2289	2.0	6	6	
females	107	9.2	2131	4.1	5	5	
median inter sexes	**120**		**2210**				
Epomophorus gambianus	**309**		**4330**		1	1	8
Epomophorus labiatus	**63.9**	12.8	**1586**	9.4	22	22	S
Epomophorus wahlbergi	**74.1**	9.1	**1814**	9.9	10	10	S
	94.4		2220		1	1	13
Micropteropus pusillus	**25.1**	8.6	**826**	5.8	8	8	S
Scotonycteris zenkeri	**21.3**	4.8	**710**	3.3	3	3	S
Casinycteris argynnis *)	26.8	14.3	832	11.2	13	13	S
males	24.5	5.2	756	5.0	7	7	
females	29.5	14.0	921	4.2	6	6	
median inter sexes	**27.0**		**838**				

> Table 5 <

	BoW in g	CV in %	BrW in mg	CV in %	n BoW	n BrW	Source of data
PTEROPODIDAE (cont.)							
PTEROPODINAE (cont.)							
Cynopterus brachyotis	**33.9**	14.0	**981**	9.4	22	22	S
Cynopterus horsfieldi	**59.3**	9.5	**1373**	5.5	14	14	S
Cynopterus sphinx sphinx	**51.7**	11.0	**1199**	5.9	16	16	S
Cynopterus sph. angulatus	**33.1**	0.9	**923**	2.5	2	2	S
Megaerops ecaudatus	**24.0**	8.7	**797**	8.3	10	10	S
Megaerops niphanae	**32.6**	15.8	**958**	12.1	3	3	S
Chironax melanocephalus	**17.7**	4.4	**606**	3.6	10	10	S
Penthetor lucasi	**35.0**		**980**		1	1	12
Sphaerias blanfordi	**29.5**	6.7	**832**	6.2	11	11	S
Balionycteris maculata	**14.4**	5.8	**512**	5.9	3	3	S
NYCTIMENINAE							
Nyctimene albiventer	**29.7**	0.5	**825**	14.7	2	2	S
Nyctimene robinsoni	**47.1**	5.5	**1232**	6.6	8	8	S
Nyctimene vizcaccia	**35.4**	6.5	**988**	7.1	6	6	S
Paranyctimene raptor	**26.1**	9.9	**731**	5.4	7	7	S
MACROGLOSSINAE							
Eonycteris spelaea	**58.7**	17.7	**1310**	9.3	11	11	S
Megaloglossus woermanni	**17.9**	8.1	**679**	3.5	10	10	S
Macroglossus minimus	**14.6**	9.1	**561**	7.0	17	17	S
Macroglossus sobrinus	**21.8**	7.1	**694**	7.8	21	21	S
Syconycteris a. australis §)	**14.7**		**570**		1	1	S
S. a. papuana + finschi §)	**17.5**	8.4	**627**	5.8	16	16	S
Melonycteris melanops	**47.3**	10.5	**1291**	3.9	9	9	S
Nesonycteris woodfordi	**35.4**	16.0	**1017**	4.9	7	7	S
*Notopteris macdonaldi *)*	69.3	12.3	1444	6.0	10	10	S
males	76.5	11.5	1514	3.8	4	4	
females	64.5	6.3	1397	5.1	6	6	
median inter sexes	**70.5**		**1456**				
RHINOPOMATIDAE							
Rhinopoma hardwickei	**12.9**	21.1	**275**	4.9	11	11	S
Rhinopoma microphyllum	**29.4**	12.9	**387**	5.0	13	13	S
EMBALLONURIDAE							
EMBALLONURINAE							
Emballonura monticola	**5.3**	7.1	**166**	6.1	15	15	S
Emballonura raffrayana	**5.9**	7.7	**174**	4.8	6	6	S
Emballonura semicaudata	**7.0**	9.9	**184**	6.5	13	13	S
Mosia nigrescens §)							
Coleura afra	**11.5**	5.4	**257**	5.0	9	9	S
Rhynchonycteris naso	**3.8**	9.0	**118**	6.7	11	11	S
*Saccopteryx bilineata *)*	8.4	16.4	231	5.7	16	16	S
males	6.8	18.6	222	5.7	3	3	
females	8.7	13.5	233	5.5	13	13	
median inter sexes	**7.8**		**228**				

	BoW in g	CV in %	BrW in mg	CV in %	n BoW	n BrW	Source of data
EMBALLONURIDAE (cont.)							
EMBALLONURINAE (cont.)							
Saccopteryx canescens	**3.4**		**130**		1	1	S
Saccopteryx leptura	**5.1**	10.7	**161**	2.9	7	7	S
Cormura brevirostris	**8.2**	10.0	**219**	3.5	3	3	S
*Peropteryx macrotis *)*	5.6	12.6	160	7.6	18	18	S
males	5.2	5.9	156	4.8	11	11	
females	6.3	11.3	167	9.0	7	7	
median inter sexes	**5.8**		**162**				
Peropteryx trinitatis	**4.2**	10.3	**137**	6.1	6	6	S
Taphozous australis	**24.0**	7.9	**517**	1.1	3	3	S
Taphozous hildegardeae	**27.4**	7.9	**545**	4.1	7	7	S
Taphozous longimanus	**23.2**		**470**		1	1	S
Taphozous mauritianus	**29.0**	2.2	**551**	13.7	2	2	S
Taphozous melanopogon	**22.7**	5.1	**536**	5.8	10	10	S
Taphozous theobaldi	**39.8**	2.7	**681**	2.8	8	8	S
Taphozous saccolaimus	**43.0**	7.5	**671**	7.0	13	13	S
Taphozous saccolaimus ssp.	**51.4**	11.3	**672**	3.6	10	10	S
DICLIDURINAE							
Cyttarops alecto	**5.3**		**175**		1	1	S
CRASEONYCTERIDAE							
Craseonycteris thonglongyai	**2.56**	5.9	**87**	4.9	5	5	S
NYCTERIDAE							
Nycteris arge	**10.6**	7.8	**347**	2.7	6	6	S
Nycteris grandis	**29.8**	4.6	**709**	4.0	10	10	S
Nycteris hispida	**7.8**		**284**		1	1	S
Nycteris javanica	**18.4**		**470**		1	1	S
Nycteris macrotis	**14.5**	8.3	**388**	5.9	3	3	S
Nycteris nana	**6.9**	8.8	**238**	3.3	4	4	S
Nycteris thebaica	**8.9**	12.6	**323**	5.6	3	3	S
Nycteris tragata	**14.4**	5.4	**460**	7.8	2	2	S
MEGADERMATIDAE							
Megaderma lyra	**38.6**	13.4	**907**	6.8	20	20	S
Megaderma spasma	**22.6**	13.6	**644**	8.2	15	15	S
Macroderma gigas	**119.8**	3.0	**1704**	4.0	4	4	S
Cardioderma cor	**26.0**	13.2	**670**	5.6	11	11	S
Lavia frons	**23.4**	2.8	**644**	3.0	3	3	S
RHINOLOPHIDAE							
Rhinolophus acuminatus	**12.1**	15.2	**302**	11.3	10	10	S
Rhinolophus affinis	**13.7**	8.0	**355**	7.9	16	16	S
Rhinolophus capensis	12.8	4.7	256	3.9	7	7	13
Rhinolophus clivosus	20.1	2.5	242	12.4	5	5	13
Rhinolophus coelophyllus	**7.7**	10.8	**259**	5.4	8	8	S
Rhinolophus cornutus	**6.6**	4.5	**194**	5.3	4	4	S

	BoW in g	CV in %	BrW in mg	CV in %	n BoW	n BrW	Source of data
RHINOLOPHIDAE (cont.)							
Rhinolophus eloquens	**19.1**	4.1	**404**	11.0	2	2	S
Rhinolophus ferrumequinum	23.5		350		1	1	2
Rhinolophus hildebrandti	**29.1**		**630**		1	1	S
Rhinolophus hipposideros	**4.6**	10.4	**149**	9.3	4	4	S
Rhinolophus landeri	**9.7**	5.7	**277**	5.7	12	12	S
Rhinolophus lepidus	**6.0**	3.6	**185**	3.8	6	6	S
Rhinolophus luctus	**32.8**	11.4	**621**	17.1	3	3	S
Rhinolophus macrotis	**6.3**	8.8	**208**	4.2	7	7	S
Rhinolophus malayanus	**6.4**	5.6	**202**	3.6	8	8	S
Rhinolophus megaphyllus	**9.8**	6.2	**261**	3.4	7	7	S
Rhinolophus megaphyllus ssp.	**7.0**	13.3	**230**	4.3	3	3	S
Rhinolophus mehelyi	**9.7**	7.1	**310**	2.7	6	6	S
Rhinolophus paradoxolophus	**9.3**		**267**		1	1	S
Rhinolophus pearsoni	**13.9**	15.9	**332**	5.3	7	7	S
Rhinolophus pusillus	**4.7**	8.8	**164**	7.6	8	8	S
Rhinolophus refulgens	**5.4**		**190**		1	1	S
Rhinolophus robinsoni	**8.4**	3.4	**277**	1.8	2	2	S
Rhinolophus stheno ssp.	**7.0**	6.3	**244**	6.5	7	7	S
Rhinolophus stheno stheno	**8.5**	4.5	**285**	6.0	6	6	S
Rhinolophus thomasi	**9.0**	7.8	**233**	3.2	8	8	S
Rhinolophus trifoliatus	**14.6**	14.3	**352**	7.9	9	9	S
Rhinolophus yunanensis	**23.5**	12.4	**439**	3.4	2	2	S
HIPPOSIDERIDAE							
Hipposideros armiger	**56.0**	8.4	**800**	5.4	10	10	S
Hipposideros ater	**6.9**	11.4	**213**	4.6	5	5	S
Hipposideros ater aruensis	**5.5**	2.6	**190**	1.9	2	2	S
Hipposideros ater saevus	**4.2**	1.9	**179**	4.9	4	4	S
Hipposideros ater ssp.	**7.2**		**245**		1	1	S
Hipposideros bicolor gentilis	**6.2**	8.5	**233**	4.7	6	6	S
Hipposideros bicolor atrox	**8.3**	4.2	**245**	6.9	3	3	S
Hipposideros caffer	**11.0**	5.2	**306**	4.6	14	14	S
Hipposideros caffer ssp.	**10.4**	14.2	**303**	5.6	22	22	S
Hipp. calcar. calcaratus §)	**13.1**	4.4	**402**	4.1	7	7	S
Hipp. calcar. cupidus §)	**9.6**	12.3	**336**	3.2	6	6	S
Hipposideros cervinus	**7.3**	8.7	**214**	4.3	17	17	S
*Hipposideros commersoni *)*	99.2	29.4	749	10.8	9	9	S
males	125.7	9.1	757	4.6	4	4	
females	78.1	24.0	743	14.8	5	5	
median inter sexes	**101.9**		**750**				
*Hipposideros diadema *)*	41.9	13.4	705	7.3	21	21	S
males	45.5	12.3	735	8.6	7	7	
females	40.0	11.9	690	5.6	14	14	
median inter sexes	**42.8**		**713**				
Hipposideros fulvus	**7.0**	9.4	**236**	5.7	6	6	S

	BoW in g	CV in %	BrW in mg	CV in %	n BoW	n BrW	Source of data
HIPPOSIDERIDAE (cont.)							
Hipposideros galeritus	**10.4**	6.7	**278**	1.0	3	3	S
Hipposideros halophyllus	**4.0**	9.0	**152**	1.4	2	2	S
Hipposideros lankadiva	**48.5**	5.1	**733**	2.7	9	9	S
Hipposideros larvatus	**20.5**	11.3	**413**	6.8	20	20	S
Hipposideros lekaguli	**30.3**	5.6	**584**	2.9	10	10	S
H. magg. maggietaylorae §)	**24.9**		**634**		1	1	S
H. maggietaylorae erroris §)	**16.1**	4.9	**451**	3.1	4	4	S
Hipposideros ridleyi	**9.6**	4.0	**268**	7.4	10	10	S
Hipposideros ruber	**7.3**		**265**		1	1	S
Hipposideros speoris	**11.5**	5.1	**284**	2.6	6	6	S
Hipposideros turpis	**34.8**	12.1	**599**	2.1	4	4	S
Asellia tridens	**12.8**		**240**		1	1	11
Aselliscus stoliczkanus	**4.9**	3.5	**150**	6.6	3	3	S
Aselliscus tricuspidatus	**4.4**	2.8	**134**	3.7	10	10	S
Rhinonycteris aurantius	**8.6**	3.3	**248**	3.0	4	4	S
Triaenops persicus	**13.7**	5.6	**271**	6.9	10	10	S
NOCTILIONIDAE							
Noctilio albiventris	**32.7**	23.0	**597**	7.0	17	17	10,12,S
*Noctilio leporinus *)*	**58.0**	16.6	**1183**	11.1	17	17	10,S
males	69.6	17.1	1200	3.5	3	3	
females	51.2	5.6	1072	3.0	6	6	
median inter sexes	60.4		1136				
MORMOOPIDAE							
Pteronotus davyi	**10.6**		**310**		1	1	10
Pteronotus gymnonotus	**13.6**	2.2	**343**	2.4	6	6	S
Pteronotus personatus	**8.1**	10.7	**228**	6.0	18	18	S
Pteronotus parnelli	**20.2**	15.2	**543**	7.3	15	15	S
Mormoops megalophylla	**15.7**	9.2	**386**	6.8	9	9	S
PHYLLOSTOMIDAE							
PHYLLOSTOMINAE							
Micronycteris megalotis	**6.5**	7.2	**274**	4.5	9	9	S
Micronycteris minuta	**6.9**		**286**		1	1	S
Micronycteris schmidtorum	**7.8**	5.1	**308**	2.9	10	10	S
Micronycteris brachyotis	**14.7**		**408**		1	1	S
Lonchorhina aurita	**18.0**		**377**		1	1	12
Macrophyllum macrophyllum	**8.6**	7.5	**321**	4.9	10	10	S
Tonatia bidens	**29.4**	7.0	**788**	7.4	10	10	S
Tonatia schulzi	**21.3**		**510**		1	1	S
Tonatia sylvicola	**32.3**		**756**		1	1	S
Mimon crenulatum	**11.8**	1.8	**326**	0.2	2	2	S
	14.8	9.7	340	11.5	4	4	12
Phyllostomus discolor	**36.6**	10.9	**1090**	4.5	15	15	S
Phyllostomus elongatus	**44.9**	8.8	**885**	3.1	9	9	S

	BoW in g	CV in %	BrW in mg	CV in %	n BoW	n BrW	Source of data
PHYLLOSTOMIDAE (cont.)							
PHYLLOSTOMINAE (cont.)							
*Phyllostomus hastatus *)*	91.8	11.3	1534	7.4	16	16	S
males	97.1	8.7	1587	5.8	10	10	
females	83.1	8.1	1446	6.3	6	6	
median inter sexes	**90.1**		**1517**				
Phylloderma stenops	**46.1**	4.6	**1338**	2.5	2	2	S
Trachops cirrhosus	**36.9**	10.3	**1003**	1.6	2	2	S
	36.0		820		1	1	12
Vampyrum spectrum	**173**	9.0	**2587**	2.7	2	2	S
	189		3150		1	1	12
GLOSSOPHAGINAE							
Glossophaga longirostris	**11.3**	5.7	**435**	11.4	2	2	S
Glossophaga soricina	**9.9**	12.2	**393**	4.7	20	20	S
Monophyllus plethodon	**15.0**	9.0	**445**	2.2	2	2	S
Monophyllus luciae	**14.8**		**380**		1	1	12
Leptonycteris curasoae	**24.5**		**610**		1	1	S
	24.4	11.5	556	8.8	5	5	10
Leptonycteris nivalis	**22.8**	7.1	**586**	5.5	4	4	S
Lonchophylla mordax	**10.7**		**432**		1	1	S
Lonchophylla thomasi	**6.9**		**337**		1	1	S
Lionycteris spurrelli	**7.55**	11.1	**352**	4.8	4	4	S
Anoura caudifer	**10.6**	7.1	**405**	5.7	8	8	S
Anoura geoffroyi	**16.0**	9.5	**586**	4.0	12	12	S
Choeroniscus minor	**9.6**		**389**		1	1	S
CAROLLIINAE							
Carollia castanea	**13.1**	11.2	**452**	9.6	9	9	S
Carollia perspicillata	**17.8**	13.3	**546**	5.8	26	26	S
Rhinophylla pumilio	**8.9**	4.0	**356**	8.1	10	10	S
STURNIRINAE							
Sturnira lilium	**20.2**	10.7	**618**	6.5	29	29	S
Sturnira ludovici	**25.9**	9.9	**676**	7.9	4	4	S
Sturnira tildae	**24.4**	7.1	**701**	5.2	7	7	S
STENODERMINAE							
Uroderma bilobatum	**16.2**	15.2	**612**	6.1	22	22	S
Vampyrops brachycephalus	**12.9**	10.2	**583**	4.5	4	4	S
Vampyrops brachyceph. ssp.	**19.3**	7.0	**633**	5.9	2	2	S
Vampyrops helleri	**13.8**	11.7	**519**	8.0	12	12	S
Vampyrops infuscus	**50.2**		**1110**		1	1	S
Vampyrops lineatus	**22.6**	7.6	**741**	4.1	5	5	S
Vampyrops vittatus	**36.6**	0.0	**1175**	3.0	2	2	S
Vampyrodes caraccioloi	**39.7**		**947**		1	1	S
Vampyressa pusilla	**8.6**	8.0	**378**	8.8	9	9	S
Chiroderma salvini	**26.1**	15.8	**814**	10.1	4	4	S
Chiroderma trinitatum	**13.7**	5.2	**537**	5.8	2	2	S
Chiroderma villosum	**24.8**	8.9	**706**	0.4	3	3	S

	BoW in g	CV in %	BrW in mg	CV in %	n BoW	n BrW	Source of data
PHYLLOSTOMIDAE (cont.)							
STENODERMINAE (cont.)							
Chiroderma villosum ssp.	**31.7**		**835**		1	1	S
Ectophylla macconnelli	**7.5**	4.7	**344**	6.7	5	5	S
Artibeus cinereus	**12.6**	9.7	**467**	6.0	18	18	S
Artibeus cinereus toltecus	**16.6**	11.0	**543**	2.8	3	3	S
Artibeus concolor	**19.4**	6.2	**633**	2.7	2	2	S
Artibeus glaucus	**12.2**		**476**		1	1	S
Artibeus jamaicensis	**41.0**	10.0	**1016**	6.6	16	16	S
Artibeus lituratus	**56.0**	12.7	**1233**	7.4	25	25	S
Artibeus phaeotis	**11.8**	14.2	**448**	8.6	8	8	S
Enchisthenes harti	**16.1**	16.8	**511**	7.3	8	8	S
Ardops spp.	**22.5**	24.4	**565**	9.0	4	4	12,S
Ametrida centurio	**10.5**		**290**		1	1	12
Sphaeronycteris toxophyllum	**18.4**	12.7	**524**	7.2	2	2	12,S
BRACHYPHYLLINAE							
Brachyphylla cavernarum	**44.5**	4.1	**1196**	6.0	6	6	S
	49.0		1360		1	1	12
DESMODONTIDAE							
Desmodus rotundus	**36.3**	13.9	**999**	5.3	25	25	S
	29.7	14.6	843	9.9	12	12	10
Diaemus youngi	**34.6**	8.8	**978**	4.9	6	6	10
Diphylla ecaudata	**30.9**	10.4	**798**	5.5	9	9	S
NATALIDAE							
Natalus tumidirostris	**6.9**	5.5	**245**	2.9	10	10	S
FURIPTERIDAE							
Furipterus horrens	**3.43**	3.0	**127**	2.6	6	6	S
THYROPTERIDAE							
Thyroptera tricolor	**4.5**		**230**		1	1	12
VESPERTILIONIDAE							
VESPERTILIONINAE							
Myotis adversus	**11.2**	3.4	**254**	1.3	5	5	S
Myotis albescens	**4.0**	2.9	**126**	2.6	3	3	S
Myotis altarium	**11.0**	9.9	**240**	2.1	4	4	S
Myotis annectans	**9.8**	1.4	**221**	7.0	2	2	S
Myotis bechsteini	9.5	7.4	265	8.0	2	2	S
	11				1		S
	10		**265**		3	2	
Myotis bocagei	**7.0**	4.9	**190**	3.8	6	6	S
Myotis dasycneme	**15.0**		**310**		1	1	S
Myotis daubentoni	7.0		230		1	1	S
	10				1		S
	8.5		**230**		2	1	
Myotis fortidens	**4.3**	9.6	**130**	12.6	9	9	S
Myotis hasselti	**7.5**		**188**		1	1	S

	BoW in g	CV in %	BrW in mg	CV in %	n BoW	n BrW	Source of data
VESPERTILIONIDAE (cont.)							
VESPERTILIONINAE (cont.)							
Myotis horsfieldi	**6.3**	6.8	**197**	4.2	5	5	S
Myotis lucifugus	**7.9**	13.0	**169**	4.4	8	8	S
Myotis montivagus	**8.3**	4.8	**170**	3.3	5	5	S
Myotis muricola	**4.8**	13.1	**124**	5.6	18	18	S
Myotis myotis	**25.6**	14.4	**485**	4.9	7	7	9,S
Myotis mystacinus	**5.25**	27.5	**154**	3.1	4	4	2,S
Myotis nattereri	7.0		220		1	1	S
	8.0				1		S
	7.5		**220**		2	1	
Myotis nigricans	**4.6**	8.5	**136**	7.8	12	12	S
	4.45	25.8	115	18.0	6	6	10
Myotis nigricans ssp.	**6.6**	11.9	**156**	2.9	7	7	S
Myotis riparius	**7.95**	5.9	**173**	10.7	4	4	S
Myotis siligorensis	**3.46**	11.6	**103**	10.0	7	7	S
Myotis sp.?	**4.2**		**148**		1	1	S
Myotis tricolor	**13.6**	5.1	**259**	11.6	6	6	13
Pipistrellus babu	**4.8**	12.8	**97**	7.2	4	4	S
Pipistrellus ceylonicus	**10.1**	6.2	**182**	7.5	8	8	S
Pipistrellus circumdatus	**10.4**	11.1	**230**	6.3	10	10	S
Pipistrellus coromandra	**4.6**	14.1	**105**	7.0	7	7	S
Pipistrellus crassulus	**6.1**	4.9	**132**	4.4	4	4	S
Pipistrellus imbricatus	**6.3**	15.2	**137**	5.2	15	15	S
Pipistrellus javanicus	**4.5**	8.0	**119**	4.7	3	3	S
Pipistrellus kuhli	**6.3**		**166**		1	1	S
Pipistrellus mimus	**3.9**	21.6	**94**	10.5	10	10	S
Pipistrellus nanus	**3.9**	9.0	**100**	6.8	19	19	S
P. papuanus + angulatus §)	**5.2**	13.5	**113**	6.9	12	12	S
Pipistrellus pulveratus	**5.3**	8.8	**143**	4.8	6	6	S
Pipistrellus subflavus	**5.3**	10.7	**125**	2.8	3	3	S
Scotozous dormeri	**6.8**	8.3	**143**	7.1	10	10	S
Nyctalus noctula	**27.0**	8.3	**364**	7.6	9	9	7,9,S
	15.3		275		1	1	12
Nyctalus stenopterus	**16.1**	20.4	**244**	4.4	5	5	S
Glischropus tylopus	**4.9**	15.9	**100**	10.8	4	4	S
Eptesicus brasiliensis	**9.2**	9.2	**200**	6.4	12	12	S
Eptesicus br. melanopterus	**8.8**	6.3	**195**	9.8	4	4	S
	6.5		165		1	1	12
Eptesicus demissus	**15.0**		**231**		1	1	S
Eptesicus flavescens	**10.5**	18.3	**190**	1.1	2	2	S
Eptesicus fuscus	**13.6**		**238**		1	1	S
	14.0		330		1	1	12
Eptesicus gaumeri	**7.6**	2.3	**189**	1.3	4	4	S
Eptesicus pachyotis	**6.6**	11.4	**158**	6.6	3	3	S
Eptesicus pumilus	**5.4**	7.9	**122**	12.2	4	4	S

	BoW in g	CV in %	BrW in mg	CV in %	n BoW	n BrW	Source of data
VESPERTILIONIDAE (cont.)							
VESPERTILIONINAE (cont.)							
Eptesicus serotinus	**23.4**		**310**		1	1	4
	12.2		168		1	1	3
Eptesicus s. andersoni	**20.9**	7.5	**344**	10.1	5	5	S
Ia io	**53.3**		**761**		1	1	S
Vespertilio murinus **)	**33.3**		**445**		1	1	2
Tylonycteris pachypus	**4.8**	14.1	**80**	11.2	28	28	S
	5.4		90		1	1	12
Tylonycteris robustula	**7.2**	16.4	**115**	8.3	25	25	S
	8.0		136				12
Hesperoptenus blanfordi	**7.3**	7.0	**158**	4.2	10	10	S
Glauconycteris humeralis	**5.5**	1.8	**144**	6.3	3	3	S
Glauconycteris poensis	**8.6**		**205**		1	1	S
Chalinolobus gouldi	**11.7**	0.9	**240**	4.8	3	3	S
Chalinolobus morio	**8.3**	3.7	**184**	4.9	5	5	S
Scotorepens sanborni	**9.1**		**158**		1	1	S
Rhogeessa parvula	**4.5**	18.3	**112**	12.4	8	8	S
	3.8		90		1	1	12
Rhogeessa tumida	**5.1**	9.8	**106**	4.8	3	3	S
Scotomanes ornatus	**21.8**	14.0	**384**	4.4	2	2	S
Scotomanes sp.?	**16.4**	10.4	**338**	9.1	3	3	S
Scotophilus dinganii	**24.3**	13.9	**437**	5.7	13	13	S
Scotophilus heathi	**31.5**	6.0	**476**	4.1	10	10	S
Scotophilus kuhli	**20.4**	10.0	**341**	5.8	22	22	S
Scotophilus temmincki	18.4		356		1	1	12
Scotophilus nigrita (gigas)	**101.3**	9.0	**1046**	10.7	3	3	8
Lasiurus borealis	**7.8**		**170**		1	1	S
Barbastella barbastellus	**5.8**		**183**		2	2	4,5
Plecotus auritus	**6.6**	16.6	**238**	5.5	3	3	3,5
MINIOPTERINAE							
Miniopterus australis	**7.1**	2.7	**204**	5.1	6	6	S
Miniopterus haradai	**8.3**	8.5	**223**	7.0	8	8	S
Miniopterus inflatus	**14.9**	6.8	**325**	4.8	8	8	S
Miniopterus magnater	**15.0**	9.4	**324**	5.5	14	14	S
Miniopterus medius	**11.4**	6.0	**265**	4.8	10	10	S
Miniopterus pusillus	**8.6**	2.5	**216**	2.8	10	10	S
Miniopterus schreibersi	11.4	1.8	254	6.3	5	5	13
Miniopterus s. schreibersi	**12.5**	9.8	**291**	2.8	6	6	S
Miniopterus s. arenarius	**10.2**	6.0	**242**	4.2	12	12	S
Miniopterus s. blepotis	**14.3**	4.5	**309**	4.2	7	7	S
Miniopterus tristis	**15.6**	12.6	**333**	1.9	8	8	S

**) Doubtful determination; probably *Myotis myotis*

	BoW in g	CV in %	BrW in mg	CV in %	n BoW	n BrW	Source of data
VESPERTILIONIDAE (cont.)							
MURININAE							
Murina cyclotis cyclotis	**7.1**	19.9	**249**	6.0	2	2	S
Murina c. peninsularis	**8.65**	2.5	**281**	3.8	2	2	S
Murina huttoni rubella	**8.8**	2.3	**258**	2.8	4	4	S
Murina tubinaris	**5.45**	24.7	**217**	7.8	2	2	S
Harpiocephalus harpia	**13.5**		**442**		1	1	S
KERIVOULINAE							
Kerivoula papillosa	**10.1**		**310**		1	1	S
	13.3		305		1	1	12
Kerivoula pellucida	**4.1**		**203**		1	1	S
Kerivoula phalaena	**3.0**		**116**		1	1	S
Phoniscus atrox	**4.2**		**188**		1	1	S
NYCTOPHILINAE							
Nyctophilus geoffroyi	**7.2**	12.6	**167**	5.8	5	5	S
Nyctophilus timoriensis	**10.9**	9.6	**254**	7.8	3	3	S
MOLOSSIDAE							
Tadarida aegyptiaca	**17.7**	5.0	**393**	3.4	6	6	S
Tadarida condylura	**27.2**	17.1	**457**	6.4	9	9	S
Tadarida mops	**30.9**	5.7	**475**	6.2	10	10	S
Tadarida niveiventer	**21.7**	6.7	**432**	5.5	8	8	S
Tadarida beccarii	**14.2**	5.6	**233**	6.5	6	6	S
Tadarida jugularis	**11.5**		**220**		1	1	S
Tadarida jobensis	**16.5**		**360**		1	1	S
Tadarida leucostigma	**26.5**	5.7	**392**	2.5	6	6	S
Tadarida limbatus	**13.3**	0.4	**293**	6.5	4	4	S
Tadarida plicata	23.6		445		1	1	12
Tadarida plicata plicata	**17.4**	1.8	**325**	6.9	3	3	S
Tadarida pumila	**10.3**	5.1	**243**	5.7	18	18	S
Tadarida pumila ssp.	**11.0**	2.8	**253**	3.5	5	5	S
	11.2	17.0	229	1.3	3	3	8
Otomops martiensseni	**41.5**	7.9	**756**	3.7	14	14	S
Molossops abrasus	**33.3**	2.1	**418**	4.9	4	4	S
Molossops greenhalli	**17.5**		**300**		1	1	S
Molossops planirostris	**7.25**	2.9	**185**	3.8	2	2	S
Eumops auripendulus	**33.8**	7.6	**612**	3.5	6	6	S
Eumops glaucinus	**36.5**	5.9	**653**	3.2	3	3	S
Eumops perotis	**47.5**		**802**		1	1	12
Molossus ater	**33.6**	17.8	**526**	9.7	19	19	S
Molossus major	**13.0**	11.7	**267**	7.6	77	77	10
Molossus molossus	**13.7**	11.7	**317**	7.1	24	24	S
Molossus trinitatis	**19.2**	0.7	**435**	2.1	2	2	S
Cheiromeles torquatus	**167**	6.5	**1362**	3.8	8	8	S

Notes to Table 5: If there is more than one line for a species, the first line includes the pairs of body and brain weights used to establish the standards. The other lines include either single values (either body weights or brain weights), or separate data on males, females and median inter sexes in cases of sex differences, or the combined data from the preceding lines used to establish the standards, and/or data from the literature which more strongly deviate and were therefore not included in the standards. The standards are bold-faced.

*) Median values between males and females are given in the case of statistically significant sex differences (99% for BoW, $P < .01$; 95% for BrW, $P < .05$).

§) See addendum at end of Volume 3, pp. 1592-1595

Key to the source of body and brain weight data:
1: M Weber (1896); **2**: Dubois (1897); **3**: Ziehen (1899); **4**: Dräseke (1903d); **5**: Welcker und Brandt (1903); **6**: Warncke (1908); **7**: Kolmer (1910); **8**: Waterlot (1912); **9**: Mangold-Wirz (1966); **10**: collection Paul Pirlot in Pirlot and Stephan (1970); **11**: collection P. Pirlot in Stephan and Pirlot (1970a, b); **12**: collection P. Pirlot in Pirlot and Pottier (1977); **13**: RTF Bernard et al. (1988); **S**: collection Heinz Stephan in cooperation with Georg Baron (Canada 1980, 1984; French Guiana 1985; Trinidad 1985; Martinique 1985), Roland Bauchot (Pyrenees 1960; Madagascar 1962), Kunwar Bhatnagar (India 1986; Thailand 1986), Dennis Black (Papua New Guinea 1981/82), Fergus Clunie (Fiji 1982), Hans Erkert (Columbia 1971), Heiko D. Frahm (Sumatra 1976; Malaysia 1976), Peter Germroth (Ecuador 1989), John Gibbons (Fiji 1982), A. Gopalakrishna (India 1986), Leslie Hall (Australia 1976), Kabongo Ka Mubalamata (Zaire 1983/84), Dieter Kock (French Guiana 1985), Preecha Leucha (Thailand 1986, 1987), Linjanja Lulyo (Zaire 1983/84), Phil Martin (Australia 1976), Roger Martin (Australia 1976), A.N. McWilliam (Kenia 1979), Hans Meier (Bougainvillea 1982), Jarujin Nabhitabhata (Thailand 1986, 1987), Nivesh Nadee (Thailand 1986, 1987), John Nelson (Australia 1976; Papua New Guinea 1981; Fiji 1982), Harold Parnaby (Australia 1976), Gerd Rehkämper (Burundi 1979; Kenia 1979), Rupert Russel (Australia 1976), Dermot Smyth (Australia 1976), Wolfbernhard Spatz (Belgian Congo 1956/57), Michael Stephan (Sumatra 1976; Mexiko 1978; Burundi 1979; Zaire 1983/84), Roland Stephan (Canada 1980), Songsakdi Yenbutra (Thailand 1986, 1987), Alan Young (Australia 1976), and others. For institutional and laboratory accomodations see Acknowledgements (p. 16).

Table 6. Standards of body and brain weight, percentages of brain weight relative to body weight, encephalization indices (EIs) derived from the standards, and size indices (SIs) of NET brain volumes. EIs and SIs are related to the average of the four species of Tenrecinae (Insectivora).

	Standards		BrW	BrW	NET
	BoW	BrW	BoW	EI	SI
slope				.66	.66
y-intercept				1.544	1.506
PTEROPODIDAE					
PTEROPODINAE					
Eidolon helvum	262	4290	1.64	311	318
Rousettus aegyptiacus	136	2276	1.67	254	257
R. amplexicaud. brachyotis §)	55.4	1352	2.44	273	282
R. a. stresem. + R. leschen. §)	89.0	1721	1.93	254	262
Lissonycteris angolensis	84.3	1774	2.10	272	---
Myonycteris torquata	36.4	1157	3.18	308	319
Pteropus alecto	595	7039	1.18	297	304
Pteropus conspicillatus	750	8350	1.11	302	310
Pteropus hypomelanus	390	5302	1.36	295	303
Pteropus lylei	314	6129	1.95	394	400
Pteropus edwardsi	361	6900	1.91	404	---
Pteropus macrotis	366	6127	1.67	356	---
Pteropus mahaganus	465	5280	1.14	262	270
Pteropus neohibernicus	840	9114	1.09	306	311
Pteropus poliocephalus	695	7230	1.04	275	281
Pteropus samoensis	310	5792	1.87	375	385
Pteropus scapulatus	375	5360	1.43	306	317
Pteropus temmincki	250	4890	1.96	365	376
Pteropus tonganus	436	6082	1.39	315	324
Pteropus vampyrus	1014	9121	0.90	270	275
Dobsonia inermis	149	2700	1.81	284	293
D. moluccensis magna §)	465	5342	1.15	265	272
D. molucc. anderseni §)	230	3829	1.66	302	312
Dobsonia praedatrix	184	3028	1.65	277	283
Hypsignathus monstrosus	310	3479	1.12	226	230
Epomops franqueti	120	2210	1.84	268	275
Epomophorus gambianus	309	4330	1.40	281	---
Epomophorus labiatus	63.9	1586	2.48	292	299
Epomophorus wahlbergi	74.1	1814	2.45	302	---
Micropteropus pusillus	25.1	826	3.29	281	290
Scotonycteris zenkeri	21.3	710	3.33	269	277
Casinycteris argynnis	27.0	838	3.10	272	280

§) see addendum at end of Volume 3.

Abbreviations:

BoW	body weight	NET	net brain volume
BrW	brain weight	SI	size index
EI	encephalization index		

	Standards		BrW	BrW	NET
	BoW	BrW	BoW	EI	SI
slope				.66	.66
y-intercept				1.544	1.506
PTEROPODIDAE (cont.)					
PTEROPODINAE (cont.)					
Cynopterus brachyotis	33.9	981	2.89	274	283
Cynopterus horsfieldi	59.3	1373	2.32	265	274
Cynopterus sphinx sphinx	51.7	1199	2.32	253	261
Cynopterus sph. angulatus	33.1	923	2.79	262	---
Megaerops ecaudatus	24.0	797	3.32	280	289
Megaerops niphanae	32.6	958	2.94	275	---
Chironax melanocephalus	17.7	606	3.42	260	268
Penthetor lucasi	35.0	980	2.80	268	---
Sphaerias blanfordi	29.5	832	2.82	255	260
Balionycteris maculata	14.4	512	3.56	252	260
NYCTIMENINAE					
Nyctimene albiventer	29.7	825	2.78	251	259
Nyctimene robinsoni	47.1	1232	2.62	277	285
Nyctimene vizcaccia	35.4	988	2.79	268	277
Paranyctimene raptor	26.1	731	2.80	243	250
MACROGLOSSINAE					
Eonycteris spelaea	58.7	1310	2.23	255	263
Megaloglossus woermanni	17.9	679	3.79	289	300
Macroglossus minimus	14.6	561	3.84	273	282
Macroglossus sobrinus	21.8	694	3.18	259	268
Syconycteris a. australis §)	14.7	570	3.88	276	286
S. a. papuana + finschi §)	17.5	627	3.58	271	281
Melonycteris melanops	47.3	1291	2.73	289	300
Nesonycteris woodfordi	35.4	1017	2.87	276	286
Notopteris macdonaldi	70.5	1456	2.07	251	260
RHINOPOMATIDAE					
Rhinopoma hardwickei	12.9	275	2.13	145	150
Rhinopoma microphyllum	29.4	387	1.32	119	122
EMBALLONURIDAE					
EMBALLONURINAE					
Emballonura monticola	5.3	166	3.13	158	163
Emballonura raffrayana	5.9	174	2.95	154	159
Emballonura semicaudata	7.0	184	2.63	146	150
Mosia nigrescens §)	3.7	120	3.24	145	---
Coleura afra	11.5	257	2.23	147	152
Rhynchonycteris naso	3.8	118	3.11	140	144
Saccopteryx bilineata	7.8	228	2.92	168	174
Saccopteryx canescens	3.4	130	3.82	166	171
Saccopteryx leptura	5.1	161	3.16	157	162
Cormura brevirostris	8.2	219	2.67	156	161

> Table 6 <

	Standards		BrW	BrW	NET
	BoW	BrW	BoW	EI	SI
slope				.66	.66
y-intercept				1.544	1.506
EMBALLONURIDAE (cont.)					
EMBALLONURINAE (cont.)					
Peropteryx macrotis	5.8	162	2.79	145	151
Peropteryx trinitatis	4.2	137	3.26	152	157
Taphozous australis	24.0	517	2.15	181	187
Taphozous hildegardeae	27.4	545	1.99	175	181
Taphozous longimanus	23.2	470	2.03	169	---
Taphozous mauritianus	29.0	551	1.90	171	177
Taphozous melanopogon	22.7	536	2.36	195	201
Taphozous theobaldi	39.8	681	1.71	171	176
Taphozous saccolaimus	43.0	671	1.56	160	164
Taphozous saccolaimus ssp.	51.4	672	1.31	143	---
DICLIDURINAE					
Cyttarops alecto	5.3	175	3.30	166	172
CRASEONYCTERIDAE					
Craseonycteris thonglongyai	2.56	87	3.40	134	137
NYCTERIDAE					
Nycteris arge	10.6	347	3.27	209	217
Nycteris grandis	29.8	709	2.38	216	224
Nycteris hispida	7.8	284	3.64	209	217
Nycteris javanica	18.4	470	2.55	196	204
Nycteris macrotis	14.5	388	2.68	190	196
Nycteris nana	6.9	238	3.45	190	197
Nycteris thebaica	8.9	323	3.63	218	227
Nycteris tragata	14.4	460	3.19	226	233
MEGADERMATIDAE					
Megaderma lyra	38.6	907	2.35	233	240
Megaderma spasma	22.6	644	2.85	235	242
Macroderma gigas	119.8	1704	1.42	207	212
Cardioderma cor	26.0	670	2.58	223	229
Lavia frons	23.4	644	2.75	230	238
RHINOLOPHIDAE					
Rhinolophus acuminatus	12.1	302	2.50	166	---
Rhinolophus affinis	13.7	355	2.59	180	---
Rhinolophus coelophyllus	7.7	259	3.36	192	---
Rhinolophus cornutus	6.6	194	2.94	160	165
Rhinolophus eloquens	19.1	404	2.12	165	170
Rhinolophus hildebrandti	29.1	630	2.16	195	---
Rhinolophus hipposideros	4.6	149	3.24	156	160
Rhinolophus landeri	9.7	277	2.86	177	183
Rhinolophus lepidus	6.0	185	3.08	162	167

	Standards		BrW	BrW	NET
	BoW	BrW	BoW	EI	SI
slope				.66	.66
y-intercept				1.544	1.506
RHINOLOPHIDAE (cont.)					
Rhinolophus luctus	32.8	621	1.89	177	182
Rhinolophus macrotis	6.3	208	3.30	176	181
Rhinolophus malayanus	6.4	202	3.16	170	175
Rhinolophus megaphyllus	9.8	261	2.66	165	171
Rhinolophus megaphyllus ssp.	7.0	230	3.29	182	188
Rhinolophus mehelyi	9.7	310	3.20	198	---
Rhinolophus paradoxolophus	9.3	267	2.87	175	179
Rhinolophus pearsoni	13.9	332	2.39	167	172
Rhinolophus pusillus	4.7	164	3.49	169	174
Rhinolophus refulgens	5.4	190	3.52	178	---
Rhinolophus robinsoni	8.4	277	3.30	194	---
Rhinolophus stheno ssp.	7.0	244	3.49	193	---
Rhinolophus stheno stheno	8.5	285	3.35	198	---
Rhinolophus thomasi	9.0	233	2.59	156	---
Rhinolophus trifoliatus	14.6	352	2.41	171	176
Rhinolophus yunanensis	23.5	439	1.87	156	160
HIPPOSIDERIDAE					
Hipposideros armiger	56.0	800	1.43	160	164
Hipposideros ater	6.9	213	3.09	170	176
Hipposideros ater aruensis	5.5	190	3.45	176	181
Hipposideros ater saevus	4.2	179	4.26	198	205
Hipposideros ater ssp.	7.2	245	3.40	190	---
Hipposideros bicolor gentilis	6.2	233	3.76	200	---
Hipposideros bicolor atrox	8.3	245	2.95	173	---
Hipposideros caffer	11.0	306	2.78	180	186
Hipposideros caffer ssp.	10.4	303	2.91	185	---
Hipposideros c. calcaratus §)	13.1	402	3.07	210	217
Hipposideros calc. cupidus §)	9.6	336	3.50	216	222
Hipposideros cervinus.	7.3	214	2.93	165	169
Hipposideros commersoni	101.9	750	0.74	101	104
Hipposideros diadema	42.8	713	1.67	171	175
Hipposideros fulvus	7.0	236	3.37	187	192
Hipposideros galeritus	10.4	278	2.67	169	---
Hipposideros halophyllus	4.0	152	3.80	174	178
Hipposideros lankadiva	48.5	733	1.51	162	164
Hipposideros larvatus	20.5	413	2.01	161	166
Hipposideros lekaguli	30.3	584	1.93	176	179
H. magg. maggietaylorae §)	24.9	634	2.55	217	224
H. maggietaylorae erroris §)	16.1	451	2.80	206	211
Hipposideros ridleyi	9.6	268	2.79	172	177
Hipposideros ruber	7.3	265	3.63	204	---

> Table 6 <

	Standards		BrW	BrW	NET
	BoW	BrW	BoW	EI	SI
slope				.66	.66
y-intercept				1.544	1.506
HIPPOSIDERIDAE (cont.)					
Hipposideros speoris	11.5	284	2.47	162	167
Hipposideros turpis	34.8	599	1.72	164	168
Asellia tridens	12.8	240	1.87	127	---
Aselliscus stoliczkanus	4.9	150	3.06	150	153
Aselliscus tricuspidatus	4.4	134	3.05	144	148
Rhinonycteris aurantius	8.6	248	2.88	171	177
Triaenops persicus	13.7	271	1.98	138	143
NOCTILIONIDAE					
Noctilio albiventris	32.7	597	1.83	171	175
Noctilio leporinus	58.0	1183	2.04	232	236
MORMOOPIDAE					
Pteronotus davyi	10.6	310	2.92	186	---
Pteronotus gymnonotus	13.6	343	2.52	175	180
Pteronotus personatus	8.1	228	2.81	164	169
Pteronotus parnelli	20.2	543	2.69	213	220
Mormoops megalophylla	15.7	386	2.46	179	185
PHYLLOSTOMIDAE					
PHYLLOSTOMINAE					
Micronycteris megalotis	6.5	274	4.22	228	235
Micronycteris minuta	6.9	286	4.14	228	235
Micronycteris schmidtorum	7.8	308	3.95	227	233
Micronycteris brachyotis	14.7	408	2.78	198	203
Lonchorhina aurita	18.0	377	2.09	160	---
Macrophyllum macrophyllum	8.6	321	3.73	222	227
Tonatia bidens	29.4	788	2.68	242	250
Tonatia schulzi	21.3	510	2.39	194	197
Tonatia sylvicola	32.3	756	2.34	218	225
Mimon crenulatum	11.8	326	2.76	183	188
Phyllostomus discolor	36.6	1090	2.98	289	299
Phyllostomus elongatus	44.9	885	1.97	205	211
Phyllostomus hastatus	90.1	1517	1.68	222	230
Phylloderma stenops	46.1	1338	2.90	305	316
Trachops cirrhosus	36.9	1003	2.72	265	274
Vampyrum spectrum	173	2587	1.50	246	255
GLOSSOPHAGINAE					
Glossophaga longirostris	11.3	435	3.85	251	261
Glossophaga soricina	9.9	393	3.97	247	257
Monophyllus plethodon	15.0	445	2.97	213	218
Monophyllus luciae	14.8	380	2.57	183	---

	Standards		BrW	BrW	NET
	BoW	BrW	BoW	EI	SI
slope				.66	.66
y-intercept				1.544	1.506
PHYLLOSTOMIDAE (cont.)					
GLOSSOPHAGINAE (cont.)					
Leptonycteris curasoae	24.5	610	2.49	211	218
Leptonycteris nivalis	22.8	586	2.57	213	218
Lonchophylla mordax	10.7	432	4.04	258	268
Lonchophylla thomasi	6.9	337	4.88	269	276
Lionycteris spurrelli	7.55	352	4.66	265	275
Anoura caudifer	10.6	405	3.82	244	251
Anoura geoffroyi	16.0	586	3.66	269	279
Choeroniscus minor	9.6	389	4.05	250	258
CAROLLIINAE					
Carollia castanea	13.1	452	3.45	236	243
Carollia perspicillata	17.8	546	3.07	233	241
Rhinophylla pumilio	8.9	356	4.00	240	246
STURNIRINAE					
Sturnira lilium	20.2	618	3.06	243	251
Sturnira ludovici	25.9	676	2.61	226	230
Sturnira tildae	24.4	701	2.87	243	251
STENODERMINAE					
Uroderma bilobatum	16.2	612	3.78	278	288
Vampyrops brachycephalus	12.9	583	4.52	308	320
Vampyrops brachyceph. ssp.	19.3	633	3.28	256	---
Vampyrops helleri	13.8	519	3.76	262	272
Vampyrops infuscus	50.2	1110	2.21	239	246
Vampyrops lineatus	22.6	741	3.28	270	276
Vampyrops vittatus	36.6	1175	3.21	312	323
Vampyrodes caraccioloi	39.7	947	2.39	238	244
Vampyressa pusilla	8.6	378	4.40	261	271
Chiroderma salvini	26.1	814	3.12	270	279
Chiroderma trinitatum	13.7	537	3.92	273	281
Chiroderma villosum	24.8	706	2.85	242	250
Chiroderma villosum ssp.	31.7	835	2.63	244	251
Ectophylla macconnelli	7.5	344	4.59	260	270
Artibeus cinereus	12.6	467	3.71	251	260
Artibeus cinereus toltecus	16.6	543	3.27	243	---
Artibeus concolor	19.4	633	3.26	256	261
Artibeus glaucus	12.2	476	3.90	261	---
Artibeus jamaicensis	41.0	1016	2.48	250	259
Artibeus lituratus	56.0	1233	2.20	247	254
Artibeus phaeotis	11.8	448	3.80	251	---

	Standards		BrW	BrW	NET
	BoW	BrW	BoW	EI	SI
slope				.66	.66
y-intercept				1.544	1.506
PHYLLOSTOMIDAE (cont.)					
STENODERMINAE (cont.)					
Enchisthenes harti	16.1	511	3.17	233	239
Ardops nichollsi	22.5	565	2.51	207	210
Ametrida centurio	10.5	290	2.76	176	---
Sphaeronycteris toxophyllum	18.4	524	2.85	219	226
BRACHYPHYLLINAE					
Brachyphylla cavernarum	44.5	1196	2.69	279	288
DESMODONTIDAE					
Desmodus rotundus	36.3	999	2.75	267	275
Diaemus youngi	34.6	978	2.83	269	---
Diphylla ecaudata	30.9	798	2.58	237	246
NATALIDAE					
Natalus tumidirostris	6.9	245	3.55	196	203
FURIPTERIDAE					
Furipterus horrens	3.43	127	3.70	161	166
THYROPTERIDAE					
Thyroptera tricolor	4.5	230	5.11	244	---
VESPERTILIONIDAE					
VESPERTILIONINAE					
Myotis adversus	11.2	254	2.27	147	151
Myotis albescens	4.0	126	3.15	144	149
Myotis altarium	11.0	240	2.18	141	145
Myotis annectans	9.8	221	2.26	140	143
Myotis bechsteini	10.0	265	2.65	166	170
Myotis bocagei	7.0	190	2.71	150	155
Myotis dasycneme	15.0	310	2.07	148	152
Myotis daubentoni	8.5	230	2.71	160	---
Myotis fortidens	4.3	130	3.02	142	---
Myotis hasselti	7.5	188	2.51	142	---
Myotis horsfieldi	6.3	197	3.13	167	---
Myotis lucifugus	7.9	169	2.14	123	---
Myotis montivagus	8.3	170	2.05	120	122
Myotis muricola	4.8	124	2.58	126	---
Myotis myotis	25.6	485	1.89	163	167
Myotis mystacinus	5.25	154	2.93	147	---
Myotis nattereri	7.5	220	2.93	166	172
Myotis nigricans	4.6	136	2.96	142	147
Myotis nigricans ssp.	6.6	156	2.36	128	132

	Standards		BrW	BrW	NET
	BoW	BrW	BoW	EI	SI
slope				.66	.66
y-intercept				1.544	1.506
VESPERTILIONIDAE (cont.)					
VESPERTILIONINAE (cont.)					
Myotis riparius	7.95	173	2.18	126	---
Myotis siligorensis	3.46	103	2.98	130	133
Myotis tricolor	13.6	259	1.90	132	---
Myotis sp.?	4.2	148	3.52	164	---
Pipistrellus babu	4.8	97	2.02	98	100
Pipistrellus ceylonicus	10.1	182	1.80	113	114
Pipistrellus circumdatus	10.4	230	2.21	140	142
Pipistrellus coromandra	4.6	105	2.28	110	---
Pipistrellus crassulus	6.1	132	2.16	114	118
Pipistrellus imbricatus	6.3	137	2.17	116	120
Pipistrellus javanicus	4.5	119	2.64	126	129
Pipistrellus kuhli	6.3	166	2.63	141	---
Pipistrellus mimus	3.9	94	2.41	109	112
Pipistrellus nanus	3.9	100	2.56	116	120
Pipistrellus pulveratus	5.3	143	2.70	136	139
Pipistrellus subflavus	5.3	125	2.36	119	122
P. papuanus + P. angulatus §)	5.2	113	2.17	109	112
Scotozous dormeri	6.8	143	2.10	115	117
Nyctalus noctula	27.0	364	1.35	118	120
Nyctalus stenopterus	16.1	244	1.52	111	114
Glischropus tylopus	4.9	100	2.04	100	103
Eptesicus brasiliensis	9.2	200	2.17	132	136
Eptesicus br. melanopterus	8.8	195	2.22	133	136
Eptesicus demissus	15.0	231	1.54	111	---
Eptesicus flavescens	10.5	190	1.81	115	119
Eptesicus fuscus	13.6	238	1.75	121	125
Eptesicus gaumeri	7.6	189	2.49	142	---
Eptesicus pachyotis	6.6	158	2.39	130	---
Eptesicus pumilus	5.4	122	2.26	115	118
Eptesicus serotinus	23.4	310	1.32	111	---
Eptesicus serotinus andersoni	20.9	344	1.65	132	135
Ia io	53.3	761	1.43	158	163
Vespertilio murinus *)	33.3	445	1.34	126	---
Tylonycteris pachypus	4.8	80	1.67	81	84
Tylonycteris robustula	7.2	115	1.60	89	92
Hesperoptenus blanfordi	7.3	158	2.16	122	124
Glauconycteris humeralis	5.5	144	2.62	134	138
Glauconycteris poensis	8.6	205	2.38	142	147
Chalinolobus gouldi	11.7	240	2.05	135	138
Chalinolobus morio	8.3	184	2.22	130	134

*) Doubtful determination; probably *Myotis myotis*

> Table 6 <

	Standards		BrW	BrW	NET
	BoW	BrW	BoW	EI	SI
slope				.66	.66
y-intercept				1.544	1.506
VESPERTILIONIDAE (cont.)					
VESPERTILIONINAE (cont.)					
Scotorepens sanborni	9.1	158	1.74	105	108
Rhogeessa parvula	4.5	112	2.49	119	123
Rhogeessa tumida	5.1	106	2.08	103	---
Scotomanes ornatus	21.8	384	1.76	144	146
Scotomanes sp.?	16.4	338	2.06	152	155
Scotophilus dinganii	24.3	437	1.80	152	157
Scotophilus heathi	31.5	476	1.51	140	142
Scotophilus kuhli	20.4	341	1.67	133	137
Scotophilus nigrita (gigas)	101.3	1046	1.03	142	---
Lasiurus borealis	7.8	170	2.18	125	130
Barbastella barbastellus	5.8	183	3.16	164	---
Plecotus auritus	6.6	238	3.61	196	---
MINIOPTERINAE					
Miniopterus australis	7.1	204	2.87	160	166
Miniopterus haradai	8.3	223	2.69	158	161
Miniopterus inflatus	14.9	325	2.18	156	161
Miniopterus magnater	15.0	324	2.16	155	160
Miniopterus medius	11.4	265	2.32	152	157
Miniopterus pusillus	8.6	216	2.51	149	154
Miniopterus s. schreibersi	12.5	291	2.33	157	---
Miniopterus s. arenarius	10.2	242	2.37	149	155
Miniopterus s. blepotis	14.3	309	2.16	153	157
Miniopterus tristis	15.6	333	2.13	155	159
MURININAE					
Murina cyclotis cyclotis	7.1	249	3.51	195	202
Murina c. peninsularis	8.65	281	3.25	193	199
Murina huttoni rubella	8.8	258	2.93	175	180
Murina tubinaris	5.45	217	3.98	203	209
Harpiocephalus harpia	13.5	442	3.27	227	234
KERIVOULINAE					
Kerivoula papillosa	10.1	310	3.07	193	200
Kerivoula pellucida	4.1	203	4.95	229	236
Kerivoula phalaena	3.0	116	3.87	161	167
Phoniscus atrox	4.2	188	4.48	208	216
NYCTOPHILINAE					
Nyctophilus geoffroyi	7.2	167	2.32	130	133
Nyctophilus timoriensis	10.9	254	2.33	150	155

	Standards		BrW	BrW	NET
	BoW	BrW	BoW	EI	SI
slope				.66	.66
y-intercept				1.544	1.506
MOLOSSIDAE					
Tadarida aegyptiaca	17.7	393	2.22	169	173
Tadarida condylura	27.2	457	1.68	148	152
Tadarida mops	30.9	475	1.54	141	145
Tadarida niveiventer	21.7	432	1.99	162	167
Tadarida beccarii	14.2	233	1.64	116	119
Tadarida jugularis	11.5	220	1.91	125	---
Tadarida jobensis	16.5	360	2.18	162	167
Tadarida leucostigma	26.5	392	1.48	129	132
Tadarida limbatus	13.3	293	2.20	152	---
Tadarida plicata plicata	17.4	325	1.87	141	145
Tadarida pumila	10.3	243	2.36	149	153
Tadarida pumila ssp.	11.0	253	2.30	149	---
Otomops martiensseni	41.5	756	1.82	185	191
Molossops abrasus	33.3	418	1.26	118	121
Molossops greenhalli	17.5	300	1.71	130	133
Molossops planirostris	7.25	185	2.55	143	147
Eumops auripendulus	33.8	612	1.81	171	176
Eumops glaucinus	36.5	653	1.79	174	179
Eumops perotis	47.5	802	1.69	179	---
Molossus ater	33.6	526	1.57	148	153
Molossus major	13.0	267	2.05	140	---
Molossus molossus	13.7	317	2.31	161	165
Molossus trinitatis	19.2	435	2.27	177	183
Cheiromeles torquatus	167	1362	0.82	133	136

Abbreviations:

BoW	body weight	NET	net brain volume
BrW	brain weight	SI	size index
EI	encephalization index		

§) see addendum at end of Volume 3, pp. 1592-1595

Table 7. Average percentages of brain weight related to body weight, and averages of the encephalization indices (EI) and size indices (SI) of the NET brain volumes in supraspecific taxonomic units related to the average of the four Tenrecinae (AvTn)

For encephalization indices and NET indices in species see Table 6

	N BrW	N NET	BrW / BoW	BrW EI	NET SI
PTEROPODIDAE	55	47	2.30	284	289
PTEROPODINAE *	42	34	2.08	289	294
Rousettus	3	3	2.02	260	267
Pteropus	14	12	1.43	323	321
Dobsonia	4	4	1.57	282	290
Epomophorus	3	1	2.11	292	299
Cynopterus	4	3	2.58	264	272
NYCTIMENINAE *	4	4	2.75	260	268
MACROGLOSSINAE *	9	9	3.13	271	281
RHINOPOMATIDAE *	2	2	1.72	132	136
EMBALLONURIDAE	20	18	2.55	161	167
EMBALLONURINAE *	19	17	2.51	161	166
NON-*TAPHOZOUS* EMB.	11	11	2.97	153	159
Emballonura	3	3	2.90	152	158
Saccopteryx	3	3	3.30	164	169
TAPHOZOUS	8	6	1.88	171	181
DICLIDURINAE *	1	1	3.30	166	172
CRASEONYCTERIDAE *	1	1	3.40	134	137
NYCTERIDAE *	8	8	3.10	207	214
MEGADERMATIDAE *	5	5	2.39	225	232
RHINOLOPHIDAE *	25	15	2.86	175	174
HIPPOSIDERIDAE	31	24	2.71	174	177
HIX: *H. commersoni* excluded *	30	23	2.78	176	180
Hipposideros	26	20	2.74	179	181
H. commersoni excluded	25	19	2.82	182	185
NOCTILIONIDAE *	2	2	1.93	201	206
MORMOOPIDAE *	5	4	2.68	184	189
Pteronotus	4	3	2.74	185	190
PHYLLOSTOMIDAE	60	53	3.20	242	252
PHYLLOSTOMINAE *	16	15	2.80	227	238
Micronycteris	4	4	3.77	220	227
Tonatia	3	3	2.47	218	224
Phyllostomus	3	3	2.21	239	246
GLOSSOPHAGINAE *	12	11	3.63	239	253
CAROLLIINAE *	3	3	3.51	237	243
STURNIRINAE *	3	3	2.85	237	244
STENODERMINAE *	25	20	3.27	252	264
Vampyrops	6	5	3.38	275	287
Chiroderma	4	4	3.13	257	265
Artibeus	7	4	3.23	251	259

	N BrW	N NET	BrW / BoW	BrW EI	NET SI
PHYLLOSTOMIDAE (cont.)					
BRACHYPHYLLINAE *	1	1	2.69	279	288
DESMODONTIDAE *	3	2	2.72	258	260
NATALIDAE *	1	1	3.55	196	203
FURIPTERIDAE *	1	1	3.70	161	166
THYROPTERIDAE *	1	--	5.11	244	---
VESPERTILIONIDAE	92	70	2.39	141	145
VESPERTILIONINAE	71	50	2.23	132	132
VEX: *Tylonycteris* excluded *	69	48	2.25	133	134
Myotis	23	13	2.57	144	149
Pipistrellus	13	11	2.32	119	121
Eptesicus	10	6	1.96	124	128
Tylonycteris	2	2	1.63	85	88
Scotophilus	4	3	1.50	142	145
MINIOPTERINAE *	10	9	2.37	154	159
MURININAE *	5	5	3.39	199	205
KERIVOULINAE *	4	4	4.09	198	205
NYCTOPHILINAE *	2	2	2.32	140	144
MOLOSSIDAE *	24	19	1.88	150	154
Tadarida	12	9	1.92	145	150
Molossops	3	3	1.84	130	134
Eumops	3	2	1.76	175	178
Molossus	4	3	2.05	156	167
CHIROPTERA (AvCS)	336	272	2.59	195	203
CHIROPTERA (AvCF)	NN=29		2.92	203	207§
MEGACHIR. (AvMegS)	55	47	2.30	284	289
MEGACHIR. (AvMegF)	NN= 3		2.65	273	281
MICROCHIR. (AvMicS)	281	225	2.65	178	185
MICROCHIR. (AvMicF)	NN=26		2.95	195	199§

Number of families and/or subfamilies marked by * = NN; § = NN - 1
Number of species = N; Abbreviations:

AvCF	average of Chiroptera families and/or subfamilies		
AvCS	average of Chiroptera species		
AvMegF	average of Megachiroptera families and/or subfamilies		
AvMegS	average of Megachiroptera species		
AvMicF	average of Microchiroptera families and/or subfamilies		
AvMicS	average of Microchiroptera species		
BoW	body weight	NET	net brain volume
BrW	brain weight	SI	size index
EI	encephalization index		

Table 8. Volumes of net brain (NET), ventricles (VENT) and rests (REST) and their percentages relative to standard brain volumes (= 100%)

For standard brain volumes (BrV) see Table 3

For average percentages in supraspecific taxonomic units see Table 9

	n	NET		VENT		REST	
		mm^3	%	mm^3	%	mm^3	%
PTEROPODIDAE							
PTEROPODINAE							
Eidolon helvum	3	4018.6	97.0	68.5	1.7	53.8	1.3
Rousettus aegyptiacus	4	2112.6	96.2	21.0	1.0	63.3	2.9
R. amplexicaud. brachyot. §)	3	1278.9	98.0	12.7	1.0	13.4	1.0
R. amplexic. stresemanni §)	3	1624.7	97.8	17.5	1.1	19.1	1.1
Myonycteris torquata	3	1097.2	98.2	9.5	0.9	10.1	0.9
Pteropus alecto	2	6617.8	97.4	90.3	1.3	86.3	1.3
Pteropus conspicillatus	1	7848.6	97.4	117.6	1.5	93.7	1.2
Pteropus hypomelanus	1	4978.1	97.3	70.3	1.4	69.4	1.4
Pteropus lylei	4	5700.7	96.4	124.0	2.1	91.3	1.5
Pteropus mahaganus	2	4980.1	97.7	59.4	1.2	57.1	1.1
Pteropus neohibernicus	2	8486.6	96.5	190.9	2.2	119.8	1.4
Pteropus poliocephalus	2	6762.2	96.9	124.8	1.8	91.8	1.3
Pteropus samoensis	1	5448.9	97.5	76.9	1.4	64.9	1.2
Pteropus scapulatus	2	5074.1	98.1	49.9	1.0	49.8	1.0
Pteropus temmincki	2	4607.4	97.6	63.7	1.4	49.0	1.0
Pteropus tonganus	3	5730.2	97.6	82.2	1.4	58.2	1.0
Pteropus vampyrus	2	8501.0	96.6	201.8	2.3	101.2	1.1
Dobsonia inermis	3	2555.3	98.0	26.5	1.0	24.4	0.9
D. moluccensis magna §)	2	5018.1	97.3	81.5	1.6	56.8	1.1
D. molucc. anderseni §)	3	3616.1	97.8	39.2	1.1	40.7	1.1
Dobsonia praedatrix	3	2836.3	97.0	38.1	1.3	48.3	1.7
Hypsignathus monstrosus	3	3249.1	96.8	36.9	1.1	72.2	2.1
Epomops franqueti	3	2077.2	97.4	32.2	1.5	23.8	1.1
Epomophorus labiatus	3	1489.5	97.3	14.2	0.9	27.1	1.8
Micropteropus pusillus	3	779.5	97.8	7.6	1.0	10.1	1.3
Scotonycteris zenkeri	2	669.7	97.7	7.4	1.1	8.3	1.2
Casinycteris argynnis	5	791.1	97.8	8.5	1.0	9.4	1.2
Cynopterus brachyotis	3	926.8	97.9	9.0	1.0	11.1	1.2
Cynopterus horsfieldi	3	1301.0	98.2	10.8	0.8	13.5	1.0
Cynopterus sphinx sphinx	3	1129.9	97.6	13.4	1.2	14.0	1.2
Megaerops ecaudatus	2	755.8	98.2	6.8	0.9	6.7	0.9
Chironax melanocephalus	3	572.5	97.9	5.8	1.0	6.6	1.1
Sphaerias blanfordi	4	777.5	96.8	13.9	1.7	11.6	1.5
Balionycteris maculata	2	484.5	98.0	4.3	0.9	5.4	1.1

Abbreviations and notes:

NET net volume of brain VENT ventricles

REST comprising rests of meninges, stumps of nerves, hypophysis, etc.

§) see addendum at end of Volume 3, pp. 1592-1595; n = number of individuals

	n	NET mm³	NET %	VENT mm³	VENT %	REST mm³	REST %
PTEROPODIDAE (cont.)							
NYCTIMENINAE							
Nyctimene albiventer	2	779.1	97.8	8.8	1.1	8.5	1.1
Nyctimene robinsoni	2	1161.8	97.7	13.8	1.2	13.6	1.1
Nyctimene vizcaccia	3	933.7	97.9	8.1	0.9	11.9	1.2
Paranyctimene raptor	3	690.6	97.9	8.6	1.2	6.4	0.9
MACROGLOSSINAE							
Eonycteris spelaea	3	1239.9	98.1	12.0	0.9	12.6	1.0
Megaloglossus woermanni	3	645.0	98.4	5.3	0.8	5.1	0.8
Macroglossus minimus	4	530.8	98.0	5.3	1.0	5.4	1.0
Macroglossus sobrinus	3	657.7	98.2	5.5	0.8	6.7	1.0
Syconycteris a. australis §)	1	541.0	98.3	3.9	0.7	5.2	1.0
S. australis papuana §)	3	595.3	98.4	5.1	0.8	4.8	0.8
Melonycteris melanops	3	1225.7	98.4	9.8	0.8	10.7	0.9
Nesonycteris woodfordi	3	965.5	98.4	8.0	0.8	8.1	0.8
Notopteris macdonaldi	3	1385.0	98.6	8.1	0.6	12.3	0.9
RHINOPOMATIDAE							
Rhinopoma hardwickei	3	260.0	98.0	2.0	0.8	3.4	1.3
Rhinopoma microphyllum	3	364.5	97.6	4.1	1.1	5.0	1.3
EMBALLONURIDAE							
EMBALLONURINAE							
Emballonura monticola	3	157.4	98.2	1.0	0.7	1.8	1.1
Emballonura raffrayana	3	165.0	98.2	1.3	0.8	1.6	1.0
Emballonura semicaudata	3	173.8	97.9	1.4	0.8	2.4	1.3
Coleura afra	3	243.8	98.3	1.5	0.6	2.7	1.1
Rhynchonycteris naso	3	111.6	98.0	0.7	0.6	1.6	1.4
Saccopteryx bilineata	4	216.3	98.3	0.9	0.4	2.9	1.3
Saccopteryx canescens	1	122.8	97.9	0.8	0.6	1.9	1.5
Saccopteryx leptura	3	152.5	98.2	1.0	0.6	1.9	1.2
Cormura brevirostris	3	206.7	97.8	2.3	1.1	2.4	1.1
Peropteryx macrotis	3	154.0	98.5	0.9	0.6	1.4	0.9
Peropteryx trinitatis	3	129.7	98.1	1.0	0.8	1.6	1.2
Taphozous australis	2	488.9	98.0	3.8	0.8	6.4	1.3
Taphozous hildegardeae	3	514.6	97.8	6.3	1.2	5.2	1.0
Taphozous mauritianus	2	522.3	98.2	3.7	0.7	5.9	1.1
Taphozous melanopogon	3	506.7	97.9	3.8	0.7	7.0	1.4
Taphozous theobaldi	3	642.3	97.7	6.0	0.9	9.0	1.4
Taphozous saccolaimus	3	631.2	97.5	6.6	1.0	9.9	1.5
DICLIDURINAE							
Cyttarops alecto	1	165.6	98.0	1.4	0.8	1.9	1.1
CRASEONYCTERIDAE							
Craseonycteris thonglongyai	3	81.7	97.2	0.6	0.7	1.7	2.0

	n	NET mm³	NET %	VENT mm³	VENT %	REST mm³	REST %
NYCTERIDAE							
Nycteris arge	2	331.0	98.8	1.0	0.3	3.0	0.9
Nycteris grandis	4	674.2	98.5	3.7	0.5	6.5	1.0
Nycteris hispida	1	270.2	98.6	1.4	0.5	2.5	0.9
Nycteris javanica	1	447.3	98.6	2.7	0.6	3.7	0.8
Nycteris macrotis	3	367.9	98.2	2.6	0.7	4.0	1.1
Nycteris nana	3	226.2	98.5	1.0	0.4	2.6	1.1
Nycteris thebaica	3	307.6	98.6	1.2	0.4	3.0	1.0
Nycteris tragata	2	434.1	97.8	4.2	0.9	5.7	1.3
MEGADERMATIDAE							
Megaderma lyra	6	856.2	97.8	6.3	0.7	13.1	1.5
Megaderma spasma	3	607.6	97.8	7.0	1.1	6.9	1.1
Macroderma gigas	3	1603.4	97.5	15.4	0.9	26.0	1.6
Cardioderma cor	3	631.1	97.6	6.8	1.1	8.8	1.4
Lavia frons	2	611.4	98.4	2.5	0.4	7.8	1.2
RHINOLOPHIDAE							
Rhinolophus cornutus	2	183.5	98.0	1.4	0.7	2.4	1.3
Rhinolophus eloquens	2	382.3	98.0	2.4	0.6	5.3	1.4
Rhinolophus hipposideros	2	140.4	97.6	1.2	0.9	2.2	1.5
Rhinolophus landeri	3	262.5	98.2	2.0	0.7	2.9	1.1
Rhinolophus lepidus	3	174.8	97.9	1.5	0.8	2.3	1.3
Rhinolophus luctus	3	583.5	97.3	5.9	1.0	10.0	1.7
Rhinolophus macrotis	2	195.9	97.6	2.4	1.2	2.5	1.2
Rhinolophus malayanus	2	190.9	97.9	1.2	0.6	2.8	1.5
Rhinolophus megaphyllus	2	247.2	98.1	1.6	0.7	3.1	1.2
Rhinolophus megaph. ssp.	3	217.8	98.1	1.5	0.7	2.8	1.2
Rhinolophus paradoxolophus	1	250.1	97.0	3.3	1.3	4.3	1.7
Rhinolophus pearsoni	1	313.8	97.9	2.9	0.9	3.8	1.2
Rhinolophus pusillus	3	155.2	98.0	1.0	0.6	2.1	1.3
Rhinolophus trifoliatus	3	330.8	97.4	3.1	0.9	5.8	1.7
Rhinolophus yunanensis	2	412.9	97.4	4.8	1.1	6.0	1.4
HIPPOSIDERIDAE							
Hipposideros armiger	3	750.9	97.2	7.2	0.9	14.1	1.8
Hipposideros ater	2	202.0	98.3	1.2	0.6	2.4	1.2
Hipposideros ater aruensis	2	179.2	97.7	1.6	0.9	2.6	1.4
Hipposideros ater saevus	3	169.2	97.9	1.5	0.8	2.2	1.3
Hipposideros caffer	3	290.1	98.2	1.7	0.6	3.6	1.2
Hipp. c. calcaratus §)	3	379.4	97.8	3.7	1.0	5.0	1.3
Hipp. calc. cupidus §)	3	316.5	97.6	2.9	0.9	4.9	1.5
Hipposideros cervinus	3	201.4	97.5	2.0	0.9	3.2	1.6
Hipposideros commersoni	3	702.6	97.1	7.7	1.1	13.6	1.9
Hipposideros diadema	4	669.3	97.2	6.5	1.0	12.4	1.8
Hipposideros fulvus	2	222.6	97.7	2.2	1.0	3.0	1.3

	n	NET mm^3	NET %	VENT mm^3	VENT %	REST mm^3	REST %
HIPPOSIDERIDAE (cont.)							
Hipposideros halophyllus	2	142.4	97.0	1.6	1.1	2.8	1.9
Hipposideros lankadiva	3	682.1	96.4	8.8	1.2	16.7	2.4
Hipposideros larvatus	3	389.8	97.8	3.2	0.8	5.7	1.4
Hipposideros lekaguli	2	545.8	96.8	9.3	1.7	8.6	1.5
Hipp. m. maggietaylorae §)	1	598.2	97.8	5.2	0.8	8.6	1.4
Hipp. m. erroris §)	3	423.3	97.2	5.4	1.2	6.7	1.5
Hipposideros ridleyi	3	253.0	97.8	1.8	0.7	3.9	1.5
Hipposideros speoris	2	268.0	97.8	2.7	1.0	3.4	1.2
Hipposideros turpis	3	562.1	97.2	5.9	1.0	10.2	1.8
Aselliscus stoliczkanus	3	140.2	96.8	2.0	1.4	2.6	1.8
Aselliscus tricuspidatus	3	125.8	97.3	1.2	1.0	2.3	1.8
Rhinonycteris aurantius	3	234.3	97.9	1.6	0.7	3.5	1.5
Triaenops persicus	3	257.2	98.3	1.3	0.5	3.0	1.2
NOCTILIONIDAE							
Noctilio albiventris	3	561.9	97.5	5.2	0.9	9.2	1.6
Noctilio leporinus	3	1105.4	96.8	18.3	1.6	18.3	1.6
MORMOOPIDAE							
Pteronotus gymnonotus	3	323.1	97.6	1.9	0.6	6.0	1.8
Pteronotus personatus	3	216.0	98.1	1.5	0.7	2.6	1.2
Pteronotus parnelli	3	513.9	98.0	4.8	0.9	5.5	1.0
Mormoops megalophylla	3	365.3	98.1	2.7	0.7	4.6	1.2
PHYLLOSTOMIDAE							
PHYLLOSTOMINAE							
Micronycteris megalotis	3	258.8	97.8	2.1	0.8	3.6	1.4
Micronycteris minuta	1	269.9	97.8	1.9	0.7	4.3	1.6
Micronycteris schmidtorum	3	290.2	97.6	2.7	0.9	4.5	1.5
Micronycteris brachyotis	1	383.4	97.4	2.1	0.5	8.3	2.1
Macrophyllum macrophyllum	3	301.3	97.3	3.1	1.0	5.4	1.8
Tonatia bidens	3	745.2	98.0	4.9	0.6	10.5	1.4
Tonatia schulzi	1	475.3	96.5	6.2	1.3	10.8	2.2
Tonatia sylvicola	1	713.4	97.8	3.7	0.5	12.6	1.7
Mimon crenulatum	2	307.3	97.7	2.4	0.7	5.0	1.6
Phyllostomus discolor	3	1031.0	98.0	7.3	0.7	13.8	1.3
Phyllostomus elongatus	3	831.7	97.4	7.0	0.8	15.5	1.8
Phyllostomus hastatus	3	1438.5	98.2	11.4	0.8	14.4	1.0
Phylloderma stenops	2	1267.8	98.2	8.2	0.6	15.5	1.2
Trachops cirrhosus	2	949.2	98.0	7.0	0.7	11.9	1.2
Vampyrum spectrum	2	2449.5	98.1	23.0	0.9	24.7	1.0

	n	NET mm³	NET %	VENT mm³	VENT %	REST mm³	REST %
PHYLLOSTOMIDAE (cont.)							
GLOSSOPHAGINAE							
Glossophaga longirostris	1	414.3	98.7	2.2	0.5	3.4	0.8
Glossophaga soricina	3	374.8	98.8	1.9	0.5	2.6	0.7
Monophyllus plethodon	2	417.4	97.2	5.7	1.3	6.5	1.5
Leptonycteris curasoae	1	575.9	97.8	7.4	1.3	5.5	0.9
Leptonycteris nivalis	3	550.1	97.3	6.7	1.2	8.8	1.5
Lonchophylla mordax	1	410.0	98.3	2.5	0.6	4.5	1.1
Lonchophylla thomasi	1	316.3	97.3	3.0	0.9	6.0	1.8
Lionycteris spurrelli	3	334.7	98.5	1.5	0.5	3.6	1.0
Anoura caudifer	3	382.5	97.8	3.2	0.8	5.2	1.3
Anoura geoffroyi	3	558.4	98.7	2.8	0.5	4.4	0.8
Choeroniscus minor	1	367.9	98.0	2.6	0.7	5.0	1.3
CAROLLIINAE							
Carollia castanea	5	424.9	97.4	3.6	0.8	7.8	1.8
Carollia perspicillata	3	517.0	98.1	3.5	0.7	6.4	1.2
Rhinophylla pumilio	3	334.2	97.3	3.8	1.1	5.6	1.6
STURNIRINAE							
Sturnira lilium	3	584.0	97.9	5.0	0.8	7.5	1.3
Sturnira ludovici	2	630.8	96.7	9.9	1.5	11.8	1.8
Sturnira tildae	3	661.8	97.8	6.8	1.0	8.1	1.2
STENODERMINAE							
Uroderma bilobatum	3	580.1	98.2	3.8	0.6	6.8	1.2
Vampyrops brachycephalus	2	554.1	98.5	3.3	0.6	5.3	0.9
Vampyrops helleri	3	492.7	98.3	3.5	0.7	4.8	1.0
Vampyrops infuscus	1	1045.2	97.6	11.1	1.0	15.1	1.4
Vampyrops lineatus	3	692.9	96.9	9.2	1.3	13.1	1.8
Vampyrops vittatus	2	1115.8	98.4	6.9	0.6	11.4	1.0
Vampyrodes caraccioloi	1	888.0	97.1	6.7	0.7	19.4	2.1
Vampyressa pusilla	2	359.9	98.6	1.5	0.4	3.5	1.0
Chiroderma salvini	4	770.1	98.0	5.7	0.7	9.9	1.3
Chiroderma trinitatum	2	507.7	98.0	3.8	0.7	6.8	1.3
Chiroderma villosum	3	666.1	97.7	4.9	0.7	10.5	1.5
Chiroderma villosum ssp.	1	788.4	97.8	5.6	0.7	12.0	1.5
Ectophylla macconnelli	2	327.1	98.5	1.8	0.5	3.2	1.0
Artibeus cinereus	3	443.5	98.4	2.9	0.6	4.4	1.0
Artibeus concolor	2	592.8	97.0	6.9	1.1	11.3	1.8
Artibeus jamaicensis	3	962.5	98.1	8.7	0.9	9.5	1.0
Artibeus lituratus	4	1161.9	97.6	12.9	1.1	15.4	1.3
Enchisthenes harti	4	478.7	97.0	6.2	1.3	8.4	1.7
Ardops nichollsi	3	526.3	96.5	6.7	1.2	12.4	2.3
Sphaeronycteris toxophyllum	1	496.1	98.1	3.4	0.7	6.3	1.2
BRACHYPHYLLINAE							
Brachyphylla cavernarum	4	1129.5	97.8	14.8	1.3	10.2	0.9

	n	NET mm³	NET %	VENT mm³	VENT %	REST mm³	REST %
DESMODONTIDAE							
Desmodus rotundus	3	944.4	97.9	8.8	0.9	11.1	1.2
Diphylla ecaudata	4	758.1	98.4	5.2	0.7	6.9	0.9
NATALIDAE							
Natalus tumidirostris	3	233.4	98.7	1.2	0.5	1.9	0.8
FURIPTERIDAE							
Furipterus horrens	5	120.0	97.9	0.6	0.5	1.9	1.6
VESPERTILIONIDAE							
VESPERTILIONINAE							
Myotis adversus	2	238.3	97.2	3.7	1.5	3.2	1.3
Myotis albescens	3	119.6	98.4	0.8	0.6	1.2	1.0
Myotis altarium	3	226.1	97.6	2.5	1.1	3.1	1.3
Myotis annectans	2	207.5	97.3	1.7	0.8	4.1	1.9
Myotis bechsteini	1	248.9	97.3	2.8	1.1	4.2	1.6
Myotis bocagei	3	179.2	97.7	1.5	0.8	2.7	1.5
Myotis dasycneme	1	291.0	97.3	2.8	0.9	5.4	1.8
Myotis montivagus	3	157.8	96.2	2.4	1.4	3.9	2.4
Myotis myotis	3	456.1	97.4	3.4	0.7	8.6	1.8
Myotis nattereri	1	208.3	98.1	1.8	0.9	2.2	1.1
Myotis nigricans	3	129.1	98.3	0.7	0.5	1.5	1.1
Myotis nigricans ssp.	3	147.0	97.6	1.3	0.8	2.3	1.5
Myotis siligorensis	3	96.6	97.2	1.0	1.0	1.8	1.8
Pipistrellus babu	3	90.6	96.8	1.1	1.2	1.9	2.0
Pipistrellus ceylonicus	3	168.6	96.0	2.4	1.3	4.7	2.7
Pipistrellus circumdatus	3	214.0	96.4	2.8	1.3	5.2	2.3
Pipistrellus crassulus	3	125.2	98.3	0.6	0.4	1.7	1.3
Pipistrellus imbricatus	2	129.2	97.7	1.0	0.7	2.1	1.6
Pipistrellus javanicus	3	111.5	97.0	1.4	1.2	2.1	1.8
Pipistrellus mimus	3	88.1	97.1	0.8	0.9	1.9	2.0
Pipistrellus nanus	3	94.5	97.9	0.6	0.6	1.4	1.4
Pipistrellus papuanus §)	2	106.4	97.6	1.0	0.9	1.7	1.5
Pipistrellus pulveratus	3	133.7	96.9	1.4	1.0	2.9	2.1
Pipistrellus subflavus	2	117.2	97.1	1.3	1.1	2.2	1.8
Scotozous dormeri	3	133.4	96.7	1.4	1.0	3.2	2.3
Nyctalus noctula	3	338.3	96.3	4.9	1.4	8.1	2.3
Nyctalus stenopterus	2	228.9	97.2	1.7	0.7	5.0	2.1
Glischropus tylopus	3	94.1	97.4	0.7	0.7	1.8	1.9
Eptesicus brasiliensis	3	189.0	97.9	1.6	0.8	2.4	1.3
Eptesicus br. melanopterus	3	183.3	97.4	1.8	1.0	3.1	1.7
Eptesicus flavescens	2	179.5	97.9	1.2	0.7	2.7	1.5
Eptesicus fuscus	1	224.6	97.8	1.7	0.7	3.4	1.5
Eptesicus pumilus	2	115.2	97.8	0.6	0.5	2.0	1.7
Eptesicus s. andersoni	3	320.9	96.6	4.4	1.3	6.8	2.0
Ia io	1	719.3	97.9	6.1	0.8	9.1	1.2

	n	NET mm³	NET %	VENT mm³	VENT %	REST mm³	REST %
VESPERTILIONIDAE (cont.)							
VESPERTILIONINAE (cont.)							
Tylonycteris pachypus	4	75.5	97.8	0.4	0.5	1.4	1.8
Tylonycteris robustula	3	108.5	97.7	0.8	0.7	1.8	1.6
Hesperoptenus blanfordi	3	148.0	97.0	1.4	0.9	3.1	2.0
Glauconycteris humeralis	3	136.4	98.1	1.0	0.7	1.6	1.2
Glauconycteris poensis	1	194.4	98.2	1.2	0.6	2.2	1.1
Chalinolobus gouldi	2	224.8	97.0	2.1	0.9	4.8	2.1
Chalinolobus morio	2	174.0	97.9	0.9	0.5	2.7	1.5
Scotorepens sanborni	1	149.1	97.7	0.8	0.6	2.6	1.7
Rhogeessa parvula	4	106.0	98.1	0.6	0.5	1.5	1.4
Scotomanes ornatus	2	358.7	96.8	5.1	1.4	6.9	1.9
Scotomanes sp.?	3	315.8	96.8	5.3	1.6	5.2	1.6
Scotophilus dingani	3	412.2	97.7	4.1	1.0	5.5	1.3
Scotophilus heathi	3	444.8	96.8	6.7	1.5	7.9	1.7
Scotophilus kuhli	3	320.6	97.4	3.8	1.2	4.7	1.4
Lasiurus borealis	1	161.4	98.3	0.5	0.3	2.2	1.3
MINIOPTERINAE							
Miniopterus australis	2	194.0	98.5	0.7	0.4	2.2	1.1
Miniopterus haradei	3	208.6	96.9	2.7	1.3	3.9	1.8
Miniopterus inflatus	3	307.8	98.1	2.3	0.7	3.6	1.1
Miniopterus magnater	3	305.7	97.8	2.2	0.7	4.8	1.5
Miniopterus medius	3	250.3	97.9	1.9	0.7	3.6	1.4
Miniopterus pusillus	3	204.7	98.2	1.2	0.6	2.6	1.2
M. schreibersi arenarius	3	229.4	98.2	1.6	0.7	2.6	1.1
Miniopterus s. blepotis	2	291.6	97.8	2.3	0.8	4.4	1.5
Miniopterus tristis	3	313.3	97.5	3.3	1.0	4.9	1.5
MURININAE							
Murina cyclotis cyclotis	2	235.8	98.1	1.5	0.6	3.1	1.3
Murina c. peninsularis	2	265.6	97.9	1.7	0.6	3.9	1.4
Murina huttoni rubella	3	243.0	97.6	2.0	0.8	3.9	1.6
Murina tubinaris	2	205.1	97.9	1.4	0.7	2.9	1.4
Harpiocephalus harpia	1	417.7	97.9	3.9	0.9	5.0	1.2
KERIVOULINAE							
Kerivoula papillosa	1	295.1	98.6	1.6	0.5	2.5	0.8
Kerivoula pellucida	1	192.1	98.0	0.7	0.4	2.2	1.6
Kerivoula phalaena	1	110.3	98.6	0.5	0.5	1.1	1.0
Phoniscus atrox	1	178.9	98.6	0.7	0.4	1.9	1.1
NYCTOPHILINAE							
Nyctophilus geoffroyi	2	156.9	97.3	1.7	1.0	2.7	1.7
Nyctophilus timoriensis	2	240.6	98.1	1.1	0.5	3.4	1.4

	n	NET mm^3	NET %	VENT mm^3	VENT %	REST mm^3	REST %
MOLOSSIDAE							
Tadarida aegyptiaca	3	369.6	97.4	4.7	1.2	5.1	1.3
Tadarida condylura	3	430.7	97.6	4.5	1.0	6.0	1.4
Tadarida mops	3	447.2	97.5	5.4	1.2	5.9	1.3
Tadarida niveiventer	3	407.0	97.6	5.4	1.3	4.6	1.1
Tadarida beccarii	2	219.1	97.4	1.8	0.8	4.0	1.8
Tadarida jobensis	1	339.6	97.7	3.4	1.0	4.5	1.3
Tadarida leucostigma	1	369.1	97.6	4.4	1.1	4.9	1.3
Tadarida plicata plicata	3	306.1	97.6	3.3	1.0	4.3	1.4
Tadarida pumila	3	228.9	97.6	2.8	1.2	2.9	1.2
Otomops martiensseni	3	714.9	98.0	8.6	1.2	6.3	0.9
Molossops abrasus	3	391.8	97.1	4.7	1.2	7.0	1.7
Molossops greenhalli	1	282.8	97.7	2.0	0.7	4.8	1.7
Molossops planirostris	2	174.2	97.6	1.3	0.7	3.0	1.7
Eumops auripendulus	3	575.7	97.4	7.0	1.2	8.1	1.4
Eumops glaucinus	2	618.0	98.0	3.9	0.6	8.4	1.3
Molossus ater	3	497.9	98.1	3.4	0.7	6.4	1.2
Molossus molossus	3	298.3	97.5	3.5	1.1	4.1	1.4
Molossus trinitatis	2	411.7	98.0	3.6	0.9	4.6	1.1
Cheiromeles torquatus	3	1275.4	97.0	14.6	1.1	24.7	1.9

Abbreviations and notes:

NET net volume of brain
REST comprising rests of meninges, stumps of nerves, hypophysis, etc.
VENT ventricles

§) see addendum at end of Volume 3, pp. 1592-1595

n = number of individuals

Table 9. Average percentages of NET brain, ventricles (VENT), and rests (REST) relative to standard brain volumes (BrV = 100%)

For volumes and percentages see Table 8

	N	NET	VENT	REST
PTEROPODIDAE	47	97.6	1.2	1.2
PTEROPODINAE *	34	97.4	1.3	1.3
Rousettus	3	97.3	1.0	1.7
Pteropus	12	97.2	1.6	1.2
Dobsonia	4	97.6	1.2	1.2
Cynopterus	3	97.9	1.0	1.1
NYCTIMENINAE *	4	97.8	1.1	1.1
MACROGLOSSINAE *	9	98.3	0.8	0.9
RHINOPOMATIDAE *	2	97.8	0.9	1.3
EMBALLONURIDAE	18	98.0	0.8	1.2
EMBALLONURINAE *	17	98.0	0.8	1.2
NON-*TAPHOZOUS* EMB.	11	98.1	0.7	1.2
Emballonura	3	98.1	0.7	1.2
Saccopteryx	3	98.1	0.6	1.3
TAPHOZOUS	6	97.8	0.9	1.3
DICLIDURINAE *	1	98.1	0.8	1.1
CRASEONYCTERIDAE *	1	97.3	0.7	2.0
NYCTERIDAE *	8	98.5	0.5	1.0
MEGADERMATIDAE *	5	97.8	0.8	1.4
RHINOLOPHIDAE *	15	97.8	0.8	1.4
HIPPOSIDERIDAE	24	97.5	1.0	1.5
HIX: *H. commersoni* excluded *	23	97.5	1.0	1.5
Hipposideros	20	97.5	1.0	1.5
H. commersoni excluded	19	97.5	1.0	1.5
NOCTILIONIDAE *	2	97.2	1.2	1.6
MORMOOPIDAE *	4	98.0	0.7	1.3
Pteronotus	3	97.9	0.7	1.4
PHYLLOSTOMIDAE	53	97.8	0.8	1.4
PHYLLOSTOMINAE *	15	97.7	0.8	1.5
Micronycteris	4	97.7	0.7	1.6
Tonatia	3	97.4	0.8	1.8
Phyllostomus	3	97.9	0.7	1.4
GLOSSOPHAGINAE *	11	98.0	0.8	1.2
CAROLLIINAE *	3	97.6	0.9	1.5
STURNIRINAE *	3	97.5	1.1	1.4
STENODERMINAE *	20	97.8	0.8	1.4
Vampyrops	5	97.9	0.9	1.2
Chiroderma	4	97.9	0.7	1.4
Artibeus	4	97.8	0.9	1.3
BRACHYPHYLLINAE *	1	97.8	1.3	0.9

	N	NET	VENT	REST
DESMODONTIDAE *	2	98.2	0.8	1.0
NATALIDAE *	1	98.7	0.5	0.8
FURIPTERIDAE *	1	97.9	0.5	1.6
VESPERTILIONIDAE	70	97.6	0.8	1.6
VESPERTILIONINAE	50	97.4	0.9	1.7
VEX: *Tylonycteris* excluded*	48	97.4	0.9	1.7
Myotis	13	97.5	0.9	1.6
Pipistrellus	11	97.1	1.0	1.9
Eptesicus	6	97.6	0.8	1.6
Tylonycteris	2	97.7	0.6	1.7
Scotophilus	3	97.3	1.2	1.5
MINIOPTERINAE *	9	97.8	0.8	1.4
MURININAE *	5	97.9	0.7	1.4
KERIVOULINAE *	4	98.5	0.4	1.1
NYCTOPHILINAE *	2	97.7	0.8	1.5
MOLOSSIDAE *	18	97.6	1.0	1.4
Tadarida	8	97.6	1.1	1.3
Molossops	3	97.4	0.9	1.7
Molossus	3	97.9	0.9	1.2
CHIROPTERA (AvCS)	272	97.7	0.9	1.4
CHIROPTERA (AvCF)	NN=28	97.9	0.8	1.3
MEGACHIR. (AvMegS)	47	97.6	1.2	1.2
MEGACHIR. (AvMegF)	NN= 3	97.8	1.1	1.1
MICROCHIR. (AvMicS)	225	97.7	0.9	1.4
MICROCHIR. (AvMicF)	NN=25	97.9	0.8	1.3

Number of species = N
Number of families and/or subfamilies marked by * = NN
Abbreviations:

AvCF	average of Chiroptera families and/or subfamilies
AvCS	average of Chiroptera species
AvMegF	average of Megachiroptera families and/or subfamilies
AvMegS	average of Megachiroptera species
AvMicF	average of Microchiroptera families and/or subfamilies
AvMicS	average of Microchiroptera species
NET	net volume of brain
REST	comprising rest of meninges, stumps of nerves, hypophysis, etc.
VENT	ventricles

Table 10. Volumes of the five fundamental brain parts (in mm^3)

For size indices in species see Table 11
For average size indices in supraspecific taxonomic units see Table 12

	n	OBL	MES	CER	DIE	TEL
PTEROPODIDAE						
PTEROPODINAE						
Eidolon helvum	3	273.7	200.7	463.3	366.5	2714
Rousettus aegyptiacus	4	152.1	109.1	288.2	192.4	1371
R. amplexicaud. brachyotis §)	3	93.3	68.0	164.2	110.7	842.7
R. ampl. stresemanni §)	3	126.9	95.9	215.2	137.1	1050
Myonycteris torquata	3	82.2	66.7	155.8	99.2	693.2
Pteropus alecto	2	392.1	316.3	843.7	574.1	4492
Pteropus conspicillatus	1	413.1	334.6	989.4	685.4	5426
Pteropus hypomelanus	1	317.6	239.3	614.8	433.2	3373
Pteropus lylei	4	325.9	240.6	711.4	514.2	3909
Pteropus mahaganus	2	317.2	223.3	720.5	406.5	3313
Pteropus neohibernicus	2	477.2	361.6	1007.6	720.0	5920
Pteropus poliocephalus	2	431.6	337.1	880.3	607.8	4505
Pteropus samoensis	1	326.0	250.9	723.8	468.6	3680
Pteropus scapulatus	2	284.3	236.1	682.0	436.7	3435
Pteropus temmincki	2	273.0	225.3	578.1	419.0	3112
Pteropus tonganus	3	362.9	248.0	731.8	516.1	3871
Pteropus vampyrus	2	448.7	380.7	1019.9	687.3	5964
Dobsonia inermis	3	172.8	110.0	357.0	201.2	1714
D. moluccensis magna §)	2	327.3	208.5	733.9	409.0	3339
D. molucc. anderseni §)	3	213.1	156.3	482.7	287.8	2476
Dobsonia praedatrix	3	193.2	127.1	375.5	239.6	1901
Hypsignathus monstrosus	3	282.7	216.1	496.5	299.5	1954
Epomops franqueti	3	173.5	138.4	318.5	192.4	1254
Epomophorus labiatus	3	114.6	92.5	206.9	136.3	939.2
Micropteropus pusillus	3	63.5	48.4	108.1	67.7	491.8
Scotonycteris zenkeri	2	56.4	42.0	89.3	58.1	423.8
Casinycteris argynnis	5	69.3	54.2	116.7	71.5	479.3
Cynopterus brachyotis	3	68.3	51.4	122.7	78.1	606.3
Cynopterus horsfieldi	3	92.7	74.8	180.5	109.1	843.9
Cynopterus sphinx sphinx	3	83.5	66.0	158.6	94.7	727.1
Megaerops ecaudatus	2	53.9	44.1	89.9	65.7	502.3
Chironax melanocephalus	3	44.3	31.3	72.9	48.0	375.9
Sphaerias blanfordi	4	69.2	49.4	114.8	67.9	476.2
Balionycteris maculata	2	38.9	29.7	57.7	40.6	317.5

Abbreviations and notes:

CER cerebellum; for volumes of components see Table 28
DIE diencephalon; for volumes of components see Table 30
MES mesencephalon; for volumes of components see Table 25, 30
OBL medulla oblongata; for volumes of components see Tables 13, 16, 19, 22
TEL telencephalon; for volumes of components see Table 32, 35

§) see addendum at end of Volume 3, pp. 1592-1595; n = number of individuals

	n	OBL	MES	CER	DIE	TEL
PTEROPODIDAE (cont.)						
NYCTIMENINAE						
Nyctimene albiventer	2	61.2	47.3	111.9	65.6	493.0
Nyctimene robinsoni	2	83.7	72.3	162.0	94.8	749.0
Nyctimene vizcaccia	3	71.0	56.0	135.6	79.4	591.7
Paranyctimene raptor	3	51.6	42.3	101.2	57.8	437.6
MACROGLOSSINAE						
Eonycteris spelaea	3	95.7	71.4	172.7	105.5	794.6
Megaloglossus woermanni	3	51.2	38.3	103.2	53.5	398.7
Macroglossus minimus	4	41.3	33.7	79.2	46.6	329.9
Macroglossus sobrinus	3	52.2	43.2	94.2	61.9	406.2
Syconycteris a. australis §)	1	37.9	31.4	64.9	45.2	361.6
S. australis papuana §)	3	46.1	34.3	69.3	51.5	394.0
Melonycteris melanops	3	83.1	60.6	151.1	105.4	825.5
Nesonycteris woodfordi	3	71.3	48.3	133.6	81.4	630.9
Notopteris macdonaldi	3	106.5	66.2	211.3	118.2	882.9
RHINOPOMATIDAE						
Rhinopoma hardwickei	3	35.8	30.3	42.6	24.6	126.8
Rhinopoma microphyllum	3	51.3	40.3	60.4	32.1	180.5
EMBALLONURIDAE						
EMBALLONURINAE						
Emballonura monticola	3	21.0	20.0	31.8	14.8	69.8
Emballonura raffrayana	3	23.1	21.4	31.2	14.9	74.4
Emballonura semicaudata	3	25.7	19.9	36.0	14.1	78.0
Coleura afra	3	33.1	26.3	49.1	20.4	114.9
Rhynchonycteris naso	3	15.7	14.6	22.6	9.4	49.3
Saccopteryx bilineata	4	27.2	28.6	37.8	18.6	104.2
Saccopteryx canescens	1	17.3	17.6	21.4	11.0	55.5
Saccopteryx leptura	3	21.1	22.2	27.4	13.3	68.5
Cormura brevirostris	3	27.4	27.3	38.9	18.4	94.7
Peropteryx macrotis	3	20.3	19.3	30.4	12.9	71.1
Peropteryx trinitatis	3	19.4	17.4	22.8	11.5	58.5
Taphozous australis	2	65.8	54.6	104.1	36.8	227.6
Taphozous hildegardeae	3	68.5	53.1	114.9	37.9	240.2
Taphozous mauritianus	2	70.4	64.7	114.0	42.3	230.9
Taphozous melanopogon	3	67.0	51.9	112.7	38.3	236.7
Taphozous theobaldi	3	96.5	66.5	152.7	48.2	278.4
Taphozous saccolaimus	3	78.7	72.9	130.6	51.6	297.4
DICLIDURINAE						
Cyttarops alecto	1	22.7	21.4	45.4	13.7	62.5
CRASEONYCTERIDAE						
Craseonycteris thonglongyai	3	13.4	10.6	13.1	7.5	37.1

	n	OBL	MES	CER	DIE	TEL
NYCTERIDAE						
Nycteris arge	2	40.9	37.5	62.0	25.3	165.3
Nycteris grandis	4	80.3	72.8	139.0	53.2	328.8
Nycteris hispida	1	30.5	30.1	55.2	20.8	133.7
Nycteris javanica	1	55.5	54.1	90.2	36.3	211.2
Nycteris macrotis	3	47.7	38.5	70.9	30.5	180.2
Nycteris nana	3	27.8	24.9	47.9	17.9	107.7
Nycteris thebaica	3	39.6	32.8	58.0	24.6	152.6
Nycteris tragata	2	51.1	45.1	96.2	33.6	208.1
MEGADERMATIDAE						
Megaderma lyra	6	105.3	80.5	162.5	73.0	434.9
Megaderma spasma	3	66.7	61.9	117.8	47.7	313.5
Macroderma gigas	3	186.7	144.5	301.4	137.8	833.0
Cardioderma cor	3	71.5	66.6	111.7	54.6	326.6
Lavia frons	2	67.7	66.0	138.8	50.3	288.6
RHINOLOPHIDAE						
Rhinolophus cornutus	2	23.8	18.0	40.5	14.1	87.0
Rhinolophus eloquens	2	50.0	36.1	80.6	29.9	185.6
Rhinolophus hipposideros	2	23.0	15.2	30.4	12.4	59.4
Rhinolophus landeri	3	35.3	27.9	56.3	21.1	121.9
Rhinolophus lepidus	3	25.3	19.1	32.0	14.1	84.3
Rhinolophus luctus	3	67.5	56.5	141.6	46.4	271.5
Rhinolophus macrotis	2	28.6	20.2	42.4	14.9	89.9
Rhinolophus malayanus	2	27.3	20.4	40.2	14.7	88.2
Rhinolophus megaphyllus	2	32.0	24.1	52.4	17.9	120.8
Rhinolophus megaphyllus ssp.	3	28.2	22.1	43.4	18.4	105.6
Rhinolophus paradoxolophus	1	34.4	25.1	53.1	18.5	119.0
Rhinolophus pearsoni	1	38.9	29.8	63.9	25.5	155.6
Rhinolophus pusillus	3	21.7	16.0	29.9	12.4	75.1
Rhinolophus trifoliatus	3	42.8	38.7	70.0	28.1	151.1
Rhinolophus yunanensis	2	51.5	41.1	80.3	32.8	207.3
HIPPOSIDERIDAE						
Hipposideros armiger	3	105.8	77.7	142.2	56.1	369.2
Hipposideros ater	2	28.5	23.4	41.0	15.6	93.5
Hipposideros ater aruensis	2	26.5	22.5	33.2	14.7	82.4
Hipposideros ater saevus	3	22.9	18.6	35.3	12.3	80.2
Hipposideros caffer	3	39.1	30.0	64.2	21.2	135.6
H. calcaratus calcaratus §)	3	47.9	37.2	78.6	29.6	186.0
H. calcaratus cupidus §)	3	36.2	30.9	66.8	23.0	159.6
Hipposideros cervinus	3	29.0	22.3	42.0	15.5	92.6
Hipposideros commersoni	3	86.5	58.7	128.4	54.2	374.8
Hipposideros diadema	4	87.7	69.2	123.8	53.4	335.1
Hipposideros fulvus	2	33.8	25.2	41.7	17.8	104.0

	n	OBL	MES	CER	DIE	TEL
HIPPOSIDERIDAE (cont.)						
Hipposideros halophyllus	2	24.5	16.9	28.3	10.5	62.2
Hipposideros lankadiva	3	93.0	62.9	127.3	53.4	345.5
Hipposideros larvatus	3	50.5	40.7	69.7	30.9	198.0
Hipposideros lekaguli	2	74.0	53.5	121.9	38.8	257.7
H. magg. maggietaylorae §)	1	69.1	48.7	119.4	44.4	316.6
H. maggietaylorae erroris §)	3	54.9	40.2	86.7	33.1	208.4
Hipposideros ridleyi	3	36.2	28.4	47.1	21.1	120.3
Hipposideros speoris	2	41.7	28.7	50.7	20.4	126.5
Hipposideros turpis	3	73.3	57.3	110.0	42.6	278.9
Aselliscus stoliczkanus	3	22.7	14.2	34.3	9.8	59.1
Aselliscus tricuspidatus	3	19.7	14.0	23.1	9.8	59.1
Rhinonycteris aurantius	3	31.1	25.3	50.2	17.7	110.0
Triaenops persicus	3	38.7	26.0	66.1	17.8	108.7
NOCTILIONIDAE						
Noctilio albiventris	3	66.0	49.0	96.9	46.2	303.9
Noctilio leporinus	3	124.4	89.8	194.1	88.5	608.6
MORMOOPIDAE						
Pteronotus gymnonotus	3	44.5	25.7	70.4	22.9	159.6
Pteronotus personatus	3	28.7	18.9	45.3	15.6	107.5
Pteronotus parnelli	3	62.7	40.9	113.0	37.4	259.9
Mormoops megalophylla	3	46.0	30.8	105.7	24.4	158.4
PHYLLOSTOMIDAE						
PHYLLOSTOMINAE						
Micronycteris megalotis	3	31.1	23.4	49.4	19.8	135.0
Micronycteris minuta	1	28.4	20.7	54.8	18.0	148.0
Micronycteris schmidtorum	3	30.1	23.1	57.5	21.1	158.3
Micronycteris brachyotis	1	44.0	28.6	76.3	27.8	206.7
Macrophyllum macrophyllum	3	38.9	28.6	55.1	24.2	154.6
Tonatia bidens	3	78.8	63.3	133.6	53.7	415.7
Tonatia schulzi	1	59.5	51.3	84.2	37.0	243.3
Tonatia sylvicola	1	86.1	61.9	126.1	55.3	384.0
Mimon crenulatum	2	35.4	30.2	53.7	26.0	162.1
Phyllostomus discolor	3	91.2	63.4	170.4	82.4	623.6
Phyllostomus elongatus	3	91.9	64.0	171.1	64.6	440.0
Phyllostomus hastatus	3	140.2	95.7	281.3	111.0	810.4
Phylloderma stenops	2	98.8	71.1	204.1	89.7	804.0
Trachops cirrhosus	2	97.4	81.6	181.3	67.4	521.4
Vampyrum spectrum	2	233.5	154.9	532.7	196.7	1331.6

	n	OBL	MES	CER	DIE	TEL
PHYLLOSTOMIDAE (cont.)						
GLOSSOPHAGINAE						
Glossophaga longirostris	1	42.1	27.8	92.4	31.8	220.3
Glossophaga soricina	3	37.9	25.8	81.6	28.5	201.1
Monophyllus plethodon	2	45.2	28.5	89.0	31.7	222.9
Leptonycteris curasoae	1	62.2	35.2	119.4	44.0	315.0
Leptonycteris nivalis	3	56.0	35.9	104.4	43.3	310.5
Lonchophylla mordax	1	45.3	31.2	79.2	32.9	221.4
Lonchophylla thomasi	1	31.2	21.5	61.6	23.6	178.4
Lionycteris spurrelli	3	35.0	24.7	70.6	23.7	180.8
Anoura caudifer	3	39.0	24.8	74.2	29.9	214.7
Anoura geoffroyi	3	52.0	34.5	109.7	41.5	320.8
Choeroniscus minor	1	35.2	24.1	78.3	29.8	200.5
CAROLLIINAE						
Carollia castanea	5	43.2	30.4	71.1	32.4	247.9
Carollia perspicillata	3	49.1	36.4	92.1	37.8	301.7
Rhinophylla pumilio	3	31.8	22.7	52.0	23.7	204.0
STURNIRINAE						
Sturnira lilium	3	50.9	33.2	96.1	45.4	358.4
Sturnira ludovici	2	63.4	40.1	108.4	50.4	368.6
Sturnira tildae	3	54.2	36.5	103.6	50.5	417.0
STENODERMINAE						
Uroderma bilobatum	3	50.9	41.1	103.0	43.2	341.9
Vampyrops brachycephalus	2	50.5	39.2	116.9	40.2	283.4
Vampyrops helleri	3	44.3	34.5	94.8	35.6	283.4
Vampyrops infuscus	1	99.7	70.6	212.7	79.2	583.1
Vampyrops lineatus	3	60.8	44.4	130.8	50.7	406.2
Vampyrops vittatus	2	94.1	67.1	217.4	79.5	657.8
Vampyrodes caraccioloi	1	90.3	70.8	164.5	71.3	491.1
Vampyressa pusilla	2	32.4	25.9	66.5	27.0	207.9
Chiroderma salvini	4	71.0	55.1	153.9	59.4	430.7
Chiroderma trinitatum	2	45.7	35.4	97.6	38.4	290.7
Chiroderma villosum	3	62.3	51.6	132.5	49.8	369.9
Chiroderma villosum ssp.	1	80.9	66.2	158.2	61.6	421.5
Ectophylla macconnelli	2	33.4	24.3	62.8	24.5	182.1
Artibeus cinereus	3	41.5	29.7	81.6	32.6	258.2
Artibeus concolor	2	55.2	43.1	114.4	44.6	335.5
Artibeus jamaicensis	3	83.7	58.9	173.8	72.6	573.4
Artibeus lituratus	4	94.7	68.9	221.9	85.7	690.7
Enchisthenes harti	4	44.7	32.4	81.2	37.8	282.5
Ardops nichollsi	3	49.7	33.7	84.8	40.6	317.5
Sphaeronycteris toxophyllum	1	66.6	41.7	82.7	41.2	263.8
BRACHYPHYLLINAE						
Brachyphylla cavernarum	4	106.1	68.3	176.7	96.4	682.1

	n	OBL	MES	CER	DIE	TEL
DESMODONTIDAE						
Desmodus rotundus	3	80.0	58.3	166.4	74.9	564.8
Diphylla ecaudata	4	63.1	53.0	132.0	63.6	446.4
NATALIDAE						
Natalus tumidirostris	3	26.6	19.2	40.0	18.3	129.3
FURIPTERIDAE						
Furipterus horrens	5	16.3	12.7	28.1	9.2	53.8
VESPERTILIONIDAE						
VESPERTILIONINAE						
Myotis adversus	2	35.3	26.1	44.0	19.9	113.0
Myotis albescens	3	18.1	12.9	24.3	10.0	54.3
Myotis altarium	3	34.0	23.3	44.2	18.2	106.5
Myotis annectans	2	30.8	20.4	36.3	17.9	102.1
Myotis bechsteini	1	39.4	22.8	42.5	22.6	121.5
Myotis bocagei	3	24.8	20.1	33.7	15.8	84.9
Myotis dasycneme	1	49.3	32.6	49.8	26.4	133.0
Myotis montivagus	3	24.7	18.1	29.2	13.7	72.1
Myotis myotis	3	70.3	46.7	87.1	40.2	211.8
Myotis nattereri	1	31.6	20.5	37.4	17.1	101.7
Myotis nigricans	3	17.1	14.9	23.8	10.0	63.2
Myotis nigricans ssp.	3	20.3	16.3	27.7	12.4	70.3
Myotis siligorensis	3	15.3	11.1	17.5	8.7	44.1
Pipistrellus babu	3	15.7	10.7	16.8	7.7	39.6
Pipistrellus ceylonicus	3	26.7	19.2	30.7	13.5	78.6
Pipistrellus circumdatus	3	30.4	23.3	40.7	17.5	102.2
Pipistrellus crassulus	3	19.9	14.6	21.1	12.0	57.5
Pipistrellus imbricatus	2	21.6	16.3	20.3	13.7	57.3
Pipistrellus javanicus	3	18.2	13.2	19.5	9.4	51.2
Pipistrellus mimus	3	15.1	10.4	15.6	7.6	39.3
Pipistrellus nanus	3	15.2	10.2	18.5	7.9	42.8
Pipistrellus papuanus §)	3	19.3	13.6	18.3	9.4	45.7
Pipistrellus pulveratus	3	20.8	15.4	22.3	11.6	63.6
Pipistrellus subflavus	2	18.3	13.1	21.3	10.4	54.2
Scotozous dormeri	3	25.3	15.2	22.9	11.7	58.4
Nyctalus noctula	3	57.6	37.7	80.5	25.4	137.1
Nyctalus stenopterus	2	37.2	26.5	40.6	22.0	102.6
Glischropus tylopus	3	16.3	11.2	18.0	7.8	40.7
Eptesicus brasiliensis	3	26.1	21.7	36.1	15.3	89.9
Eptesicus br. melanopterus	3	28.1	19.6	36.0	16.1	83.5
Eptesicus flavescens	2	29.9	21.2	34.2	15.1	79.2
Eptesicus fuscus	1	31.1	22.6	38.7	19.0	113.2
Eptesicus pumilus	2	18.8	12.7	21.5	9.7	52.4
Eptesicus serotinus andersoni	3	47.1	32.6	55.2	26.8	159.3
Ia io	1	95.3	70.4	145.6	59.4	348.7

	n	OBL	MES	CER	DIE	TEL
VESPERTILIONIDAE (cont.)						
VESPERTILIONINAE (cont.)						
Tylonycteris pachypus	4	14.9	9.7	11.7	6.7	32.6
Tylonycteris robustula	3	19.8	13.4	20.6	9.2	45.4
Hesperoptenus blanfordi	3	24.8	19.1	25.0	12.7	66.3
Glauconycteris humeralis	3	21.6	16.9	31.3	10.4	56.2
Glauconycteris poensis	1	30.3	22.9	43.3	15.2	82.7
Chalinolobus gouldi	2	33.7	23.6	49.9	17.0	100.6
Chalinolobus morio	2	28.0	19.3	34.8	13.7	78.2
Scotorepens sanborni	1	23.6	16.8	27.0	13.8	68.0
Rhogeessa parvula	4	17.3	12.9	18.0	9.3	48.6
Scotomanes ornatus	2	55.0	43.4	55.0	33.0	172.3
Scotomanes spec?	3	43.1	34.8	57.6	26.5	153.9
Scotophilus dingani	3	58.2	45.5	81.5	35.0	192.1
Scotophilus heathi	3	65.6	49.9	89.0	36.4	203.9
Scotophilus kuhli	3	48.8	37.6	58.2	28.7	147.4
Lasiurus borealis	1	23.4	18.7	40.9	12.5	65.8
MINIOPTERINAE						
Miniopterus australis	2	27.2	18.0	39.7	14.6	94.6
Miniopterus haradei	3	30.3	19.5	45.7	15.9	97.2
Miniopterus inflatus	3	43.8	27.4	79.5	21.9	135.2
Miniopterus magnater	3	41.6	26.9	73.3	21.8	142.1
Miniopterus medius	3	37.2	24.3	61.1	18.3	109.5
Miniopterus pusillus	3	28.6	18.2	47.1	15.8	95.0
M. schreibersi arenarius	3	31.7	20.5	49.8	17.2	110.2
Miniopterus s. blepotis	2	41.3	26.6	73.5	20.4	129.8
Miniopterus tristis	3	44.2	30.0	71.7	23.1	144.2
MURININAE						
Murina cyclotis cyclotis	2	28.8	21.2	39.9	18.3	127.7
Murina c. peninsularis	2	29.9	21.3	48.0	19.5	146.9
Murina huttoni rubella	3	27.0	20.4	40.8	18.9	136.0
Murina tubinaris	2	23.6	16.1	35.7	15.9	113.9
Harpiocephalus harpia	1	43.1	30.3	68.7	31.7	243.9
KERIVOULINAE						
Kerivoula papillosa	1	34.2	22.9	65.9	21.4	150.8
Kerivoula pellucida	1	23.9	19.9	36.5	14.6	97.1
Kerivoula phalaena	1	17.3	13.1	23.2	9.1	47.8
Phoniscus atrox	1	18.3	13.3	29.5	12.1	105.6
NYCTOPHILINAE						
Nyctophilus geoffroyi	2	21.6	15.5	29.2	12.6	78.1
Nyctophilus timoriensis	2	31.3	26.7	43.9	19.2	119.5

	n	OBL	MES	CER	DIE	TEL
MOLOSSIDAE						
Tadarida aegyptiaca	3	54.0	35.1	70.3	30.4	179.9
Tadarida condylura	3	58.6	41.5	74.4	37.9	218.2
Tadarida mops	3	60.6	43.0	74.6	43.3	225.7
Tadarida niveiventer	3	60.8	42.9	79.1	35.5	188.7
Tadarida beccarii	2	34.4	23.1	41.4	18.5	101.8
Tadarida jobensis	1	44.8	32.5	71.4	27.6	163.3
Tadarida leucostigma	1	55.9	39.7	70.0	28.2	175.3
Tadarida plicata plicata	3	46.3	31.2	61.4	24.4	142.8
Tadarida pumila	3	36.3	27.1	41.8	19.9	103.8
Otomops martiensseni	3	89.6	59.9	201.6	53.4	310.4
Molossops abrasus	3	58.7	44.9	69.7	34.3	184.2
Molossops greenhalli	1	45.9	34.8	48.8	25.5	127.8
Molossops planirostris	2	26.7	19.9	29.7	15.8	82.2
Eumops auripendulus	3	80.2	60.1	136.6	46.6	252.1
Eumops glaucinus	2	82.2	64.4	148.8	46.2	276.3
Molossus ater	3	66.0	53.8	93.8	40.8	243.6
Molossus molossus	3	42.6	32.8	58.7	23.5	140.7
Molossus trinitatis	2	59.0	43.4	85.3	33.2	190.8
Cheiromeles torquatus	3	161.6	111.6	213.4	108.2	680.6

Abbreviations and notes:

CER	cerebellum;	for volumes of components see Table 28
DIE	diencephalon;	for volumes of components see Table 30
MES	mesencephalon;	for volumes of components see Table 25, 30
OBL	medulla oblongata;	for volumes of components see Tables 13, 16, 19, 22
TEL	telencephalon;	for volumes of components see Table 32, 35

§) see addendum at end of Volume 3, pp. 1592-1595

n = number of individuals

Table 11. Size indices of the five fundamental brain parts related to the average of the four species of Tenrecinae (AvTn = 100)

For volumes in species see Table 10
For average size indices in supraspecific taxonomic units see Table 12

	n	OBL	MES	CER	DIE	TEL
slope		.61	.58	.68	.65	.68
y-intercept		.776	.481	.552	.350	1.242
PTEROPODIDAE						
PTEROPODINAE						
Eidolon helvum	3	154	262	295	439	353
Rousettus aegyptiacus	4	127	209	286	353	278
R. amplexicaud. brachyotis §)	3	135	219	300	364	315
R. ampl. stresemanni §)	3	137	235	285	331	284
Myonycteris torquata	3	154	274	379	429	345
Pteropus alecto	2	133	257	307	403	334
Pteropus conspicillatus	1	122	238	308	414	345
Pteropus hypomelanus	1	140	248	298	400	334
Pteropus lylei	4	164	283	400	547	449
Pteropus mahaganus	2	125	209	310	335	291
Pteropus neohibernicus	2	131	241	290	404	348
Pteropus poliocephalus	2	133	250	288	386	301
Pteropus samoensis	1	165	297	411	503	426
Pteropus scapulatus	2	128	251	340	414	350
Pteropus temmincki	2	158	303	380	517	417
Pteropus tonganus	3	149	241	329	444	356
Pteropus vampyrus	2	110	227	259	341	309
Dobsonia inermis	3	137	199	333	348	327
D. moluccensis magna §)	2	129	195	316	337	294
D. molucc. anderseni §)	3	129	220	335	375	351
Dobsonia praedatrix	3	134	204	304	361	314
Hypsignathus monstrosus	3	143	256	282	321	226
Epomops franqueti	3	157	285	345	383	277
Epomophorus labiatus	3	152	274	344	408	318
Micropteropus pusillus	3	149	247	339	372	315
Scotonycteris zenkeri	2	146	236	313	355	303
Casinycteris argynnis	5	155	265	348	375	292
Cynopterus brachyotis	3	133	220	314	353	316
Cynopterus horsfieldi	3	129	232	315	343	301

Abbreviations and notes:

CER cerebellum; for indices of components see Table 29
DIE diencephalon; for indices of components see Table 31
MES mesencephalon; for indices of components see Table 26, 31
OBL medulla oblongata; for indices of components see Tables 14, 17, 20, 23
TEL telencephalon; for indices of components see Tables 33, 35

§) see addendum at end of Volume 3, pp. 1592-1595; n = number of individuals

	n	OBL	MES	CER	DIE	TEL
slope		.61	.58	.68	.65	.68
y-intercept		.776	.481	.552	.350	1.242
PTEROPODIDAE (cont.)						
PTEROPODINAE (cont.)						
Cynopterus sphinx sphinx	3	126	221	304	326	285
Megaerops ecaudatus	2	130	230	291	372	331
Chironax melanocephalus	3	129	196	290	331	305
Sphaerias blanfordi	4	147	229	323	336	273
Balionycteris maculata	2	128	209	264	320	297
NYCTIMENINAE						
Nyctimene albiventer	2	129	219	313	323	281
Nyctimene robinsoni	2	134	256	331	346	312
Nyctimene vizcaccia	3	135	234	336	349	300
Paranyctimene raptor	3	118	211	309	310	273
MACROGLOSSINAE						
Eonycteris spelaea	3	134	222	304	334	285
Megaloglossus woermanni	3	148	238	407	367	321
Macroglossus minimus	4	135	235	359	364	305
Macroglossus sobrinus	3	133	239	325	373	286
Syconycteris a. australis §)	1	123	218	293	352	333
S. australis papuana §)	3	135	215	278	358	322
Melonycteris melanops	3	132	214	308	384	343
Nesonycteris woodfordi	3	136	201	332	358	320
Notopteris macdonaldi	3	133	185	328	332	280
RHINOPOMATIDAE						
Rhinopoma hardwickei	3	126	227	210	208	128
Rhinopoma microphyllum	3	109	187	170	159	104
EMBALLONURIDAE						
EMBALLONURINAE						
Emballonura monticola	3	127	251	287	224	129
Emballonura raffrayana	3	131	252	262	210	128
Emballonura semicaudata	3	132	213	269	178	119
Coleura afra	3	125	210	262	186	125
Rhynchonycteris naso	3	117	223	256	176	114
Saccopteryx bilineata	4	130	287	262	218	148
Saccopteryx canescens	1	138	286	261	222	138
Saccopteryx leptura	3	131	285	254	205	130
Cormura brevirostris	3	127	266	261	209	130
Peropteryx macrotis	3	117	230	258	184	123
Peropteryx trinitatis	3	136	250	241	203	126
Taphozous australis	2	159	286	336	208	150
Taphozous hildegardeae	3	152	257	339	197	145
Taphozous mauritianus	2	151	303	324	212	134
Taphozous melanopogon	3	167	280	378	225	162
Taphozous theobaldi	3	171	259	350	196	130
Taphozous saccolaimus	3	133	272	284	200	132

	n	OBL	MES	CER	DIE	TEL
slope		.61	.58	.68	.65	.68
y-intercept		.776	.481	.552	.350	1.242
EMBALLONURIDAE (cont.)						
DICLIDURINAE						
Cyttarops alecto	1	137	268	410	207	115
CRASEONYCTERIDAE						
Craseonycteris thonglongyai	3	126	202	194	183	112
NYCTERIDAE						
Nycteris arge	2	162	315	349	244	190
Nycteris grandis	4	170	336	388	262	187
Nycteris hispida	1	146	302	383	244	190
Nycteris javanica	1	157	330	349	244	167
Nycteris macrotis	3	156	269	323	240	168
Nycteris nana	3	143	268	362	228	166
Nycteris thebaica	3	175	305	368	265	198
Nycteris tragata	2	168	317	440	265	194
MEGADERMATIDAE						
Megaderma lyra	6	190	320	380	304	208
Megaderma spasma	3	167	335	396	281	216
Macroderma gigas	3	169	298	326	274	184
Cardioderma cor	3	164	333	342	293	204
Lavia frons	2	166	350	456	289	194
RHINOLOPHIDAE						
Rhinolophus cornutus	2	126	199	315	185	138
Rhinolophus eloquens	2	139	216	304	197	143
Rhinolophus hipposideros	2	152	207	302	205	120
Rhinolophus landeri	3	148	247	337	215	149
Rhinolophus lepidus	3	142	223	265	197	143
Rhinolophus luctus	3	134	247	370	215	145
Rhinolophus macrotis	2	156	230	340	201	147
Rhinolophus malayanus	2	147	230	319	197	143
Rhinolophus megaphyllus	2	133	212	311	181	147
Rhinolophus megaphyllus ssp.	3	144	237	324	232	161
Rhinolophus paradoxolophus	1	148	227	327	194	150
Rhinolophus pearsoni	1	131	214	300	206	149
Rhinolophus pusillus	3	142	216	293	203	150
Rhinolophus trifoliatus	3	140	270	317	220	140
Rhinolophus yunanensis	2	126	218	263	188	139
HIPPOSIDERIDAE						
Hipposideros armiger	3	152	249	258	183	137
Hipposideros ater	2	147	252	309	199	144
Hipposideros ater aruensis	2	157	276	292	216	148
Hipposideros ater saevus	3	160	267	373	216	173
Hipposideros caffer	3	152	246	353	200	152
Hipp. c. calcaratus §)	3	167	276	384	248	185
Hipp. calc. cupidus §)	3	153	275	403	237	196

	n	OBL	MES	CER	DIE	TEL
slope		.61	.58	.68	.65	.68
y-intercept		.776	.481	.552	.350	1.242
HIPPOSIDERIDAE (cont.)						
Hipposideros cervinus	3	144	233	305	190	137
Hipposideros commersoni	3	86	133	155	120	93
Hipposideros diadema	4	149	259	270	208	149
Hipposideros fulvus	2	173	270	312	224	159
Hipposideros halophyllus	2	176	250	309	190	139
Hipposideros lankadiva	3	146	219	255	191	141
Hipposideros larvatus	3	134	233	251	194	145
Hipposideros lekaguli	2	155	244	336	189	145
H. magg. maggietaylorae §)	1	163	249	376	246	204
H. maggietaylorae erroris §)	3	169	265	368	243	180
Hipposideros ridleyi	3	152	252	284	216	148
Hipposideros speoris	2	157	230	270	186	138
Hipposideros turpis	3	141	241	276	189	143
Aselliscus stoliczkanus	3	144	187	327	156	115
Aselliscus tricuspidatus	3	134	196	237	168	124
Rhinonycteris aurantius	3	140	240	326	195	146
Triaenops persicus	3	131	188	313	145	105
NOCTILIONIDAE						
Noctilio albiventris	3	132	214	254	214	163
Noctilio leporinus	3	175	281	344	282	220
MORMOOPIDAE						
Pteronotus gymnonotus	3	152	187	335	187	155
Pteronotus personatus	3	134	186	307	178	148
Pteronotus parnelli	3	168	236	411	237	193
Mormoops megalophylla	3	144	206	456	182	139
PHYLLOSTOMIDAE						
PHYLLOSTOMINAE						
Micronycteris megalotis	3	166	261	388	262	217
Micronycteris minuta	1	147	223	413	229	228
Micronycteris schmidtorum	3	144	232	399	248	224
Micronycteris brachyotis	1	143	199	344	217	190
Macrophyllum macrophyllum	3	176	272	358	267	205
Tonatia bidens	3	168	294	376	266	239
Tonatia schulzi	1	154	288	295	226	174
Tonatia sylvicola	1	173	273	333	258	207
Mimon crenulatum	2	132	239	281	233	173
Phyllostomus discolor	3	170	259	413	355	309
Phyllostomus elongatus	3	151	233	361	244	190
Phyllostomus hastatus	3	151	232	370	266	218
Phylloderma stenops	2	160	255	423	332	340
Trachops cirrhosus	2	181	333	437	289	257
Vampyrum spectrum	2	169	258	449	308	229

> Table 11 <

	n	OBL	MES	CER	DIE	TEL
slope		.61	.58	.68	.65	.68
y-intercept		.776	.481	.552	.350	1.242
PHYLLOSTOMIDAE (cont.)						
GLOSSOPHAGINAE						
Glossophaga longirostris	1	161	225	499	293	243
Glossophaga soricina	3	157	226	482	286	242
Monophyllus plethodon	2	145	196	396	244	203
Leptonycteris curasoae	1	148	182	380	246	205
Leptonycteris nivalis	3	139	193	350	253	212
Lonchophylla mordax	1	179	261	444	315	253
Lonchophylla thomasi	1	161	232	465	301	275
Lionycteris spurrelli	3	171	252	501	285	262
Anoura caudifer	3	155	208	418	288	247
Anoura geoffroyi	3	161	228	467	305	279
Choeroniscus minor	1	148	214	472	306	247
CAROLLIINAE						
Carollia castanea	5	151	226	347	272	247
Carollia perspicillata	3	142	226	365	260	244
Rhinophylla pumilio	3	140	211	330	255	264
STURNIRINAE						
Sturnira lilium	3	136	192	349	288	266
Sturnira ludovici	2	146	200	333	271	231
Sturnira tildae	3	129	189	331	283	272
STENODERMINAE						
Uroderma bilobatum	3	156	270	435	316	295
Vampyrops brachycephalus	2	178	294	576	341	309
Vampyrops helleri	3	150	249	446	289	272
Vampyrops infuscus	1	153	240	416	277	233
Vampyrops lineatus	3	152	241	440	298	279
Vampyrops vittatus	2	175	275	527	342	326
Vampyrodes caraccioloi	1	160	277	378	291	230
Vampyressa pusilla	2	146	246	432	298	276
Chiroderma salvini	4	163	274	470	318	268
Chiroderma trinitatum	2	155	256	462	313	281
Chiroderma villosum	3	147	265	419	276	239
Chiroderma villosum ssp.	1	165	295	423	291	230
Ectophylla macconnelli	2	164	250	448	296	265
Artibeus cinereus	3	148	225	409	280	264
Artibeus concolor	2	152	255	427	290	256
Artibeus jamaicensis	3	146	226	390	290	263
Artibeus lituratus	4	136	220	403	280	256
Enchisthenes harti	4	137	214	344	277	245
Ardops nichollsi	3	124	183	286	240	219
Sphaeronycteris toxophyllum	1	189	255	320	277	209
BRACHYPHYLLINAE						
Brachyphylla cavernarum	4	175	250	375	365	296

	n	OBL	MES	CER	DIE	TEL
slope		.61	.58	.68	.65	.68
y-intercept		.776	.481	.552	.350	1.242
DESMODONTIDAE						
Desmodus rotundus	3	150	240	406	324	281
Diphylla ecaudata	4	130	239	359	306	248
NATALIDAE						
Natalus tumidirostris	3	137	207	301	233	199
FURIPTERIDAE						
Furipterus horrens	5	129	205	341	184	133
VESPERTILIONIDAE						
VESPERTILIONINAE						
Myotis adversus	2	136	212	239	184	125
Myotis albescens	3	130	191	266	182	121
Myotis altarium	3	132	191	243	171	119
Myotis annectans	2	128	180	216	181	124
Myotis bechsteini	1	162	198	249	226	145
Myotis bocagei	3	127	214	252	199	129
Myotis dasycneme	1	158	224	222	203	121
Myotis montivagus	3	114	176	194	154	98
Myotis myotis	3	163	235	269	218	134
Myotis nattereri	1	155	210	267	206	148
Myotis nigricans	3	113	203	237	166	128
Myotis nigricans ssp.	3	108	180	215	162	112
Myotis siligorensis	3	120	178	211	173	109
Pipistrellus babu	3	101	143	163	125	78
Pipistrellus ceylonicus	3	109	166	179	134	93
Pipistrellus circumdatus	3	122	198	232	170	119
Pipistrellus crassulus	3	111	169	173	166	96
Pipistrellus imbricatus	2	118	185	163	184	94
Pipistrellus javanicus	3	121	182	196	158	105
Pipistrellus mimus	3	110	157	174	141	89
Pipistrellus nanus	3	111	153	205	145	97
Pipistrellus papuanus §)	2	119	173	167	145	85
Pipistrellus pulveratus	3	126	194	201	175	117
Pipistrellus subflavus	2	111	165	192	157	100
Scotozous dormeri	3	131	165	174	150	91
Nyctalus noctula	3	129	184	240	133	84
Nyctalus stenopterus	2	114	174	172	161	89
Glischropus tylopus	3	104	147	172	124	79
Eptesicus brasiliensis	3	113	198	224	162	114
Eptesicus br. melanopterus	3	125	183	230	175	109
Eptesicus flavescens	2	119	179	194	146	92
Eptesicus fuscus	1	106	164	184	156	110
Eptesicus pumilus	2	113	158	192	145	95
Eptesicus serotinus andersoni	3	124	184	196	166	115
Ia io	1	141	232	273	200	134

	n	OBL	MES	CER	DIE	TEL
slope		.61	.58	.68	.65	.68
y-intercept		.776	.481	.552	.350	1.242
VESPERTILIONIDAE (cont.)						
VESPERTILIONINAE (cont.)						
Tylonycteris pachypus	4	96	129	113	108	64
Tylonycteris robustula	3	100	141	151	114	68
Hesperoptenus blanfordi	3	123	200	182	156	98
Glauconycteris humeralis	3	128	207	275	154	101
Glauconycteris poensis	1	137	218	281	167	110
Chalinolobus gouldi	2	126	187	263	153	108
Chalinolobus morio	2	129	187	231	154	106
Scotorepens sanborni	1	103	154	169	146	87
Rhogeessa parvula	4	116	178	182	156	100
Scotomanes ornatus	2	141	240	190	199	121
Scotomanes sp?	3	131	227	241	192	132
Scotophilus dingani	3	139	236	261	196	126
Scotophilus heathi	3	134	223	239	173	112
Scotophilus kuhli	3	130	216	210	181	109
Lasiurus borealis	1	112	188	284	147	93
MINIOPTERINAE						
Miniopterus australis	2	138	191	293	182	143
Miniopterus haradei	3	140	189	304	180	132
Miniopterus inflatus	3	141	189	355	169	123
Miniopterus magnater	3	134	185	326	168	129
Miniopterus medius	3	141	195	327	168	120
Miniopterus pusillus	3	129	172	306	175	126
M. schreibersi arenarius	3	129	177	288	169	130
Miniopterus s. blepotis	2	136	188	338	162	122
Miniopterus tristis	3	139	202	311	173	128
MURININAE						
Murina cyclotis cyclotis	2	146	224	295	229	193
Murina c. peninsularis	2	134	202	310	214	194
Murina huttoni rubella	3	120	191	261	205	178
Murina tubinaris	2	141	199	316	235	206
Harpiocephalus harpia	1	147	221	328	261	238
KERIVOULINAE						
Kerivoula papillosa	1	140	197	384	212	179
Kerivoula pellucida	1	169	290	392	261	213
Kerivoula phalaena	1	148	228	308	199	130
Phoniscus atrox	1	128	191	312	213	228
NYCTOPHILINAE						
Nyctophilus geoffroyi	2	109	163	214	156	117
Nyctophilus timoriensis	2	122	221	243	182	135

	n	OBL	MES	CER	DIE	TEL
slope		.61	.58	.68	.65	.68
y-intercept		.776	.481	.552	.350	1.242
MOLOSSIDAE						
Tadarida aegyptiaca	3	157	219	279	209	146
Tadarida condylura	3	131	202	221	198	132
Tadarida mops	3	125	194	203	208	125
Tadarida niveiventer	3	156	238	274	215	133
Tadarida beccarii	2	114	164	191	147	96
Tadarida jobensis	1	136	212	298	199	139
Tadarida leucostigma	1	127	196	211	150	108
Tadarida plicata plicata	3	136	196	247	170	117
Tadarida pumila	3	147	232	240	195	122
Otomops martiensseni	3	155	228	449	212	141
Molossops abrasus	3	116	194	180	157	97
Molossops greenhalli	1	134	219	196	177	104
Molossops planirostris	2	133	208	217	194	122
Eumops auripendulus	3	157	258	350	211	132
Eumops glaucinus	2	153	264	362	199	137
Molossus ater	3	130	231	241	186	128
Molossus molossus	3	145	238	278	192	136
Molossus trinitatis	2	163	258	321	217	147
Cheiromeles torquatus	3	119	189	184	174	120

Abbreviations and notes:

CER cerebellum; for indices of components see Table 29
DIE diencephalon; for indices of components see Table 31
MES mesencephalon; for indices of components see Table 26, 31
OBL medulla oblongata; for indices of components see Tables 14, 17, 20, 23
TEL telencephalon; for indices of components see Tables 33, 35

§) see addendum at end of Volume 3, pp. 1592-1595

n = number of individuals

Table 12. Average size indices of the five fundamental brain parts related to the average of the four Tenrecinae (AvTn = 100)

For volumes in species see Table 10
For size indices in species see Table 11

	N	NET	OBL	MES	CER	DIE	TEL
slope		.66	.61	.58	.68	.65	.68
y-intercept		1.506	.776	.481	.552	.350	1.242
PTEROPODIDAE	47	289	137	235	320	374	318
PTEROPODINAE *	34	294	139	240	318	384	322
Rousettus	3	267	133	221	291	349	292
Pteropus	12	321	138	254	327	426	355
Dobsonia	4	290	132	205	322	355	321
Cynopterus	3	272	129	224	311	341	301
NYCTIMENINAE *	4	268	129	230	322	332	292
MACROGLOSSINAE *	9	281	134	219	326	358	311
RHINOPOMATIDAE *	2	136	118	207	190	184	116
EMBALLONURIDAE	18	167	138	260	294	203	132
EMBALLONURINAE *	17	166	138	259	287	203	133
NON-*TAPHOZOUS* EMB.	11	159	128	250	261	201	128
Emballonura	3	158	130	239	273	204	125
Saccopteryx	3	169	133	286	259	215	139
TAPHOZOUS	6	181	155	276	335	206	142
DICLIDURINAE *	1	172	137	268	410	207	115
CRASEONYCTERIDAE *	1	137	126	202	194	183	112
NYCTERIDAE *	8	214	160	305	370	249	182
MEGADERMATIDAE *	5	232	171	327	380	288	201
RHINOLOPHIDAE *	15	174	141	226	313	202	144
HIPPOSIDERIDAE	24	177	149	239	306	198	148
HIX: *H. commersoni* excl. *	23	180	152	243	312	201	150
Hipposideros	20	181	152	246	307	204	153
H. commersoni excluded	19	185	155	252	315	209	156
NOCTILIONIDAE *	2	206	153	248	299	248	191
MORMOOPIDAE *	4	189	149	204	377	196	159
Pteronotus	3	190	151	203	351	201	165
PHYLLOSTOMIDAE	53	252	155	241	404	283	247
PHYLLOSTOMINAE *	15	238	159	257	376	267	227
Micronycteris	4	227	150	229	386	239	215
Tonatia	3	224	165	285	335	250	207
Phyllostomus	3	246	157	242	381	288	239
GLOSSOPHAGINAE *	11	253	157	220	443	284	242
CAROLLIINAE *	3	243	144	221	347	262	252
STURNIRINAE *	3	244	137	194	338	281	256
STENODERMINAE *	20	264	155	250	423	294	261
Vampyrops	5	287	162	260	481	310	284
Chiroderma	4	265	157	272	443	300	255
Artibeus	4	259	145	232	407	285	260
BRACHYPHYLLINAE *	1	288	175	250	375	365	296

	N	NET	OBL	MES	CER	DIE	TEL
slope		.66	.61	.58	.68	.65	.68
y-intercept		1.506	.776	.481	.552	.350	1.242
DESMODONTIDAE *	2	260	140	239	382	315	265
NATALIDAE *	1	203	137	207	301	233	199
FURIPTERIDAE *	1	166	129	205	341	184	133
VESPERTILIONIDAE	70	145	127	191	241	174	121
VESPERTILIONINAE	50	132	123	188	213	165	107
VEX: *Tylonycteris* excl.*	48	134	124	190	216	167	108
Myotis	13	149	134	199	237	187	124
Pipistrellus	11	121	114	171	186	155	98
Eptesicus	6	128	117	178	203	158	106
Tylonycteris	2	88	98	135	132	111	66
Scotophilus	3	145	134	225	237	183	115
MINIOPTERINAE *	9	159	136	187	317	172	128
MURININAE *	5	205	138	207	302	229	202
KERIVOULINAE *	4	205	146	227	349	221	187
NYCTOPHILINAE *	2	144	115	192	228	169	126
MOLOSSIDAE *	19	154	139	218	260	190	125
Tadarida	9	150	136	206	240	188	124
Molossops	3	134	128	207	198	176	108
Molossus	3	167	146	242	280	198	137
CHIROPTERA (AvCS)	272	203	141	228	311	243	190
CHIROPTERA (AvCF)	NN=28	207	142	230	325	245	194
MEGACHIR. (AvMegS)	47	289	137	235	320	374	318
MEGACHIR. (AvMegF)	NN= 3	281	134	230	322	358	308
MICROCHIR. (AvMicS)	225	185	142	227	309	216	163
MICROCHIR. (AvMicF)	NN=25	199	143	230	325	232	180

Number of families and/or subfamilies marked by * = NN
Number of species = N
Abbreviations:

AvCF	average of Chiroptera families and/or subfamilies		
AvCS	average of Chiroptera species		
AvMegF	average of Megachiroptera families and/or subfamilies		
AvMegS	average of Megachiroptera species		
AvMicF	average of Microchiroptera families and/or subfamilies		
AvMicS	average of Microchiroptera species		
CER	cerebellum	NET	net volume
DIE	diencephalon	OBL	medulla oblongata
MES	mesencephalon	TEL	telencephalon

Table 13. Volumes (in mm^3) of the sensory trigeminal complex and of the tractus solitarii and funicular nuclei of the medulla oblongata

For size indices in species see Table 14

For average size indices in supraspecific taxonomic units see Table 15

	TR	TSO	FUN	FGR	FCM	FCE
PTEROPODIDAE						
PTEROPODINAE						
Eidolon helvum	24.143	4.976	6.896	1.309	4.094	1.492
Rousettus aegyptiacus	15.519	2.951	4.968	0.778	2.959	1.231
Myonycteris torquata	6.912	1.723	1.961	0.347	1.002	0.612
Pteropus conspicillatus	39.538	8.623	13.382	2.521	8.139	2.723
Pteropus poliocephalus	39.751	8.147	14.420	2.296	8.847	3.277
Pteropus scapulatus	26.509	5.855	9.658	1.742	5.185	2.732
Pteropus tonganus	34.970	6.378	11.861	2.026	7.386	2.449
Pteropus vampyrus	40.134	11.535	15.101	3.092	8.423	3.586
Dobsonia inermis	17.850	3.822	4.603	0.875	2.627	1.101
D. moluccensis magna	33.737	7.165	9.722	1.859	5.436	2.427
Dobsonia praedatrix	19.788	4.409	5.502	0.845	2.948	1.708
Epomops franqueti	14.345	3.568	4.400	0.590	2.474	1.336
Epomophorus labiatus	9.689	2.566	2.808	0.417	1.649	0.742
Micropteropus pusillus	5.960	1.321	1.670	0.252	1.018	0.400
Scotonycteris zenkeri	5.445	1.193	1.393	0.244	0.737	0.413
Casinycteris argynnis	6.506	1.542	1.826	0.310	0.983	0.534
Cynopterus brachyotis	6.400	1.205	1.961	0.305	1.214	0.442
Cynopterus horsfieldi	8.381	1.791	2.401	0.410	1.456	0.535
Megaerops ecaudatus	5.050	1.175	1.381	0.272	0.725	0.385
Chironax melanocephalus	4.749	1.006	1.247	0.212	0.681	0.354
Balionycteris maculata	4.458	0.914	1.098	0.176	0.622	0.300
NYCTIMENINAE						
Nyctimene robinsoni	7.680	1.749	2.002	0.311	1.151	0.540
Nyctimene vizcaccia	6.283	1.465	1.541	0.200	0.855	0.486
MACROGLOSSINAE						
Eonycteris spelaea	9.540	1.791	2.861	0.519	1.738	0.605
Macroglossus minimus **	4.420	0.637	1.032	0.157	0.653	0.222
Syconycteris a. australis	3.576	0.872	1.159	0.194	0.678	0.287
Notopteris macdonaldi	10.414	1.728	3.770	0.547	2.430	0.793

** = two specimens measured

Abbreviations:

FCE	ncl. fasc. cuneatus externus	FUN	funicular nuclei
FCM	ncl. fasc. cuneatus medialis	TR	complexus sensorius trigeminalis
FGR	nucleus fascicularis gracilis	TSO	nucleus tractus solitarii

	TR	TSO	FUN	FGR	FCM	FCE
RHINOPOMATIDAE						
Rhinopoma hardwickei	4.173	0.424	0.961	0.243	0.558	0.161
Rhinopoma microphyllum **	5.145	0.676	1.229	0.307	0.718	0.204
EMBALLONURIDAE						
EMBALLONURINAE						
Emballonura monticola	1.815	0.364	0.369	0.042	0.219	0.107
Emballonura raffrayana	1.829	0.371	0.372	0.050	0.190	0.132
Emballonura semicaudata	2.097	0.389	0.486	0.056	0.280	0.151
Coleura afra	3.177	0.561	0.559	0.087	0.319	0.152
Rhynchonycteris naso	1.627	0.259	0.231	0.034	0.133	0.064
Saccopteryx leptura	1.829	0.320	0.348	0.051	0.210	0.087
Cormura brevirostris	2.601	0.389	0.521	0.097	0.264	0.160
Peropteryx trinitatis	1.731	0.242	0.302	0.054	0.153	0.095
Taphozous australis	6.408	0.947	1.326	0.199	0.750	0.377
Taphozous hildegardeae	6.753	0.975	1.361	0.168	0.768	0.425
Taphozous theobaldi	10.633	1.305	1.797	0.251	1.087	0.459
Taphozous saccolaimus	8.203	1.291	1.843	0.252	1.192	0.400
DICLIDURINAE						
Cyttarops alecto	1.819	0.330	0.432	0.070	0.205	0.156
CRASEONYCTERIDAE						
Craseonycteris thonglongyai	1.251	0.162	0.230	0.051	0.127	0.053
NYCTERIDAE						
Nycteris grandis	8.608	1.043	1.230	0.206	0.708	0.315
Nycteris macrotis	6.100	0.703	0.962	0.145	0.585	0.232
Nycteris nana	3.461	0.438	0.514	0.073	0.304	0.137
Nycteris thebaica	4.753	0.628	0.740	0.100	0.452	0.187
MEGADERMATIDAE						
Megaderma lyra	13.818	1.635	2.127	0.440	1.142	0.545
Megaderma spasma	6.802	0.896	1.464	0.266	0.857	0.340
Macroderma gigas	20.144	2.522	3.799	0.733	2.248	0.819
Cardioderma cor	8.486	1.049	1.077	0.227	0.601	0.250
Lavia frons	6.380	1.123	1.359	0.269	0.741	0.349
RHINOLOPHIDAE						
Rhinolophus landeri	3.799	0.548	0.799	0.158	0.373	0.269
Rhinolophus luctus	6.566	0.979	1.215	0.277	0.531	0.406
Rhinolophus pusillus	2.505	0.293	0.454	0.096	0.226	0.132
HIPPOSIDERIDAE						
Hipposideros ater	2.892	0.431	0.575	0.092	0.355	0.128
Hipposideros ater aruensis	2.940	0.395	0.497	0.094	0.303	0.100
Hipposideros commersoni	7.968	1.170	2.164	0.429	1.223	0.512
Hipposideros diadema	9.178	1.178	1.587	0.337	0.868	0.383
Aselliscus stoliczkanus	2.308	0.297	0.345	0.072	0.206	0.067
Aselliscus tricuspidatus	2.258	0.282	0.419	0.092	0.217	0.110
Rhinonycteris aurantius	2.709	0.394	0.532	0.110	0.268	0.154
Triaenops persicus	3.752	0.502	0.607	0.085	0.357	0.166

> Table 13 <

	TR	TSO	FUN	FGR	FCM	FCE
NOCTILIONIDAE						
Noctilio albiventris	8.155	1.222	1.516	0.384	0.805	0.328
Noctilio leporinus	11.595	1.932	2.839	0.869	1.281	0.689
MORMOOPIDAE						
Pteronotus gymnonotus	5.973	0.721	1.012	0.182	0.579	0.251
Pteronotus parnelli	8.663	0.856	1.281	0.210	0.768	0.304
Mormoops megalophylla	5.088	0.680	0.743	0.112	0.441	0.191
PHYLLOSTOMIDAE						
PHYLLOSTOMINAE						
Micronycteris megalotis	3.658	0.481	0.652	0.121	0.369	0.163
Micronycteris brachyotis	4.667	0.676	0.904	0.190	0.464	0.251
Macrophyllum macrophyllum	4.168	0.586	0.870	0.204	0.447	0.220
Tonatia bidens	6.861	0.926	1.436	0.253	0.827	0.356
Mimon crenulatum	4.501	0.473	0.758	0.165	0.404	0.190
Phyllostomus discolor	10.977	1.628	2.652	0.597	1.395	0.661
Phyllostomus hastatus	15.814	2.363	3.765	0.764	2.018	0.984
Phylloderma stenops	11.679	1.501	2.659	0.532	1.633	0.494
Trachops cirrhosus	12.046	1.480	2.051	0.409	1.119	0.522
Vampyrum spectrum	22.347	3.311	5.501	0.892	3.269	1.340
GLOSSOPHAGINAE						
Glossophaga soricina	4.042	0.658	0.824	0.148	0.533	0.143
Leptonycteris curasoae	7.072	0.825	1.554	0.335	0.907	0.312
Lionycteris spurrelli	3.854	0.602	0.826	0.168	0.470	0.187
Anoura geoffroyi	5.320	0.740	1.507	0.291	0.906	0.310
CAROLLIINAE						
Carollia perspicillata	6.231	0.842	1.254	0.230	0.667	0.356
Rhinophylla pumilio	3.960	0.568	0.660	0.103	0.391	0.166
STURNIRINAE						
Sturnira lilium	5.548	0.945	1.128	0.246	0.672	0.211
STENODERMINAE						
Uroderma bilobatum	5.155	0.799	1.208	0.267	0.710	0.231
Vampyrops vittatus	10.646	1.517	2.181	0.405	1.257	0.520
Chiroderma villosum	5.501	1.059	1.573	0.270	0.918	0.385
Artibeus cinereus	3.978	0.551	0.898	0.195	0.510	0.193
Artibeus lituratus	8.885	1.495	2.133	0.482	1.177	0.474
BRACHYPHYLLINAE						
Brachyphylla cavernarum	11.632	1.787	2.513	0.527	1.474	0.512

	TR	TSO	FUN	FGR	FCM	FCE
DESMODONTIDAE						
Desmodus rotundus	6.901	1.076	1.887	0.335	1.065	0.487
Diphylla ecaudata	6.138	0.974	1.596	0.293	0.929	0.375
NATALIDAE						
Natalus tumidirostris	2.715	0.447	0.506	0.118	0.274	0.115
FURIPTERIDAE						
Furipterus horrens	1.563	0.241	0.266	0.068	0.134	0.065
VESPERTILIONIDAE						
VESPERTILIONINAE						
Myotis albescens	1.881	0.291	0.397	0.064	0.198	0.135
Myotis annectans	3.151	0.458	0.658	0.093	0.367	0.198
Myotis dasycneme	4.042	0.880	1.041	0.137	0.658	0.246
Myotis myotis	6.755	1.000	1.258	0.209	0.744	0.306
Myotis siligorensis	1.633	0.251	0.313	0.049	0.192	0.072
Pipistrellus ceylonicus	2.790	0.437	0.552	0.081	0.315	0.156
Pipistrellus crassulus	1.776	0.305	0.372	0.056	0.206	0.109
Pipistrellus javanicus	1.819	0.287	0.280	0.036	0.159	0.086
Pipistrellus mimus	1.594	0.251	0.310	0.043	0.169	0.098
Scotozous dormeri	2.783	0.442	0.543	0.115	0.295	0.133
Nyctalus noctula	5.927	0.983	1.224	0.168	0.637	0.419
Glischropus tylopus	1.609	0.292	0.318	0.045	0.166	0.107
Eptesicus brasiliensis	2.763	0.469	0.527	0.100	0.277	0.149
Eptesicus br. melanopterus	3.336	0.499	0.609	0.124	0.288	0.197
Eptesicus flavescens	3.666	0.523	0.617	0.140	0.302	0.175
Eptesicus pumilus	2.136	0.328	0.391	0.075	0.213	0.103
Eptesicus serotinus andersoni	4.676	0.714	0.903	0.188	0.412	0.304
Ia io	9.454	1.612	1.611	0.204	0.840	0.567
Tylonycteris pachypus	1.646	0.240	0.307	0.049	0.176	0.083
Tylonycteris robustula	2.383	0.331	0.464	0.075	0.265	0.124
Hesperoptenus blanfordi	2.323	0.323	0.579	0.092	0.328	0.159
Glauconycteris humeralis	2.040	0.391	0.395	0.060	0.233	0.103
Chalinolobus gouldi	3.255	0.545	0.722	0.092	0.425	0.206
Scotorepens sanborni	2.074	0.360	0.503	0.085	0.281	0.137
Rhogeessa parvula	1.478	0.215	0.275	0.051	0.161	0.064
Scotomanes ornatus	5.647	1.024	1.300	0.225	0.704	0.371
Scotomanes sp.?	3.789	0.668	0.917	0.154	0.507	0.256
Scotophilus dinganii	5.894	0.953	1.209	0.226	0.705	0.278
Scotophilus heathi	7.086	1.132	1.441	0.271	0.849	0.322
Lasiurus borealis	2.154	0.414	0.577	0.074	0.337	0.167
MINIOPTERINAE						
Miniopterus australis	2.499	0.452	0.609	0.101	0.376	0.132
Miniopterus magnater	4.036	0.621	0.804	0.159	0.439	0.206
Miniopterus medius	3.404	0.546	0.704	0.113	0.406	0.184
Miniopterus tristis	4.406	0.747	0.770	0.129	0.451	0.190

	TR	TSO	FUN	FGR	FCM	FCE
VESPERTILIONIDAE (cont.)						
MURININAE						
Murina cyclotis cyclotis	3.001	0.473	0.643	0.123	0.318	0.201
Murina c. peninsularis	3.278	0.523	0.647	0.129	0.325	0.193
Murina huttoni rubella	3.257	0.532	0.491	0.089	0.255	0.147
Murina tubinaris	2.772	0.424	0.512	0.096	0.255	0.161
Harpiocephalus harpia	4.516	0.728	0.832	0.139	0.443	0.249
KERIVOULINAE						
Kerivoula papillosa	2.548	0.414	0.520	0.101	0.257	0.162
Kerivoula pellucida	1.730	0.247	0.383	0.079	0.184	0.121
Kerivoula phalaena	1.479	0.187	0.283	0.059	0.146	0.078
Phoniscus atrox	1.937	0.314	0.291	0.041	0.158	0.093
NYCTOPHILINAE						
Nyctophilus geoffroyi	2.320	0.329	0.390	0.076	0.223	0.091
Nyctophilus timoriensis	3.306	0.497	0.651	0.127	0.372	0.152
MOLOSSIDAE						
Tadarida condylura	5.797	0.937	1.432	0.261	0.898	0.273
Tadarida beccarii	3.396	0.512	0.802	0.173	0.465	0.164
Tadarida jobensis	5.124	0.708	1.155	0.222	0.709	0.223
Tadarida pumila	3.896	0.625	0.857	0.173	0.500	0.183
Otomops martiensseni	9.060	1.020	2.347	0.430	1.259	0.659
Molossops abrasus	6.339	0.866	1.555	0.336	0.797	0.422
Molossops planirostris	2.790	0.379	0.668	0.157	0.340	0.170
Molossus ater	6.935	0.937	1.648	0.325	0.926	0.397
Molossus molossus	4.537	0.632	0.934	0.167	0.528	0.238
Cheiromeles torquatus	17.602	1.760	3.936	1.097	2.142	0.697

** = two specimens measured

Abbreviations:

FCE	ncl. fasc. cuneatus externus	FUN	funicular nuclei
FCM	ncl. fasc. cuneatus medialis	TR	complexus sensorius trigeminalis
FGR	nucleus fascicularis gracilis	TSO	nucleus tractus solitarii

Table 14. Size indices of the sensory trigeminal complex and of the tractus solitarii and funicular nuclei of the medulla oblongata related to the average of the four Tenrecinae (AvTn = 100)

For volumes in species see Table 13

For average size indices in supraspecific taxonomic units see Table 15

	TR	TSO	FUN	FGR	FCM	FCE
slope	.59	.58	.62	.62	.62	.61
y-intercept	-.001	-.926	-.927	-1.515	-1.300	-1.400
PTEROPODIDAE						
PTEROPODINAE						
Eidolon helvum	91	166	185	136	259	125
Rousettus aegyptiacus	86	144	200	121	281	154
Myonycteris torquata	83	181	178	122	215	172
Pteropus conspicillatus	80	156	187	136	268	121
Pteropus poliocephalus	84	154	211	130	305	152
Pteropus scapulatus	80	159	207	145	262	185
Pteropus tonganus	97	158	232	153	340	151
Pteropus vampyrus	68	176	175	139	230	132
Dobsonia inermis	93	177	175	129	236	131
D. moluccensis magna	90	171	182	135	241	144
Dobsonia praedatrix	91	181	183	109	232	178
Epomops franqueti	85	187	191	99	254	181
Epomophorus labiatus	84	194	180	104	250	148
Micropteropus pusillus	89	172	191	112	275	141
Scotonycteris zenkeri	90	171	177	120	221	160
Casinycteris argynnis	93	192	200	131	254	180
Cynopterus brachyotis	80	132	187	112	272	129
Cynopterus horsfieldi	76	141	161	107	231	111
Megaerops ecaudatus	78	157	163	124	202	139
Chironax melanocephalus	87	160	177	117	229	154
Balionycteris maculata	93	164	178	110	237	148
NYCTIMENINAE						
Nyctimene robinsoni	79	158	155	94	211	129
Nyctimene vizcaccia	77	156	143	72	187	138
MACROGLOSSINAE						
Eonycteris spelaea	87	142	194	136	278	127
Macroglossus minimus **	91	113	166	97	247	109
Syconycteris a. australis	73	155	185	120	256	140
Notopteris macdonaldi	85	123	228	128	346	149

** = two specimens measured

Abbreviations:

FCE	ncl. fasc. cuneatus externus	FUN	funicular nuclei
FCM	ncl. fasc. cuneatus medialis	TR	complexus sensorius trigeminalis
FGR	nucleus fascicularis gracilis	TSO	nucleus tractus solitarii

	TR	TSO	FUN	FGR	FCM	FCE
slope	.59	.58	.62	.62	.62	.61
y-intercept	-.001	-.926	-.927	-1.515	-1.300	-1.400
RHINOPOMATIDAE						
Rhinopoma hardwickei	93	81	166	163	228	85
Rhinopoma microphyllum **	70	80	128	123	176	65
EMBALLONURIDAE						
EMBALLONURINAE						
Emballonura monticola	68	117	111	49	156	97
Emballonura raffrayana	64	112	105	55	126	113
Emballonura semicaudata	67	106	123	55	167	116
Coleura afra	75	115	104	63	140	86
Rhynchonycteris naso	74	101	85	49	116	71
Saccopteryx leptura	70	105	107	61	152	81
Cormura brevirostris	75	97	119	86	143	112
Peropteryx trinitatis	74	89	105	72	126	100
Taphozous australis	98	126	156	91	209	136
Taphozous hildegardeae	96	120	148	71	197	142
Taphozous theobaldi	121	130	155	84	221	122
Taphozous saccolaimus	89	123	151	80	231	101
DICLIDURINAE						
Cyttarops alecto	68	106	130	82	146	142
CRASEONYCTERIDAE						
Craseonycteris thonglongyai	72	79	109	92	141	75
NYCTERIDAE						
Nycteris grandis	116	123	127	82	172	100
Nycteris macrotis	126	126	155	91	222	114
Nycteris nana	111	120	131	73	183	106
Nycteris thebaica	131	149	161	85	233	124
MEGADERMATIDAE						
Megaderma lyra	160	166	187	150	237	147
Megaderma spasma	108	124	179	126	247	128
Macroderma gigas	120	133	165	123	231	111
Cardioderma cor	124	134	121	98	159	86
Lavia frons	100	152	163	125	209	128
RHINOLOPHIDAE						
Rhinolophus landeri	100	124	165	126	182	169
Rhinolophus luctus	84	109	118	104	122	121
Rhinolophus pusillus	101	101	147	121	173	129
HIPPOSIDERIDAE						
Hipposideros ater	93	119	147	91	214	99
Hipposideros ater aruensis	108	124	146	107	210	89
Hipposideros commersoni	52	68	104	80	139	77
Hipposideros diadema	100	112	131	107	169	97
Aselliscus stoliczkanus	91	100	109	88	153	64
Aselliscus tricuspidatus	94	101	141	121	173	112

	TR	TSO	FUN	FGR	FCM	FCE
slope	.59	.58	.62	.62	.62	.61
y-intercept	-.001	-.926	-.927	-1.515	-1.300	-1.400
HIPPOSIDERIDAE (cont.)						
Rhinonycteris aurantius	76	95	119	95	141	104
Triaenops persicus	80	93	101	55	140	84
NOCTILIONIDAE						
Noctilio albiventris	104	136	147	145	185	98
Noctilio leporinus	106	155	194	229	206	145
MORMOOPIDAE						
Pteronotus gymnonotus	128	134	170	118	229	129
Pteronotus parnelli	147	126	168	107	238	122
Mormoops megalophylla	100	116	114	66	159	89
PHYLLOSTOMIDAE						
PHYLLOSTOMINAE						
Micronycteris megalotis	122	137	173	124	230	131
Micronycteris brachyotis	96	120	144	117	175	122
Macrophyllum macrophyllum	117	142	194	176	235	148
Tonatia bidens	94	110	149	102	203	114
Mimon crenulatum	105	95	139	117	174	106
Phyllostomus discolor	132	170	241	210	299	185
Phyllostomus hastatus	111	146	195	154	247	159
Phylloderma stenops	122	137	209	162	303	120
Trachops cirrhosus	144	154	185	143	238	145
Vampyrum spectrum	107	141	190	120	267	145
GLOSSOPHAGINAE						
Glossophaga soricina	105	147	168	117	257	89
Leptonycteris curasoae	107	109	181	151	249	111
Lionycteris spurrelli	117	157	199	157	268	137
Anoura geoffroyi	104	125	228	170	324	144
CAROLLIINAE						
Carollia perspicillata	114	134	178	126	223	154
Rhinophylla pumilio	109	135	144	87	201	110
STURNIRINAE						
Sturnira lilium	94	139	148	125	208	85
STENODERMINAE						
Uroderma bilobatum	100	134	182	155	252	106
Vampyrops vittatus	128	159	198	142	269	145
Chiroderma villosum	83	139	182	121	250	137
Artibeus cinereus	89	107	158	133	211	104
Artibeus lituratus	83	122	149	130	194	102
BRACHYPHYLLINAE						
Brachyphylla cavernarum	124	167	202	164	280	127

	TR	TSO	FUN	FGR	FCM	FCE
slope	.59	.58	.62	.62	.62	.61
y-intercept	-.001	-.926	-.927	-1.515	-1.300	-1.400
DESMODONTIDAE						
Desmodus rotundus	83	113	172	118	229	137
Diphylla ecaudata	81	112	161	114	221	116
NATALIDAE						
Natalus tumidirostris	87	123	129	116	165	89
FURIPTERIDAE						
Furipterus horrens	76	99	105	103	124	77
VESPERTILIONIDAE						
VESPERTILIONINAE						
Myotis albescens	83	110	142	88	167	146
Myotis annectans	82	103	135	74	178	124
Myotis dasycneme	82	154	164	84	245	118
Myotis myotis	100	129	142	92	199	106
Myotis siligorensis	79	103	123	74	177	85
Pipistrellus ceylonicus	71	96	111	63	150	95
Pipistrellus crassulus	61	90	102	60	134	91
Pipistrellus javanicus	75	101	93	46	125	86
Pipistrellus mimus	72	96	113	61	145	107
Scotozous dormeri	90	123	140	114	179	104
Nyctalus noctula	85	123	134	71	165	141
Glischropus tylopus	63	98	100	55	124	102
Eptesicus brasiliensis	75	109	112	83	140	97
Eptesicus br. melanopterus	93	119	134	105	149	131
Eptesicus flavescens	92	113	121	106	140	105
Eptesicus pumilus	79	104	116	86	150	93
Eptesicus serotinus andersoni	78	103	116	93	125	119
Ia io	91	135	116	57	142	126
Tylonycteris pachypus	65	81	98	61	132	80
Tylonycteris robustula	75	89	115	72	155	93
Hesperoptenus blanfordi	72	86	143	88	191	119
Glauconycteris humeralis	75	123	116	68	161	91
Chalinolobus gouldi	76	110	133	65	185	115
Scotorepens sanborni	56	84	108	71	143	89
Rhogeessa parvula	61	76	92	66	126	64
Scotomanes ornatus	92	145	163	109	208	142
Scotomanes sp.?	73	111	137	89	179	117
Scotophilus dinganii	90	126	141	102	195	100
Scotophilus heathi	93	129	143	104	199	99
Lasiurus borealis	64	106	137	67	188	120
MINIOPTERINAE						
Miniopterus australis	79	122	153	98	222	101
Miniopterus magnater	82	109	127	97	163	99
Miniopterus medius	81	112	132	82	179	105
Miniopterus tristis	87	128	118	77	164	89

	TR	TSO	FUN	FGR	FCM	FCE
slope	.59	.58	.62	.62	.62	.61
y-intercept	-.001	-.926	-.927	-1.515	-1.300	-1.400
VESPERTILIONIDAE (cont.)						
MURININAE						
Murina cyclotis cyclotis	95	128	161	120	188	153
Murina c. peninsularis	92	126	144	111	170	130
Murina huttoni rubella	90	127	108	76	132	98
Murina tubinaris	102	134	151	110	178	143
Harpiocephalus harpia	97	136	140	91	176	128
KERIVOULINAE						
Kerivoula papillosa	65	91	105	79	122	99
Kerivoula pellucida	75	92	135	108	153	128
Kerivoula phalaena	78	83	121	98	148	100
Phoniscus atrox	83	115	101	55	130	97
NYCTOPHILINAE						
Nyctophilus geoffroyi	73	88	97	73	131	68
Nyctophilus timoriensis	81	105	125	94	169	89
MOLOSSIDAE						
Tadarida condylura	83	116	156	110	231	92
Tadarida beccarii	71	93	131	110	179	82
Tadarida jobensis	98	117	172	128	249	101
Tadarida pumila	99	136	171	134	235	111
Otomops martiensseni	101	99	197	140	249	170
Molossops abrasus	80	96	150	125	181	125
Molossops planirostris	87	101	165	151	199	128
Molossus ater	87	103	158	120	209	117
Molossus molossus	97	117	156	108	208	121
Cheiromeles torquatus	86	76	139	150	179	77

** = two specimens measured

Abbreviations:

FCE	ncl. fasc. cuneatus externus	FUN	funicular nuclei
FCM	ncl. fasc. cuneatus medialis	TR	complexus sensorius trigeminalis
FGR	nucleus fascicularis gracilis	TSO	nucleus tractus solitarii

Table 15. Average size indices of the sensory trigeminal complex and of the tractus solitarii and funicular nuclei related to the average of the four Tenrecinae (AvTn = 100)

For volumes in species see Table 13
For size indices in species see Table 14

	N	TR	TSO	FUN	FGR	FCM	FCE
slope		.59	.58	.62	.62	.62	.61
y-intercept		-.001	-.926	-.927	-1.515	-1.300	-1.400
PTEROPODIDAE	27	85	161	185	120	253	145
PTEROPODINAE *	21	86	166	187	123	252	149
Pteropus	5	82	161	202	140	281	148
Dobsonia	3	92	176	180	124	236	151
Cynopterus	2	78	137	174	110	252	120
NYCTIMENINAE *	2	78	157	149	83	199	134
MACROGLOSSINAE *	4	84	133	193	120	282	131
RHINOPOMATIDAE *	2	81	81	147	143	202	75
EMBALLONURIDAE	13	80	111	123	69	164	109
EMBALLONURINAE *	12	81	112	122	68	165	106
NON-*TAPHOZOUS* EMB.	8	71	105	107	61	141	97
TAPHOZOUS	4	101	125	153	81	214	125
DICLIDURINAE *	1	68	106	130	82	146	142
CRASEONYCTERIDAE *	1	72	79	109	92	141	75
NYCTERIDAE *	4	121	129	143	82	203	111
MEGADERMATIDAE *	5	123	142	163	124	217	120
RHINOLOPHIDAE *	3	95	111	143	117	159	140
HIPPOSIDERIDAE	8	87	101	125	93	167	91
HIX: *H. commersoni* excl.*	7	92	106	128	95	171	93
Hipposideros	4	88	106	132	96	183	90
H. commersoni excluded	3	100	118	141	102	197	95
NOCTILIONIDAE *	2	105	145	171	187	195	122
MORMOOPIDAE *	3	125	125	151	97	209	113
Pteronotus	2	138	130	169	112	233	125
PHYLLOSTOMIDAE	23	109	136	180	139	242	127
PHYLLOSTOMINAE *	10	115	135	182	142	237	137
Micronycteris	2	109	128	159	120	203	126
Phyllostomus	2	121	158	218	182	273	172
GLOSSOPHAGINAE *	4	108	134	194	149	274	120
CAROLLIINAE *	2	112	134	161	107	212	132
STURNIRINAE *	1	94	139	148	125	208	85
STENODERMINAE *	5	97	132	174	136	235	119
Artibeus	2	86	114	153	131	203	103
BRACHYPHYLLINAE *	1	124	167	202	164	280	127
DESMODONTIDAE *	2	82	113	166	116	225	127

	N	TR	TSO	FUN	FGR	FCM	FCE
slope		.59	.58	.62	.62	.62	.61
y-intercept		-.001	-.926	-.927	-1.515	-1.300	-1.400
NATALIDAE *	1	87	123	129	116	165	89
FURIPTERIDAE *	1	76	99	105	103	124	77
VESPERTILIONIDAE	45	80	111	126	83	163	107
VESPERTILIONINAE	30	78	109	125	79	163	107
VEX: *Tylonycteris* excl.*	28	79	111	126	80	165	108
Myotis	5	85	120	141	82	193	116
Pipistrellus	4	70	96	105	57	138	95
Eptesicus	5	83	110	120	95	141	109
Tylonycteris	2	70	85	107	67	144	87
Scotophilus	2	91	128	142	103	197	99
MINIOPTERINAE *	4	82	118	132	88	182	98
MURININAE *	5	95	130	141	102	169	130
KERIVOULINAE *	4	75	96	115	85	138	106
NYCTOPHILINAE *	2	77	97	111	84	150	79
MOLOSSIDAE *	10	89	105	159	128	212	112
Tadarida	4	88	116	157	120	224	96
Molossops	2	84	98	157	138	190	126
Molossus	2	92	110	157	114	209	119
CHIROPTERA (AvCS)	150	90	124	150	106	200	118
CHIROPTERA (AvCF)	NN=28	93	122	149	112	197	113
MEGACHIR. (AvMegS)	27	85	161	185	120	253	145
MEGACHIR. (AvMegF)	NN= 3	83	152	176	109	244	138
MICROCHIR. (AvMicS)	123	92	116	143	103	188	111
MICROCHIR. (AvMicF)	NN=25	94	119	146	112	191	110

Number of families and/or subfamilies marked by * = NN
Number of species = N
Abbreviations:

AvCF	average of Chiroptera families and/or subfamilies		
AvCS	average of Chiroptera species		
AvMegF	average of Megachiroptera families and/or subfamilies		
AvMegS	average of Megachiroptera species		
AvMicF	average of Microchiroptera families and/or subfamilies		
AvMicS	average of Microchiroptera species		
FCE	ncl. fasc. cuneatus externus	FUN	funicular nuclei
FCM	ncl. fasc. cuneatus medialis	TR	complexus sensorius trigeminalis
FGR	nucleus fascicularis gracilis	TSO	nucleus tractus solitarii

Table 16. Volumes of some motor and relay nuclei of the medulla oblongata

Size indices in species see Table 17
Average size indices in supraspecific taxonomic units see Table 18

	mot X	mot XII	REL	INO	PRP
PTEROPODIDAE					
PTEROPODINAE					
Eidolon helvum	0.566	1.328	2.161	3.673	1.278
Rousettus aegyptiacus	0.444	1.003	1.468	2.337	0.829
Myonycteris torquata	0.193	0.432	0.663	1.191	0.484
Pteropus conspicillatus	1.095	2.724	3.142	7.267	2.397
Pteropus poliocephalus	1.116	2.791	3.849	7.890	2.148
Pteropus scapulatus	0.617	1.987	2.620	5.753	1.620
Pteropus tonganus	0.965	2.582	3.358	6.240	1.553
Pteropus vampyrus	1.233	2.713	3.510	8.328	2.741
Dobsonia inermis	0.616	1.404	1.400	2.971	0.693
Dobs. moluccensis magna	0.889	2.451	2.930	6.457	1.907
Dobsonia praedatrix	0.580	1.474	1.708	3.143	1.244
Epomops franqueti	0.490	1.234	1.451	2.677	1.079
Epomophorus labiatus	0.281	0.966	0.966	1.989	0.740
Micropteropus pusillus	0.132	0.550	0.518	1.014	0.437
Scotonycteris zenkeri	0.163	0.414	0.450	0.767	0.396
Casinycteris argynnis	0.208	0.493	0.587	1.055	0.613
Cynopterus brachyotis	0.177	0.511	0.525	0.841	0.331
Cynopterus horsfieldi	0.266	0.673	0.719	1.307	0.677
Megaerops ecaudatus	0.135	0.327	0.476	0.626	0.370
Chironax melanocephalus	0.112	0.280	0.425	0.632	0.362
Balionycteris maculata	0.111	0.246	0.334	0.527	0.241
NYCTIMENINAE					
Nyctimene robinsoni	0.238	0.385	0.777	1.154	0.586
Nyctimene vizcaccia	0.154	0.475	0.645	1.115	0.484
MACROGLOSSINAE					
Eonycteris spelaea	0.221	0.576	0.874	1.419	0.619
Macroglossus minimus **	0.095	0.269	0.308	0.686	0.312
Syconycteris a. australis	0.116	0.243	0.431	0.706	0.322
Notopteris macdonaldi	0.233	0.706	0.834	1.806	0.677

** = two specimens measured

Abbreviations:

INO	complexus olivaris inferior	PRP	nucleus prepositus hypoglossi
mot X	ncl. dorsalis mot. nervi vagi	REL	nucleus reticularis lateralis
mot XII	nucleus nervi hypoglossi		

	mot X	mot XII	REL	INO	PRP
RHINOPOMATIDAE					
Rhinopoma hardwickei	0.092	0.172	0.373	0.485	0.195
Rhinopoma microphyllum **	0.113	0.192	0.400	0.674	0.239
EMBALLONURIDAE					
EMBALLONURINAE					
Emballonura monticola	0.046	0.121	0.165	0.289	0.170
Emballonura raffrayana	0.050	0.145	0.222	0.349	0.162
Emballonura semicaudata	0.053	0.162	0.238	0.389	0.195
Coleura afra	0.055	0.188	0.296	0.495	0.137
Rhynchonycteris naso	0.028	0.073	0.129	0.254	0.081
Saccopteryx leptura	0.040	0.096	0.159	0.306	0.132
Cormura brevirostris	0.059	0.134	0.186	0.387	0.134
Peropteryx trinitatis	0.034	0.085	0.136	0.320	0.106
Taphozous australis	0.131	0.318	0.394	0.840	0.286
Taphozous hildegardeae	0.154	0.314	0.646	1.150	0.364
Taphozous theobaldi	0.229	0.406	0.851	1.486	0.429
Taphozous saccolaimus	0.127	0.331	0.734	1.640	0.391
DICLIDURINAE					
Cyttarops alecto	0.046	0.116	0.234	0.459	0.196
CRASEONYCTERIDAE					
Craseonycteris thonglongyai	0.020	0.046	0.073	0.161	0.059
NYCTERIDAE					
Nycteris grandis	0.129	0.293	0.431	1.473	0.626
Nycteris macrotis	0.056	0.197	0.347	0.996	0.446
Nycteris nana	0.042	0.128	0.208	0.553	0.258
Nycteris thebaica	0.058	0.179	0.248	0.717	0.353
MEGADERMATIDAE					
Megaderma lyra	0.315	0.378	1.152	1.843	0.581
Megaderma spasma	0.144	0.183	0.440	1.230	0.472
Macroderma gigas	0.404	0.588	0.995	3.157	1.221
Cardioderma cor	0.120	0.248	0.451	1.027	0.510
Lavia frons	0.161	0.271	0.556	1.234	0.529
RHINOLOPHIDAE					
Rhinolophus landeri	0.086	0.150	0.274	0.432	0.226
Rhinolophus luctus	0.172	0.272	0.660	1.039	0.464
Rhinolophus pusillus	0.046	0.107	0.156	0.269	0.125
HIPPOSIDERIDAE					
Hipposideros ater	0.063	0.088	0.233	0.352	0.199
Hipposideros ater aruensis	0.054	0.079	0.219	0.284	0.173
Hipposideros commersoni	0.214	0.290	0.657	0.958	0.389
Hipposideros diadema	0.230	0.356	0.649	0.929	0.469
Aselliscus stoliczkanus	0.038	0.071	0.161	0.400	0.180
Aselliscus tricuspidatus	0.036	0.084	0.104	0.239	0.126
Rhinonycteris aurantius	0.058	0.086	0.251	0.487	0.196
Triaenops persicus	0.076	0.134	0.315	0.658	0.246

> Table 16 <

	mot X	mot XII	REL	INO	PRP
NOCTILIONIDAE					
Noctilio albiventris	0.219	0.303	0.452	0.941	0.293
Noctilio leporinus	0.293	0.589	0.738	1.871	0.658
MORMOOPIDAE					
Pteronotus personatus	0.092	0.170	0.252	0.710	0.230
Pteronotus parnelli	0.107	0.267	0.344	0.884	0.233
Mormoops megalophylla	0.077	0.174	0.302	0.881	0.274
PHYLLOSTOMIDAE					
PHYLLOSTOMINAE					
Micronycteris megalotis	0.065	0.116	0.262	0.680	0.235
Micronycteris brachyotis	0.086	0.182	0.351	0.756	0.245
Macrophyllum macrophyllum	0.073	0.145	0.377	0.674	0.296
Tonatia bidens	0.153	0.357	0.524	1.252	0.532
Mimon crenulatum	0.068	0.203	0.214	0.728	0.236
Phyllostomus discolor	0.216	0.607	0.818	1.389	0.474
Phyllostomus hastatus	0.262	0.793	1.090	2.133	0.751
Phylloderma stenops	0.191	0.473	0.932	1.394	0.546
Trachops cirrhosus	0.197	0.431	0.704	1.778	1.002
Vampyrum spectrum	0.611	0.841	1.948	3.052	1.331
GLOSSOPHAGINAE					
Glossophaga soricina	0.075	0.222	0.291	0.620	0.274
Leptonycteris curasoae	0.110	0.428	0.369	1.077	0.340
Lionycteris spurrelli	0.077	0.211	0.251	0.588	0.245
Anoura geoffroyi	0.100	0.324	0.344	0.897	0.337
CAROLLIINAE					
Carollia perspicillata	0.096	0.286	0.394	0.900	0.356
Rhinophylla pumilio	0.075	0.188	0.232	0.450	0.239
STURNIRINAE					
Sturnira lilium	0.084	0.345	0.314	0.766	0.309
STENODERMINAE					
Uroderma bilobatum	0.112	0.378	0.355	0.867	0.365
Vampyrops vittatus	0.159	0.546	0.599	1.821	0.565
Chiroderma villosum	0.138	0.420	0.595	1.118	0.489
Artibeus cinereus	0.086	0.203	0.360	0.669	0.283
Artibeus lituratus	0.217	0.534	0.765	1.668	0.548
BRACHYPHYLLINAE					
Brachyphylla cavernarum	0.203	0.558	0.827	1.668	0.482

	mot X	mot XII	REL	INO	PRP
DESMODONTIDAE					
Desmodus rotundus	0.159	0.542	0.646	1.394	0.423
Diphylla ecaudata	0.122	0.443	0.497	1.225	0.395
NATALIDAE					
Natalus tumidirostris	0.058	0.182	0.153	0.458	0.207
FURIPTERIDAE					
Furipterus horrens	0.035	0.086	0.161	0.270	0.133
VESPERTILIONIDAE					
VESPERTILIONINAE					
Myotis albescens	0.049	0.096	0.132	0.337	0.124
Myotis annectans	0.081	0.164	0.197	0.401	0.145
Myotis dasycneme	0.118	0.241	0.274	0.957	0.260
Myotis myotis	0.125	0.289	0.353	0.974	0.384
Myotis siligorensis	0.033	0.074	0.095	0.249	0.107
Pipistrellus ceylonicus	0.050	0.135	0.208	0.335	0.161
Pipistrellus crassulus	0.048	0.086	0.128	0.260	0.140
Pipistrellus javanicus	0.040	0.088	0.113	0.233	0.132
Pipistrellus mimus	0.025	0.063	0.115	0.230	0.105
Scotozous dormeri	0.054	0.111	0.192	0.327	0.185
Nyctalus noctula	0.153	0.298	0.458	0.899	0.396
Glischropus tylopus	0.035	0.079	0.094	0.228	0.126
Eptesicus brasiliensis	0.062	0.141	0.158	0.399	0.229
Eptesicus br. melanopterus	0.068	0.139	0.202	0.420	0.237
Eptesicus flavescens	0.088	0.172	0.255	0.498	0.247
Eptesicus pumilus	0.038	0.090	0.103	0.279	0.145
Eptesicus serotinus andersoni	0.105	0.230	0.274	0.597	0.305
Ia io	0.222	0.400	0.575	1.584	0.564
Tylonycteris pachypus	0.033	0.097	0.083	0.168	0.103
Tylonycteris robustula	0.043	0.104	0.113	0.257	0.136
Hesperoptenus blanfordi	0.051	0.103	0.185	0.447	0.138
Glauconycteris humeralis	0.059	0.121	0.138	0.380	0.179
Chalinolobus gouldi	0.069	0.142	0.224	0.540	0.274
Scotorepens sanborni	0.045	0.106	0.144	0.307	0.173
Rhogeessa parvula	0.027	0.081	0.111	0.194	0.068
Scotomanes ornatus	0.121	0.302	0.520	0.787	0.376
Scotomanes sp.?	0.079	0.204	0.387	0.576	0.297
Scotophilus dinganii	0.133	0.263	0.492	0.674	0.358
Scotophilus heathi	0.185	0.357	0.658	0.791	0.422
Lasiurus borealis	0.049	0.146	0.169	0.395	0.181
MINIOPTERINAE					
Miniopterus australis	0.066	0.127	0.218	0.389	0.195
Miniopterus magnater	0.113	0.194	0.310	0.689	0.266
Miniopterus medius	0.078	0.157	0.295	0.597	0.270
Miniopterus tristis	0.119	0.238	0.352	0.671	0.311

	mot X	mot XII	REL	INO	PRP
VESPERTILIONIDAE (cont.)					
MURININAE					
Murina cyclotis cyclotis	0.070	0.143	0.163	0.483	0.283
Murina c. peninsularis	0.075	0.150	0.155	0.559	0.273
Murina huttoni rubella	0.082	0.172	0.211	0.570	0.244
Murina tubinaris	0.075	0.145	0.178	0.441	0.229
Harpiocephalus harpia	0.074	0.226	0.210	0.712	0.290
KERIVOULINAE					
Kerivoula papillosa	0.045	0.163	0.188	0.517	0.319
Kerivoula pellucida	0.022	0.075	0.164	0.337	0.207
Kerivoula phalaena	0.023	0.088	0.117	0.281	0.183
Phoniscus atrox	0.024	0.105	0.116	0.308	0.185
NYCTOPHILINAE					
Nyctophilus geoffroyi	0.035	0.097	0.107	0.344	0.138
Nyctophilus timoriensis	0.041	0.123	0.155	0.499	0.236
MOLOSSIDAE					
Tadarida condylura	0.152	0.178	0.500	0.716	0.246
Tadarida beccarii	0.066	0.128	0.244	0.451	0.161
Tadarida jobensis	0.120	0.169	0.374	0.604	0.247
Tadarida pumila	0.061	0.129	0.333	0.526	0.191
Otomops martiensseni	0.198	0.373	0.730	1.586	0.402
Molossops abrasus	0.148	0.207	0.539	0.783	0.398
Molossops planirostris	0.049	0.108	0.200	0.383	0.158
Molossus ater	0.131	0.211	0.544	1.068	0.398
Molossus molossus	0.066	0.148	0.373	0.585	0.231
Cheiromeles torquatus	0.243	0.538	1.318	1.658	0.776

** = two specimens measured

Abbreviations:

INO	complexus olivaris inferior	PRP	nucleus prepositus hypoglossi
mot X	ncl. dorsalis mot. nervi vagi	REL	nucleus reticularis lateralis
mot XII	nucleus nervi hypoglossi		

Table 17. Size indices of some motor and relay nuclei of the medulla oblongata related to the average of the four Tenrecinae (AvTn = 100)

For volumes in species see Table 16
For average size indices in supraspecific taxonomic units see Table 18

	OBL	mot X	mot XII	REL	INO	PRP
slope	.61	.62	.59	.58	.61	.52
y-intercept	.776	-1.844	-1.476	-1.379	-1.340	-1.250
PTEROPODIDAE						
PTEROPODINAE						
Eidolon helvum	154	125	149	205	269	126
Rousettus aegyptiacus	127	147	165	203	255	115
Myonycteris torquata	154	145	155	197	291	133
Pteropus conspicillatus	122	126	164	162	280	136
Pteropus poliocephalus	133	135	176	207	319	127
Pteropus scapulatus	128	109	180	202	339	132
Pteropus tonganus	149	156	214	237	335	117
Pteropus vampyrus	110	118	137	152	267	133
Dobsonia inermis	137	193	219	184	307	91
D. moluccensis magna	129	138	196	199	333	139
Dobsonia praedatrix	134	160	203	199	286	147
Epomops franqueti	157	176	219	216	316	159
Epomophorus labiatus	152	149	249	207	345	151
Micropteropus pusillus	149	125	246	191	311	146
Scotonycteris zenkeri	146	171	204	183	260	143
Casinycteris argynnis	155	188	211	208	309	196
Cynopterus brachyotis	133	139	191	163	215	94
Cynopterus horsfieldi	129	148	181	161	237	144
Megaerops ecaudatus	130	131	150	180	197	126
Chironax melanocephalus	129	132	154	192	240	145
Balionycteris maculata	128	149	153	170	226	107
NYCTIMENINAE						
Nyctimene robinsoni	134	153	119	199	241	141
Nyctimene vizcaccia	140	117	173	195	277	135
MACROGLOSSINAE						
Eonycteris spelaea	134	123	156	197	259	132
Macroglossus minimus **	135	126	165	156	293	138
Syconycteris a. australis	123	153	149	217	300	142
Notopteris macdonaldi	133	116	172	169	295	132

** = two specimens measured

Abbreviations:

INO	complexus olivaris inferior	PRP	nucleus prepositus hypoglossi
mot X	ncl. dorsalis mot. nervi vagi	REL	nucleus reticularis lateralis
mot XII	nucleus nervi hypoglossi		

	OBL	mot X	mot XII	REL	INO	PRP
slope	.61	.62	.59	.58	.61	.52
y-intercept	.776	-1.844	-1.476	-1.379	-1.340	-1.250
RHINOPOMATIDAE						
Rhinopoma hardwickei	126	132	114	203	223	92
Rhinopoma microphyllum **	109	97	78	135	188	73
EMBALLONURIDAE						
EMBALLONURINAE						
Emballonura monticola	127	114	135	150	229	127
Emballonura raffrayana	131	116	152	190	258	115
Emballonura semicaudata	132	111	154	184	260	126
Coleura afra	125	84	133	172	244	68
Rhynchonycteris naso	117	84	100	143	247	72
Saccopteryx leptura	131	102	110	148	247	101
Cormura brevirostris	127	112	116	132	234	80
Peropteryx trinitatis	136	99	109	141	291	90
Taphozous australis	159	127	146	149	265	97
Taphozous hildegardeae	152	138	133	227	334	116
Taphozous theobaldi	171	163	138	240	344	112
Taphozous saccolaimus	133	86	108	198	362	98
DICLIDURINAE						
Cyttarops alecto	137	113	130	212	363	146
CRASEONYCTERIDAE						
Craseonycteris thonglongyai	126	78	78	101	199	64
NYCTERIDAE						
Nycteris grandis	170	110	118	144	406	191
Nycteris macrotis	159	75	122	176	426	197
Nycteris nana	148	87	123	163	372	168
Nycteris thebaica	175	104	147	167	414	202
MEGADERMATIDAE						
Megaderma lyra	190	229	131	331	434	155
Megaderma spasma	167	146	87	172	402	166
Macroderma gigas	169	145	105	148	373	180
Cardioderma cor	164	111	108	163	308	167
Lavia frons	166	159	126	214	394	182
RHINOLOPHIDAE						
Rhinolophus landeri	148	147	117	176	236	123
Rhinolophus luctus	133	138	104	208	270	134
Rhinolophus pusillus	142	123	128	152	229	99
HIPPOSIDERIDAE						
Hipposideros ater	147	133	84	182	237	129
Hipposideros ater aruensis	157	130	87	195	220	127
Hipposideros commersoni	86	85	57	108	125	62
Hipposideros diadema	149	157	116	176	205	118
Aselliscus stoliczkanus	144	99	83	154	332	140
Aselliscus tricuspidatus	134	101	105	106	211	104

	OBL	mot X	mot XII	REL	INO	PRP
slope	.61	.62	.59	.58	.61	.52
y-intercept	.776	-1.844	-1.476	-1.379	-1.340	-1.250
HIPPOSIDERIDAE (cont.)						
Rhinonycteris aurantius	140	107	72	172	287	114
Triaenops persicus	131	104	86	165	292	112
NOCTILIONIDAE						
Noctilio albiventris	132	176	116	143	245	85
Noctilio leporinus	175	165	161	168	344	142
MORMOOPIDAE						
Pteronotus gymnonotus	152	127	109	133	316	105
Pteronotus parnelli	168	116	136	144	309	87
Mormoops megalophylla	144	98	102	146	359	116
PHYLLOSTOMIDAE						
PHYLLOSTOMINAE						
Micronycteris megalotis	166	141	115	212	475	158
Micronycteris brachyotis	143	113	111	177	321	108
Macrophyllum macrophyllum	176	135	122	259	397	172
Tonatia bidens	168	131	145	176	348	163
Mimon crenulatum	132	103	141	122	353	116
Phyllostomus discolor	170	161	217	243	338	130
Phyllostomus hastatus	151	112	167	192	300	128
Phylloderma stenops	160	124	148	242	295	133
Trachops cirrhosus	181	147	154	208	430	273
Vampyrum spectrum	169	175	120	235	288	162
GLOSSOPHAGINAE						
Glossophaga soricina	157	127	172	184	335	148
Leptonycteris curasoae	148	106	194	138	335	115
Lionycteris spurrelli	171	154	192	186	375	152
Anoura geoffroyi	161	125	189	165	362	142
CAROLLIINAE						
Carollia perspicillata	142	112	157	177	340	142
Rhinophylla pumilio	140	135	155	156	259	136
STURNIRINAE						
Sturnira lilium	136	91	175	131	268	115
STENODERMINAE						
Uroderma bilobatum	156	139	219	169	347	152
Vampyrops vittatus	175	119	195	178	443	155
Chiroderma villosum	147	131	189	221	345	164
Artibeus cinereus	148	125	136	198	312	135
Artibeus lituratus	136	125	149	177	313	120
BRACHYPHYLLINAE						
Brachyphylla cavernarum	175	135	178	219	360	119

	OBL	mot X	mot XII	REL	INO	PRP
slope	.61	.62	.59	.58	.61	.52
y-intercept	.776	-1.844	-1.476	-1.379	-1.340	-1.250
DESMODONTIDAE						
Desmodus rotundus	150	120	195	193	341	116
Diphylla ecaudata	130	102	175	163	331	118
NATALIDAE						
Natalus tumidirostris	137	122	174	119	308	134
FURIPTERIDAE						
Furipterus horrens	129	113	125	188	279	125
VESPERTILIONIDAE						
VESPERTILIONINAE						
Myotis albescens	130	144	126	142	316	107
Myotis annectans	128	138	127	125	218	78
Myotis dasycneme	158	154	146	136	401	113
Myotis myotis	163	117	128	129	295	126
Myotis siligorensis	120	106	106	111	256	100
Pipistrellus ceylonicus	109	83	103	130	179	86
Pipistrellus crassulus	111	108	89	107	189	97
Pipistrellus javanicus	121	110	109	113	204	107
Pipistrellus mimus	110	76	84	125	219	92
Scotozous dormeri	131	115	107	151	222	121
Nyctalus noctula	129	138	128	162	264	127
Glischropus tylopus	104	90	92	89	189	98
Eptesicus brasiliensis	113	110	114	104	225	128
Eptesicus br. melanopterus	125	123	115	137	244	136
Eptesicus flavescens	119	143	128	156	260	129
Eptesicus pumilus	113	94	100	93	219	107
Eptesicus serotinus andersoni	124	111	114	112	204	112
Ia io	141	132	115	137	306	127
Tylonycteris pachypus	96	87	115	80	141	81
Tylonycteris robustula	100	88	97	86	169	87
Hesperoptenus blanfordi	123	105	95	139	291	87
Glauconycteris humeralis	128	144	133	123	294	131
Chalinolobus gouldi	126	105	100	129	264	136
Scotorepens sanborni	103	80	86	96	175	98
Rhogeessa parvula	116	74	100	111	169	55
Scotomanes ornatus	141	125	147	208	263	135
Scotomanes sp.?	131	98	117	183	229	123
Scotophilus dinganii	139	128	120	185	210	121
Scotophilus heathi	134	152	139	213	211	125
Lasiurus borealis	112	95	130	123	247	110
MINIOPTERINAE						
Miniopterus australis	138	136	119	167	258	125
Miniopterus magnater	134	147	118	154	289	116
Miniopterus medius	141	121	111	172	296	135
Miniopterus tristis	139	152	141	171	275	132

	OBL	mot X	mot XII	REL	INO	PRP
slope	.61	.62	.59	.58	.61	.52
y-intercept	.776	-1.844	-1.476	-1.379	-1.340	-1.250
VESPERTILIONIDAE (cont.)						
MURININAE						
Murina cyclotis cyclotis	146	146	135	125	320	182
Murina c. peninsularis	134	137	126	106	328	158
Murina huttoni rubella	120	149	143	143	331	140
Murina tubinaris	141	182	160	159	343	168
Harpiocephalus harpia	147	103	146	111	318	133
KERIVOULINAE						
Kerivoula papillosa	140	75	125	118	276	170
Kerivoula pellucida	176	64	97	173	311	177
Kerivoula phalaena	148	82	138	148	315	184
Phoniscus atrox	128	69	135	120	281	156
NYCTOPHILINAE						
Nyctophilus geoffroyi	109	71	91	81	225	88
Nyctophilus timoriensis	122	65	90	93	254	121
MOLOSSIDAE						
Tadarida condylura	131	137	76	176	209	79
Tadarida beccarii	114	88	80	125	196	72
Tadarida jobensis	136	147	97	176	239	102
Tadarida pumila	147	101	97	206	277	101
Otomops martiensseni	155	137	124	201	357	103
Molossops abrasus	116	117	78	169	202	114
Molossops planirostris	133	100	101	152	250	100
Molossus ater	130	104	79	170	274	114
Molossus molossus	145	91	95	196	259	105
Cheiromeles torquatus	119	71	79	162	160	96

** = two specimens measured

Abbreviations:

INO	complexus olivaris inferior	PRP	nucleus prepositus hypoglossi
mot X	ncl. dorsalis mot. nervi vagi	REL	nucleus reticularis lateralis
mot XII	nucleus nervi hypoglossi		

Table 18. Average size indices of some motor and relay nuclei related to the average of the four Tenrecinae (AvTn = 100)

For volumes in species see Table 16
For size indices in species see Table 17

	N	OBL	mot X	mot XII	REL	INO	PRP
slope		.61	.62	.59	.58	.61	.52
y-intercept		.776	-1.844	-1.476	-1.379	-1.340	-1.250
PTEROPODIDAE	27	136	142	180	191	281	134
PTEROPODINAE *	21	137	146	186	191	283	134
Pteropus	5	129	129	174	192	308	129
Dobsonia	3	134	164	206	194	309	126
Cynopterus	2	131	144	186	162	226	119
NYCTIMENINAE *	2	137	135	146	197	259	138
MACROGLOSSINAE *	4	131	130	160	185	286	136
RHINOPOMATIDAE *	2	118	114	96	169	205	82
EMBALLONURIDAE	13	137	112	128	176	283	104
EMBALLONURINAE *	12	137	111	128	173	276	100
NON-*TAPHOZOUS* EMB.	8	128	103	126	157	251	97
TAPHOZOUS	4	154	128	131	204	326	106
DICLIDURINAE *	1	137	113	130	212	363	146
CRASEONYCTERIDAE *	1	126	78	78	101	199	64
NYCTERIDAE *	4	163	94	127	162	405	189
MEGADERMATIDAE *	5	171	158	111	206	382	170
RHINOLOPHIDAE *	3	141	136	116	179	245	119
HIPPOSIDERIDAE	8	136	115	86	157	239	113
HIX: *H. commersoni* excl. *	7	143	119	90	164	255	121
Hipposideros	4	135	126	86	165	197	109
H. commersoni excluded	3	151	140	95	184	221	125
NOCTILIONIDAE *	2	153	170	138	155	295	113
MORMOOPIDAE *	3	154	114	116	141	328	103
Pteronotus	2	160	122	122	138	313	96
PHYLLOSTOMIDAE	23	157	129	163	190	345	145
PHYLLOSTOMINAE *	10	161	134	144	207	355	154
Micronycteris	2	155	127	113	194	398	133
Phyllostomus	2	160	137	192	217	319	129
GLOSSOPHAGINAE *	4	159	128	187	168	352	139
CAROLLIINAE *	2	141	124	156	167	300	139
STURNIRINAE *	1	136	91	175	131	268	115
STENODERMINAE *	5	153	128	178	189	352	145
Artibeus	2	142	125	142	188	313	127
BRACHYPHYLLINAE *	1	175	135	178	219	360	119
DESMODONTIDAE *	2	140	111	185	178	336	117

	N	OBL	mot X	mot XII	REL	INO	PRP
slope		.61	.62	.59	.58	.61	.52
y-intercept		.776	-1.844	-1.476	-1.379	-1.340	-1.250
NATALIDAE *	1	137	122	174	119	308	134
FURIPTERIDAE *	1	129	113	125	188	279	125
VESPERTILIONIDAE	45	128	113	117	133	255	121
VESPERTILIONINAE	30	123	112	114	131	236	109
VEX: *Tylonycteris* excl. *	28	125	114	114	135	242	111
Myotis	5	140	132	127	129	297	105
Pipistrellus	4	113	94	96	119	198	96
Eptesicus	5	119	116	114	120	230	122
Tylonycteris	2	98	87	106	83	155	84
Scotophilus	2	137	140	130	199	211	123
MINIOPTERINAE *	4	138	139	122	166	279	127
MURININAE *	5	138	143	142	129	328	156
KERIVOULINAE *	4	148	72	124	140	296	172
NYCTOPHILINAE *	2	115	68	90	87	240	104
MOLOSSIDAE *	10	132	109	91	173	242	99
Tadarida	4	132	118	87	171	230	88
Molossops	2	125	109	89	160	226	107
Molossus	2	137	97	87	183	267	109
CHIROPTERA (AvCS)	150	139	122	134	165	285	126
CHIROPTERA (AvCF)	NN=28	142	120	136	165	297	128
MEGACHIR. (AvMegS)	27	136	142	180	191	281	134
MEGACHIR. (AvMegF)	NN= 3	135	137	164	191	276	136
MICROCHIR. (AvMicS)	123	140	118	124	160	286	124
MICROCHIR. (AvMicF)	NN=25	143	118	133	162	300	127

Number of families and/or subfamilies marked by * = NN
Number of species = N
Abbreviations:

AvCF	average of Chiroptera families and/or subfamilies		
AvCS	average of Chiroptera species		
AvMegF	average of Megachiroptera families and/or subfamilies)		
AvMegS	average of Megachiroptera species		
AvMicF	average of Microchiroptera families and/or subfamilies		
AvMicS	average of Microchiroptera species		
INO	complexus olivaris inferior	OBL	medulla oblongata
mot X	ncl. dorsalis mot. nervi vagi	PRP	nucleus prepositus hypoglossi
mot XII	nucleus nervi hypoglossi	REL	nucleus reticularis lateralis

Table 19. Volumes of the vestibular nuclei of the medulla oblongata

Size indices in species see Table 20
Average size indices in supraspecific taxonomic units see Table 21

	VC	VM	VI	VL	VS
PTEROPODIDAE					
PTEROPODINAE					
Eidolon helvum	16.465	6.435	4.979	2.397	2.654
Rousettus aegyptiacus	9.596	3.767	2.672	1.666	1.491
Myonycteris torquata	6.759	2.560	1.850	1.095	1.254
Pteropus conspicillatus	21.897	8.434	6.132	3.679	3.652
Pteropus poliocephalus	22.302	8.607	6.072	4.554	3.069
Pteropus scapulatus	16.133	6.068	4.330	3.102	2.633
Pteropus tonganus	17.893	6.704	4.986	3.476	2.727
Pteropus vampyrus	26.380	10.458	6.796	4.695	4.430
Dobsonia inermis	10.414	4.071	2.901	1.659	1.783
D. moluccensis magna	17.985	7.146	5.121	3.006	2.712
Dobsonia praedatrix	10.274	4.179	2.540	1.905	1.649
Epomops franqueti	12.394	5.075	3.619	1.993	1.707
Epomophorus labiatus	8.457	3.370	2.182	1.641	1.265
Micropteropus pusillus	4.098	1.692	1.121	0.717	0.569
Scotonycteris zenkeri	4.364	1.789	1.026	0.768	0.781
Casinycteris argynnis	4.600	1.973	1.273	0.728	0.626
Cynopterus brachyotis	4.611	1.951	1.307	0.655	0.699
Cynopterus horsfieldi	6.972	2.853	1.835	1.167	1.116
Megaerops ecaudatus	3.724	1.602	0.879	0.622	0.622
Chironax melanocephalus	3.245	1.343	0.825	0.510	0.566
Balionycteris maculata	2.736	1.151	0.667	0.449	0.469
NYCTIMENINAE					
Nyctimene robinsoni	6.486	2.604	1.715	1.148	1.020
Nyctimene vizcaccia	5.917	2.350	1.432	1.178	0.957
MACROGLOSSINAE					
Eonycteris spelaea	5.650	2.492	1.448	0.804	0.906
Macroglossus minimus **	3.080	1.217	0.779	0.480	0.605
Syconycteris a. australis	2.548	1.107	0.615	0.360	0.467
Notopteris macdonaldi	6.741	2.680	1.556	1.199	1.306

** = two specimens measured

Abbreviations:

VC	vestibular complex	VM	nucleus vestibularis medialis
VI	nucleus vestibularis inferior	VS	nucleus vestibularis superior
VL	nucleus vestibularis lateralis		

	VC	VM	VI	VL	VS
RHINOPOMATIDAE					
Rhinopoma hardwickei	2.137	0.830	0.545	0.405	0.357
Rhinopoma microphyllum **	2.939	1.077	0.823	0.522	0.518
EMBALLONURIDAE					
EMBALLONURINAE					
Emballonura monticola	1.539	0.565	0.441	0.327	0.206
Emballonura raffrayana	1.716	0.622	0.441	0.416	0.237
Emballonura semicaudata	1.525	0.578	0.393	0.330	0.224
Coleura afra	1.996	0.844	0.521	0.355	0.276
Rhynchonycteris naso	1.100	0.460	0.295	0.192	0.153
Saccopteryx leptura	1.389	0.541	0.363	0.268	0.216
Cormura brevirostris	1.703	0.604	0.483	0.315	0.301
Peropteryx trinitatis	1.089	0.398	0.278	0.200	0.214
Taphozous australis	3.788	1.393	1.022	0.762	0.612
Taphozous hildegardeae	3.846	1.612	1.032	0.636	0.567
Taphozous theobaldi	5.130	1.967	1.238	0.995	0.931
Taphozous saccolaimus	4.399	1.690	1.132	0.832	0.745
DICLIDURINAE					
Cyttarops alecto	1.602	0.623	0.461	0.305	0.213
CRASEONYCTERIDAE					
Craseonycteris thonglongyai	0.739	0.295	0.197	0.135	0.112
NYCTERIDAE					
Nycteris grandis	5.162	1.966	1.606	0.776	0.815
Nycteris macrotis	3.396	1.333	0.931	0.536	0.596
Nycteris nana	2.069	0.894	0.561	0.283	0.331
Nycteris thebaica	2.594	1.074	0.761	0.354	0.404
MEGADERMATIDAE					
Megaderma lyra	6.102	2.490	1.638	1.021	0.953
Megaderma spasma	4.208	1.667	1.131	0.782	0.628
Macroderma gigas	10.448	4.165	2.936	1.847	1.499
Cardioderma cor	5.164	1.943	1.299	0.903	1.020
Lavia frons	4.462	1.640	1.174	0.942	0.706
RHINOLOPHIDAE					
Rhinolophus landeri	2.122	0.827	0.579	0.404	0.312
Rhinolophus luctus	4.165	1.690	1.109	0.755	0.611
Rhinolophus pusillus	1.144	0.427	0.299	0.241	0.177
HIPPOSIDERIDAE					
Hipposideros ater	1.599	0.647	0.464	0.274	0.214
Hipposideros ater aruensis	1.523	0.564	0.438	0.285	0.236
Hipposideros commersoni	4.075	1.620	1.080	0.801	0.575
Hipposideros diadema	4.831	1.910	1.366	0.842	0.714
Aselliscus stoliczkanus	1.505	0.566	0.417	0.291	0.231
Aselliscus tricuspidatus	0.962	0.390	0.241	0.169	0.162
Rhinonycteris aurantius	1.536	0.595	0.394	0.281	0.266
Triaenops persicus	2.177	0.835	0.609	0.476	0.257

	VC	VM	VI	VL	VS
NOCTILIONIDAE					
Noctilio albiventris	2.762	0.928	0.660	0.666	0.508
Noctilio leporinus	5.841	2.208	1.389	1.188	1.057
MORMOOPIDAE					
Pteronotus personatus	2.171	0.936	0.550	0.360	0.326
Pteronotus parnelli	2.970	1.271	0.876	0.450	0.374
Mormoops megalophylla	2.859	1.191	0.767	0.517	0.384
PHYLLOSTOMIDAE					
PHYLLOSTOMINAE					
Micronycteris megalotis	2.131	0.788	0.651	0.311	0.381
Micronycteris brachyotis	2.415	0.920	0.711	0.367	0.417
Macrophyllum macrophyllum	2.369	0.910	0.786	0.380	0.292
Tonatia bidens	4.316	1.677	1.235	0.668	0.736
Mimon crenulatum	2.104	0.794	0.561	0.361	0.387
Phyllostomus discolor	5.165	2.048	1.480	0.850	0.787
Phyllostomus hastatus	7.174	2.790	2.171	1.168	1.046
Phylloderma stenops	4.861	2.105	1.145	0.943	0.668
Trachops cirrhosus	5.983	2.470	1.767	0.860	0.886
Vampyrum spectrum	12.158	5.250	3.132	2.011	1.764
GLOSSOPHAGINAE					
Glossophaga soricina	2.441	0.998	0.691	0.438	0.315
Leptonycteris curasoae	3.655	1.395	1.074	0.699	0.487
Lionycteris spurrelli	2.440	0.978	0.696	0.392	0.373
Anoura geoffroyi	3.107	1.230	0.853	0.573	0.451
CAROLLIINAE					
Carollia perspicillata	2.792	1.082	0.953	0.395	0.362
Rhinophylla pumilio	2.050	0.743	0.591	0.291	0.426
STURNIRINAE					
Sturnira lilium	3.012	1.238	0.995	0.367	0.412
STENODERMINAE					
Uroderma bilobatum	3.244	1.316	0.984	0.459	0.485
Vampyrops vittatus	4.997	1.995	1.673	0.634	0.695
Chiroderma villosum	3.668	1.432	1.032	0.670	0.534
Artibeus cinereus	2.463	1.022	0.730	0.308	0.403
Artibeus lituratus	4.909	2.013	1.304	0.722	0.870
BRACHYPHYLLINAE					
Brachyphylla cavernarum	5.405	1.910	1.295	1.225	0.975

	VC	VM	VI	VL	VS
DESMODONTIDAE					
Desmodus rotundus	4.814	1.831	1.378	0.850	0.755
Diphylla ecaudata	4.203	1.690	1.175	0.787	0.551
NATALIDAE					
Natalus tumidirostris	1.646	0.665	0.449	0.291	0.241
FURIPTERIDAE					
Furipterus horrens	1.184	0.482	0.306	0.220	0.175
VESPERTILIONIDAE					
VESPERTILIONINAE					
Myotis albescens	1.117	0.450	0.336	0.186	0.145
Myotis annectans	1.701	0.694	0.463	0.296	0.247
Myotis dasycneme	3.103	1.192	0.994	0.454	0.462
Myotis myotis	4.014	1.544	1.290	0.532	0.648
Myotis siligorensis	0.939	0.382	0.276	0.133	0.149
Pipistrellus ceylonicus	1.441	0.580	0.422	0.239	0.200
Pipistrellus crassulus	1.052	0.424	0.331	0.159	0.137
Pipistrellus javanicus	0.961	0.369	0.288	0.177	0.127
Pipistrellus mimus	0.806	0.319	0.227	0.136	0.125
Scotozous dormeri	1.355	0.535	0.341	0.265	0.215
Nyctalus noctula	3.280	1.298	0.814	0.693	0.476
Glischropus tylopus	0.867	0.343	0.255	0.114	0.156
Eptesicus brasiliensis	1.549	0.585	0.424	0.291	0.248
Eptesicus br. melanopterus	1.532	0.580	0.433	0.271	0.248
Eptesicus flavescens	1.741	0.693	0.485	0.295	0.267
Eptesicus pumilus	1.016	0.395	0.278	0.184	0.158
Eptesicus serotinus andersoni	2.456	0.907	0.652	0.456	0.441
Ia io	5.216	2.011	1.514	0.906	0.785
Tylonycteris pachypus	0.758	0.302	0.202	0.145	0.109
Tylonycteris robustula	1.185	0.461	0.331	0.217	0.176
Hesperoptenus blanfordi	1.251	0.460	0.343	0.251	0.197
Glauconycteris humeralis	1.310	0.483	0.372	0.253	0.203
Chalinolobus gouldi	2.061	0.755	0.583	0.392	0.331
Scotorepens sanborni	1.193	0.433	0.338	0.226	0.195
Rhogeessa parvula	0.713	0.264	0.190	0.132	0.127
Scotomanes ornatus	3.536	1.200	0.993	0.647	0.696
Scotomanes sp.?	2.652	0.920	0.713	0.564	0.456
Scotophilus dinganii	3.188	1.199	0.934	0.534	0.522
Scotophilus heathi	3.767	1.443	1.072	0.571	0.681
Lasiurus borealis	1.459	0.545	0.453	0.242	0.219
MINIOPTERINAE					
Miniopterus australis	1.569	0.597	0.399	0.266	0.307
Miniopterus magnater	2.411	0.863	0.708	0.407	0.433
Miniopterus medius	2.343	0.895	0.605	0.434	0.409
Miniopterus tristis	2.468	0.890	0.657	0.488	0.433

	VC	VM	VI	VL	VS
VESPERTILIONIDAE (cont.)					
MURININAE					
Murina cyclotis cyclotis	1.824	0.748	0.419	0.340	0.317
Murina c. peninsularis	1.971	0.792	0.458	0.389	0.332
Murina huttoni rubella	2.203	0.862	0.509	0.428	0.403
Murina tubinaris	1.647	0.630	0.413	0.318	0.286
Harpiocephalus harpia	2.793	1.044	0.759	0.551	0.439
KERIVOULINAE					
Kerivoula papillosa	2.256	0.867	0.665	0.325	0.399
Kerivoula pellucida	1.568	0.621	0.434	0.240	0.273
Kerivoula phalaena	1.194	0.456	0.356	0.200	0.183
Phoniscus atrox	1.329	0.510	0.363	0.259	0.198
NYCTOPHILINAE					
Nyctophilus geoffroyi	1.324	0.508	0.391	0.220	0.205
Nyctophilus timoriensis	1.912	0.752	0.612	0.288	0.260
MOLOSSIDAE					
Tadarida condylura	3.310	1.235	0.900	0.553	0.622
Tadarida beccarii	1.896	0.735	0.493	0.309	0.359
Tadarida jobensis	2.521	0.948	0.629	0.490	0.454
Tadarida pumila	2.146	0.814	0.635	0.345	0.352
Otomops martiensseni	4.945	1.829	1.335	1.035	0.745
Molossops abrasus	2.966	1.214	0.753	0.543	0.456
Molossops planirostris	1.549	0.538	0.397	0.315	0.300
Molossus ater	3.679	1.490	0.922	0.697	0.570
Molossus molossus	2.370	0.932	0.596	0.442	0.400
Cheiromeles torquatus	6.848	2.861	1.979	1.178	0.829

** = two specimens measured

Abbreviations:

VC	vestibular complex	VM	nucleus vestibularis medialis
VI	nucleus vestibularis inferior	VS	nucleus vestibularis superior
VL	nucleus vestibularis lateralis		

Table 20. Size indices of the vestibular nuclei related to the average of the four Tenrecinae (AvTn = 100)

For volumes in species see Table 19
For average size indices in supraspecific taxonomic units see Table 21

	VC	VM	VI	VL	VS
slope	.53	.53	.55	.54	.51
y-intercept	-.393	-.822	-.966	-1.179	-1.149
PTEROPODIDAE					
PTEROPODINAE					
Eidolon helvum	213	223	215	179	219
Rousettus aegyptiacus	176	185	166	177	172
Myonycteris torquata	249	253	237	237	283
Pteropus conspicillatus	162	168	149	156	176
Pteropus poliocephalus	172	178	154	201	154
Pteropus scapulatus	172	174	154	191	181
Pteropus tonganus	177	178	163	197	173
Pteropus vampyrus	166	177	140	169	183
Dobsonia inermis	181	191	171	168	196
D. moluccensis magna	171	183	162	165	167
Dobsonia praedatrix	160	175	133	172	163
Epomops franqueti	242	266	240	227	209
Epomophorus labiatus	231	247	205	262	214
Micropteropus pusillus	184	203	176	190	155
Scotonycteris zenkeri	213	235	176	222	231
Casinycteris argynnis	198	228	192	185	164
Cynopterus brachyotis	176	200	174	147	163
Cynopterus horsfieldi	198	218	180	194	196
Megaerops ecaudatus	171	197	142	169	173
Chironax melanocephalus	175	194	157	163	184
Balionycteris maculata	164	186	142	161	169
NYCTIMENINAE					
Nyctimene robinsoni	208	224	191	216	201
Nyctimene vizcaccia	221	236	186	259	219
MACROGLOSSINAE					
Eonycteris spelaea	161	191	143	135	160
Macroglossus minimus **	184	195	165	170	217
Syconycteris a. australis	152	177	130	127	167
Notopteris macdonaldi	175	186	138	182	210

** = two specimens measured

Abbreviations:

VC	vestibular complex	VM	nucleus vestibularis medialis
VI	nucleus vestibularis inferior	VS	nucleus vestibularis superior
VL	nucleus vestibularis lateralis		

	VC	VM	VI	VL	VS
slope	.53	.53	.55	.54	.51
y-intercept	-.393	-.822	-.966	-1.179	-1.149
RHINOPOMATIDAE					
Rhinopoma hardwickei	136	142	123	154	136
Rhinopoma microphyllum **	121	119	118	127	130
EMBALLONURIDAE					
EMBALLONURINAE					
Emballonura monticola	157	155	163	201	124
Emballonura raffrayana	166	161	154	241	135
Emballonura semicaudata	134	137	125	174	117
Coleura afra	135	154	126	143	112
Rhynchonycteris naso	134	150	131	141	109
Saccopteryx leptura	145	151	137	168	133
Cormura brevirostris	138	131	140	153	145
Peropteryx trinitatis	126	124	117	139	145
Taphozous australis	174	172	165	207	170
Taphozous hildegardeae	164	185	154	161	148
Taphozous theobaldi	180	185	151	205	200
Taphozous saccolaimus	148	153	132	165	154
DICLIDURINAE					
Cyttarops alecto	164	171	170	187	128
CRASEONYCTERIDAE					
Craseonycteris thonglongyai	111	119	109	122	98
NYCTERIDAE					
Nycteris grandis	211	216	230	187	203
Nycteris macrotis	203	214	198	191	215
Nycteris nana	184	213	179	150	174
Nycteris thebaica	201	224	212	164	187
MEGADERMATIDAE					
Megaderma lyra	218	238	203	214	208
Megaderma spasma	199	212	188	219	180
Macroderma gigas	204	219	195	210	184
Cardioderma cor	227	229	200	235	273
Lavia frons	207	205	192	259	199
RHINOLOPHIDAE					
Rhinolophus landeri	157	165	153	179	138
Rhinolophus luctus	162	176	150	173	145
Rhinolophus pusillus	124	125	118	158	113
HIPPOSIDERIDAE					
Hipposideros ater	142	154	148	146	113
Hipposideros ater aruensis	152	152	159	171	139
Hipposideros commersoni	87	93	79	100	77
Hipposideros diadema	163	173	160	167	148
Aselliscus stoliczkanus	160	162	161	186	145
Aselliscus tricuspidatus	108	118	99	115	107

	VC	VM	VI	VL	VS
slope	.53	.53	.55	.54	.51
y-intercept	-.393	-.822	-.966	-1.179	-1.149
HIPPOSIDERIDAE (cont.)					
Rhinonycteris aurantius	121	126	112	133	125
Triaenops persicus	134	138	134	175	95
NOCTILIONIDAE					
Noctilio albiventris	108	97	90	153	121
Noctilio leporinus	168	170	138	200	188
MORMOOPIDAE					
Pteronotus gymnonotus	135	156	121	133	121
Pteronotus parnelli	149	172	155	134	114
Mormoops megalophylla	164	184	156	176	133
PHYLLOSTOMIDAE					
PHYLLOSTOMINAE					
Micronycteris megalotis	195	194	215	171	207
Micronycteris brachyotis	144	147	150	130	149
Macrophyllum macrophyllum	187	193	223	180	137
Tonatia bidens	178	185	178	162	185
Mimon crenulatum	141	143	134	144	155
Phyllostomus discolor	189	202	189	184	177
Phyllostomus hastatus	163	170	169	155	148
Phylloderma stenops	158	183	129	180	133
Trachops cirrhosus	218	242	225	185	198
Vampyrum spectrum	196	227	170	188	180
GLOSSOPHAGINAE					
Glossophaga soricina	179	196	181	192	138
Leptonycteris curasoae	166	170	171	188	134
Lionycteris spurrelli	207	222	212	199	188
Anoura geoffroyi	177	188	172	194	154
CAROLLIINAE					
Carollia perspicillata	150	156	181	126	117
Rhinophylla pumilio	159	155	164	135	197
STURNIRINAE					
Sturnira lilium	151	167	176	109	125
STENODERMINAE					
Uroderma bilobatum	183	200	197	154	165
Vampyrops vittatus	183	196	214	137	156
Chiroderma villosum	165	173	163	179	146
Artibeus cinereus	159	177	168	118	156
Artibeus lituratus	144	158	132	124	157
BRACHYPHYLLINAE					
Brachyphylla cavernarum	179	170	148	238	198

> Table 20 <

	VC	VM	VI	VL	VS
slope	.53	.53	.55	.54	.51
y-intercept	-.393	-.822	-.966	-1.179	-1.149
DESMODONTIDAE					
Desmodus rotundus	177	181	177	185	170
Diphylla ecaudata	169	182	165	186	135
NATALIDAE					
Natalus tumidirostris	146	159	144	155	127
FURIPTERIDAE					
Furipterus horrens	152	167	144	171	132
VESPERTILIONIDAE					
VESPERTILIONINAE					
Myotis albescens	132	143	145	133	101
Myotis annectans	125	137	122	130	109
Myotis dasycneme	183	188	207	159	164
Myotis myotis	178	184	201	139	175
Myotis siligorensis	120	131	129	103	111
Pipistrellus ceylonicus	105	113	109	104	87
Pipistrellus crassulus	100	108	113	91	77
Pipistrellus javanicus	107	110	117	118	83
Pipistrellus mimus	97	103	99	99	88
Scotozous dormeri	121	129	110	142	114
Nyctalus noctula	141	150	123	176	125
Glischropus tylopus	92	98	99	73	97
Eptesicus brasiliensis	118	120	116	133	113
Eptesicus br. melanopterus	120	122	121	127	115
Eptesicus flavescens	124	132	123	125	114
Eptesicus pumilus	103	107	102	112	94
Eptesicus serotinus andersoni	121	120	113	133	132
Ia io	157	162	157	160	146
Tylonycteris pachypus	82	87	79	94	69
Tylonycteris robustula	103	108	103	113	91
Hesperoptenus blanfordi	108	106	106	130	101
Glauconycteris humeralis	131	130	135	152	120
Chalinolobus gouldi	138	136	139	157	133
Scotorepens sanborni	91	89	93	104	89
Rhogeessa parvula	79	79	77	89	83
Scotomanes ornatus	171	156	169	185	204
Scotomanes sp.?	149	139	141	188	154
Scotophilus dinganii	145	147	149	144	145
Scotophilus heathi	150	154	149	134	165
Lasiurus borealis	121	122	135	121	108
MINIOPTERINAE					
Miniopterus australis	137	140	126	140	159
Miniopterus magnater	142	136	148	143	153
Miniopterus medius	159	164	147	176	167
Miniopterus tristis	142	138	134	167	150

	VC	VM	VI	VL	VS
slope	.53	.53	.55	.54	.51
y-intercept	-.393	-.822	-.966	-1.179	-1.149
VESPERTILIONIDAE (cont.)					
MURININAE					
Murina cyclotis cyclotis	160	176	132	178	164
Murina c. peninsularis	155	168	129	183	156
Murina huttoni rubella	172	181	142	200	187
Murina tubinaris	166	170	150	192	170
Harpiocephalus harpia	174	174	168	204	164
KERIVOULINAE					
Kerivoula papillosa	164	169	172	141	173
Kerivoula pellucida	183	195	185	169	188
Kerivoula phalaena	165	169	180	167	147
Phoniscus atrox	154	158	152	180	134
NYCTOPHILINAE					
Nyctophilus geoffroyi	115	118	122	115	105
Nyctophilus timoriensis	133	141	152	120	108
MOLOSSIDAE					
Tadarida condylura	142	142	135	140	163
Tadarida beccarii	115	120	106	111	131
Tadarida jobensis	141	142	124	163	153
Tadarida pumila	154	157	163	148	151
Otomops martiensseni	170	169	159	209	157
Molossops abrasus	114	126	101	123	108
Molossops planirostris	134	125	123	163	154
Molossus ater	141	153	123	158	134
Molossus molossus	146	154	131	162	148
Cheiromeles torquatus	112	126	110	112	86

** = two specimens measured

Abbreviations:

VC	vestibular complex	VM	nucleus vestibularis medialis
VI	nucleus vestibularis inferior	VS	nucleus vestibularis superior
VL	nucleus vestibularis lateralis		

Table 21. Average size indices of the vestibular nuclei related to the average of the four Tenrecinae (AvTn = 100)

For volumes in species see Table 19
For size indices in species see Table 20

	N	VC	VM	VI	VL	VS
slope		.53	.53	.55	.54	.51
y-intercept		-.393	-.822	-.966	-1.179	-1.149
PTEROPODIDAE	27	187	203	170	186	189
PTEROPODINAE *	21	188	203	173	187	187
Pteropus	5	170	175	152	183	173
Dobsonia	3	171	183	155	168	175
Cynopterus	2	187	209	177	171	180
NYCTIMENINAE *	2	214	230	188	238	210
MACROGLOSSINAE *	4	168	187	144	153	189
RHINOPOMATIDAE *	2	129	131	121	140	133
EMBALLONURIDAE	13	151	156	143	176	140
EMBALLONURINAE *	12	150	155	141	175	141
NON-*TAPHOZOUS* EMB.	8	142	145	137	170	127
TAPHOZOUS	4	167	174	151	184	168
DICLIDURINAE *	1	164	171	170	187	128
CRASEONYCTERIDAE *	1	111	119	109	122	98
NYCTERIDAE *	4	200	217	205	173	195
MEGADERMATIDAE *	5	211	221	196	228	209
RHINOLOPHIDAE *	3	148	155	141	170	132
HIPPOSIDERIDAE	8	134	140	131	149	119
HIX: *H. commersoni* excl. *	7	140	146	139	156	125
Hipposideros	4	136	143	136	146	119
H. commersoni excluded	3	153	160	156	161	133
NOCTILIONIDAE *	2	138	134	114	177	154
MORMOOPIDAE *	3	149	170	144	148	123
Pteronotus	2	142	164	138	133	118
PHYLLOSTOMIDAE	23	173	183	176	164	161
PHYLLOSTOMINAE *	10	177	189	178	168	167
Micronycteris	2	169	170	183	150	178
Phyllostomus	2	176	186	179	169	163
GLOSSOPHAGINAE *	4	182	194	184	193	154
CAROLLIINAE *	2	155	155	173	130	157
STURNIRINAE *	1	151	167	176	109	125
STENODERMINAE *	5	167	181	175	142	156
Artibeus	2	151	168	150	121	157
BRACHYPHYLLINAE *	1	179	170	148	238	198
DESMODONTIDAE *	2	173	182	171	185	153

	N	VC	VM	VI	VL	VS
slope		.53	.53	.55	.54	.51
y-intercept		-.393	-.822	-.966	-1.179	-1.149
NATALIDAE *	1	146	159	144	155	127
FURIPTERIDAE *	1	152	167	144	171	132
VESPERTILIONIDAE	45	134	138	134	141	130
VESPERTILIONINAE	30	124	127	126	129	117
VEX: *Tylonycteris* excl.*	28	126	129	129	131	119
Myotis	5	148	157	161	133	132
Pipistrellus	4	102	109	110	103	84
Eptesicus	5	117	120	115	126	114
Tylonycteris	2	92	97	91	104	80
Scotophilus	2	147	150	149	139	155
MINIOPTERINAE *	4	145	144	138	156	157
MURININAE *	5	165	174	144	191	168
KERIVOULINAE *	4	166	173	172	164	160
NYCTOPHILINAE *	2	124	130	137	117	107
MOLOSSIDAE *	10	137	141	128	149	138
Tadarida	4	138	140	132	141	149
Molossops	2	124	125	112	143	131
Molossus	2	144	154	127	160	141
CHIROPTERA (AvCS)	150	157	165	151	162	151
CHIROPTERA (AvCF)	NN=28	159	168	155	166	152
MEGACHIR. (AvMegS)	27	187	203	170	186	189
MEGACHIR. (AvMegF)	NN= 3	190	207	168	193	195
MICROCHIR. (AvMicS)	123	150	157	147	157	142
MICROCHIR. (AvMicF)	NN=25	155	163	153	163	146

Number of families and/or subfamilies marked by * = NN
Number of species = N
Abbreviations:

AvCF	average of Chiroptera families and/or subfamilies		
AvCS	average of Chiroptera species		
AvMegF	average of Megachiroptera families and/or subfamilies		
AvMegS	average of Megachiroptera species		
AvMicF	average of Microchiroptera families and/or subfamilies		
AvMicS	average of Microchiroptera species		
VC	vestibular complex	VM	nucleus vestibularis medialis
VI	nucleus vestibularis inferior	VS	nucleus vestibularis superior
VL	nucleus vestibularis lateralis		

Table 22. Volumes of the auditory nuclei of the medulla oblongata

Size indices in species see Table 23
Average size indices in supraspecific taxonomic units see Table 24

	AUD	CON	DCO	VCO	OLS
PTEROPODIDAE					
PTEROPODINAE					
Eidolon helvum	12.771	9.244	5.223	4.022	3.527
Rousettus aegyptiacus	9.883	6.444	3.177	3.267	3.438
Myonycteris torquata	5.201	3.647	1.993	1.654	1.554
Pteropus conspicillatus	13.574	9.001	4.136	4.865	4.573
Pteropus poliocephalus	13.241	9.198	4.316	4.881	4.043
Pteropus scapulatus	11.491	7.858	3.693	4.164	3.633
Pteropus tonganus	11.927	7.417	3.521	3.897	4.510
Pteropus vampyrus	16.926	10.637	4.746	5.891	6.289
Dobsonia inermis	6.132	4.000	1.890	2.110	2.131
D. moluccensis magna	13.500	8.790	4.171	4.619	4.710
Dobsonia praedatrix	7.094	4.370	1.978	2.392	2.725
Epomops franqueti	10.435	7.543	4.357	3.186	2.892
Epomophorus labiatus	6.903	4.997	2.942	2.055	1.906
Micropteropus pusillus	3.515	2.328	1.363	0.965	1.187
Scotonycteris zenkeri	2.974	2.081	1.167	0.914	0.894
Casinycteris argynnis	3.749	2.794	1.557	1.237	0.955
Cynopterus brachyotis	3.474	2.456	1.274	1.182	1.018
Cynopterus horsfieldi	6.066	4.668	2.568	2.100	1.399
Megaerops ecaudatus	3.546	2.688	1.566	1.122	0.858
Chironax melanocephalus	2.643	1.899	1.053	0.846	0.744
Balionycteris maculata	1.817	1.305	0.691	0.615	0.512
NYCTIMENINAE					
Nyctimene robinsoni	4.764	3.141	1.535	1.606	1.623
Nyctimene vizcaccia	4.351	2.635	1.224	1.411	1.716
MACROGLOSSINAE					
Eonycteris spelaea	5.484	3.796	2.317	1.479	1.688
Macroglossus minimus **	2.401	1.649	0.890	0.760	0.752
Syconycteris a. australis	2.128	1.411	0.738	0.673	0.717
Notopteris macdonaldi	5.071	3.181	1.672	1.509	1.890

** = two specimens measured

Abbreviations:

AUD	auditory (acoustic) nuclei	OLS	complexus olivaris superior
CON	cochlear nuclei	VCO	nucleus cochlearis ventralis
DCO	nucleus cochlearis dorsalis		

	AUD	CON	DCO	VCO	OLS
RHINOPOMATIDAE					
Rhinopoma hardwickei	4.410	2.333	0.720	1.614	2.077
Rhinopoma microphyllum **	5.982	3.175	0.877	2.298	2.807
EMBALLONURIDAE					
EMBALLONURINAE					
Emballonura monticola	2.915	1.617	0.392	1.224	1.298
Emballonura raffrayana	3.119	1.696	0.413	1.283	1.423
Emballonura semicaudata	3.456	1.876	0.377	1.499	1.581
Coleura afra	4.082	2.159	0.439	1.720	1.923
Rhynchonycteris naso	2.339	1.167	0.203	0.964	1.172
Saccopteryx leptura	3.475	2.038	0.555	1.483	1.437
Cormura brevirostris	4.369	2.472	0.760	1.712	1.897
Peropteryx trinitatis	2.814	1.598	0.381	1.217	1.215
Taphozous australis	8.245	4.222	0.900	3.321	4.024
Taphozous hildegardeae	8.493	4.679	1.320	3.358	3.814
Taphozous theobaldi	11.753	6.281	1.518	4.762	5.472
Taphozous saccolaimus	10.090	5.515	1.584	3.931	4.575
DICLIDURINAE					
Cyttarops alecto	3.507	1.873	0.373	1.500	1.634
CRASEONYCTERIDAE					
Craseonycteris thonglongyai	1.838	0.995	0.236	0.759	0.842
NYCTERIDAE					
Nycteris grandis	12.589	4.990	1.014	3.977	7.599
Nycteris macrotis	6.989	2.836	0.637	2.199	4.154
Nycteris nana	4.475	1.809	0.412	1.397	2.665
Nycteris thebaica	5.794	2.489	0.611	1.878	3.305
MEGADERMATIDAE					
Megaderma lyra	13.655	6.253	2.089	4.164	7.402
Megaderma spasma	10.564	5.187	1.791	3.396	5.377
Macroderma gigas	22.358	10.266	3.598	6.668	12.092
Cardioderma cor	10.808	5.411	1.645	3.766	5.397
Lavia frons	10.919	5.070	1.714	3.356	5.849
RHINOLOPHIDAE					
Rhinolophus landeri	3.941	2.022	0.413	1.609	1.919
Rhinolophus luctus	8.293	4.452	0.873	3.579	3.840
Rhinolophus pusillus	2.409	1.153	0.231	0.922	1.257
HIPPOSIDERIDAE					
Hipposideros ater	3.735	1.965	0.385	1.581	1.769
Hipposideros ater aruensis	3.098	1.643	0.356	1.287	1.455
Hipposideros commersoni	8.786	4.431	0.736	3.695	4.355
Hipposideros diadema	10.760	5.524	0.947	4.577	5.236
Aselliscus stoliczkanus	2.926	1.271	0.290	0.981	1.655
Aselliscus tricuspidatus	2.516	1.325	0.262	1.063	1.192
Rhinonycteris aurantius	3.499	1.675	0.335	1.340	1.824
Triaenops persicus	4.072	1.721	0.318	1.403	2.350

> Table 22 <

	AUD	CON	DCO	VCO	OLS
NOCTILIONIDAE					
Noctilio albiventris	7.269	3.526	0.293	3.233	3.743
Noctilio leporinus	15.911	6.616	0.696	5.920	9.295
MORMOOPIDAE					
Pteronotus personatus	4.877	1.816	0.290	1.526	3.061
Pteronotus parnelli	6.156	2.395	0.435	1.959	3.761
Mormoops megalophylla	5.332	2.279	0.447	1.832	3.053
PHYLLOSTOMIDAE					
PHYLLOSTOMINAE					
Micronycteris megalotis	4.197	1.851	0.427	1.425	2.345
Micronycteris brachyotis	4.187	1.986	0.466	1.520	2.201
Macrophyllum macrophyllum	6.223	2.605	0.275	2.330	3.618
Tonatia bidens	13.371	6.428	2.001	4.427	6.943
Mimon crenulatum	5.922	3.156	0.819	2.337	2.766
Phyllostomus discolor	9.669	4.759	1.416	3.343	4.910
Phyllostomus hastatus	15.816	7.964	2.238	5.726	7.852
Phylloderma stenops	10.204	4.859	1.438	3.422	5.345
Trachops cirrhosus	16.338	7.791	1.877	5.914	8.547
Vampyrum spectrum	27.596	14.219	4.606	9.613	13.377
GLOSSOPHAGINAE					
Glossophaga soricina	3.744	2.022	0.613	1.409	1.722
Leptonycteris curasoae	5.573	2.726	0.631	2.095	2.847
Lionycteris spurrelli	3.705	1.737	0.433	1.304	1.968
Anoura geoffroyi	5.199	2.635	0.747	1.888	2.563
CAROLLIINAE					
Carollia perspicillata	5.267	2.804	0.948	1.856	2.463
Rhinophylla pumilio	4.567	2.252	0.622	1.630	2.315
STURNIRINAE					
Sturnira lilium	4.765	2.406	0.797	1.609	2.359
STENODERMINAE					
Uroderma bilobatum	5.979	2.947	1.000	1.947	3.032
Vampyrops vittatus	11.558	5.535	1.511	4.024	6.023
Chiroderma villosum	7.945	4.000	1.357	2.643	3.945
Artibeus cinereus	4.741	2.337	0.570	1.767	2.404
Artibeus lituratus	9.680	5.001	1.345	3.656	4.678
BRACHYPHYLLINAE					
Brachyphylla cavernarum	8.631	4.747	1.107	3.640	3.884

	AUD	CON	DCO	VCO	OLS
DESMODONTIDAE					
Desmodus rotundus	9.770	4.306	1.156	3.150	5.464
Diphylla ecaudata	8.300	3.883	1.284	2.599	4.417
NATALIDAE					
Natalus tumidirostris	3.284	1.340	0.174	1.167	1.943
FURIPTERIDAE					
Furipterus horrens	2.776	1.259	0.191	1.067	1.518
VESPERTILIONIDAE					
VESPERTILIONINAE					
Myotis albescens	2.262	1.067	0.189	0.878	1.195
Myotis annectans	3.561	1.664	0.226	1.438	1.896
Myotis dasycneme	6.360	3.022	0.314	2.708	3.338
Myotis myotis	7.496	3.268	0.384	2.884	4.227
Myotis siligorensis	2.015	0.875	0.100	0.776	1.140
Pipistrellus ceylonicus	3.408	1.601	0.354	1.247	1.807
Pipistrellus crassulus	2.273	1.043	0.199	0.844	1.230
Pipistrellus javanicus	2.279	1.087	0.242	0.845	1.192
Pipistrellus mimus	1.891	0.918	0.179	0.740	0.973
Scotozous dormeri	3.037	1.251	0.200	1.051	1.786
Nyctalus noctula	8.611	3.651	0.606	3.046	4.960
Glischropus tylopus	2.124	0.909	0.127	0.783	1.215
Eptesicus brasiliensis	3.470	1.414	0.260	1.153	2.056
Eptesicus br. melanopterus	3.250	1.399	0.230	1.169	1.852
Eptesicus flavescens	4.050	1.787	0.274	1.513	2.263
Eptesicus pumilus	2.463	1.129	0.211	0.918	1.334
Eptesicus serotinus andersoni	6.079	2.610	0.590	2.019	3.470
Ia io	14.192	6.354	1.615	4.739	7.838
Tylonycteris pachypus	1.749	0.817	0.164	0.653	0.933
Tylonycteris robustula	2.261	1.071	0.218	0.853	1.190
Hesperoptenus blanfordi	3.646	1.678	0.295	1.383	1.968
Glauconycteris humeralis	3.325	1.484	0.278	1.206	1.841
Chalinolobus gouldi	4.520	1.974	0.344	1.630	2.546
Scotorepens sanborni	2.770	1.188	0.187	1.000	1.582
Rhogeessa parvula	1.787	0.779	0.138	0.641	1.008
Scotomanes ornatus	8.017	3.462	0.589	2.873	4.555
Scotomanes sp.?	6.361	2.850	0.515	2.335	3.512
Scotophilus dinganii	7.582	3.095	0.480	2.615	4.486
Scotophilus heathi	8.809	3.668	0.584	3.084	5.141
Lasiurus borealis	3.049	1.263	0.180	1.083	1.785
MINIOPTERINAE					
Miniopterus australis	3.541	1.536	0.251	1.285	2.005
Miniopterus magnater	5.985	2.431	0.355	2.077	3.554
Miniopterus medius	4.765	2.132	0.349	1.783	2.633
Miniopterus tristis	5.654	2.255	0.323	1.932	3.399

	AUD	CON	DCO	VCO	OLS
VESPERTILIONIDAE (cont.)					
MURININAE					
Murina cyclotis cyclotis	3.759	1.769	0.371	1.399	1.990
Murina c. peninsularis	4.282	1.957	0.383	1.574	2.325
Murina huttoni rubella	4.331	1.995	0.409	1.586	2.336
Murina tubinaris	2.829	1.281	0.260	1.021	1.548
Harpiocephalus harpia	5.115	2.454	0.500	1.954	2.661
KERIVOULINAE					
Kerivoula papillosa	5.975	2.135	0.162	1.973	3.840
Kerivoula pellucida	4.122	1.779	0.143	1.637	2.343
Kerivoula phalaena	3.315	1.391	0.098	1.293	1.925
Phoniscus atrox	3.569	1.448	0.094	1.354	2.121
NYCTOPHILINAE					
Nyctophilus geoffroyi	3.017	1.505	0.493	1.011	1.512
Nyctophilus timoriensis	4.640	2.305	0.712	1.593	2.336
MOLOSSIDAE					
Tadarida condylura	6.200	3.160	0.596	2.564	3.040
Tadarida beccarii	4.121	2.305	0.467	1.839	1.816
Tadarida jobensis	5.261	2.610	0.421	2.189	2.651
Tadarida pumila	4.188	2.195	0.444	1.750	1.993
Otomops martiensseni	11.406	5.548	1.256	4.292	5.858
Molossops abrasus	7.193	3.566	0.744	2.822	3.627
Molossops planirostris	3.564	1.712	0.315	1.397	1.851
Molossus ater	8.625	4.504	0.834	3.669	4.121
Molossus molossus	5.507	2.682	0.506	2.175	2.825
Cheiromeles torquatus	13.203	6.423	1.372	5.052	6.780

** = two specimens measured

Abbreviations:

AUD	auditory (acoustic) nuclei	OLS	complexus olivaris superior
CON	cochlear nuclei	VCO	nucleus cochlearis ventralis
DCO	nucleus cochlearis dorsalis		

Table 23. Size indices of the auditory nuclei related to the average of the four Tenrecinae (AvTn = 100)

For volumes in species see Table 22
For average size indices in supraspecific taxonomic units see Table 24

	AUD	CON	DCO	VCO	OLS
slope	.51	.50	.50	.51	.54
y-intercept	-.267	-.461	-.867	-.700	-.742
PTEROPODIDAE					
PTEROPODINAE					
Eidolon helvum	138	165	238	118	96
Rousettus aegyptiacus	149	160	201	134	134
Myonycteris torquata	154	175	243	133	123
Pteropus conspicillatus	86	95	111	83	71
Pteropus poliocephalus	87	101	121	87	65
Pteropus scapulatus	103	117	140	102	82
Pteropus tonganus	99	103	124	88	93
Pteropus vampyrus	92	97	110	87	83
Dobsonia inermis	88	95	114	82	79
D. moluccensis magna	109	118	142	101	94
Dobsonia praedatrix	92	93	107	84	90
Epomops franqueti	168	199	293	139	120
Epomophorus labiatus	153	181	271	124	111
Micropteropus pusillus	126	134	200	93	115
Scotonycteris zenkeri	116	130	186	96	95
Casinycteris argynnis	129	155	221	115	89
Cynopterus brachyotis	107	122	161	98	84
Cynopterus horsfieldi	140	175	245	131	85
Megaerops ecaudatus	130	159	235	111	85
Chironax melanocephalus	113	130	184	98	87
Balionycteris maculata	86	99	134	79	67
NYCTIMENINAE					
Nyctimene robinsoni	124	132	165	113	112
Nyctimene vizcaccia	131	128	151	115	138
MACROGLOSSINAE					
Eonycteris spelaea	127	143	223	93	103
Macroglossus minimus **	113	125	171	97	98
Syconycteris a. australis	100	106	142	86	93
Notopteris macdonaldi	107	110	147	86	105

** = two specimens measured

Abbreviations:

AUD	auditory (acoustic) nuclei	OLS	complexus olivaris superior
CON	cochlear nuclei	VCO	nucleus cochlearis ventralis
DCO	nucleus cochlearis dorsalis		

> Table 23 <

	AUD	CON	DCO	VCO	OLS
slope	.51	.50	.50	.51	.54
y-intercept	-.267	-.461	-.867	-.700	-.742
RHINOPOMATIDAE					
Rhinopoma hardwickei	221	188	147	219	288
Rhinopoma microphyllum **	197	169	119	205	250
EMBALLONURIDAE					
EMBALLONURINAE					
Emballonura monticola	230	203	125	262	291
Emballonura raffrayana	233	202	125	260	301
Emballonura semicaudata	237	205	105	279	305
Coleura afra	217	184	95	248	284
Rhynchonycteris naso	219	173	77	244	315
Saccopteryx leptura	280	261	181	324	329
Cormura brevirostris	276	250	195	293	336
Peropteryx trinitatis	250	225	137	293	309
Taphozous australis	302	249	135	329	399
Taphozous hildegardeae	290	258	186	311	352
Taphozous theobaldi	332	288	177	365	413
Taphozous saccolaimus	274	243	178	289	331
DICLIDURINAE					
Cyttarops alecto	277	235	119	321	367
CRASEONYCTERIDAE					
Craseonycteris thonglongyai	210	180	109	236	280
NYCTERIDAE					
Nycteris grandis	412	264	137	353	671
Nycteris macrotis	330	215	123	282	541
Nycteris nana	309	199	115	261	519
Nycteris thebaica	351	241	151	309	560
MEGADERMATIDAE					
Megaderma lyra	392	291	248	324	568
Megaderma spasma	398	315	277	347	551
Macroderma gigas	360	271	242	291	504
Cardioderma cor	379	307	237	358	513
Lavia frons	404	303	261	337	588
RHINOLOPHIDAE					
Rhinolophus landeri	229	188	98	253	311
Rhinolophus luctus	259	225	112	302	322
Rhinolophus pusillus	202	154	78	210	301
HIPPOSIDERIDAE					
Hipposideros ater	258	216	108	296	344
Hipposideros ater aruensis	240	203	112	270	320
Hipposideros commersoni	154	127	54	175	198
Hipposideros diadema	293	244	107	338	380
Aselliscus stoliczkanus	241	166	97	219	387
Aselliscus tricuspidatus	219	183	92	250	296

	AUD	CON	DCO	VCO	OLS
slope	.51	.50	.50	.51	.54
y-intercept	-.267	-.461	-.867	-.700	-.742
HIPPOSIDERIDAE (cont.)					
Rhinonycteris aurantius	216	165	84	224	315
Triaenops persicus	198	134	63	185	316
NOCTILIONIDAE					
Noctilio albiventris	227	178	38	274	314
Noctilio leporinus	371	251	67	374	573
MORMOOPIDAE					
Pteronotus gymnonotus	238	142	58	202	413
Pteronotus parnelli	246	154	71	212	410
Mormoops megalophylla	242	166	83	225	381
PHYLLOSTOMIDAE					
PHYLLOSTOMINAE					
Micronycteris megalotis	299	210	123	275	471
Micronycteris brachyotis	197	150	89	193	285
Macrophyllum macrophyllum	384	257	69	390	625
Tonatia bidens	441	343	272	396	618
Mimon crenulatum	311	266	175	333	403
Phyllostomus discolor	285	227	172	267	388
Phyllostomus hastatus	295	243	174	289	381
Phylloderma stenops	267	207	156	243	373
Trachops cirrhosus	480	371	228	471	672
Vampyrum spectrum	369	312	258	348	457
GLOSSOPHAGINAE					
Glossophaga soricina	215	186	143	219	276
Leptonycteris curasoae	202	159	94	205	279
Lionycteris spurrelli	244	183	116	233	365
Anoura geoffroyi	234	190	138	230	317
CAROLLIINAE					
Carollia perspicillata	224	192	165	214	287
Rhinophylla pumilio	277	218	153	268	392
STURNIRINAE					
Sturnira lilium	190	155	131	174	257
STENODERMINAE					
Uroderma bilobatum	267	212	183	236	372
Vampyrops vittatus	341	264	184	322	476
Chiroderma villosum	286	232	201	258	385
Artibeus cinereus	241	190	118	243	338
Artibeus lituratus	230	193	132	235	294
BRACHYPHYLLINAE					
Brachyphylla cavernarum	230	206	122	263	276

	AUD	CON	DCO	VCO	OLS
slope	.51	.50	.50	.51	.54
y-intercept	-.267	-.461	-.867	-.700	-.742
DESMODONTIDAE					
Desmodus rotundus	289	207	141	253	434
Diphylla ecaudata	267	202	170	226	382
NATALIDAE					
Natalus tumidirostris	227	148	49	218	378
FURIPTERIDAE					
Furipterus horrens	274	196	76	285	431
VESPERTILIONIDAE					
VESPERTILIONINAE					
Myotis albescens	206	154	69	217	312
Myotis annectans	206	154	53	225	305
Myotis dasycneme	296	226	60	341	427
Myotis myotis	265	187	56	277	405
Myotis siligorensis	198	136	39	206	322
Pipistrellus ceylonicus	194	146	82	192	286
Pipistrellus crassulus	167	122	59	168	256
Pipistrellus javanicus	196	148	84	197	292
Pipistrellus mimus	175	134	67	185	258
Scotozous dormeri	211	139	56	198	350
Nyctalus noctula	297	203	86	284	462
Glischropus tylopus	175	119	42	174	284
Eptesicus brasiliensis	207	135	63	186	342
Eptesicus br. melanopterus	198	136	57	193	316
Eptesicus flavescens	226	159	62	229	351
Eptesicus pumilus	193	140	67	195	296
Eptesicus serotinus andersoni	239	165	95	215	371
Ia io	345	252	163	313	506
Tylonycteris pachypus	145	108	55	147	221
Tylonycteris robustula	153	115	60	156	226
Hesperoptenus blanfordi	245	180	80	252	371
Glauconycteris humeralis	258	183	87	253	405
Chalinolobus gouldi	238	167	74	233	372
Scotorepens sanborni	166	114	46	163	265
Rhogeessa parvula	153	106	48	149	247
Scotomanes ornatus	308	214	93	299	476
Scotomanes sp.?	282	203	94	281	428
Scotophilus dinganii	275	182	72	258	442
Scotophilus heathi	280	189	77	266	440
Lasiurus borealis	198	131	48	190	325
MINIOPTERINAE					
Miniopterus australis	241	167	69	237	384
Miniopterus magnater	278	181	67	262	455
Miniopterus medius	255	183	76	258	391
Miniopterus tristis	258	165	60	239	426

	AUD	CON	DCO	VCO	OLS
slope	.51	.50	.50	.51	.54
y-intercept	-.267	-.461	-.867	-.700	-.742
VESPERTILIONIDAE (cont.)					
MURININAE					
Murina cyclotis cyclotis	256	192	102	258	381
Murina c. peninsularis	263	192	96	263	400
Murina huttoni rubella	264	194	101	262	399
Murina tubinaris	220	159	82	215	342
Harpiocephalus harpia	251	193	100	260	360
KERIVOULINAE					
Kerivoula papillosa	340	194	37	304	608
Kerivoula pellucida	371	254	52	399	604
Kerivoula phalaena	350	232	41	370	587
Phoniscus atrox	317	204	34	326	539
NYCTOPHILINAE					
Nyctophilus geoffroyi	204	162	135	185	288
Nyctophilus timoriensis	254	202	159	236	355
MOLOSSIDAE					
Tadarida condylura	213	175	84	238	282
Tadarida beccarii	197	177	91	238	239
Tadarida jobensis	233	186	76	263	322
Tadarida pumila	236	198	102	267	312
Otomops martiensseni	315	249	144	322	433
Molossops abrasus	223	179	95	237	302
Molossops planirostris	240	184	86	255	351
Molossus ater	266	225	106	306	341
Molossus molossus	268	209	101	287	379
Cheiromeles torquatus	180	144	78	186	236

** = two specimens measured

Abbreviations:

AUD	auditory (acoustic) nuclei	OLS	complexus olivaris superior
CON	cochlear nuclei	VCO	nucleus cochlearis ventralis
DCO	nucleus cochlearis dorsalis		

Table 24. Average size indices of the auditory nuclei related to the average of the four Tenrecinae (AvTn = 100)

For volumes in species see Table 22
For size indices in species see Table 23

	N	AUD	CON	DCO	VCO	OLS
slope		.51	.50	.50	.51	.54
y-intercept		-.267	-.461	-.867	-.700	-.742
PTEROPODIDAE	27	117	131	177	103	96
PTEROPODINAE *	21	117	133	180	104	93
Pteropus	5	93	102	121	89	79
Dobsonia	3	96	102	121	89	88
Cynopterus	2	123	149	203	115	85
NYCTIMENINAE *	2	127	130	158	114	125
MACROGLOSSINAE *	4	112	121	171	90	100
RHINOPOMATIDAE *	2	209	179	133	212	269
EMBALLONURIDAE	13	263	229	141	294	333
EMBALLONURINAE *	12	262	228	143	292	331
NON-*TAPHOZOUS* EMB.	8	243	213	130	276	309
TAPHOZOUS	4	299	260	169	324	374
DICLIDURINAE *	1	277	235	119	321	367
CRASEONYCTERIDAE *	1	210	180	109	236	280
NYCTERIDAE *	4	351	230	132	301	573
MEGADERMATIDAE *	5	387	297	253	331	545
RHINOLOPHIDAE *	3	230	189	96	255	311
HIPPOSIDERIDAE	8	227	180	89	245	320
HIX: *H. commersoni* excl.*	7	238	187	95	255	337
Hipposideros	4	236	197	95	270	311
H. commersoni excluded	3	264	221	109	301	348
NOCTILIONIDAE *	2	299	215	52	324	444
MORMOOPIDAE *	3	242	154	71	213	401
Pteronotus	2	242	148	65	207	411
PHYLLOSTOMIDAE	23	283	225	156	274	391
PHYLLOSTOMINAE *	10	333	258	172	320	467
Micronycteris	2	248	180	106	234	378
Phyllostomus	2	290	235	173	278	385
GLOSSOPHAGINAE *	4	224	180	123	222	309
CAROLLIINAE *	2	251	205	159	241	340
STURNIRINAE *	1	190	155	131	174	257
STENODERMINAE *	5	273	218	164	259	373
Artibeus	2	235	192	125	239	316
BRACHYPHYLLINAE *	1	230	206	122	263	276
DESMODONTIDAE *	2	278	204	156	240	408

	N	AUD	CON	DCO	VCO	OLS
slope		.51	.50	.50	.51	.54
y-intercept		-.267	-.461	-.867	-.700	-.742
NATALIDAE *	1	227	148	49	218	378
FURIPTERIDAE *	1	274	196	76	285	431
VESPERTILIONIDAE	45	240	169	73	238	375
VESPERTILIONINAE	30	223	158	70	221	345
VEX: *Tylonycteris* excl.*	28	228	162	71	226	354
Myotis	5	234	171	56	253	354
Pipistrellus	4	183	138	73	186	273
Eptesicus	5	212	147	69	204	335
Tylonycteris	2	149	112	57	152	224
Scotophilus	2	278	185	74	262	441
MINIOPTERINAE *	4	258	174	68	249	414
MURININAE *	5	251	186	96	252	376
KERIVOULINAE *	4	345	221	41	350	585
NYCTOPHILINAE *	2	229	182	147	211	321
MOLOSSIDAE *	10	237	192	96	260	320
Tadarida	4	220	184	88	252	289
Molossops	2	231	181	91	246	326
Molossus	2	267	217	103	297	360
CHIROPTERA (AvCS)	150	234	186	123	232	327
CHIROPTERA (AvCF)	NN=28	246	192	121	243	349
MEGACHIR. (AvMegS)	27	117	131	177	103	96
MEGACHIR. (AvMegF)	NN= 3	119	128	170	103	106
MICROCHIR. (AvMicS)	123	260	197	111	260	378
MICROCHIR. (AvMicF)	NN=25	261	199	115	260	379

Number of families and/or subfamilies marked by * = NN
Number of species = N
Abbreviations:

AvCF	average of Chiroptera families and/or subfamilies		
AvCS	average of Chiroptera species		
AvMegF	average of Megachiroptera families and/or subfamilies		
AvMegS	average of Megachiroptera species		
AvMicF	average of Microchiroptera families and/or subfamilies		
AvMicS	average of Microchiroptera species		
AUD	auditory (acoustic) nuclei	OLS	complexus olivaris superior
CON	cochlear nuclei	VCO	nucleus cochlearis ventralis
DCO	nucleus cochlearis dorsalis		

Table 25. Volumes of mesencephalic components

Size indices in species see Table 26
Average size indices in supraspecific taxonomic units see Table 27

	MES	MTG	MTC	INC	SUC
PTEROPODIDAE					
PTEROPODINAE					
Eidolon helvum	200.71	120.77	79.94	28.59	51.35
Rousettus aegyptiacus	109.12	67.99	41.13	15.47	25.66
Myonycteris torquata	66.67	41.31	25.36	7.45	17.91
Pteropus conspicillatus	334.60	207.25	127.36	37.41	89.95
Pteropus poliocephalus	337.07	220.00	117.07	36.84	80.23
Pteropus scapulatus	236.15	165.05	71.10	21.15	49.95
Pteropus tonganus	248.02	155.31	92.72	28.94	63.78
Pteropus vampyrus	380.74	235.60	145.14	40.28	104.86
Dobsonia inermis	109.96	73.96	35.99	12.83	23.17
D. moluccensis magna	208.55	134.74	73.80	25.55	48.26
Dobsonia praedatrix	127.13	82.76	44.37	16.40	27.97
Epomops franqueti	138.37	80.64	57.72	22.60	35.13
Epomophorus labiatus	92.48	53.12	39.36	14.87	24.48
Micropteropus pusillus	48.40	28.03	20.38	7.04	13.34
Scotonycteris zenkeri	42.03	24.37	17.66	6.39	11.27
Casinycteris argynnis	54.17	30.57	23.60	8.50	15.10
Cynopterus brachyotis	51.44	26.81	24.64	8.80	15.84
Cynopterus horsfieldi	74.85	42.27	32.58	11.18	21.40
Megaerops ecaudatus	44.08	26.31	17.77	7.47	10.30
Chironax melanocephalus	31.34	17.65	13.68	5.01	8.68
Balionycteris maculata	29.73	16.78	12.95	4.80	8.15
NYCTIMENINAE					
Nyctimene robinsoni	72.29	47.33	24.96	9.24	15.72
Nyctimene vizcaccia	55.98	31.81	24.17	8.31	15.86
MACROGLOSSINAE					
Eonycteris spelaea	71.37	42.64	28.73	11.38	17.35
Macroglossus minimus	33.69	18.15	15.55	6.33	9.21
Syconycteris a. australis	31.42	16.81	14.60	5.32	9.28
Notopteris macdonaldi	66.18	42.22	23.95	9.71	14.24

** = two specimens measured
Abbreviations:
INC colliculi inferiores
MES mesencephalon
MTC tectum mesencephali (colliculi inferiores + superiores)
MTG tegmentum mesencephali
SUC colliculi superiores

	MES	MTG	MTC	INC	SUC
RHINOPOMATIDAE					
Rhinopoma hardwickei	30.34	13.75	16.59	12.05	4.54
Rhinopoma microphyllum **	40.26	18.16	22.10	16.77	5.33
EMBALLONURIDAE					
EMBALLONURINAE					
Emballonura monticola	19.99	8.36	11.63	8.26	3.38
Emballonura raffrayana	21.36	8.56	12.80	9.31	3.49
Emballonura semicaudata	19.89	8.70	11.19	7.91	3.28
Coleura afra	26.26	10.60	15.66	10.71	4.94
Rhynchonycteris naso	14.63	6.85	7.78	5.21	2.57
Saccopteryx leptura	22.21	8.59	13.62	8.89	4.73
Cormura brevirostris	27.26	10.67	16.59	11.32	5.27
Peropteryx trinitatis	17.42	6.82	10.60	7.32	3.28
Taphozous australis	54.62	27.36	27.26	19.92	7.34
Taphozous hildegardeae	53.06	25.02	28.05	21.27	6.78
Taphozous theobaldi	66.53	31.83	34.70	25.45	9.25
Taphozous saccolaimus	72.94	32.80	40.14	31.01	9.13
DICLIDURINAE					
Cyttarops alecto	21.36	9.43	11.93	7.64	4.29
CRASEONYCTERIDAE					
Craseonycteris thonglongyai	10.55	4.97	5.59	3.91	1.68
NYCTERIDAE					
Nycteris grandis	72.82	36.55	36.27	24.59	11.68
Nycteris macrotis	38.46	18.64	19.82	12.83	6.99
Nycteris nana	24.90	12.72	12.18	7.67	4.51
Nycteris thebaica	32.80	15.86	16.94	10.99	5.95
MEGADERMATIDAE					
Megaderma lyra	80.54	32.94	47.59	35.51	12.08
Megaderma spasma	61.94	26.79	35.16	26.95	8.21
Macroderma gigas	144.55	71.98	72.56	48.68	23.89
Cardioderma cor	66.63	30.56	36.06	25.16	10.90
Lavia frons	66.00	31.39	34.61	24.70	9.92
RHINOLOPHIDAE					
Rhinolophus landeri	27.94	13.31	14.63	10.72	3.91
Rhinolophus luctus	56.51	23.11	33.40	24.85	8.55
Rhinolophus pusillus	16.01	8.27	7.73	5.41	2.33
HIPPOSIDERIDAE					
Hipposideros ater	23.39	10.47	12.92	8.95	3.97
Hipposideros ater aruensis	22.46	9.85	12.61	8.92	3.70
Hipposideros commersoni	58.73	28.33	30.40	20.44	9.96
Hipposideros diadema	69.18	32.80	36.37	26.59	9.78
Aselliscus stoliczkanus	14.23	6.63	7.60	4.87	2.72
Aselliscus tricuspidatus	14.00	7.27	6.73	4.49	2.24
Rhinonycteris aurantius	25.29	11.99	13.29	8.68	4.61
Triaenops persicus	25.95	12.83	13.12	8.44	4.69

> Table 25 <

	MES	MTG	MTC	INC	SUC
NOCTILIONIDAE					
Noctilio albiventris	48.96	23.00	25.96	18.95	7.01
Noctilio leporinus	89.79	46.14	43.65	32.67	10.98
MORMOOPIDAE					
Pteronotus gymnonotus	25.71	12.17	13.54	9.27	4.27
Pteronotus parnelli	40.88	20.66	20.22	14.34	5.89
Mormoops megalophylla	30.83	14.00	16.84	11.00	5.84
PHYLLOSTOMIDAE					
PHYLLOSTOMINAE					
Micronycteris megalotis	23.41	11.01	12.40	8.09	4.31
Micronycteris brachyotis	28.61	14.70	13.91	8.81	5.10
Macrophyllum macrophyllum	28.63	14.83	13.80	9.31	4.49
Tonatia bidens	63.35	25.47	37.87	29.06	8.81
Mimon crenulatum	30.21	13.19	17.02	13.12	3.90
Phyllostomus discolor	63.38	33.91	29.46	18.53	10.93
Phyllostomus hastatus	95.74	51.02	44.72	30.80	13.92
Phylloderma stenops	71.14	38.75	32.39	20.11	12.28
Trachops cirrhosus	81.65	40.71	40.94	30.56	10.38
Vampyrum spectrum	154.89	76.69	78.20	55.65	22.55
GLOSSOPHAGINAE					
Glossophaga soricina	25.81	15.04	10.77	6.37	4.40
Leptonycteris curasoae	35.24	19.33	15.91	9.72	6.19
Lionycteris spurrelli	24.66	13.41	11.25	6.87	4.38
Anoura geoffroyi	34.51	21.31	13.20	8.01	5.19
CAROLLIINAE					
Carollia perspicillata	36.38	20.68	15.70	9.59	6.10
Rhinophylla pumilio	22.74	12.19	10.56	7.04	3.52
STURNIRINAE					
Sturnira lilium	33.23	19.70	13.53	7.47	6.06
STENODERMINAE					
Uroderma bilobatum	41.13	20.77	20.37	12.74	7.63
Vampyrops vittatus	67.08	35.15	31.93	19.82	12.11
Chiroderma villosum	51.57	24.39	27.18	18.12	9.06
Artibeus cinereus	29.66	15.79	13.87	8.85	5.02
Artibeus lituratus	68.88	39.10	29.78	18.83	10.95
BRACHYPHYLLINAE					
Brachyphylla cavernarum	68.32	40.33	27.99	17.90	10.09

	MES	MTG	MTC	INC	SUC
DESMODONTIDAE					
Desmodus rotundus	58.27	34.77	23.50	15.66	7.84
Diphylla ecaudata	52.96	28.63	24.33	15.30	9.04
NATALIDAE					
Natalus tumidirostris	19.22	10.14	9.08	6.33	2.74
FURIPTERIDAE					
Furipterus horrens	12.69	6.64	6.05	3.96	2.09
VESPERTILIONIDAE					
VESPERTILIONINAE					
Myotis albescens	12.94	5.58	7.36	5.22	2.14
Myotis annectans	20.42	8.22	12.20	8.70	3.50
Myotis dasycneme	32.55	13.72	18.83	13.40	5.43
Myotis myotis	46.69	24.01	22.68	16.52	6.16
Myotis siligorensis	11.09	4.84	6.25	4.50	1.75
Pipistrellus ceylonicus	19.17	8.36	10.81	7.69	3.13
Pipistrellus crassulus	14.61	6.47	8.14	5.56	2.58
Pipistrellus javanicus	13.21	6.23	6.98	5.16	1.82
Pipistrellus mimus	10.45	4.49	5.96	4.27	1.69
Scotozous dormeri	15.16	6.57	8.59	6.05	2.54
Nyctalus noctula	37.71	17.47	20.23	14.25	5.98
Glischropus tylopus	11.20	4.90	6.30	4.35	1.95
Eptesicus brasiliensis	21.66	10.21	11.45	7.72	3.73
Eptesicus br. melanopterus	19.59	8.46	11.13	7.63	3.50
Eptesicus flavescens	21.15	8.53	12.62	8.35	4.27
Eptesicus pumilus	12.73	5.51	7.22	4.86	2.35
Eptesicus serotinus andersoni	32.55	13.88	18.67	13.40	5.27
Ia io	70.42	27.06	43.36	31.30	12.06
Tylonycteris pachypus	9.73	4.23	5.50	3.84	1.66
Tylonycteris robustula	13.43	5.74	7.70	5.37	2.32
Hesperoptenus blanfordi	19.14	8.69	10.45	7.59	2.86
Glauconycteris humeralis	16.87	6.97	9.89	7.49	2.40
Chalinolobus gouldi	23.58	11.25	12.32	8.88	3.44
Scotorepens sanborni	16.75	7.62	9.13	6.53	2.60
Rhogeessa parvula	12.91	7.76	5.15	3.66	1.49
Scotomanes ornatus	43.40	18.17	25.23	19.40	5.83
Scotomanes sp.?	34.76	16.29	18.47	14.38	4.10
Scotophilus dinganii	45.48	21.00	24.48	17.96	6.52
Scotophilus heathi	49.88	24.34	25.54	18.45	7.09
Lasiurus borealis	18.70	9.12	9.58	6.50	3.08
MINIOPTERINAE					
Miniopterus australis	17.99	9.00	8.98	6.50	2.48
Miniopterus magnater	26.89	12.06	14.82	11.05	3.77
Miniopterus medius	24.26	12.40	11.86	8.64	3.22
Miniopterus tristis	30.04	14.40	15.64	11.63	4.01

	MES	MTG	MTC	INC	SUC
VESPERTILIONIDAE (cont.)					
MURININAE					
Murina cyclotis cyclotis	21.17	9.57	11.59	7.62	3.98
Murina c. peninsularis	21.34	9.81	11.53	7.77	3.76
Murina huttoni rubella	20.38	9.78	10.60	7.09	3.51
Murina tubinaris	16.07	8.20	7.87	4.95	2.91
Harpiocephalus harpia	30.32	15.83	14.49	9.73	4.76
KERIVOULINAE					
Kerivoula papillosa	22.86	12.49	10.37	7.41	2.96
Kerivoula pellucida	19.93	10.25	9.69	7.24	2.44
Kerivoula phalaena	13.05	6.49	6.57	4.65	1.91
Phoniscus atrox	13.26	6.75	6.51	4.635	1.88
NYCTOPHILINAE					
Nyctophilus geoffroyi	15.46	7.20	8.26	6.17	2.09
Nyctophilus timoriensis	26.72	11.20	15.52	11.72	3.80
MOLOSSIDAE					
Tadarida condylura	41.54	18.04	23.50	15.99	7.51
Tadarida beccarii	23.10	11.30	11.80	7.86	3.94
Tadarida jobensis	32.55	14.77	17.78	12.26	5.51
Tadarida pumila	27.12	12.30	14.82	10.58	4.23
Otomops martiensseni	59.87	28.14	31.73	23.11	8.62
Molossops abrasus	44.93	18.87	26.06	17.27	8.79
Molossops planirostris	19.89	8.64	11.25	7.48	3.77
Molossus ater	53.77	23.83	29.94	22.03	7.91
Molossus molossus	32.81	15.64	17.17	12.50	4.68
Cheiromeles torquatus	111.59	47.99	63.60	40.44	23.16

** = two specimens measured

Abbreviations:

INC	colliculi inferiores	MTG	tegmentum mesencephali
MES	mesencephalon	SUC	colliculi superiores
MTC	tectum mesencephali (colliculi inferiores + superiores)		

Table 26. Size indices of mesencephalic components related to the average of the four Tenrecinae (AvTn = 100)

For volumes in species see Table 25
For average size indices in supraspecific taxonomic units see Table 27

	MES	MTG	MTC	INC	SUC
slope	.58	.58	.58	.58	.58
y-intercept	.481	.213	.141	-.061	-.287
PTEROPODIDAE					
PTEROPODINAE					
Eidolon helvum	262	293	229	130	394
Rousettus aegyptiacus	209	241	172	103	288
Myonycteris torquata	274	315	228	107	431
Pteropus conspicillatus	238	273	198	93	375
Pteropus poliocephalus	250	303	190	95	349
Pteropus scapulatus	251	325	165	78	311
Pteropus tonganus	241	280	197	98	364
Pteropus vampyrus	227	260	189	84	367
Dobsonia inermis	199	249	143	81	246
D. moluccensis magna	195	234	151	83	265
Dobsonia praedatrix	204	246	156	92	263
Epomops franqueti	285	307	260	162	423
Epomophorus labiatus	274	292	255	154	425
Micropteropus pusillus	247	265	227	125	398
Scotonycteris zenkeri	236	253	217	125	370
Casinycteris argynnis	265	277	252	145	432
Cynopterus brachyotis	220	213	231	131	397
Cynopterus horsfieldi	232	242	221	120	388
Megaerops ecaudatus	230	255	203	136	316
Chironax melanocephalus	196	204	187	109	317
Balionycteris maculata	209	219	199	118	336
NYCTIMENINAE					
Nyctimene robinsoni	256	310	193	114	326
Nyctimene vizcaccia	234	246	221	121	388
MACROGLOSSINAE					
Eonycteris spelaea	222	246	196	123	317
Macroglossus minimus	235	235	237	154	377
Syconycteris a. australis	218	217	222	129	378
Notopteris macdonaldi	185	219	147	95	234

** = two specimens measured

Abbreviations:

INC	colliculi inferiores	MTG	tegmentum mesencephali
MES	mesencephalon	SUC	colliculi superiores
MTC	tectum mesencephali (colliculi inferiores + superiores)		

	MES	MTG	MTC	INC	SUC
slope	.58	.58	.58	.58	.58
y-intercept	.481	.213	.141	-.061	-.287
RHINOPOMATIDAE					
Rhinopoma hardwickei	227	191	272	315	200
Rhinopoma microphyllum **	187	156	225	272	145
EMBALLONURIDAE					
EMBALLONURINAE					
Emballonura monticola	251	194	320	361	248
Emballonura raffrayana	252	187	331	383	242
Emballonura semicaudata	213	172	262	294	205
Coleura afra	210	157	274	299	232
Rhynchonycteris naso	223	193	259	277	229
Saccopteryx leptura	285	204	383	398	356
Cormura brevirostris	266	193	354	384	301
Peropteryx trinitatis	250	182	333	366	276
Taphozous australis	286	265	312	363	225
Taphozous hildegardeae	257	225	297	359	192
Taphozous theobaldi	259	230	296	346	211
Taphozous saccolaimus	272	227	327	403	200
DICLIDURINAE					
Cyttarops alecto	268	219	328	335	315
CRASEONYCTERIDAE					
Craseonycteris thonglongyai	202	176	234	261	189
NYCTERIDAE					
Nycteris grandis	336	312	366	395	316
Nycteris macrotis	269	242	304	313	287
Nycteris nana	268	254	287	288	285
Nycteris thebaica	305	273	345	356	324
MEGADERMATIDAE					
Megaderma lyra	320	242	413	491	281
Megaderma spasma	335	269	417	508	261
Macroderma gigas	298	275	327	349	288
Cardioderma cor	333	283	394	438	319
Lavia frons	350	309	402	457	309
RHINOLOPHIDAE					
Rhinolophus landeri	247	218	283	330	203
Rhinolophus luctus	247	187	319	378	219
Rhinolophus pusillus	216	206	228	254	184
HIPPOSIDERIDAE					
Hipposideros ater	252	209	305	336	251
Hipposideros ater aruensis	276	224	339	382	266
Hipposideros commersoni	133	119	150	161	132
Hipposideros diadema	259	227	298	346	214
Aselliscus stoliczkanus	187	162	218	223	210
Aselliscus tricuspidatus	196	188	206	219	184

	MES	MTG	MTC	INC	SUC
slope	.58	.58	.58	.58	.58
y-intercept	.481	.213	.141	-.061	-.287
HIPPOSIDERIDAE (cont.)					
Rhinonycteris aurantius	240	211	276	287	256
Triaenops persicus	188	172	208	213	199
NOCTILIONIDAE					
Noctilio albiventris	214	186	248	289	180
Noctilio leporinus	281	268	299	357	202
MORMOOPIDAE					
Pteronotus gymnonotus	187	164	215	235	182
Pteronotus parnelli	236	221	256	289	199
Mormoops megalophylla	206	174	246	256	229
PHYLLOSTOMIDAE					
PHYLLOSTOMINAE					
Micronycteris megalotis	261	228	303	314	282
Micronycteris brachyotis	199	189	211	213	208
Macrophyllum macrophyllum	272	261	286	308	249
Tonatia bidens	294	220	385	471	240
Mimon crenulatum	239	193	294	361	181
Phyllostomus discolor	259	257	264	264	262
Phyllostomus hastatus	232	230	238	261	198
Phylloderma stenops	255	257	254	251	258
Trachops cirrhosus	333	307	365	434	248
Vampyrum spectrum	258	236	285	322	220
GLOSSOPHAGINAE					
Glossophaga soricina	226	244	206	194	226
Leptonycteris curasoae	182	185	180	175	188
Lionycteris spurrelli	252	254	252	245	263
Anoura geoffroyi	228	261	191	185	201
CAROLLIINAE					
Carollia perspicillata	226	238	214	208	223
Rhinophylla pumilio	211	210	215	228	192
STURNIRINAE					
Sturnira lilium	192	211	171	150	205
STENODERMINAE					
Uroderma bilobatum	270	253	293	292	294
Vampyrops vittatus	275	267	286	283	291
Chiroderma villosum	265	232	305	324	273
Artibeus cinereus	225	222	231	234	224
Artibeus lituratus	220	232	208	210	205
BRACHYPHYLLINAE					
Brachyphylla cavernarum	250	273	224	228	216

	MES	MTG	MTC	INC	SUC
slope	.58	.58	.58	.58	.58
y-intercept	.481	.213	.141	-.061	-.287
DESMODONTIDAE					
Desmodus rotundus	240	265	212	224	189
Diphylla ecaudata	239	240	240	241	239
NATALIDAE					
Natalus tumidirostris	207	203	214	238	173
FURIPTERIDAE					
Furipterus horrens	205	199	214	223	198
VESPERTILIONIDAE					
VESPERTILIONINAE					
Myotis albescens	191	153	238	269	186
Myotis annectans	180	134	235	266	181
Myotis dasycneme	224	175	283	321	219
Myotis myotis	235	224	250	290	182
Myotis siligorensis	178	144	220	252	165
Pipistrellus ceylonicus	166	134	204	231	158
Pipistrellus crassulus	169	139	206	224	175
Pipistrellus javanicus	182	159	211	248	148
Pipistrellus mimus	157	125	196	223	149
Scotozous dormeri	165	132	204	229	162
Nyctalus noctula	184	158	216	243	171
Glischropus tylopus	147	119	181	199	150
Eptesicus brasiliensis	198	173	228	245	199
Eptesicus br. melanopterus	183	147	228	249	192
Eptesicus flavescens	179	134	233	246	211
Eptesicus pumilus	158	127	196	210	171
Eptesicus serotinus andersoni	184	146	231	265	175
Ia io	232	165	312	359	233
Tylonycteris pachypus	129	104	160	178	129
Tylonycteris robustula	141	112	177	197	143
Hesperoptenus blanfordi	200	168	238	276	175
Glauconycteris humeralis	207	159	266	321	173
Chalinolobus gouldi	187	165	214	245	160
Scotorepens sanborni	154	130	183	209	140
Rhogeessa parvula	178	199	155	176	120
Scotomanes ornatus	240	186	305	374	189
Scotomanes sp.?	227	197	264	327	157
Scotophilus dinganii	236	202	278	325	199
Scotophilus heathi	223	202	250	287	186
Lasiurus borealis	188	170	210	227	181
MINIOPTERINAE					
Miniopterus australis	191	177	208	240	154
Miniopterus magnater	185	154	223	264	152
Miniopterus medius	195	185	209	242	152
Miniopterus tristis	202	179	230	272	158

	MES	MTG	MTC	INC	SUC
slope	.58	.58	.58	.58	.58
y-intercept	.481	.213	.141	-.061	-.287
VESPERTILIONIDAE (cont.)					
MURININAE					
Murina cyclotis cyclotis	224	188	269	281	247
Murina c. peninsularis	202	172	238	256	209
Murina huttoni rubella	191	170	217	231	193
Murina tubinaris	199	188	213	213	211
Harpiocephalus harpia	221	214	231	247	204
KERIVOULINAE					
Kerivoula papillosa	197	200	196	223	150
Kerivoula pellucida	290	277	309	368	209
Kerivoula phalaena	228	210	251	283	196
Phoniscus atrox	191	180	205	232	158
NYCTOPHILINAE					
Nyctophilus geoffroyi	163	140	190	226	129
Nyctophilus timoriensis	221	172	281	338	184
MOLOSSIDAE					
Tadarida condylura	202	163	250	271	214
Tadarida beccarii	164	149	183	194	164
Tadarida jobensis	212	178	253	278	210
Tadarida pumila	232	195	277	315	212
Otomops martiensseni	228	199	264	306	192
Molossops abrasus	194	151	247	260	223
Molossops planirostris	208	168	258	273	231
Molossus ater	231	190	282	330	199
Molossus molossus	238	210	272	315	198
Cheiromeles torquatus	189	151	236	239	230

** = two specimens measured

Abbreviations:

INC	colliculi inferiores	MTG	tegmentum mesencephali
MES	mesencephalon	SUC	colliculi superiores
MTC	tectum mesencephali (colliculi inferiores + superiores)		

Table 27. Average size indices of mesencephalic components related to the average of the four Tenrecinae (AvTn = 100)

For volumes in species see Table 25
For size indices in species see Table 26

	N	MES	MTG	MTC	INC	SUC
slope		.58	.58	.58	.58	.58
y-intercept		.481	.213	.141	-.061	-.287
PTEROPODIDAE	27	233	260	203	115	351
PTEROPODINAE *	21	235	264	203	113	355
Pteropus	5	241	288	188	90	353
Dobsonia	3	200	243	150	85	258
Cynopterus	2	226	228	226	126	393
NYCTIMENINAE *	2	245	278	207	118	357
MACROGLOSSINAE *	4	215	229	200	125	326
RHINOPOMATIDAE *	2	207	174	248	293	172
EMBALLONURIDAE	13	253	204	313	351	249
EMBALLONURINAE *	12	252	203	312	353	243
NON-*TAPHOZOUS* EMB.	8	244	185	314	345	261
TAPHOZOUS	4	269	237	308	368	207
DICLIDURINAE *	1	268	219	328	335	315
CRASEONYCTERIDAE *	1	202	176	234	261	189
NYCTERIDAE *	4	295	270	326	338	303
MEGADERMATIDAE *	5	327	275	390	449	291
RHINOLOPHIDAE *	3	237	204	277	321	202
HIPPOSIDERIDAE	8	216	189	250	271	214
HIX: *H. commersoni* excl.*	7	228	199	264	287	226
Hipposideros	4	230	195	273	306	216
H. commersoni excluded	3	262	220	314	355	244
NOCTILIONIDAE *	2	248	227	274	323	191
MORMOOPIDAE *	3	210	186	239	260	203
Pteronotus	2	212	193	236	262	191
PHYLLOSTOMIDAE	22	244	236	256	269	233
PHYLLOSTOMINAE *	10	260	238	288	320	235
Micronycteris	2	230	209	257	264	245
Phyllostomus	2	246	243	251	262	230
GLOSSOPHAGINAE *	4	222	236	207	200	219
CAROLLIINAE *	2	219	224	214	218	207
STURNIRINAE *	1	192	211	171	150	205
STENODERMINAE *	5	251	241	265	268	257
Artibeus	2	223	227	220	222	214
BRACHYPHYLLINAE *	1	250	273	224	228	216
DESMODONTIDAE *	2	240	253	226	233	214

	N	MES	MTG	MTC	INC	SUC
slope		.58	.58	.58	.58	.58
y-intercept		.481	.213	.141	-.061	-.287
NATALIDAE *	1	207	203	214	238	173
FURIPTERIDAE *	1	205	199	214	223	198
VESPERTILIONIDAE	45	194	166	228	258	175
VESPERTILIONINAE	30	187	156	226	257	173
VEX: *Tylonycteris* excl.*	28	191	159	230	262	175
Myotis	5	202	166	245	280	186
Pipistrellus	4	168	139	204	232	157
Eptesicus	5	180	145	223	243	190
Tylonycteris	2	135	108	168	187	136
Scotophilus	2	229	202	264	306	192
MINIOPTERINAE *	4	193	174	217	255	154
MURININAE *	5	207	186	234	246	213
KERIVOULINAE *	4	227	217	240	276	178
NYCTOPHILINAE *	2	192	156	235	282	156
MOLOSSIDAE *	10	210	175	252	278	207
Tadarida	4	203	171	241	265	200
Molossops	2	201	159	252	267	227
Molossus	2	234	200	277	323	199
CHIROPTERA (AvCS)	150	226	209	248	255	236
CHIROPTERA (AvCF)	NN=28	230	216	248	259	228
MEGACHIR. (AvMegS)	27	233	260	203	115	351
MEGACHIR. (AvMegF)	NN= 3	232	257	203	119	346
MICROCHIR. (AvMicS)	123	225	198	258	285	210
MICROCHIR. (AvMicF)	NN=25	230	211	253	276	214

Number of families and/or subfamilies marked by * = NN
Number of species = N
Abbreviations:

AvCF	average of Chiroptera families and/or subfamilies		
AvCS	average of Chiroptera species		
AvMegF	average of Megachiroptera families and/or subfamilies		
AvMegS	average of Megachiroptera species		
AvMicF	average of Microchiroptera families and/or subfamilies		
AvMicS	average of Microchiroptera species		
INC	colliculi inferiores	MTG	tegmentum mesencephali
MES	mesencephalon	SUC	colliculi superiores
MTC	tectum mesencephali (colliculi inferiores + superiores)		

Table 28. Volumes of cerebellar nuclei in mm^3

For size indices in species and average size indices in supraspecific taxonomic units see Table 29

	CER	TCN	MCN	ICN	LCN
PTEROPODIDAE					
Cynopterus brachyotis	122.690	2.676	0.666	1.000	1.010
Cynopterus horsfieldi	180.460	3.350	0.892	1.214	1.244
Eonycteris spelaea	172.665	3.434	1.006	1.088	1.340
RHINOLOPHIDAE					
Rhinolophus hipposideros	30.411	0.746	0.166	0.258	0.322
HIPPOSIDERIDAE					
Hipposideros bicolor	42.400	0.842	0.174	0.274	0.394
NOCTILIONIDAE					
Noctilio leporinus	194.082	5.254	1.076	2.056	2.122
PHYLLOSTOMIDAE					
Phyllostomus discolor	170.445	2.490	0.572	0.872	1.046
Glossophaga soricina	81.597	1.712	0.458	0.576	0.678
Carollia perspicillata	92.059	2.022	0.400	0.764	0.858
Sturnira lilium	96.057	1.782	0.362	0.572	0.848
Vampyrops helleri	97.232	1.710	0.358	0.532	0.820
Artibeus jamaicensis	173.820	3.012	0.556	1.076	1.380
Artibeus lituratus	221.877	4.304	0.810	1.478	2.016
DESMODONTIDAE					
Desmodus rotundus	166.361	2.448	0.422	0.718	1.308
VESPERTILIONIDAE					
Myotis bechsteini	42.532	1.090	0.306	0.268	0.516
Myotis myotis	87.119	2.046	0.558	0.640	0.848
Myotis nattereri	37.440	1.064	0.218	0.336	0.510
MOLOSSIDAE					
Tadarida leucostigma	69.986	1.382	0.388	0.504	0.490

Abbreviations:

CER cerebellum
ICN nucleus interpositus cerebelli
LCN nucleus lateralis cerebelli
MCN nucleus medialis cerebelli
TCN total cerebellar nuclei

Table 29. Size indices and average size indices of cerebellar nuclei related to the average of the four Tenrecinae (AvTn = 100)

For volumes in species see Table 28

	CER	TCN	MCN	ICN	LCN
slope	.68	.68	.68	.68	.68
y-intercept	.552	-.979	-1.638	-1.314	-1.476
PTEROPODIDAE					
Cynopterus brachyotis	314	232	264	188	275
Cynopterus horsfieldi	315	199	241	156	232
Eonycteris spelaea	304	205	274	141	251
RHINOLOPHIDAE					
Rhinolophus hipposideros	302	252	256	188	341
HIPPOSIDERIDAE					
Hipposideros bicolor	280	189	178	133	277
NOCTILIONIDAE					
Noctilio leporinus	344	316	296	268	401
PHYLLOSTOMIDAE					
Phyllostomus discolor	413	205	215	155	271
Glossophaga soricina	482	343	419	250	427
Carollia perspicillata	365	272	245	222	362
Sturnira lilium	349	220	204	153	329
Vampyrops helleri	458	273	261	184	412
Artibeus jamaicensis	390	230	193	177	331
Artibeus lituratus	403	266	228	197	391
DESMODONTIDAE					
Desmodus rotundus	406	203	159	129	340
VESPERTILIONIDAE					
Myotis bechsteini	249	217	278	115	323
Myotis myotis	269	215	267	145	280
Myotis nattereri	267	258	241	176	388
MOLOSSIDAE					
Tadarida leucostigma	211	142	182	112	158
CHIROPTERA (AvCS; N = 18)	340	235	244	172	322
MEGACHIR. (AvMegS; N = 3)	311	212	260	161	253
MICROCHIR. (AvMicS; N = 15)	346	240	241	174	335
PHYLLOSTOMIDAE (N = 7)	409	258	252	191	360
STENODERMINAE (N = 3)	417	256	227	186	378
VESPERTILIONIDAE (N = 3)	262	230	262	146	330

Abbreviations:

CER cerebellum
ICN nucleus interpositus cerebelli
LCN nucleus lateralis cerebelli
MCN nucleus medialis cerebelli
TCN total cerebellar nuclei

Table 30. Volumes of some mesencephalic and diencephalic components in mm^3

For size indices in species and average size indices in supraspecific taxonomic units see Table 31

	NLL	CGM	CGL	GLD	GLV
PTEROPODIDAE					
Cynopterus brachyotis	0.946	2.584	6.966	5.140	1.826
Cynopterus horsfieldi	1.248	4.050	7.740	6.178	1.562
Eonycteris spelaea	1.486	4.726	6.346	4.680	1.666
RHINOLOPHIDAE					
Rhinolophus hipposideros	1.130	1.034	0.198	0.178	0.020
HIPPOSIDERIDAE					
Hipposideros bicolor	2.184	2.692	0.420	0.312	0.108
Asellia tridens	2.196	3.040	0.876	0.534	0.342
NOCTILIONIDAE					
Noctilio leporinus	6.074	13.276	1.646	1.254	0.392
PHYLLOSTOMIDAE					
Phyllostomus discolor	2.458	4.990	2.140	1.700	0.440
Glossophaga soricina	2.148	4.386	0.916	0.682	0.234
Carollia perspicillata	2.436	3.722	1.074	0.728	0.346
Sturnira lilium	1.722	3.020	1.318	1.058	0.260
Vampyrops helleri	1.856	3.188	1.372	1.074	0.298
Artibeus jamaicensis	3.554	7.026	1.964	1.456	0.508
Artibeus lituratus	4.154	7.560	2.750	2.256	0.494
DESMODONTIDAE					
Desmodus rotundus	2.608	8.640	1.116	0.844	0.272
VESPERTILIONIDAE					
Myotis bechsteini	1.796	2.598	0.278	0.196	0.082
Myotis myotis	3.418	4.928	0.970	0.636	0.334
Myotis nattereri	2.064	2.348	0.272	0.186	0.086
MOLOSSIDAE					
Tadarida leucostigma	2.058	4.528	0.842	0.562	0.280

Abbreviations for Tables 30 and 31:

CGL	corpus geniculatum laterale	GLV	ventral part of CGL
CGM	corpus geniculatum mediale	MES	mesencephalon
DIE	diencephalon	NLL	nuclei lemnisci lateralis
GLD	dorsal part of CGL		

Table 31. Size indices and average size indices of some mesencephalic and diencephalic components related to the average of the four Tenrecinae (AvTn = 100)

For volumes in species see Table 30

	MES	NLL	DIE	CGM	CGL	GLD	GLV
slope	.58	.58	.65	.72	.60	.60	.60
y-intercept	.481	-.865	.350	-1.052	-1.080	-1.246	-1.579
PTEROPODIDAE							
Cynopterus brachyotis	220	90	353	230	1011	1094	836
Cynopterus horsfieldi	232	86	343	241	803	940	512
Eonycteris spelaea	222	103	334	284	663	716	549
RHINOLOPHIDAE							
Rhinolophus hipposideros	207	342	205	388	95	126	30
HIPPOSIDERIDAE							
Hipposideros bicolor	246	466	211	656	141	153	114
Asellia tridens	192	367	174	547	228	204	281
NOCTILIONIDAE							
Noctilio leporinus	281	422	282	804	173	193	130
PHYLLOSTOMIDAE							
Phyllostomus discolor	259	223	355	421	297	345	192
Glossophaga soricina	226	416	286	949	278	304	224
Carollia perspicillata	226	336	260	528	229	228	233
Sturnira lilium	192	221	288	391	261	307	162
Vampyrops helleri	247	297	287	543	342	392	234
Artibeus jamaicensis	226	302	290	546	254	276	208
Artibeus lituratus	220	295	280	470	295	355	167
DESMODONTIDAE							
Desmodus rotundus	240	238	324	733	155	172	120
VESPERTILIONIDAE							
Myotis bechsteini	198	346	226	558	84	87	78
Myotis myotis	235	382	218	538	167	160	181
Myotis nattereri	210	470	206	620	98	98	97
MOLOSSIDAE							
Tadarida leucostigma	196	225	150	482	142	139	149
CHIROPTERA (AvCS; N = 19)	225	296	267	523	301	331	237
MEGACHIR. (AvMegS; N = 3)	225	93	344	252	826	917	632
MICROCHIR. (AvMicS; N = 16)	225	334	253	573	202	221	163
PHYLLOSTOMIDAE (N = 7)	228	299	292	550	280	315	203
STENODERMINAE (N = 3)	231	298	286	520	297	341	203
VESPERTILIONIDAE (N = 3)	215	399	217	572	116	115	119

Abbreviations as in Table 30

Table 32. Volumes of telencephalic components (number of individuals as in Table 33)

For size indices in species see Table 33

For average size indices in supraspecific taxonomic units see Table 34

	MOB	PAL	STR	SEP	AMY	HIP	SCH	NEO
PTEROPODIDAE								
PTEROPODINAE								
Eidolon helvum	208.7	243.5	285.3	61.7	99.4	258.1	151.7	1406
Rousettus aegyptiacus	114.5	127.7	130.2	32.0	50.8	145.4	82.9	687.4
R. amplexic. brachyot. §)	85.7	93.4	74.9	20.0	35.2	101.4	35.8	396.4
R. ampl. stresemanni §)	117.1	116.0	84.9	25.8	45.5	131.1	59.6	469.6
Myonycteris torquata	56.1	67.2	58.6	17.5	24.6	85.7	47.1	336.5
Pteropus alecto	238.2	312.5	440.6	90.3	115.9	357.4	220.9	2716
Pteropus conspicillatus	306.2	377.1	510.7	97.0	141.7	394.3	260.4	3339
Pteropus hypomelanus	219.5	277.9	320.4	69.8	99.8	283.5	196.8	1906
Pteropus lylei	207.8	274.5	393.0	70.0	102.2	288.2	186.4	2386
Pteropus mahaganus	168.1	273.6	334.8	67.8	106.8	274.5	171.6	1915
Pteropus neohibernicus	344.0	396.7	583.8	96.8	139.6	400.4	283.0	3676
Pteropus poliocephalus	252.9	309.8	444.1	90.1	114.9	358.5	225.7	2710
Pteropus samoensis	213.4	275.9	345.6	67.2	92.8	323.9	216.5	2144
Pteropus scapulatus	177.5	230.1	322.9	67.5	86.7	267.8	204.0	2078
Pteropus temmincki	194.3	234.7	268.8	64.3	92.9	277.4	165.6	1814
Pteropus tonganus	224.2	304.1	417.1	74.1	108.3	305.7	200.7	2237
Pteropus vampyrus	376.4	414.1	572.0	113.5	139.7	443.9	294.9	3610
Dobsonia inermis	145.6	168.1	177.3	39.2	68.0	186.5	94.9	834.8
D. moluccensis magna §)	288.2	328.2	309.8	69.5	117.4	322.5	160.4	1743
D. molucc. anderseni §)	220.3	241.0	238.2	52.7	77.0	243.0	119.8	1284
Dobsonia praedatrix	199.6	190.3	192.9	41.8	68.8	181.2	93.1	933.0
Hypsignathus monstrosus	136.3	161.5	160.7	45.6	56.9	222.9	114.1	1056
Epomops franqueti	107.8	115.2	99.2	28.9	46.5	159.8	82.2	614.8
Epomophorus labiatus	86.9	82.8	81.7	21.4	37.2	114.3	54.1	460.8
Micropteropus pusillus	37.4	47.9	42.6	11.3	23.2	56.9	29.6	242.9
Scotonycteris zenkeri	37.0	39.9	33.0	9.23	16.8	73.8	31.6	182.5
Casinycteris argynnis	28.6	43.6	37.4	12.2	17.3	72.0	34.6	233.7
Cynopterus brachyotis	59.6	49.9	50.6	14.6	24.6	80.9	38.5	287.6
Cynopterus horsfieldi	66.0	64.6	66.5	18.5	29.0	110.3	56.8	432.3
Cynopterus sphinx sphinx	70.2	61.1	59.8	17.3	27.2	95.0	44.3	352.2
Megaerops ecaudatus	45.5	39.2	34.1	11.5	17.4	71.4	36.5	246.8
Chironax melanocephalus	35.8	35.7	28.4	9.53	14.8	56.9	26.5	168.3
Sphaerias blanfordi	33.5	42.7	39.8	14.5	19.7	68.6	33.6	223.8
Balionycteris maculata	31.2	29.2	22.4	8.44	13.0	54.7	23.6	135.0

Abbreviations:

AMY	amygdala	PAL	paleocortex
HIP	hippocampus	SCH	schizocortex
MOB	main olfactory bulb	SEP	septum
NEO	neocortex	STR	striatum

§) see addendum at end of Volume 3, pp. 1592-1595

	MOB	PAL	STR	SEP	AMY	HIP	SCH	NEO
PTEROPODIDAE (cont.)								
NYCTIMENINAE								
Nyctimene albiventer	52.5	52.9	36.4	11.8	22.0	68.5	29.8	219.1
Nyctimene robinsoni	81.1	77.2	58.6	17.2	33.3	102.7	43.8	335.1
Nyctimene vizcaccia	73.2	71.0	45.1	14.0	26.8	73.0	31.7	257.0
Paranyctimene raptor	42.0	45.8	34.9	9.45	17.8	62.4	27.0	198.3
MACROGLOSSINAE								
Eonycteris spelaea	67.0	68.6	69.5	19.9	32.5	97.7	55.4	384.0
Megaloglossus woermanni	25.3	30.8	30.8	10.2	15.6	64.4	31.6	190.1
Macroglossus minimus	26.9	28.5	28.0	8.14	13.0	47.0	22.8	155.7
Macroglossus sobrinus	33.2	36.0	36.7	11.8	19.5	58.9	23.8	186.2
Syconycteris a. australis §)	29.5	34.8	33.0	9.74	12.5	48.4	23.7	170.0
S. australis papuana §)	33.3	36.3	35.9	9.92	15.7	57.8	24.9	180.2
Melonycteris melanops	63.7	76.2	89.4	17.7	33.7	79.4	51.2	414.2
Nesonycteris woodfordi	44.6	54.5	65.7	14.3	23.2	66.8	41.9	320.0
Notopteris macdonaldi	63.0	93.9	94.2	23.6	39.4	96.6	52.2	419.9
RHINOPOMATIDAE								
Rhinopoma hardwickei	6.15	10.8	12.5	3.86	13.8	16.1	9.68	54.0
Rhinopoma microphyllum	8.80	16.9	17.1	5.03	15.4	25.8	13.9	77.5
EMBALLONURIDAE								
EMBALLONURINAE								
Emballonura monticola	1.18	4.48	8.67	1.77	9.58	7.61	4.40	32.1
Emballonura raffrayana	1.28	6.04	8.00	2.19	8.10	9.08	5.23	34.5
Emballonura semicaudata	1.43	5.78	9.03	2.05	8.17	10.1	5.50	36.0
Coleura afra	3.98	8.90	12.7	2.78	13.8	12.4	7.76	52.5
Rhynchonycteris naso	0.91	3.62	5.91	1.40	6.79	5.48	3.01	22.1
Saccopteryx bilineata	1.99	7.53	11.1	2.70	15.7	12.2	8.96	43.9
Saccopteryx canescens	1.11	3.71	6.15	1.61	8.37	6.79	4.08	23.7
Saccopteryx leptura	1.36	5.06	7.17	1.71	9.64	8.65	4.92	30.0
Cormura brevirostris	2.28	7.37	9.26	2.38	12.9	11.2	6.18	43.2
Peropteryx macrotis	1.68	4.65	7.44	1.98	10.5	8.13	6.15	30.6
Peropteryx trinitatis	1.25	4.40	6.97	1.67	6.84	6.95	4.73	25.7
Taphozous australis	8.62	16.2	24.7	6.82	26.3	25.5	17.6	101.8
Taphozous hildegardeae	11.0	20.1	22.5	6.28	27.6	25.4	16.0	111.3
Taphozous mauritianus	9.82	17.8	20.2	6.30	27.5	25.4	18.0	105.9
Taphozous melanopogon	10.1	20.3	23.0	5.58	24.7	23.7	16.2	113.2
Taphozous theobaldi	12.7	23.3	27.9	7.00	29.2	28.2	19.5	130.6
Taphozous saccolaimus	13.3	20.8	27.4	7.35	33.0	30.1	21.3	144.2
DICLIDURINAE								
Cyttarops alecto	0.74	2.92	7.31	1.57	5.90	7.96	4.11	31.9
CRASEONYCTERIDAE								
Craseonycteris thonglongyai	0.66	2.27	3.92	1.18	4.34	4.80	2.40	17.5

> Table 32 <

	MOB	PAL	STR	SEP	AMY	HIP	SCH	NEO
NYCTERIDAE								
Nycteris arge	3.92	10.1	19.5	3.57	11.8	17.9	8.75	89.8
Nycteris grandis	6.09	16.0	37.4	6.29	22.6	36.2	17.2	187.0
Nycteris hispida	2.24	7.58	15.9	2.95	10.2	17.3	8.06	69.6
Nycteris javanica	3.38	8.82	21.4	3.58	13.7	27.3	10.3	122.7
Nycteris macrotis	4.15	9.42	17.4	3.49	14.8	18.5	10.4	102.1
Nycteris nana	2.85	6.00	13.1	2.71	7.89	14.5	6.29	54.4
Nycteris thebaica	2.90	8.42	15.8	2.89	11.1	17.4	8.42	85.7
Nycteris tragata	3.65	8.48	21.1	3.87	13.6	24.1	9.17	124.1
MEGADERMATIDAE								
Megaderma lyra	18.0	25.7	41.1	7.98	30.9	40.6	26.0	244.6
Megaderma spasma	10.5	18.4	30.7	6.23	22.0	31.1	21.7	172.9
Macroderma gigas	21.6	40.3	77.7	15.0	50.4	68.9	48.6	510.5
Cardioderma cor	8.07	16.4	29.9	6.97	24.1	32.3	22.7	186.3
Lavia frons	4.35	12.5	28.7	5.85	20.4	25.8	19.7	171.4
RHINOLOPHIDAE								
Rhinolophus cornutus	3.57	5.86	9.76	2.37	7.04	14.5	6.34	37.5
Rhinolophus eloquens	9.38	15.8	20.5	5.16	15.8	28.8	13.6	76.5
Rhinolophus hipposideros	1.97	5.46	6.87	2.03	4.95	13.6	5.48	19.0
Rhinolophus landeri	5.58	9.99	14.3	3.65	10.7	19.1	8.02	50.6
Rhinolophus lepidus	4.19	7.86	10.4	2.29	8.32	12.7	5.95	32.6
Rhinolophus luctus	6.89	16.7	25.1	5.74	25.7	34.2	21.5	135.6
Rhinolophus macrotis	3.93	7.27	10.3	2.71	6.57	19.3	7.41	32.4
Rhinolophus malayanus	3.96	7.27	9.74	2.73	9.79	14.3	5.92	34.5
Rhinolophus megaphyllus	4.57	8.67	13.9	3.08	11.2	21.0	8.77	49.6
Rhinolophus megaph. ssp.	3.06	8.24	12.2	2.86	9.42	17.9	7.55	44.4
Rhinolophus paradoxoloph.	5.41	9.51	12.8	3.13	7.98	24.3	8.90	46.9
Rhinolophus pearsoni	4.38	10.1	15.6	3.80	13.9	25.4	11.6	70.9
Rhinolophus pusillus	3.41	6.32	8.94	1.87	7.18	12.5	5.24	29.6
Rhinolophus trifoliatus	3.26	9.57	13.7	3.34	15.9	23.0	10.8	71.6
Rhinolophus yunanensis	8.84	14.5	20.4	4.84	19.2	32.7	16.2	90.6
HIPPOSIDERIDAE								
Hipposideros armiger	15.2	24.8	38.6	8.55	31.0	57.4	25.1	168.5
Hipposideros ater	3.41	7.32	11.4	2.70	7.73	16.0	6.34	38.6
Hipposideros ater aruensis	2.96	7.21	10.0	2.30	6.01	14.4	5.36	34.1
Hipposideros ater saevus	3.85	7.57	9.44	2.05	6.05	13.2	5.39	32.6
Hipposideros caffer	5.38	9.55	15.6	3.52	11.6	20.1	9.01	60.9
H. calc. calcaratus §)	10.9	16.9	23.5	4.44	13.7	24.5	10.1	82.1
H. calcaratus cupidus §)	8.79	14.2	19.6	3.53	11.2	20.3	9.33	72.6
Hipposideros cervinus.	2.95	6.63	12.0	2.44	7.89	13.9	6.04	40.7
Hipposideros commersoni	23.7	30.7	38.2	8.70	27.5	47.8	26.7	171.7
Hipposideros diadema	14.2	23.2	34.7	8.43	30.3	47.9	22.0	154.5
Hipposideros fulvus	5.16	8.20	12.4	2.73	9.71	15.1	6.18	44.6

	MOB	PAL	STR	SEP	AMY	HIP	SCH	NEO
HIPPOSIDERIDAE (cont.)								
Hipposideros halophyllus	2.39	4.80	6.83	1.68	5.36	10.7	5.01	25.5
Hipposideros lankadiva	15.4	28.2	38.0	8.76	27.8	48.8	24.0	154.5
Hipposideros larvatus	9.64	14.9	24.7	5.20	18.7	28.0	11.4	85.4
Hipposideros lekaguli	10.7	16.9	26.3	6.40	19.2	35.8	19.3	123.1
H. m. maggietaylorae §)	21.1	28.9	37.0	6.53	20.9	38.4	18.3	145.5
H. maggietayl. erroris §)	13.8	22.1	27.0	4.70	15.8	24.5	10.7	90.0
Hipposideros ridleyi	4.70	9.53	15.2	3.32	10.4	18.7	8.05	50.3
Hipposideros speoris	5.30	9.33	14.8	3.39	11.9	16.8	7.24	57.6
Hipposideros turpis	10.4	20.9	31.1	6.85	21.9	41.2	17.7	128.9
Aselliscus stoliczkanus	1.68	4.39	6.87	1.67	5.07	12.7	4.23	22.5
Aselliscus tricuspidatus	2.07	4.80	7.91	1.74	5.52	10.5	4.41	22.2
Rhinonycteris aurantius	3.87	7.70	12.7	3.15	8.73	19.2	7.31	47.3
Triaenops persicus	5.22	7.30	11.7	2.62	9.61	17.4	6.66	48.1
NOCTILIONIDAE								
Noctilio albiventris	8.77	16.5	29.2	5.25	25.5	21.5	13.0	184.2
Noctilio leporinus	18.3	31.4	62.0	10.6	46.3	50.5	27.8	361.7
MORMOOPIDAE								
Pteronotus gymnonotus	7.28	11.2	17.9	3.45	13.9	19.2	8.90	77.7
Pteronotus personatus	4.85	8.70	11.9	2.62	10.4	12.9	6.07	50.0
Pteronotus parnelli	11.8	22.3	27.6	5.89	24.3	34.6	13.6	119.8
Mormoops megalophylla	6.66	11.8	17.0	4.75	12.8	27.5	10.6	67.2
PHYLLOSTOMIDAE								
PHYLLOSTOMINAE								
Micronycteris megalotis	10.3	15.4	14.3	2.45	11.2	14.1	7.59	59.7
Micronycteris minuta	12.0	15.9	14.7	2.93	10.5	17.3	9.02	65.6
Micronycteris schmidtorum	11.3	16.8	15.9	2.79	12.0	14.4	9.05	76.0
Micronycteris brachyotis	21.8	25.5	19.0	4.41	14.0	22.6	11.2	88.0
Macrophyllum macrophyll.	4.39	9.70	17.8	3.00	10.5	19.6	9.23	80.3
Tonatia bidens	30.3	42.0	35.6	8.05	28.9	41.7	22.4	206.6
Tonatia schulzi	8.99	14.3	24.0	3.60	12.6	16.1	10.1	153.5
Tonatia sylvicola	14.6	21.3	35.3	5.63	21.2	27.1	18.3	240.2
Mimon crenulatum	7.30	10.3	15.8	2.94	11.3	18.2	8.96	87.1
Phyllostomus discolor	39.8	53.3	58.1	13.3	38.8	67.9	39.2	312.6
Phyllostomus elongatus	20.4	33.9	43.1	8.74	32.9	49.1	25.2	226.3
Phyllostomus hastatus	42.8	65.2	73.9	14.7	46.4	79.8	48.4	438.9
Phylloderma stenops	87.4	104.5	68.8	15.8	36.2	91.7	51.0	348.3
Trachops cirrhosus	23.5	36.0	50.0	8.78	27.5	50.6	28.4	296.4
Vampyrum spectrum	92.0	121.3	114.2	19.8	50.7	110.4	65.4	757.7

	MOB	PAL	STR	SEP	AMY	HIP	SCH	NEO
PHYLLOSTOMIDAE (cont.)								
GLOSSOPHAGINAE								
Glossophaga longirostris	12.9	19.6	20.3	6.09	12.5	38.6	17.0	92.9
Glossophaga soricina	11.5	18.5	18.7	5.80	12.0	31.4	14.9	88.0
Monophyllus plethodon	13.6	20.9	22.8	5.89	13.6	34.3	15.3	96.4
Leptonycteris curasoae	19.2	29.4	29.6	8.38	17.1	47.7	23.1	140.3
Leptonycteris nivalis	18.0	26.9	29.4	8.92	18.1	42.2	22.4	144.3
Lonchophylla mordax	11.2	18.0	21.1	6.63	13.0	38.4	15.6	97.5
Lonchophylla thomasi	9.44	15.1	15.4	4.63	10.5	33.1	13.5	76.7
Lionycteris spurrelli	10.3	14.9	16.5	4.93	9.25	29.5	13.6	81.7
Anoura caudifer	10.1	15.8	20.1	5.74	10.5	33.8	15.7	101.1
Anoura geoffroyi	18.2	25.5	32.1	8.54	16.6	49.0	24.1	146.1
Choeroniscus minor	9.35	13.1	15.5	4.76	11.4	34.5	16.5	95.4
CAROLLIINAE								
Carollia castanea	20.8	26.3	21.7	6.00	13.4	38.6	16.9	104.0
Carollia perspicillata	26.3	32.7	27.2	7.16	16.1	42.9	20.0	129.1
Rhinophylla pumilio	18.8	21.9	17.1	4.42	13.2	30.3	12.7	85.6
STURNIRINAE								
Sturnira lilium	28.4	37.9	34.8	8.82	18.0	45.7	23.2	161.4
Sturnira ludovici	31.3	37.2	35.3	8.68	22.4	49.4	24.3	159.7
Sturnira tildae	32.6	41.1	40.8	8.97	21.5	54.1	27.8	190.0
STENODERMINAE								
Uroderma bilobatum	28.7	32.0	27.9	8.57	17.6	42.7	28.4	155.8
Vampyrops brachycephal.	20.4	25.0	25.6	6.01	15.3	36.0	20.7	158.1
Vampyrops helleri	17.2	22.9	23.6	5.62	14.1	33.6	19.4	147.1
Vampyrops infuscus	40.8	52.1	51.5	13.0	29.1	75.4	36.4	284.5
Vampyrops lineatus	26.0	33.9	32.2	7.98	20.5	45.4	26.0	213.9
Vampyrops vittatus	41.7	52.6	55.7	14.0	30.5	71.9	46.8	344.4
Vampyrodes caraccioloi	35.3	43.1	36.9	11.2	27.0	58.3	30.2	248.6
Vampyressa pusilla	14.6	19.7	16.8	5.22	10.7	27.4	14.9	98.3
Chiroderma salvini	31.1	37.8	34.0	9.20	24.1	56.2	27.6	210.5
Chiroderma trinitatum	21.1	24.7	19.9	5.96	15.9	33.6	17.2	152.2
Chiroderma villosum	27.4	32.2	27.6	7.51	20.6	44.3	22.8	187.4
Chiroderma villosum ssp.	35.4	38.0	31.6	9.32	21.6	56.2	26.4	202.8
Ectophylla macconnelli	11.4	15.0	15.6	4.30	8.93	26.3	13.5	86.9
Artibeus cinereus	18.9	25.3	22.7	6.22	11.7	32.9	20.0	120.4
Artibeus concolor	25.8	34.0	27.8	7.62	19.2	43.5	25.3	152.3
Artibeus jamaicensis	38.7	54.4	51.8	12.0	27.3	63.8	44.1	281.1
Artibeus lituratus	54.1	71.0	60.6	15.0	34.8	79.4	49.5	326.0
Enchisthenes harti	20.9	28.0	24.3	7.14	14.8	35.6	21.3	130.3
Ardops nichollsi	28.0	30.4	28.5	7.39	18.6	36.1	19.8	148.6
Sphaeronycteris toxoph.	11.8	24.1	21.2	6.63	12.0	32.3	20.5	135.2
BRACHYPHYLLINAE								
Brachyphylla cavernarum	42.2	68.1	82.2	17.2	33.8	78.8	36.8	322.9

	MOB	PAL	STR	SEP	AMY	HIP	SCH	NEO
DESMODONTIDAE								
Desmodus rotundus	33.0	42.8	60.7	10.9	30.4	42.4	31.7	312.5
Diphylla ecaudata	36.2	38.9	45.6	8.67	21.6	41.0	23.2	231.1
NATALIDAE								
Natalus tumidirostris	3.58	8.11	14.9	3.06	8.81	27.1	8.59	55.1
FURIPTERIDAE								
Furipterus horrens	1.37	3.73	6.22	1.33	4.65	9.46	3.76	23.3
VESPERTILIONIDAE								
VESPERTILIONINAE								
Myotis adversus	5.54	10.8	14.0	3.43	12.2	13.2	7.32	46.6
Myotis albescens	2.90	5.14	6.73	1.66	6.00	7.59	3.75	20.5
Myotis altarium	4.98	9.69	11.9	3.68	9.28	21.3	8.75	36.9
Myotis annectans	5.05	9.25	10.7	3.26	10.4	17.7	8.44	37.3
Myotis bechsteini	7.81	14.1	13.0	4.73	11.0	25.5	9.18	36.3
Myotis bocagei	4.09	7.38	9.84	2.53	9.60	9.75	5.67	36.0
Myotis dasycneme	6.89	11.8	13.0	4.41	14.0	25.2	11.2	46.5
Myotis montivagus	3.55	6.69	8.58	2.16	8.39	10.3	4.67	27.8
Myotis myotis	13.3	18.3	23.6	6.69	18.8	35.6	17.2	78.3
Myotis nattereri	4.77	8.83	10.6	3.32	10.2	21.4	7.79	34.8
Myotis nigricans	2.88	7.03	7.95	2.02	7.27	9.03	4.08	23.0
Myotis nigricans ssp.	4.02	7.02	8.70	2.16	8.42	7.42	4.27	28.3
Myotis siligorensis	2.17	3.74	5.54	1.42	4.83	7.14	3.06	16.2
Pipistrellus babu	2.42	3.82	3.63	1.33	4.74	6.42	3.02	14.2
Pipistrellus ceylonicus	5.44	8.30	6.74	2.36	10.1	11.8	5.61	28.3
Pipistrellus circumdatus	4.96	8.74	9.80	3.35	10.7	20.0	9.47	35.1
Pipistrellus crassulus	3.56	5.44	5.75	1.91	7.41	8.19	3.98	21.3
Pipistrellus imbricatus	3.26	5.18	6.16	2.19	6.96	9.24	3.12	21.2
Pipistrellus javanicus	3.14	4.84	4.93	1.61	6.38	8.27	3.63	18.4
Pipistrellus mimus	2.46	3.85	3.75	1.33	4.28	6.26	2.99	14.4
Pipistrellus nanus	1.99	4.37	5.31	1.43	4.89	6.27	2.90	15.7
Pipistrellus papuanus §)	2.29	4.33	4.77	1.66	6.14	6.09	2.80	17.6
Pipistrellus pulveratus	3.76	5.86	6.37	2.11	8.00	9.97	4.87	22.6
Pipistrellus subflavus	3.53	5.72	5.32	2.31	5.27	10.9	4.55	16.6
Scotozous dormeri	3.76	6.67	5.71	1.85	7.09	7.95	3.93	21.4
Nyctalus noctula	7.74	12.2	12.4	4.55	15.6	19.2	10.5	54.9
Nyctalus stenopterus	6.09	9.36	11.2	3.10	13.6	13.9	5.90	39.4
Glischropus tylopus	2.28	4.09	4.48	1.26	5.41	6.44	2.48	14.3
Eptesicus brasiliensis	4.49	9.72	9.70	3.02	10.1	11.9	6.22	34.8
Eptesicus br. melanopterus	5.43	8.16	8.76	2.55	10.2	11.6	5.82	31.0
Eptesicus flavescens	4.77	7.88	8.07	2.41	8.59	10.7	5.24	31.5
Eptesicus fuscus	6.17	10.7	10.9	3.39	9.24	19.7	9.82	43.3
Eptesicus pumilus	2.60	4.60	5.52	1.85	5.97	9.65	3.83	18.3
Eptesicus serotinus anders.	10.0	15.6	16.1	5.01	17.0	27.6	12.3	55.7
Ia io	16.9	31.6	38.5	9.76	36.9	47.1	26.6	141.3

	MOB	PAL	STR	SEP	AMY	HIP	SCH	NEO
VESPERTILIONIDAE (cont.)								
VESPERTILIONINAE (cont.)								
Tylonycteris pachypus	1.78	3.34	3.15	1.14	4.19	4.40	2.29	12.3
Tylonycteris robustula	2.88	4.60	4.25	1.44	5.53	6.21	3.60	16.7
Hesperoptenus blanfordi	4.30	6.23	5.86	2.11	8.66	9.34	4.10	25.7
Glauconycteris humeralis	1.48	4.60	6.49	1.39	5.61	9.79	3.95	22.9
Glauconycteris poensis	2.39	5.94	8.32	1.98	9.93	12.2	5.95	36.0
Chalinolobus gouldi	5.74	8.26	10.2	3.37	11.0	17.6	7.14	37.3
Chalinolobus morio	3.15	5.65	8.06	2.69	6.38	17.9	6.27	28.1
Scotorepens sanborni	4.09	6.86	6.56	1.99	7.57	10.2	5.03	25.7
Rhogeessa parvula	2.80	4.90	5.31	1.62	5.78	6.48	3.44	18.2
Scotomanes ornatus	8.21	15.9	18.6	5.99	18.1	27.4	13.3	65.0
Scotomanes sp.?	6.97	13.6	13.9	4.47	16.6	27.8	12.8	57.7
Scotophilus dingani	11.0	15.3	18.5	4.47	20.3	22.7	12.6	87.2
Scotophilus heathi	11.2	16.8	18.9	5.53	21.1	25.1	11.7	93.6
Scotophilus kuhli	6.76	11.4	13.5	4.11	16.5	18.6	9.76	66.8
Lasiurus borealis	3.97	5.71	6.30	2.11	6.27	9.00	4.60	27.9
MINIOPTERINAE								
Miniopterus australis	4.47	7.32	11.7	2.72	9.74	15.0	6.04	37.6
Miniopterus haradei	4.83	7.81	11.1	2.79	10.9	15.9	6.15	37.7
Miniopterus inflatus	6.77	11.5	16.3	3.93	12.5	19.0	9.35	55.9
Miniopterus magnater	6.01	11.3	16.8	3.71	16.2	20.2	7.99	59.7
Miniopterus medius	4.42	9.41	13.2	2.96	12.1	15.7	5.30	46.4
Miniopterus pusillus	3.98	8.22	11.5	2.64	10.1	15.0	6.28	37.2
M. schreibersi arenarius	5.24	9.71	13.1	2.98	10.8	16.6	7.74	44.0
Miniopterus s. blepotis	5.98	12.9	16.0	3.70	12.4	19.7	9.13	49.8
Miniopterus tristis	6.12	12.5	17.7	3.68	16.5	18.4	7.03	62.2
MURININAE								
Murina cyclotis cyclotis	12.8	20.1	13.1	3.33	9.60	15.9	7.91	44.4
Murina c. peninsularis	13.5	22.2	13.9	3.40	10.4	14.8	9.87	58.7
Murina huttoni rubella	13.9	20.8	13.5	3.54	10.2	17.1	7.49	49.5
Murina tubinaris	11.1	20.0	11.4	2.99	8.53	15.2	6.61	38.0
Harpiocephalus harpia	23.0	35.7	26.4	7.08	17.4	35.8	17.2	81.3
KERIVOULINAE								
Kerivoula papillosa	3.80	7.90	12.2	2.82	10.4	36.3	8.29	69.0
Kerivoula pellucida	2.13	5.14	12.3	1.77	7.46	14.0	7.17	47.2
Kerivoula phalaena	1.62	3.64	7.09	1.08	4.45	6.76	3.03	20.1
Phoniscus atrox	2.59	5.54	9.35	1.92	7.08	24.7	9.39	45.1
NYCTOPHILINAE								
Nyctophilus geoffroyi	4.49	6.56	8.06	2.13	7.01	12.3	5.68	31.8
Nyctophilus timoriensis	5.69	9.62	12.9	3.03	12.7	19.4	8.46	47.6

	MOB	PAL	STR	SEP	AMY	HIP	SCH	NEO
MOLOSSIDAE								
Tadarida aegyptiaca	10.0	16.1	15.8	5.66	18.4	22.9	13.6	77.3
Tadarida condylura	13.3	18.4	19.7	5.81	21.5	22.9	14.5	102.0
Tadarida mops	13.9	19.9	22.8	6.26	21.2	24.5	11.6	105.6
Tadarida niveiventer	10.6	15.9	16.1	5.11	19.5	20.8	13.8	86.9
Tadarida beccarii	7.20	9.51	9.38	3.28	11.0	13.0	7.10	41.4
Tadarida jobensis	9.50	14.3	15.7	4.43	15.5	14.9	10.1	78.8
Tadarida leucostigma	8.78	16.6	15.1	5.67	19.0	19.8	9.61	80.7
Tadarida plicata plicata	7.83	12.7	13.0	4.29	16.4	16.4	10.0	62.2
Tadarida pumila	6.60	9.82	9.42	3.08	11.6	12.2	6.77	44.3
Otomops martiensseni	14.3	26.2	29.0	7.07	28.8	30.6	20.7	153.6
Molossops abrasus	10.5	16.6	15.4	4.62	21.0	22.9	12.6	80.7
Molossops greenhalli	5.08	11.8	12.1	3.65	14.9	14.4	9.72	56.0
Molossops planirostris	5.20	8.23	8.46	2.97	8.91	10.3	5.93	32.2
Eumops auripendulus	12.5	22.5	22.5	6.11	27.6	25.0	18.1	117.8
Eumops glaucinus	14.2	21.9	26.0	7.91	25.0	28.9	23.2	129.0
Molossus ater	12.9	21.1	21.6	6.23	29.0	24.4	14.6	113.8
Molossus molossus	7.48	12.9	12.6	4.07	17.2	15.2	8.67	62.6
Molossus trinitatis	10.4	15.7	17.0	4.39	21.0	20.8	11.8	89.6
Cheiromeles torquatus	40.5	54.8	61.8	17.0	55.2	61.2	40.3	349.8

Abbreviations:

AMY	amygdala	PAL	paleocortex
HIP	hippocampus	SCH	schizocortex
MOB	main olfactory bulb	SEP	septum
NEO	neocortex	STR	striatum

§) see addendum at end of Volume 3, pp. 1592-1595;
Number of individuals as in Table 33

Table 33. Size indices of telencephalic components related to the average of the four Tenrecinae (AvTn = 100)

For volumes in species see Table 32
For average size indices in supraspecific taxonomic units see Table 34

	n	MOB	PAL	STR	SEP	AMY	HIP	SCH	NEO
slope		.64	.63	.67	.61	.60	.57	.64	.73
y-intercept		.682	.799	.089	-.129	.232	.581	.062	.346
PTEROPODIDAE									
PTEROPODINAE									
Eidolon helvum	3	123	116	557	278	206	283	373	1088
Rousettus aegyptiacus	4	103	92	394	215	156	232	310	859
R. amplexicaud. brachyot. §)	3	137	118	414	233	185	270	238	954
R. ampl. stresemanni §)	3	138	109	342	225	181	266	292	799
Myonycteris torquata	3	117	111	429	262	167	290	409	1100
Pteropus alecto	2	83	89	497	247	147	246	321	1155
Pteropus conspicillatus	1	92	93	493	230	156	238	326	1199
Pteropus hypomelanus	1	100	103	479	247	163	248	375	1103
Pteropus lylei	4	109	117	680	283	190	285	408	1618
Pteropus mahaganus	2	69	91	445	215	157	217	292	975
Pteropus neohibernicus	2	96	91	522	214	144	226	330	1215
Pteropus poliocephalus	2	80	80	451	224	133	226	297	1029
Pteropus samoensis	1	113	118	603	273	174	323	477	1468
Pteropus scapulatus	2	83	87	496	244	145	240	398	1238
Pteropus temmincki	2	118	115	542	298	198	313	419	1453
Pteropus tonganus	3	95	105	579	245	166	251	356	1194
Pteropus vampyrus	2	93	84	451	224	129	225	305	1040
Dobsonia inermis	3	123	114	505	249	198	282	334	975
D. moluccensis magna §)	2	118	109	412	221	173	255	273	887
D. molucc. anderseni §)	3	141	124	508	257	173	287	320	1093
Dobsonia praedatrix	3	147	113	477	234	177	243	287	934
Hypsignathus monstrosus	3	72	69	280	186	107	222	252	723
Epomops franqueti	3	105	90	327	210	154	274	333	841
Epomophorus labiatus	3	126	96	411	228	180	281	328	999
Micropteropus pusillus	3	99	100	401	213	196	238	326	1041
Scotonycteris zenkeri	2	109	92	346	192	157	339	387	882
Casinycteris argynnis	5	72	87	335	220	140	289	364	950
Cynopterus brachyotis	3	130	86	389	229	174	285	350	990
Cynopterus horsfieldi	3	101	78	351	206	147	282	361	990

Abbreviations and notes:

AMY	amygdala	PAL	paleocortex
HIP	hippocampus	SCH	schizocortex
MOB	main olfactory bulb	SEP	septum
NEO	neocortex	STR	striatum

§) see addendum at end of Volume 3, pp. 1592-1595
n = number of individuals

	n	MOB	PAL	STR	SEP	AMY	HIP	SCH	NEO
slope		.64	.63	.67	.61	.60	.57	.64	.73
y-intercept		.682	.799	.089	-.129	.232	.581	.062	.346
PTEROPODIDAE (cont.)									
PTEROPODINAE (cont.)									
Cynopterus sphinx sphinx	3	117	81	347	210	149	263	307	891
Megaerops ecaudatus	2	124	84	330	223	151	306	413	1093
Chironax melanocephalus	3	118	93	337	222	155	290	365	931
Sphaerias blanfordi	4	80	80	336	247	152	262	334	853
Balionycteris maculata	2	118	87	305	223	153	314	372	868
NYCTIMENINAE									
Nyctimene albiventer	2	125	99	306	200	168	260	295	831
Nyctimene robinsoni	2	143	108	362	221	193	300	323	908
Nyctimene vizcaccia	3	155	119	336	214	185	251	281	857
Paranyctimene raptor	3	108	93	320	174	147	255	290	826
MACROGLOSSINAE									
Eonycteris spelaea	3	103	84	370	224	166	252	354	886
Megaloglossus woermanni	3	83	80	363	236	161	327	433	1043
Macroglossus minimus	4	100	84	378	213	152	268	356	991
Macroglossus sobrinus	3	96	82	379	242	180	267	287	885
Syconycteris a. australis §)	1	110	102	443	254	146	274	368	1077
S. australis papuana §)	3	111	95	430	233	165	297	345	1005
Melonycteris melanops	3	112	107	550	227	195	231	376	1118
Nesonycteris woodfordi	3	95	92	490	219	160	229	371	1068
Notopteris macdonaldi	3	86	102	443	237	180	224	297	847
RHINOPOMATIDAE									
Rhinopoma hardwickei	3	25	34	183	109	174	98	163	376
Rhinopoma microphyllum	3	21	32	145	86	119	99	139	296
EMBALLONURIDAE									
EMBALLONURINAE									
Emballonura monticola	3	8	25	231	86	206	77	131	428
Emballonura raffrayana	3	9	31	199	100	164	87	146	426
Emballonura semicaudata	3	9	27	200	84	149	87	137	392
Coleura afra	3	17	30	202	84	187	81	141	398
Rhynchonycteris naso	3	8	25	197	84	179	67	111	377
Saccopteryx bilineata	4	11	33	229	104	269	100	209	442
Saccopteryx canescens	1	11	27	221	103	236	89	161	437
Saccopteryx leptura	3	10	29	196	85	213	90	150	412
Cormura brevirostris	3	12	31	184	89	213	89	139	419
Peropteryx macrotis	3	11	24	187	91	214	78	173	382
Peropteryx trinitatis	3	10	28	217	93	170	80	164	407
Taphozous australis	2	23	35	240	132	229	109	199	451
Taphozous hildegardeae	3	28	40	199	112	222	101	167	448
Taphozous mauritianus	2	24	34	173	109	213	98	180	409
Taphozous melanopogon	3	29	45	231	112	222	105	190	522
Taphozous theobaldi	3	25	36	193	100	188	91	160	400
Taphozous saccolaimus	3	25	31	180	100	203	93	166	417

> Table 33 <

	n	MOB	PAL	STR	SEP	AMY	HIP	SCH	NEO
slope		.64	.63	.67	.61	.60	.57	.64	.73
y-intercept		.682	.799	.089	-.129	.232	.581	.062	.346
EMBALLONURIDAE (cont.)									
DICLIDURINAE									
Cyttarops alecto	1	5	16	195	76	127	81	123	426
CRASEONYCTERIDAE									
Craseonycteris thonglongyai	3	8	20	170	90	145	74	114	398
NYCTERIDAE									
Nycteris arge	2	18	36	326	114	168	122	167	722
Nycteris grandis	4	14	30	314	107	172	137	170	707
Nycteris hispida	1	13	33	327	113	174	141	188	700
Nycteris javanica	1	11	22	248	82	140	136	138	660
Nycteris macrotis	3	16	28	237	92	174	106	162	653
Nycteris nana	3	17	28	292	112	145	126	158	599
Nycteris thebaica	3	15	34	298	102	175	131	180	783
Nycteris tragata	2	14	25	288	102	161	138	144	799
MEGADERMATIDAE									
Megaderma lyra	6	36	41	290	116	202	133	218	766
Megaderma spasma	3	30	41	310	125	199	138	255	801
Macroderma gigas	3	21	31	256	109	167	118	197	699
Cardioderma cor	3	21	33	274	129	200	132	244	779
Lavia frons	2	12	27	283	115	180	112	227	773
RHINOLOPHIDAE									
Rhinolophus cornutus	2	22	28	225	101	133	130	164	426
Rhinolophus eloquens	2	30	39	232	115	157	141	179	401
Rhinolophus hipposideros	2	15	33	201	108	116	150	179	281
Rhinolophus landeri	3	27	38	254	123	160	137	162	434
Rhinolophus lepidus	3	28	40	256	103	166	120	164	397
Rhinolophus luctus	3	15	30	197	92	186	123	199	478
Rhinolophus macrotis	2	25	36	245	119	128	178	198	381
Rhinolophus malayanus	2	25	36	229	118	188	130	156	401
Rhinolophus megaphyllus	2	22	33	246	103	167	150	176	422
Rhinolophus megaphyllus ssp.	3	18	38	269	118	172	155	188	484
Rhinolophus paradoxolophus	1	27	37	234	108	123	179	185	415
Rhinolophus pearsoni	1	17	31	218	103	168	149	186	468
Rhinolophus pusillus	3	26	38	258	98	166	136	169	432
Rhinolophus trifoliatus	3	12	28	185	88	186	131	169	456
Rhinolophus yunanensis	2	24	32	201	95	169	142	186	408
HIPPOSIDERIDAE									
Hipposideros armiger	3	24	31	212	99	163	152	165	402
Hipposideros ater	2	21	34	255	112	142	139	160	425
Hipposideros ater aruensis	2	21	39	260	110	127	143	156	443
Hipposideros ater saevus	3	32	49	294	115	150	152	187	516
Hipposideros caffer	3	24	33	254	110	161	135	168	477
Hipposideros c. calcaratus §)	3	44	53	341	124	171	148	168	566
H. calcaratus cupidus §)	3	43	54	351	119	169	147	190	628

	n	MOB	PAL	STR	SEP	AMY	HIP	SCH	NEO
slope		.64	.63	.67	.61	.60	.57	.64	.73
y-intercept		.682	.799	.089	-.129	.232	.581	.062	.346
HIPPOSIDERIDAE (cont.)									
Hipposideros cervinus	3	17	30	259	97	140	118	147	430
Hipposideros commersoni	3	26	26	140	70	100	90	119	265
Hipposideros diadema	4	27	35	228	115	186	148	172	449
Hipposideros fulvus	2	31	38	273	112	177	130	154	486
Hipposideros halophyllus	2	20	32	220	97	137	127	179	418
Hipposideros lankadiva	3	27	39	230	110	159	140	174	410
Hipposideros larvatus	3	29	35	266	111	179	131	143	425
Hipposideros lekaguli	2	25	31	218	107	145	134	189	460
H. magg. maggietaylorae §)	1	56	61	350	124	178	161	203	627
H. maggietaylorae erroris §)	3	48	61	342	116	175	132	157	533
Hipposideros ridleyi	3	23	36	272	112	158	135	164	435
Hipposideros speoris	2	23	32	235	103	162	110	131	437
Hipposideros turpis	3	22	35	235	106	152	143	158	435
Aselliscus stoliczkanus	3	13	26	193	85	115	135	133	318
Aselliscus tricuspidatus	3	17	30	239	95	133	119	148	339
Rhinonycteris aurantius	3	20	32	246	114	141	148	160	444
Triaenops persicus	3	20	22	166	72	117	103	108	321
NOCTILIONIDAE									
Noctilio albiventris	3	20	29	230	84	184	77	121	651
Noctilio leporinus	3	28	39	332	119	237	131	179	842
MORMOOPIDAE									
Pteronotus gymnonotus	3	28	34	253	94	170	114	145	521
Pteronotus personatus	3	26	37	238	98	173	103	138	489
Pteronotus parnelli	3	36	53	300	127	235	164	172	602
Mormoops megalophylla	3	24	33	218	119	144	150	158	406
PHYLLOSTOMIDAE									
PHYLLOSTOMINAE									
Micronycteris megalotis	3	65	75	332	105	213	127	199	686
Micronycteris minuta	1	73	75	329	122	193	151	227	722
Micronycteris schmidtorum	3	63	73	326	107	206	117	211	764
Micronycteris brachyotis	1	81	75	256	115	164	128	174	558
Macrophyllum macrophyllum	3	23	40	343	108	169	151	202	752
Tonatia bidens	3	72	79	301	138	223	159	223	789
Tonatia schulzi	1	26	33	252	75	118	74	124	742
Tonatia sylvicola	1	33	38	280	91	155	98	172	857
Mimon crenulatum	2	31	35	246	88	150	117	160	648
Phyllostomus discolor	3	83	88	424	199	262	229	339	1018
Phyllostomus elongatus	3	37	49	275	116	197	147	191	635
Phyllostomus hastatus	3	50	61	295	127	183	161	236	740
Phylloderma stenops	2	157	149	430	206	213	271	381	958
Trachops cirrhosus	2	48	59	363	131	185	170	245	959
Vampyrum spectrum	2	71	75	295	115	135	154	210	794

	n	MOB	PAL	STR	SEP	AMY	HIP	SCH	NEO
slope		.64	.63	.67	.61	.60	.57	.64	.73
y-intercept		.682	.799	.089	-.129	.232	.581	.062	.346
PHYLLOSTOMIDAE (cont.)									
GLOSSOPHAGINAE									
Glossophaga longirostris	1	57	68	326	187	172	254	313	714
Glossophaga soricina	3	55	69	329	193	178	223	297	744
Monophyllus plethodon	2	50	60	303	152	157	192	234	602
Leptonycteris curasoae	1	52	62	283	160	147	202	259	612
Leptonycteris nivalis	3	51	60	295	178	162	186	262	664
Lonchophylla mordax	1	51	64	352	210	181	261	297	779
Lonchophylla thomasi	1	57	71	344	192	192	288	339	844
Lionycteris spurrelli	3	59	66	347	193	161	245	323	842
Anoura caudifer	3	46	57	337	183	149	242	301	813
Anoura geoffroyi	3	64	71	408	212	185	265	355	870
Choeroniscus minor	1	46	50	278	161	172	249	336	825
CAROLLIINAE									
Carollia castanea	5	83	83	316	168	168	234	283	717
Carollia perspicillata	3	87	85	321	166	168	218	275	711
Rhinophylla pumilio	3	96	88	322	157	209	228	272	782
STURNIRINAE									
Sturnira lilium	3	86	91	379	190	174	216	293	811
Sturnira ludovici	2	81	76	325	161	186	203	263	669
Sturnira tildae	3	88	87	391	172	186	230	312	832
STENODERMINAE									
Uroderma bilobatum	3	100	88	352	211	194	229	414	920
Vampyrops brachycephalus	2	82	79	376	170	193	220	350	1102
Vampyrops helleri	3	67	70	331	153	171	198	313	976
Vampyrops infuscus	1	69	70	304	160	163	212	258	735
Vampyrops lineatus	3	73	76	325	160	185	202	307	990
Vampyrops vittatus	2	87	87	407	209	206	243	405	1121
Vampyrodes caraccioloi	1	70	67	255	160	174	188	248	763
Vampyressa pusilla	2	77	81	324	189	173	211	326	922
Chiroderma salvini	4	80	77	312	169	200	230	296	877
Chiroderma trinitatum	2	82	76	280	163	193	198	280	1015
Chiroderma villosum	3	73	68	261	143	176	186	253	811
Chiroderma villosum ssp.	1	81	68	254	152	159	206	251	733
Ectophylla macconnelli	2	65	67	330	169	156	219	323	900
Artibeus cinereus	3	78	82	338	178	150	203	342	854
Artibeus concolor	2	80	83	310	168	190	211	328	788
Artibeus jamaicensis	3	75	83	351	168	173	202	355	842
Artibeus lituratus	4	86	89	333	173	182	210	326	778
Enchisthenes harti	4	73	77	308	176	164	191	312	772
Ardops nichollsi	3	79	68	288	149	168	161	234	690
Sphaeronycteris toxophyllum	1	38	61	245	151	123	161	275	727
BRACHYPHYLLINAE									
Brachyphylla cavernarum	4	77	99	526	229	203	238	281	911

	n	MOB	PAL	STR	SEP	AMY	HIP	SCH	NEO
slope		.64	.63	.67	.61	.60	.57	.64	.73
y-intercept		.682	.799	.089	-.129	.232	.581	.062	.346
DESMODONTIDAE									
Desmodus rotundus	3	69	71	446	164	207	144	276	1024
Diphylla ecaudata	4	84	71	373	144	161	152	223	851
NATALIDAE									
Natalus tumidirostris	3	22	38	333	127	162	237	216	607
FURIPTERIDAE									
Furipterus horrens	5	13	27	222	84	130	123	148	427
VESPERTILIONIDAE									
VESPERTILIONINAE									
Myotis adversus	2	25	37	226	106	168	87	135	360
Myotis albescens	3	25	34	217	96	153	90	134	336
Myotis altarium	3	22	34	194	115	129	142	163	289
Myotis annectans	2	24	35	188	109	156	126	170	318
Myotis bechsteini	1	37	52	226	156	162	180	182	305
Myotis bocagei	3	24	34	218	104	175	84	142	392
Myotis dasycneme	1	25	34	173	114	162	141	172	290
Myotis montivagus	3	19	28	169	80	138	81	105	267
Myotis myotis	3	35	38	219	125	158	147	187	331
Myotis nattereri	1	27	39	223	131	178	178	186	361
Myotis nigricans	3	23	43	233	107	171	99	133	340
Myotis nigricans ssp.	3	25	34	200	92	159	66	111	322
Myotis siligorensis	3	20	27	197	90	134	92	120	294
Pipistrellus babu	3	18	23	103	69	108	69	96	204
Pipistrellus ceylonicus	3	26	31	117	77	147	83	111	235
Pipistrellus circumdatus	3	23	32	166	108	155	138	183	287
Pipistrellus crassulus	3	23	28	139	85	147	77	108	256
Pipistrellus imbricatus	2	21	26	146	96	135	85	83	249
Pipistrellus javanicus	3	25	30	147	87	152	92	120	277
Pipistrellus mimus	3	21	26	123	78	111	76	108	240
Pipistrellus nanus	3	17	29	174	84	127	76	105	261
Pipistrellus papuanus §)	2	17	24	129	82	134	62	85	239
Pipistrellus pulveratus	3	27	33	170	103	172	101	145	302
Pipistrellus subflavus	2	25	32	142	112	114	111	136	221
Scotozous dormeri	3	23	32	129	77	132	70	100	238
Nyctalus noctula	3	20	24	111	82	127	77	111	223
Nyctalus stenopterus	2	21	26	142	77	150	75	86	234
Glischropus tylopus	3	17	24	126	64	122	68	78	202
Eptesicus brasiliensis	3	23	38	179	105	157	88	130	310
Eptesicus br. melanopterus	3	28	33	166	91	162	88	125	285
Eptesicus flavescens	2	22	28	136	77	123	74	101	255
Eptesicus fuscus	1	24	33	154	93	113	117	160	290
Eptesicus pumilus	2	18	25	145	89	127	97	113	241
Eptesicus serotinus anders.	3	30	36	171	105	161	128	152	273
Ia io	1	28	41	218	116	199	128	181	350

	n	MOB	PAL	STR	SEP	AMY	HIP	SCH	NEO
slope		.64	.63	.67	.61	.60	.57	.64	.73
y-intercept		.682	.799	.089	-.129	.232	.581	.062	.346
VESPERTILIONIDAE (cont.)									
VESPERTILIONINAE (cont.)									
Tylonycteris pachypus	4	14	20	90	59	96	47	73	176
Tylonycteris robustula	3	17	21	92	58	99	53	88	180
Hesperoptenus blanfordi	3	25	28	126	84	154	79	100	272
Glauconycteris humeralis	3	10	25	169	66	118	97	115	298
Glauconycteris poensis	1	13	24	160	72	160	94	130	337
Chalinolobus gouldi	2	25	28	160	101	147	114	128	279
Chalinolobus morio	2	17	24	159	99	105	141	140	270
Scotorepens sanborni	1	21	27	122	70	118	76	106	231
Rhogeessa parvula	4	22	30	158	87	137	72	114	274
Scotomanes ornatus	2	24	36	192	123	167	124	161	309
Scotomanes sp.?	3	24	37	174	109	182	148	185	338
Scotophilus dingani	3	30	33	178	86	176	97	142	383
Scotophilus heathi	3	26	30	153	91	156	92	111	340
Scotophilus kuhli	3	20	27	145	88	159	87	123	333
Lasiurus borealis	1	22	25	130	81	107	73	107	280
MINIOPTERINAE									
Miniopterus australis	2	27	34	256	111	176	129	149	405
Miniopterus haradei	3	26	33	220	103	179	125	138	362
Miniopterus inflatus	3	25	33	218	102	145	107	144	351
Miniopterus magnater	3	22	33	223	96	187	113	122	373
Miniopterus pusillus	3	21	34	222	95	164	115	137	348
Miniopterus medius	3	19	32	210	90	165	103	97	354
M. schreibersi arenarius	3	25	36	225	97	157	116	152	364
Miniopterus s. blepotis	2	23	38	220	98	148	114	144	322
Miniopterus tristis	3	22	35	229	93	186	101	105	377
MURININAE									
Murina cyclotis cyclotis	2	76	93	287	136	173	137	196	484
Murina c. peninsularis	2	70	91	268	123	167	114	215	548
Murina huttoni rubella	3	72	84	256	126	163	130	161	456
Murina tubinaris	2	78	109	299	143	181	152	194	497
Harpiocephalus harpia	1	90	110	376	195	214	213	283	548
KERIVOULINAE									
Kerivoula papillosa	1	18	29	211	93	153	255	164	575
Kerivoula pellucida	1	18	34	389	101	188	165	252	759
Kerivoula phalaena	1	17	29	277	74	135	95	130	406
Phoniscus atrox	1	22	36	291	108	175	286	325	713
NYCTOPHILINAE									
Nyctophilus geoffroyi	2	26	30	175	86	126	105	139	340
Nyctophilus timoriensis	2	26	34	213	95	178	130	159	375

	n	MOB	PAL	STR	SEP	AMY	HIP	SCH	NEO
slope		.64	.63	.67	.61	.60	.57	.64	.73
y-intercept		.682	.799	.089	-.129	.232	.581	.062	.346
MOLOSSIDAE									
Tadarida aegyptiaca	3	33	42	188	132	193	117	188	428
Tadarida condylura	3	33	37	175	104	174	91	152	413
Tadarida mops	3	32	36	186	104	158	91	112	389
Tadarida niveiventer	3	31	36	167	105	181	95	167	414
Tadarida beccarii	2	27	28	129	88	131	75	113	269
Tadarida jobensis	1	33	39	196	108	169	79	146	459
Tadarida leucostigma	1	22	34	136	103	156	80	102	333
Tadarida plicata	3	26	33	156	101	173	85	139	349
Tadarida pumila	3	31	36	161	99	168	85	132	364
Otomops martiensseni	3	27	40	195	98	180	96	166	456
Molossops abrasus	3	23	29	120	73	150	81	116	281
Molossops greenhalli	1	17	31	145	86	157	74	135	313
Molossops planirostris	2	30	38	183	120	159	87	145	342
Eumops auripendulus	3	27	39	173	96	196	88	165	406
Eumops glaucinus	2	30	36	190	119	169	98	201	421
Molossus ater	3	28	37	167	98	206	86	134	394
Molossus molossus	3	29	39	177	111	210	90	141	418
Molossus trinitatis	2	33	39	191	97	209	101	154	467
Cheiromeles torquatus	3	32	35	163	101	150	87	132	376

Abbreviations and notes:

AMY	amygdala	PAL	paleocortex
HIP	hippocampus	SCH	schizocortex
MOB	main olfactory bulb	SEP	septum
NEO	neocortex	STR	striatum

§) see addendum at end of Volume 3, pp. 1592-1595

n = number of individuals

Table 34. Average size indices of telencephalic components related to the average of the four Tenrecinae (AvTn = 100)

For volumes in species see Table 32
For size indices in species see Table 33

	MOB	PAL	STR	SEP	AMY	HIP	SCH	NEO
slope	.64	.63	.67	.61	.60	.57	.64	.73
y-intercept	.682	.799	.089	-.129	.232	.581	.062	.346
PTEROPODIDAE	108	97	424	230	165	267	341	1016
PTEROPODINAE *	107	97	435	233	163	267	342	1042
Rousettus	126	106	383	224	174	256	280	870
Pteropus	94	98	520	245	159	253	359	1224
Dobsonia	132	115	476	240	180	267	303	973
Cynopterus	116	82	362	215	157	277	340	957
NYCTIMENINAE *	133	105	331	202	173	267	297	855
MACROGLOSSINAE *	100	92	427	232	167	263	354	991
RHINOPOMATIDAE *	23	33	164	98	147	98	151	336
EMBALLONURIDAE	15	30	204	97	200	89	158	422
EMBALLONURINAE *	16	31	205	98	204	89	160	422
NON-*TAPHOZOUS* EMB.	11	28	206	91	200	84	151	411
Emballonura	9	28	210	90	173	84	138	415
Saccopteryx	11	30	215	97	239	93	173	430
TAPHOZOUS	26	37	203	111	213	99	177	441
DICLIDURINAE *	5	16	195	76	127	81	123	426
CRASEONYCTERIDAE *	8	20	170	90	145	74	114	398
NYCTERIDAE *	15	30	291	103	164	130	164	703
MEGADERMATIDAE *	24	35	283	119	190	127	228	764
RHINOLOPHIDAE *	22	34	230	106	159	143	177	419
HIPPOSIDERIDAE	27	37	253	106	152	134	160	445
HIX: *H. commersoni* excl.*	27	38	258	107	154	136	161	453
Hipposideros	29	39	262	108	157	136	164	463
H. commersoni excluded	29	40	268	110	160	138	167	474
NOCTILIONIDAE *	24	34	281	102	211	104	150	746
MORMOOPIDAE *	29	39	252	110	181	133	153	505
Pteronotus	30	42	264	107	193	127	152	537
PHYLLOSTOMIDAE	68	72	325	160	177	198	279	805
PHYLLOSTOMINAE *	61	67	316	123	184	150	219	775
Micronycteris	70	75	311	112	194	131	203	683
Tonatia	44	50	278	101	165	111	173	796
Phyllostomus	57	66	331	147	214	179	255	798
GLOSSOPHAGINAE *	53	63	327	184	169	237	301	755
CAROLLIINAE *	89	85	320	164	181	227	277	737
STURNIRINAE *	85	85	365	174	182	216	289	771
STENODERMINAE *	76	76	314	169	175	204	310	866
Vampyrops	76	76	349	170	183	215	326	985
Chiroderma	79	72	277	157	182	205	270	859
Artibeus	80	84	333	172	173	206	338	816

	MOB	PAL	STR	SEP	AMY	HIP	SCH	NEO
slope	.64	.63	.67	.61	.60	.57	.64	.73
y-intercept	.682	.799	.089	-.129	.232	.581	.062	.346
PHYLLOSTOMIDAE (cont.)								
BRACHYPHYLLINAE *	77	99	526	229	203	238	281	911
DESMODONTIDAE *	76	71	410	154	184	148	249	938
NATALIDAE *	22	38	333	127	162	237	216	607
FURIPTERIDAE *	13	27	222	84	130	123	148	427
VESPERTILIONIDAE	27	36	188	97	151	110	140	331
VESPERTILIONINAE	23	31	162	93	144	98	128	284
VEX: *Tylonycteris* excl.*	23	31	165	95	146	100	130	288
Myotis	26	36	206	109	157	117	149	323
Pipistrellus	22	28	141	89	136	88	116	252
Eptesicus	24	32	159	93	141	99	130	276
Tylonycteris	15	20	91	59	97	50	81	178
Scotophilus	25	30	159	88	163	92	125	352
MINIOPTERINAE *	23	34	225	98	167	114	132	362
MURININAE *	77	97	297	144	180	149	210	507
KERIVOULINAE *	19	32	292	94	163	200	218	613
NYCTOPHILINAE *	26	32	194	90	152	118	149	357
MOLOSSIDAE *	29	36	168	102	173	89	144	384
Tadarida	30	36	166	105	167	89	139	380
Molossops	23	32	149	93	155	81	132	312
Molossus	30	38	179	102	208	92	143	426
CHIROPTERA (AvCS)	48	53	272	135	166	157	211	597
CHIROPTERA (AvCF)	46	53	286	132	169	159	209	620
MEGACHIR. (AvMegS)	108	97	424	230	165	267	341	1016
MEGACHIR. (AvMegF)	113	98	398	222	168	266	331	963
MICROCHIR. (AvMicS)	35	44	240	115	166	134	184	509
MICROCHIR. (AvMicF)	38	48	272	122	169	147	194	579

Number of species as in Table 12

Abbreviations:

AvCF	average of Chiroptera families and/or subfamilies (NN = 28)
AvCS	average of Chiroptera species (N = 272)
AvMegF	average of Megachiroptera families and/or subfamilies (NN = 3)
AvMegS	average of Megachiroptera species (N = 47)
AvMicF	average of Microchiroptera families and/or subfamilies (NN = 25)
AvMicS	average of Microchiroptera species (N = 225)

AMY	amygdala	PAL	paleocortex
HIP	hippocampus	SCH	schizocortex
MOB	main olfactory bulb	SEP	septum
NEO	neocortex	STR	striatum

Table 35. Volumes, size indices, average size indices of the accessory olfactory bulb related to the average of the four Tenrecinae (AvTn = 100) and percentage of AOB relative to the BOL complex (MOB + AOB)

	n	Volume	Index	AOB / BOL
slope			.64	%
y-intercept			-1.903	
MORMOOPIDAE				
Pteronotus gymnonotus	3	0.126	190	1.70
Pteronotus personatus	3	0.000	0	0.00
Pteronotus parnelli	3	0.120	140	1.01
Mormoops megalophylla	3	0.000	0	0.00
PHYLLOSTOMIDAE				
PHYLLOSTOMINAE				
Micronycteris megalotis	3	0.058	139	0.56
Micronycteris minuta	1	0.027	63	0.23
Micronycteris schmidtorum	3	0.075	162	0.66
Micronycteris brachyotis	1	0.086	123	0.39
Macrophyllum macrophyllum	3	0.081	164	1.82
Tonatia bidens	3	0.201	184	0.66
Tonatia schulzi	1	0.140	158	1.53
Tonatia sylvicola	1	0.211	182	1.42
Mimon crenulatum	2	0.182	301	2.44
Phyllostomus discolor	3	0.616	492	1.52
Phyllostomus elongatus	3	0.291	204	1.40
Phyllostomus hastatus	3	0.312	140	0.72
Phylloderma stenops	2	0.384	265	0.44
Trachops cirrhosus	2	0.269	214	1.14
Vampyrum spectrum	2	0.089	26	0.10
GLOSSOPHAGINAE				
Glossophaga longirostris	1	0.301	510	2.28
Glossophaga soricina	3	0.280	516	2.37
Monophyllus plethodon	2	0.141	200	1.03
Leptonycteris curasoae	1	0.360	372	1.84
Leptonycteris nivalis	3	0.316	341	1.73
Lonchophylla mordax	1	0.168	294	1.48
Lonchophylla thomasi	1	0.152	353	1.58
Lionycteris spurrelli	3	0.087	190	0.83
Anoura caudifer	3	0.186	329	1.81
Anoura geoffroyi	3	0.504	683	2.69
Choeroniscus minor	1	0.000	0	0.00

	n	Volume	Index	AOB / BOL %
slope			.64	
y-intercept			-1.903	
PHYLLOSTOMINAE (cont.)				
CAROLLIINAE				
Carollia castanea	5	0.146	225	0.70
Carollia perspicillata	3	0.241	305	0.91
Rhinophylla pumilio	3	0.000	0	0.00
STURNIRINAE				
Sturnira lilium	3	0.188	219	0.66
Sturnira ludovici	2	0.135	135	0.43
Sturnira tildae	3	0.138	143	0.42
STENODERMINAE				
Uroderma bilobatum	3	0.213	287	0.74
Vampyrops brachycephalus	2	0.164	256	0.80
Vampyrops helleri	3	0.124	185	0.72
Vampyrops infuscus	1	0.330	215	0.80
Vampyrops lineatus	3	0.197	214	0.75
Vampyrops vittatus	2	0.231	184	0.55
Vampyrodes caraccioloi	1	0.295	223	0.83
Vampyressa pusilla	2	0.155	314	1.05
Chiroderma salvini	4	0.181	179	0.58
Chiroderma trinitatum	2	0.145	218	0.68
Chiroderma villosum	3	0.128	131	0.47
Chiroderma villosum ssp.	1	0.119	104	0.34
Ectophylla macconnelli	2	0.112	247	0.97
Artibeus cinereus	3	0.183	290	0.96
Artibeus concolor	2	0.094	113	0.36
Artibeus jamaicensis	3	0.308	229	0.79
Artibeus lituratus	4	0.375	228	0.69
Enchisthenes harti	4	0.309	417	1.46
Ardops nichollsi	3	0.222	242	0.79
Sphaeronycteris toxophyllum	1	0.021	26	0.18
PHYLLONYCTERINAE				
Brachyphylla cavernarum	3	0.000	0	0.00
DESMODONTIDAE				
Desmodus rotundus	3	0.355	285	1.06
Diphylla ecaudata	4	0.111	98	0.31

> Table 35 <

	n	Volume	Index	AOB / BOL
slope			.64	%
y-intercept			-1.903	
VESPERTILIONIDAE				
MINIOPTERINAE				
Miniopterus australis	2	0.034	78	0.76
Miniopterus haradei	3	0.057	117	1.16
Miniopterus inflatus	3	0.087	124	1.28
Miniopterus magnater	3	0.104	146	1.69
Miniopterus medius	3	0.078	132	1.74
Miniopterus pusillus	3	0.059	120	1.47
M. schreibersi arenarius	3	0.068	122	1.28
Miniopterus s. blepotis	2	0.091	132	1.50
Miniopterus tristis	3	0.074	102	1.20
	N			
MICROCHIROPTERA (AvCS)	63		218	1.05
MICROCHIROPTERA (AvCF)	NN = 8		211	1.02
MORMOOPIDAE *)	2		165	1.36
PHYLLOSTOMIDAE	50		239	1.01
PHYLLOSTOMINAE *)	15		188	1.00
Micronycteris	4		122	0.46
Tonatia	3		175	1.20
Phyllostomus	3		279	1.21
GLOSSOPHAGINAE *)	10		379	1.76
Glossophaga	2		513	2.33
Leptonycteris	2		357	1.79
Lonchophylla	2		323	1.53
Anoura	2		506	2.25
CAROLLIINAE *)	2		265	0.81
STURNIRINAE *)	3		166	0.50
STENODERMINAE *)	20		215	0.73
Vampyrops	5		211	0.72
Chiroderma	4		158	0.52
Artibeus	4		215	0.70
DESMODONTIDAE *)	2		192	0.69
VESPERTILIONIDAE				
MINIOPTERINAE *)	9		119	1.34

Number of individuals = n; number of species = N; number of families and/or subfamilies marked by *) = NN

Abbreviations:

AOB	accessory olfactory bulb	MOB	main olfactory bulb
AvCS	average of Chir. species	AvCF	av. of families and/or subfam.
BOL	complex of olfactory bulbs (AOB + MOB)		

Table 36. Schneider's area measurements of cortical regions (mm^2). (PAL + SCH) + HIP + CING = (ALLO minus BOL)

	TOT	BOL	TOT minus BOL	NEO	ALLO	ALLO minus BOL	PAL plus SCH	HIP	CING
Pteropus giganteus	1678.1	246.6	1431.5	555.5	1122.6	876.0	657.2	93.9	124.8
Rousettus aegyptiacus	634.8	103.6	531.2	196.1	438.7	335.1	251.3	55.2	28.6
Saccopteryx bilineata	93.6	6.2	87.4	30.7	62.9	56.7	43.0	10.2	3.5
Rhinolophus hipposid.	97.5	10.5	87.0	33.0	64.5	54.0	37.9	14.1	2.0
Pteronotus parnelli	221.8	24.2	197.6	75.2	146.6	122.4	96.0	21.8	4.6
Pteronotus suapurens.	158.9	17.5	141.4	61.9	97.0	79.5	53.4	15.7	10.4
Phyllostomus discolor	361.6	39.2	322.4	133.0	228.6	189.4	146.2	31.8	11.3
Glossophaga soricina	182.3	20.5	161.8	60.4	121.9	101.4	72.8	21.7	6.9
Carollia perspicillata	217.6	31.3	186.3	70.6	147.0	115.7	86.2	24.2	5.3
Desmodus rotundus	364.3	35.2	329.1	120.3	244.0	208.8	173.4	22.7	12.7
Natalus mexicanus	97.7	8.5	89.2	38.1	59.6	51.1	35.0	13.3	2.8
Myotis myotis	116.0	12.2	103.8	44.7	71.3	59.1	38.4	13.2	7.5
Plecotus auritus	128.1	13.6	114.5	39.8	88.3	74.7	50.8	16.1	7.8
Miniopterus schreib.	84.3	7.5	76.8	23.3	61.0	53.5	35.1	11.3	7.1

Table 37. Percentages of the data given in Table 36: olfactory bulb relative to the total cortex (TOT = 100), and cortical regions relative to the total cortex minus olfactory bulb (TOT minus BOL = 100)

			Percentages relative to TOT minus BOL					
	TOT	BOL	TOT minus BOL	NEO	ALLO minus BOL	PAL plus SCH	HIP	CING
Pteropus giganteus	100	14.7	100	38.8	61.2	45.9	6.6	8.7
Rousettus aegyptiacus	100	16.3	100	36.9	63.1	47.3	10.4	5.4
Saccopteryx bilineata	100	6.6	100	35.1	64.9	49.2	11.7	4.0
Rhinolophus hipposid.	100	10.8	100	37.9	62.1	43.6	16.2	2.3
Pteronotus parnelli	100	10.9	100	38.0	62.0	48.6	11.1	2.3
Pteronotus suapurens.	100	11.0	100	43.8	56.2	37.8	11.1	7.3
Phyllostomus discolor	100	10.8	100	41.3	58.7	45.3	9.9	3.5
Glossophaga soricina	100	11.2	100	37.3	62.7	45.0	13.4	4.3
Carollia perspicillata	100	14.4	100	37.9	62.1	46.3	13.0	2.8
Desmodus rotundus	100	9.7	100	36.5	63.5	52.7	6.9	3.9
Natalus mexicanus	100	8.7	100	42.7	57.3	39.3	14.9	3.1
Myotis myotis	100	10.5	100	43.0	57.0	37.1	12.7	7.2
Plecotus auritus	100	10.6	100	34.7	65.3	44.4	14.0	6.9
Miniopterus schreib.	100	8.9	100	30.3	69.7	45.7	14.7	9.3

Abbreviations:

ALLO allocortex
BOL olfactory bulb
CING cingular cortex
HIP hippocampus
NEO neocortex
PAL paleocortex
SCH schizocortex
TOT total cortex

Table 38. Allometric size indices of the cortical regions, given in Table 36, related to the index of *Pteropus giganteus* (which was set = 100).
Most body weights (BoW) were taken from Table 5; BoWs of *Pteropus giganteus* and *Myotis myotis* from Lütgemeier (see Table 41).

	BoW in g	TOT	BOL	NEO	ALLO minus BOL	PAL plus SCH	HIP	CING
slope		.47	.43	.49	.41	.42	.38	.44
y-intercept		1.875	1.157	1.337	1.765	1.611	.881	.832
Pteropus giganteus	745.7	**100**	**100**	**100**	**100**	**100**	**100**	**100**
Rousettus aegyptiacus	136.0	84	87	81	77	78	112	48
Saccopteryx bilineata	7.8	48	18	52	42	44	61	21
Rhinolophus hipposid.	4.6	63	38	72	50	49	104	15
Pteronotus parnelli	20.2	72	46	79	61	67	92	18
Pteronotus suapurens.	13.5	62	40	80	47	44	77	49
Phyllostomus discolor	36.6	89	58	105	74	79	106	34
Glossophaga soricina	9.9	83	53	90	68	68	119	37
Carollia perspicillata	17.8	75	63	79	61	63	107	22
Desmodus rotundus	36.3	90	52	95	82	94	76	38
Natalus mexicanus	5.0	61	30	80	45	44	95	20
Myotis myotis	23.3	35	22	44	28	25	52	28
Plecotus auritus	6.6	70	42	73	59	56	103	50
Miniopterus schreib.	11.4	36	18	33	34	31	59	36

Abbreviations for Tables 36 - 38:

ALLO	allocortex	NEO	neocortex
BOL	olfactory bulb	PAL	paleocortex
BoW	body weight	SCH	schizocortex
CING	cingular cortex	TOT	total cortex
HIP	hippocampus		

Table 39. Lütgemeier's area measurements of cortical regions (in mm^2)

	TOT	NEO	ALLO	SEMI	SCH	BIC
Pteropus giganteus	1548	1021	527	194	81	252
Rousettus leschenaulti	446	242	204	68	30	106
Myotis daubentoni	129	47	82	22	17	43
Myotis myotis	154	66	88	27	19	42
Pipistrellus pipistrellus	58	21	37	11	7	19
Eptesicus serotinus	121	45	76	22	14	40

Table 40. Percentage composition of the above given cortical regions relative to the total cortex

	TOT	NEO TOT	ALLO TOT	SEMI TOT	SCH TOT	BIC TOT
Pteropus giganteus	100	66.0	34.0	12.5	5.2	16.3
Rousettus leschenaulti	100	54.4	45.6	15.2	6.7	23.7
Myotis daubentoni	100	36.1	63.9	17.0	13.3	33.6
Myotis myotis	100	42.7	57.3	17.3	12.5	27.5
Pipistrellus pipistrellus	100	35.6	64.4	19.1	12.0	33.3
Eptesicus serotinus	100	37.4	62.6	17.9	11.9	32.8

Table 41. Allometric size indices of the above given cortical regions related to the index of *Pteropus* (which was set = 100)

	BoW	TOT	NEO	ALLO	SEMI	SCH	BIC
slope		.47	.49	.41	.42	.43	.39
y-intercept		1.840	1.601	1.544	1.081	.673	1.281
Pteropus giganteus	745.7	**100**	**100**	**100**	**100**	**100**	**100**
Rousettus leschenaulti	90.5	78	67	92	85	92	96
Myotis daubentoni	8.4	69	42	98	75	145	98
Myotis myotis	23.3	51	35	69	60	104	64
Pipistrellus pipistrellus	4.5	41	25	57	49	78	55
Eptesicus serotinus	19.7	43	26	64	52	83	65

Abbreviations:

ALLO	allocortex	NEO	neocortex
BIC	"bicortex" (but see p. 206)	SCH	schizocortex
BoW	body weight in gram (given by Lütgemeier)	SEMI	semicortex
		TOT	total cortex (ALLO + NEO)

Table 42. Lütgemeier's area measurements of the semicortex (in mm^2)

	SEMI	PRPI	TOL	PSD	PAM
Pteropus giganteus	194	91.0	34.3	29.4	39.4
Rousettus leschenaulti	68	29.6	13.0	10.2	15.0
Myotis daubentoni	22	7.4	3.2	3.1	8.2
Myotis myotis	27	10.3	4.2	4.6	7.8
Pipistrellus pipistrellus	11	4.2	1.6	2.2	3.2
Eptesicus serotinus	22	8.5	4.1	3.4	5.7

Table 43. Percentage composition of the above given semicortical components relative to the total semicortex

	SEMI	PRPI / SEMI	TOL / SEMI	PSD / SEMI	PAM / SEMI
Pteropus giganteus	100	46.9	17.7	15.1	20.3
Rousettus leschenaulti	100	43.6	19.2	15.1	22.1
Myotis daubentoni	100	33.6	14.6	14.2	37.6
Myotis myotis	100	38.0	15.7	17.2	29.1
Pipistrellus pipistrellus	100	37.4	14.7	19.4	28.5
Eptesicus serotinus	100	39.4	18.8	15.6	26.2

Table 44. Allometric size indices of the above given semicortical components related to the index of *Pteropus* (which was set = 100)

	BoW	SEMI	PRPI	TOL	PSD	PAM
slope		.42	.42	.42	.42	.42
y-intercept		1.081	.753	.329	.262	.389
Pteropus giganteus	745.7	**100**	**100**	**100**	**100**	**100**
Rousettus leschenaulti	90.5	85	79	92	84	92
Myotis daubentoni	8.4	75	53	61	69	137
Myotis myotis	23.3	60	48	52	67	85
Pipistrellus pipistrellus	4.5	49	39	40	64	69
Eptesicus serotinus	19.7	52	43	55	53	67

Abbreviations:

BoW	body weight in gram	SEMI	semicortex
PAM	periamygdaloid cortex	PSD	periseptal cortex + diagonal band
PRPI	prepiriform cortex	TOL	olfactory tubercle

Table 45. Lütgemeier's area measurements of the "bicortex" (in mm^2)

	BIC	RB	TT	CA	FD
Pteropus giganteus	252	30.5	17.6	132.1	71.8
Rousettus leschenaulti	106	10.6	4.2	58.2	33.0
Myotis daubentoni	43	2.6	1.0	23.0	16.4
Myotis myotis	42	4.2	1.0	22.6	14.2
Pipistrellus pipistrellus	19	4.6	0.6	8.7	5.1
Eptesicus serotinus	40	2.9	1.1	22.8	13.2

Table 46. Percentage composition of the above given "bicortical" components relative to the total bicortex

	BIC	RB / BIC	TT / BIC	CA / BIC	FD / BIC
Pteropus giganteus	100	12.1	7.0	52.4	28.5
Rousettus leschenaulti	100	10.0	4.0	54.9	31.1
Myotis daubentoni	100	6.0	2.4	53.4	38.2
Myotis myotis	100	9.9	2.5	53.9	33.7
Pipistrellus pipistrellus	100	24.0	3.0	45.9	27.1
Eptesicus serotinus	100	7.3	2.7	56.9	33.1

Table 47. Allometric size indices of the above given "bicortical" components related to the index of *Pteropus* (which was set = 100)

	BoW	BIC	RB	TT	CA	FD
slope		.39	.42	.40	.38	.38
y-intercept		1.281	.278	.096	1.029	.765
Pteropus giganteus	745.7	**100**	**100**	**100**	**100**	**100**
Rousettus leschenaulti	90.5	96	84	56	98	102
Myotis daubentoni	8.4	98	56	34	96	125
Myotis myotis	23.3	64	59	23	64	74
Pipistrellus pipistrellus	4.5	55	129	26	46	49
Eptesicus serotinus	19.7	65	44	27	69	73

Abbreviations:

BIC	"bicortex" (but see p. 206)	FD	fascia dentata
BoW	body weight in gram	RB	retrobulbar region
CA	cornu ammonis + subiculum	TT	taenia tecta

Table 48. Brodmann's area measurements of cortical regions (in mm^2)

	TOT	NEO	VIS	PRC	PRF	FRL	ALLO
CHIROPTERA							
Pteropus	2194	1510	326	146	52	198	684
INSECTIVORA							
Erinaceus	1150	372	46	48	0	48	778
PROSIMIANS							
Microcebus	1842	1250	264	188	140	328	592
Lemur	8108	6654	880	906	674	1580	1454
NON-HUMAN SIMIANS							
Callithrix	3298	3060	682	334	296	630	238
Cebus	27364	26278	3572	3644	2520	6164	1086
Ateles	34531	---	4420	---	---	---	---
Macaca	30616	29728	3732	3634	3466	7100	888
Papio	40752	38728	5118	4400	4222	8622	2024
Papio hamadryas	41188	39460	4260	5796	3934	9730	1728
Mandrillus	42642	40538	7074	4388	4336	8724	2104
Cercopithecus	29282	---	4312	3952	3250	7202	---
Hylobates	32602	31164	3798	3302	3678	6980	1438
Pan	79144	76648	6426	10778	13438	24216	2496
HOMINIDAE							
Homo	270940	263296	6990	19666	78574	98240	7644

Table 49. Percentage composition of the above given cortical regions relative to the total cortex

	TOT	NEO / TOT	VIS / TOT	PRC / TOT	PRF / TOT	FRL / TOT	ALLO / TOT
CHIROPTERA							
Pteropus	100	68.8	14.9	6.7	2.4	9.0	31.2
INSECTIVORA							
Erinaceus	100	32.3	4.0	4.2	0.0	4.2	67.7
PROSIMIANS							
Microcebus	100	67.9	14.3	10.2	7.6	17.8	32.1
Lemur	100	82.1	10.9	11.2	8.3	19.5	17.9
NON-HUMAN SIMIANS							
Callithrix	100	92.8	20.7	10.1	9.0	19.1	7.2
Cebus	100	96.0	13.1	13.3	9.2	22.5	4.0
Ateles	100	-.-	12.8	-.-	-.-	-.-	-.-
Macaca	100	97.1	12.2	11.9	11.3	23.2	2.9
Papio	100	95.0	12.6	10.8	10.4	21.2	5.0
Papio hamadryas	100	95.8	10.3	14.1	9.5	23.6	4.2
Mandrillus	100	95.1	16.6	10.3	10.2	20.5	4.9
Cercopithecus	100	-.-	14.7	13.5	11.1	24.6	-.-
Hylobates	100	95.6	11.6	10.1	11.3	21.4	4.4
Pan	100	96.8	8.1	13.6	17.0	30.6	3.2
HOMINIDAE							
Homo	100	97.2	2.6	7.3	29.0	36.3	2.8

Table 50. Allometric size indices calculated from Brodmann's area measurements of cortical regions

	BoW	TOT	NEO	VIS	PRC	PRF	FRL	ALLO
slope		.47	.49	.49	.49	.49	.49	.41
y-intercept		1.972	1.752	1.086	.737	.288	.870	1.641
CHIROPTERA								
Pteropus	818	**100**	**100**	**100**	**100**	**100**	**100**	**100**
INSECTIVORA								
Erinaceus	695	52	24	14	32	0	24	112
PROSIMIANS								
Microcebus	54	301	313	307	488	1022	627	264
Lemur	1780	257	301	184	424	887	544	154
NON-HUMAN SIMIANS								
Callithrix	280	249	342	354	387	964	537	54
Cebus	3100	667	905	570	1300	2527	1618	92
Ateles	8000	539	---	444	---	---	---	---
Macaca	7800	484	652	379	825	2212	1186	51
Papio	18000	435	564	345	663	1788	956	83
Papio hamadryas	19000	428	559	280	850	1623	1051	70
Mandrillus	19000	443	574	465	644	1789	942	85
Cercopithecus	6000	523	---	498	1020	2358	1368	---
Hylobates	5700	597	797	450	874	2737	1360	95
Pan	46000	543	704	274	1025	3594	1696	70
HOMINIDAE								
Homo	65000	1581	2042	251	1579	17740	5807	186

Abbreviations:

ALLO	allocortex	PRC	precentral (motor) cortex
BoW	body weight (estimated)	PRF	prefrontal cortex
FRL	frontal lobe	TOT	total cortex
NEO	neocortex	VIS	visual cortex

Table 51. Pirlot's data of body and brain weights and of the volumes of the five fundamental brain parts and seven telencephalon components in mm^3

	BoW in g	BrW in mg	OBL	MES	CER	DIE	TEL
Penthetor lucasi (PTN)	35	980	89.29	55.13	106.24	96.22	583.60
Lonchorhina aurita (PHN)	18	377	57.82	35.87	80.71	29.89	156.24
Ardops annectans (STE)	15	610	59.88	49.88	79.96	47.82	336.85
Ametrida centurio (STE)	10.5	290	40.00	16.97	50.00	23.94	147.87
Diaemus youngi (DES)	37	1090	104.17	80.38	166.38	85.80	599.90
Thyroptera tricolor (THY)	4.5	230	26.84	25.37	39.40	14.93	112.76
	MOB	PAL +AMY	SEP	STR	HIP	SCH	NEO
Penthetor lucasi	39.90	85.16	9.92	49.08	76.05	33.86	289.63
Lonchorhina aurita	3.34	21.88	3.24	16.59	20.32	9.59	81.28
Ardops annectans	18.48	58.55	9.79	35.55	37.20	13.93	163.35
Ametrida centurio	6.91	20.93	2.88	12.00	17.41	8.93	78.81
Diaemus youngi	27.45	81.33	10.24	60.31	41.44	21.72	357.41
Thyroptera tricolor	1.88	15.21	2.37	13.38	20.47	8.54	50.91

Table 52. Allometric size indices based on Pirlot's data

	BrW	NET	OBL	MES	CER	DIE	TEL
Penthetor lucasi (PTN)	268	278	171	197	266	426	298
Lonchorhina aurita (PHN)	160	167	166	188	317	204	125
Ardops annectans (STE)	292	300	192	291	356	367	306
Ametrida centurio (STE)	176	184	160	122	284	232	171
Diaemus youngi (DES)	287	298	193	278	401	367	295
Thyroptera tricolor (THY)	244	253	180	297	397	251	232
	MOB	PAL +AMY	SEP	STR	HIP	SCH	NEO
Penthetor lucasi	85	113	153	369	263	302	974
Lonchorhina aurita	11	44	75	195	103	131	444
Ardops annectans	68	130	253	471	209	213	1020
Ametrida centurio	32	58	92	202	120	172	638
Diaemus youngi	57	104	152	437	139	187	1154
Thyroptera tricolor	15	71	127	398	228	283	766

Abbreviations:

BoW	body weight	MOB	main olf. bulb	PTN	Pteropodinae
BrW	brain weight	NEO	neocortex	SCH	schizocortex
CER	cerebellum	NET	netto volumen	SEP	septum
DES	Desmodontidae	OBL	medulla oblongata	STE	Stenoderminae
DIE	diencephalon	PAL+AMY	paleocortex + amygdala	STR	striatum
HIP	hippocampus			TEL	telencephalon
MES	mesencephalon	PHN	Phyllostominae	THY	Thyropteridae

Table 53. Size characteristics of the various brain structures in the average Megachiroptera and Microchiroptera relative to the four Tenrecinae (AvTn = 100)

					Max					
		N	Min	Max	Min	AvCS	CV	NN	AvCF	CV
NET	Chirs	272	84	400	4.8	203	30.4	28	207	24.0
	Megs	47	230	400	1.7	289	11.3	3	281	4.6
	Mics	225	84	323	3.9	185	27.1	25	199	22.6
OBL	Chirs	272	86	190	2.2	141	13.0	28	142	10.2
	Megs	47	110	165	1.5	137	8.7	3	134	3.7
	Mics	225	86	190	2.2	142	13.6	25	143	10.5
TR	Chirs	150	52	160	2.9	90	20.5	28	93	18.5
	Megs	27	73	97	1.3	85	8.3	3	83	5.0
	Mics	123	52	160	2.9	92	21.9	25	94	18.9
TSO	Chirs	150	68	194	2.9	124	21.7	28	122	18.5
	Megs	27	113	194	1.7	161	12.2	3	152	11.2
	Mics	123	68	170	2.5	116	18.3	25	119	17.4
FGR	Chirs	150	46	215	4.7	106	30.4	28	112	25.3
	Megs	27	72	153	2.1	120	14.8	3	109	20.5
	Mics	123	46	215	4.7	103	32.9	25	112	26.1
FCM	Chirs	150	116	346	3.0	200	25.1	28	197	22.1
	Megs	27	187	346	1.9	253	14.7	3	244	17.2
	Mics	123	116	324	2.8	188	23.9	25	191	21.3
FCE	Chirs	150	64	185	2.9	118	22.9	28	113	19.4
	Megs	27	109	185	1.7	145	14.2	3	138	7.0
	Mics	123	64	185	2.9	111	21.7	25	110	19.1
REL	Chirs	150	80	331	4.1	165	24.3	28	165	19.9
	Megs	27	152	237	1.6	191	11.0	3	191	3.1
	Mics	123	80	331	4.1	160	25.9	25	162	20.7
INO	Chirs	150	141	475	3.4	285	23.0	28	297	17.8
	Megs	27	197	345	1.8	281	14.0	3	276	5.4
	Mics	123	141	475	3.4	286	24.6	25	300	18.5
PRP	Chirs	150	55	273	5.0	126	25.1	28	128	21.4
	Megs	27	91	196	2.2	134	15.0	3	136	1.5
	Mics	123	55	273	5.0	124	26.9	25	127	22.7
mot X	Chirs	150	64	229	3.6	122	23.2	28	120	20.0
	Megs	27	109	193	1.8	142	15.3	3	137	6.0
	Mics	123	64	229	3.6	118	23.7	25	118	20.8
mot XII	Chirs	150	57	249	4.4	134	28.7	28	136	24.1
	Megs	27	119	249	2.1	180	18.2	3	164	12.4
	Mics	123	57	219	3.8	124	25.7	25	133	24.6

		N	Min	Max	Max Min	AvCS	CV	NN	AvCF	CV
VC	Chirs	150	79	249	3.2	157	21.5	28	159	16.0
	Megs	27	152	249	1.6	187	14.0	3	190	12.1
	Mics	123	79	227	2.9	150	20.9	25	155	15.1
VM	Chirs	150	79	266	3.4	165	22.9	28	168	17.2
	Megs	27	168	266	1.6	203	13.5	3	207	10.5
	Mics	123	79	242	3.1	157	22.1	25	163	16.0
VI	Chirs	150	77	240	3.1	151	22.5	28	155	16.3
	Megs	27	130	240	1.8	170	17.4	3	168	13.3
	Mics	123	77	230	3.0	147	22.9	25	153	16.6
VL	Chirs	150	73	262	3.6	162	22.6	28	166	19.9
	Megs	27	127	262	2.1	186	18.2	3	193	22.2
	Mics	123	73	259	3.5	157	22.4	25	163	19.2
VS	Chirs	150	69	283	4.1	151	25.7	28	152	20.0
	Megs	27	154	283	1.8	189	15.6	3	195	6.5
	Mics	123	69	273	4.0	142	24.8	25	146	18.7
AUD	Chirs	150	86	480	5.6	234	34.3	28	246	26.1
	Megs	27	86	168	2.0	117	19.8	3	119	6.4
	Mics	123	145	480	3.3	260	24.6	25	261	18.7
CON	Chirs	150	93	371	4.0	186	28.8	28	192	20.5
	Megs	27	93	199	2.1	131	23.1	3	128	4.9
	Mics	123	106	371	3.5	197	25.4	25	199	17.1
DCO	Chirs	150	34	293	8.6	123	49.3	28	121	39.8
	Megs	27	107	293	2.7	177	30.3	3	170	6.5
	Mics	123	34	277	8.1	111	49.9	25	115	41.3
VCO	Chirs	150	79	471	6.0	232	35.1	28	243	26.8
	Megs	27	79	139	1.8	103	17.5	3	103	11.7
	Mics	123	147	471	3.2	260	22.8	25	260	17.2
OLS	Chirs	150	65	672	10.3	327	43.7	28	349	34.6
	Megs	27	65	138	2.1	96	19.5	3	106	15.9
	Mics	123	198	672	3.4	378	27.1	25	379	23.8
MES	Chirs	272	129	350	2.7	228	17.6	28	230	14.5
	Megs	47	185	303	1.6	235	11.6	3	230	4.6
	Mics	225	129	350	2.7	227	18.7	25	230	15.3
MTG	Chirs	150	104	325	3.1	209	23.7	28	216	16.9
	Megs	27	204	325	1.6	260	13.1	3	257	10.0
	Mics	123	104	312	3.0	198	23.0	25	211	16.5
NLL	Chirs	19	86	470	5.5	296	39.9	-	-	-.-
	Megs	3	86	103	1.2	93	9.5	-	-	-.-
	Mics	16	221	470	2.1	334	24.9	-	-	-.-

		N	Min	Max	Max Min	AvCS	CV	NN	AvCF	CV
MTC	Chirs	150	143	417	2.9	248	23.1	28	248	19.2
	Megs	27	143	260	1.8	203	16.2	3	204	2.0
	Mics	123	150	417	2.8	258	22.1	25	253	18.8
INC	Chirs	150	78	508	6.5	255	36.3	28	259	29.0
	Megs	27	78	162	2.1	115	20.5	3	119	5.1
	Mics	123	150	508	3.4	285	24.9	25	276	21.7
SUC	Chirs	150	120	432	3.6	236	31.1	28	228	25.5
	Megs	27	234	432	1.8	351	16.4	3	347	5.2
	Mics	123	120	356	3.0	210	22.7	25	214	19.9
CER	Chirs	272	113	576	5.1	311	25.8	28	325	19.9
	Megs	47	259	411	1.6	320	10.8	3	322	1.2
	Mics	225	113	576	5.1	309	28.0	25	325	21.1
MCN	Chirs	18	159	419	2.6	244	23.8	-	-	-.-
	Megs	3	241	274	1.1	260	6.4	-	-	-.-
	Mics	15	159	419	2.6	241	26.2	-	-	-.-
ICN	Chirs	18	112	268	2.4	172	25.3	-	-	-.-
	Megs	3	141	188	1.3	161	14.9	-	-	-.-
	Mics	15	112	268	2.4	174	26.8	-	-	-.-
LCN	Chirs	18	158	427	2.7	322	22.1	-	-	-.-
	Megs	3	232	275	1.2	253	8.6	-	-	-.-
	Mics	15	158	427	2.7	335	20.7	-	-	-.-
DIE	Chirs	272	108	547	5.1	243	32.7	28	245	25.9
	Megs	47	310	547	1.8	374	13.6	3	358	7.3
	Mics	225	108	365	3.4	216	24.3	25	232	22.4
CGM	Chirs	19	230	949	4.1	523	35.1	-	-	-.-
	Megs	3	230	284	1.2	252	11.2	-	-	-.-
	Mics	16	388	949	2.4	573	26.4	-	-	-.-
CGL	Chirs	19	84	1011	12.0	301	83.7	-	-	-.-
	Megs	3	663	1011	1.5	826	21.2	-	-	-.-
	Mics	16	84	342	4.1	202	40.0	-	-	-.-
GLD	Chirs	19	87	1094	12.6	331	85.3	-	-	-.-
	Megs	3	716	1094	1.5	917	20.7	-	-	-.-
	Mics	16	87	392	4.5	221	43.8	-	-	-.-
GLV	Chirs	19	30	836	27.9	237	82.4	-	-	-.-
	Megs	3	512	836	1.6	632	28.1	-	-	-.-
	Mics	16	30	281	9.4	163	40.6	-	-	-.-

		N	Min	Max	Max/Min	AvCS	CV	NN	AvCF	CV
TEL	Chirs	272	64	449	7.0	190	42.6	28	194	34.9
	Megs	47	226	449	2.0	318	12.7	3	308	4.9
	Mics	225	64	340	5.3	163	36.1	25	180	31.9
MOB	Chirs	272	5	157	31.4	48	75.8	28	46	76.8
	Megs	47	69	155	2.2	108	19.2	3	113	15.3
	Mics	225	5	157	31.4	35	68.5	25	38	71.6
AOB	Chirs	63	26	683	26.3	218	55.8	8	211	37.8
	Megs	0	-	-	-.-	-	-.-	-	-	-.-
	Mics	63	26	683	26.3	218	55.8	8	211	37.8
PAL	Chirs	272	16	149	9.3	53	53.4	28	53	53.7
	Megs	47	69	124	1.8	97	14.1	3	98	6.7
	Mics	225	16	149	9.3	44	48.5	25	48	52.3
STR	Chirs	272	90	680	7.6	272	38.7	28	286	31.9
	Megs	47	280	680	2.4	424	21.2	3	398	14.6
	Mics	225	90	526	5.8	240	31.9	25	272	31.3
SEP	Chirs	272	58	298	5.1	135	40.2	28	132	36.2
	Megs	47	174	298	1.7	230	10.5	3	222	7.9
	Mics	225	58	229	3.9	115	29.6	25	122	31.0
AMY	Chirs	272	96	269	2.8	166	17.0	28	169	12.1
	Megs	47	107	206	1.9	165	12.5	3	168	3.0
	Mics	225	96	269	2.8	166	17.9	25	169	12.8
HIP	Chirs	272	47	339	7.2	157	44.6	28	159	39.3
	Megs	47	217	339	1.6	267	11.6	3	266	0.9
	Mics	225	47	288	6.1	134	38.8	25	147	36.3
SCH	Chirs	272	73	477	6.5	211	42.4	28	209	34.7
	Megs	47	238	477	2.0	341	14.5	3	331	9.1
	Mics	225	73	414	5.7	184	38.1	25	194	31.4
NEO	Chirs	272	176	1618	9.2	597	47.9	28	620	36.5
	Megs	47	723	1618	2.2	1016	17.6	3	963	10.0
	Mics	225	176	1121	6.4	509	42.9	25	579	34.7

Number of species = N
Number of subfamilies and/or families = NN
Abbreviations and notes:

- AvCF average of families and subfamilies
- AvCS average of species
- CV coefficient of variation (standard deviation of the mean in percent of the mean)
- Max maximum index in a group of species
- Min minimum index in a group of species
- Max/Min quotient of maximum index through minimum index